349-02789K
Feb. 2023

SHIMADZU AUTOGRAPH SOFTWARE

TRAPEZIUM X

Instruction Manual

Data Processing Reference

Read the instruction manual thoroughly before you use the product.
Keep this instruction manual for future reference.

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Before Using the Autograph System

Introduction

Read this instruction manual prior to operating the product.

Thank you for purchasing the TRAPEZIUMX.

This Guide describes the TRAPEZIUMX operating procedures, mainly showing flow of operations of this software.

If you want to execute any operation, find the corresponding item from the table of contents, and refer to the procedure described for the item.

Please read the instruction manual carefully before using the testing machine.

Keep it stored safely in an accessible location for easy reference.

NOTE

- Be sure to understand the instructions in this manual before operating the system.
  - Attach this instruction manual before lending or transferring it to another party.
  - If this instruction manual or product warning labels are lost or destroyed, promptly contact your Shimadzu representative.
  - This instruction manual lists safety precautions to ensure the system is operated safely. Be sure to read them before using the system.

Notice

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  - Information in this manual is subject to change without notice.
  - The utmost care has been taken in preparing this document, however, if any errors or omissions are detected, it may not be possible to correct them immediately.

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Documentation Conventions

In this instruction manual, the following safety-related signal words are used to indicate the degree of hazard or possible damage.

Symbol	Meaning
WARNING	Indicates a potentially hazardous situation which, if not avoided, could result in serious injury or possibly death.
CAUTION	Indicates a potentially hazardous situation which, if not avoided, may result in minor to moderate injury or equipment damage.
[ ]	Text shown in the GUI (menus, buttons, dialog boxes, and window names) Example: Select [File] - [Save as] Click [OK] the [TRAPEZIUMX Home] window
" "	Text shown temporarily in a window (such as error messages), and text entered by the user

Safety Guidelines

Autograph systems generate very large forces used to measure the strength characteristics of materials or products.

In some situations it could cause material damage or serious harm to humans (such as injury or death).

Therefore, to ensure the system is used safely, please thoroughly understand and observe the following warnings.

WARNING

Operating the system carelessly could result in serious injury or death. Keep hands, head and all other body parts away from the testing space when the crosshead is in motion.

WARNING

Moving parts can produce dangerous forces corresponding to the system capacity. All operators should read the instruction manual thoroughly to ensure they operate the system correctly. Control access to the system to ensure only trained operators operate the system.

WARNING

Due to risk of electric shock, never touch inside the controller unit, located in the testing machine.

Safety Guidelines

WARNING

Do not open the ball screw shaft protective cover unless lubricating the screw shaft. Do not leave the shaft cover open while performing tests. Keep hair, clothing, rags, and fingers from getting caught in the ball screw when lubricating the ball screw.

WARNING

Keep the face and body away from the test specimen during tests. When the specimen breaks, flying debris could injure eyes or other body parts. Always use a safety guard or other measure to provide protection from broken specimens.

WARNING

Due to risk of electric shock, injury, or fire, wiring, maintenance, and inspection work should be perform only by a person with specialized expertise.

WARNING

To prevent electric shock, always ground power cables (100 Ω Max.).

WARNING

Do not operate the system in a manner that subjects the crosshead to loads when the load cell is removed and do not operate the system in a manner that does not transmit the load to the load cell. Doing so would cause overloading to be undetected, which could result in damage to the frame or jigs.

Safety Guidelines

WARNING

Never operate the crosshead without crosshead limit switches in place. Doing so could cause the crosshead to hit the jig and damage the frame, jig, or load cell.

WARNING

Always use Shimadzu brand testing jigs on Autograph testing machines.

CAUTION

This system includes a built-in safety feature that stops jog or return movements if the load level applied to the load cell varies by more than a given amount. This feature is one of the safety circuits, but for fast movements it cannot eliminate the risk of overshooting. Furthermore, for safety and operational convenience reasons, this feature will not stop motion in the unloading direction. Therefore, this feature provides no guarantee that impacts and overloading will be prevented in the testing space. Also, do not use this feature for positioning or other control purposes.

CAUTION

For long tests, note that continuous operation is limited to 10 hours for Autograph systems.

Note

- During tests such as the compression test, load cell breakage may occur due to overshooting when the specimen or compression plates collide during high-speed movement.
- Open the ball screw protection cover only after turning off the power. Before turning on the power, always make sure to close the ball screw protection cover.
- Make sure you do not allow any water to enter inside the unit as it may cause machine failure.
- While a PC is connected, you cannot clear any errors from the operation panel. Press the error clear button displayed on the PC screen.
- Immediately stop use when you detect any unusual smell or sound coming from the testing machine. After pushing the emergency switch, disconnect the power cable.
- Running the initialization process resets the crosshead position and initializes all methods and system configurations. Always reconfigure the settings after initializing. Method files are also completely deleted.
- If the crosshead moves unexpectedly due to an error in testing setup, push the emergency switch on the main unit. To stop the testing machine in normal state while a test is running, use the stop key.
- The crosshead moves at high speed when returning to origin. Do not put your hand inside the moving parts. Ensure that the jigs do not hit any surrounding objects. If the testing machine acts in an abnormal manner, push the emergency switch immediately.
- There is a risk that the contents of the personal computer hard disk may be lost due to an unexpected accident. Be sure to create a backup to protect important data from unexpected accidents
- If the test result file size exceeds the allowable value, the measurement data may be lost. For more information on the allowable value, see "349-02787 TRAPEZIUMX User Guide" "Volume of Data Which Can Be Saved and Managed".
- Due to the risk of error, do not save method files or test files in external folders such as those on network drives or USB flash drive.
- Starting antivirus software or any software other than TRAPEZIUMX while a test is ongoing may compromise correct operation of TRAPEZIUMX. While a test is ongoing, do not start other software.
- Calculation results produced by software may contain floating-point errors.

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b. This Agreement shall be construed and governed in accordance with the laws of Japan, excluding its conflict of law rules.
c. The exclusive jurisdiction for any disputes arising out of or in connection with this Agreement shall be Kyoto District Court of Japan.
d. The invalidity or unenforceability of any provision of this Agreement shall not affect the validity or enforceability of any other provision.

Warranty

Shimadzu provides the following warranty for this product.

1. Period:

Please contact your Shimadzu representative for information about the period of this warranty.

2. Description:

If a product/part failure occurs for reasons attributable to Shimadzu during the warranty period, Shimadzu will repair or replace the product/part free of charge. However, in the case of products which are usually available on the market only for a short time, such as personal computers and their peripherals/parts, Shimadzu may not be able to provide identical replacement products.

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2. In no event will Shimadzu's liability to you, whether in contract, tort (including negligence), or otherwise, exceed the amount you paid for the product.

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Failures caused by the following are excluded from the warranty, even if they occur during the warranty period.

1. Improper product handling
2. Repairs or modifications performed by parties other than Shimadzu or Shimadzu designated companies
3. Product use in combination with hardware or software other than that designated by Shimadzu
4. Computer viruses leading to device failures and damage to data and software, including the product's basic software
5. Power failures, including power outages and sudden voltage drops, leading to device failures and damage to data and software, including the product's basic software
6. Turning OFF the product without following the proper shutdown procedure leading to device failures and damage to data and software, including the product's basic software
7. Reasons unrelated to the product itself
8. Product use in harsh environments, such as those subject to high temperatures or humidity levels, corrosive gases, or strong vibrations
9. Fires, earthquakes, or any other act of nature, contamination by radioactive or hazardous substances, or any other force majeure event, including wars, riots, and crimes
10. Product movement or transportation after installation

11. Consumables and equivalent items
    Recording media such as CD-ROMs are considered consumable items.
    * If there is a document such as a warranty provided with the product, or there is a separate contract agreed upon that includes warranty conditions, the provisions of those documents shall apply.

After-Sales Service and Availability of Replacement Parts

After-Sales Service

If any problem occurs with this product, perform an inspection and take appropriate corrective action as described in Instruction Manual.

If the problem persists, or the symptoms are not covered in the troubleshooting section, contact your Shimadzu representative.

Replacement Parts Availability

Replacement parts for this product will be available for a period of seven (7) years after the product is discontinued. Thereafter, such parts may cease to be available.

Note, however, that the availability of parts not manufactured by Shimadzu shall be determined by the relevant manufacturers.

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This Manual

This is a reference manual that describes various dialog boxes and data processing functions in the TRAPEZIUMX software (used for Autograph universal testing machines). To those using universal testing machines, note that this manual refers to pressure cells as load cells. In addition, actual windows may differ from some windows shown in this manual, depending on the testing machine used with the software.

Configuration

This reference manual is comprised of the following chapters.

Chapter 1

Data Processing Items in the Single or Control Mode

Chapter 2

Data Processing Items in the Single Mode Peel Test

Chapter 3

Data Processing Items in the Cycle Mode

Chapter 4

Data Processing Items in the Texture Mode

Chapter 5

Statistical Processing Items

Chapter 6

Calculating Stress and Strain

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Terminology

Document Notation

In this document, various items are denoted as follows.

Notation	Explanation	Descriptive Example
[ ]	Text shown in the GUI (menus, buttons, dialog boxes, and window names)	Select [File] - [Save as] Click [OK] the [TRAPEZIUMX Home] window
" "	Text shown temporarily in a window (such as error messages), and text entered by the user	

Related Documents

TRAPEZIUMX User's Guide:

This explains the procedural stages for TRAPEZIUMX. (349-02787)

TRAPEZIUMX Software Reference Manual:

This provides features and functions in the main windows and various dialog boxes.

(349-02788)

TRAPEZIUMX Hardware Selfcheck and Settings Guide:

This provides an explanation of AUTOGRAPH AG-X series selfcheck functions, and software configuration methods for AG-Xplus, AG-X,AGS-X, EZ-X and AG-IS, and EZGraph. (349-02790)

This manual describes the data processing calculations and dialog box settings used by TRAPEZIUMX to process data.

List of Values Determined by Data Processing

(1) Data Processing Items in the Single Mode

- r-Value
- Ag
- n-Value
- new-n-Value
- Energy1 to 5
- Elastic Loop
- Flexural Compliance
- Upper Yield Point (%FS)
- Upper Yield Point (Disp.)
- Upper Yield Point (Points)
- Upper Yield Point (%YP)
- Max.
- Max Disp.
- Min.
- Reduction
- Lower Yield Point
- YS1 to 2
- Elastic(Standard)
- Elastic(Chord)
- Elastic(Secant)
- Elastic(Tangent)
- Elastic(Max. Slope)
- Elastic(Devided)
- LASE1 to 10
- EASL1 to 10
- Preset1 to 10
- Fitted
- Break
- Nonlinear Elongation
- Poisson’s
- PointPicking1 to 20
- Hold Preset
- Hold Difference
- Total Hold Difference

(2) Data Processing Items in the Single Mode Peel Test.

- Node
- Node Mean
- Force Mean
- Static Friction
- All Mean
- Dynamic Friction
- PeakMax1 to 9
- PeakMax Mean
- PeakMin1 to 9
- PeakMin Mean
- Peak Median
- Peak Mean
- Peak Mean All
- First
- ValleyMax1 to 9
- ValleyMax Mean
- ValleyMin1 to 9
- ValleyMin Mean
- Valley Mean All
- Valley Mean

(3) Data Processing Items in the Cycle Mode

- Energy_I
- Energy_D
- Hysteresis
- Hysteresis Loss
- Max.
- Min.
- Elastic_I
- Elastic_D
- LASE_I1 to 10
- LASE_D1 to 10
- LASE_Ave.1 to 10
- EASL_I1 to 10
- EASL_D1 to 10
- EASL_Ave1 to 10
- Preset_I1 to 10
- Preset_D1 to 10
- Preset_Ave1 to 10
- PointPicking1 to 20

(4) Data Processing Items in the Control Mode

- Energy1 to 5
- Upper Yield Point
- Max.
- Max Disp.
- Min.
- Lower Yield Point
- Yield Strength1 to 2
- Elastic(Standard)
- Elastic(Chord)
- Elastic(Secant)
- Elastic(Tangent)
- Elastic(Max. Slope)
- Elastic(Devided)
- Preset1 to 10
- Break
- PointPicking1 to 20
- Hold Preset
- Hold Difference
- Total Hold Difference

(5) Data Processing Items in the Texture Mode

- Energy1 to 5
- Hardness
- Gumminess
- Cohesiveness
- Max Disp.
- Force_Mean
- Chewiness
- Elastic(Standard)
- Elastic(Chord)
- Elastic(Secant)
- Elastic(Standard)
- Elastic(Tangent)
- Elastic(Max. Slope)
- Elastic(Devided)
- Springness
- Preset1 to 10
- No. of convex points
- PeakMax1 to 10
- Break
- Adhesiveness
- Adhesive
- PointPicking1 to 20
- ValleyMin1 to 10
- Brittleness
- Resilience

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Chapter 1
Data Processing Items in the Single or Control Mode

This chapter describes processing data in the single and control modes. The procedure for processing tensile, compression, and bending data is the same for both modes.

Different data processing settings are enabled in different modes. The following symbols are used to indicate which modes will allow specifying the given process. (These symbols are listed to the right of the process heading.)

Single: Can be specified only in the single test mode.

Control: Can be specified only in the control test mode.

Tension: Can be specified only in the tensile test mode.

Compression: Can be specified only in the compression test mode.

Bending: Can be specified only in the bending test mode.

Metal: Can be specified only if the specimen material is metal.

Mark: Allows displaying a mark on the graph.

Example 1:

Single

Indicates process can be specified only in the single mode (for any test type or material)

Example 2:

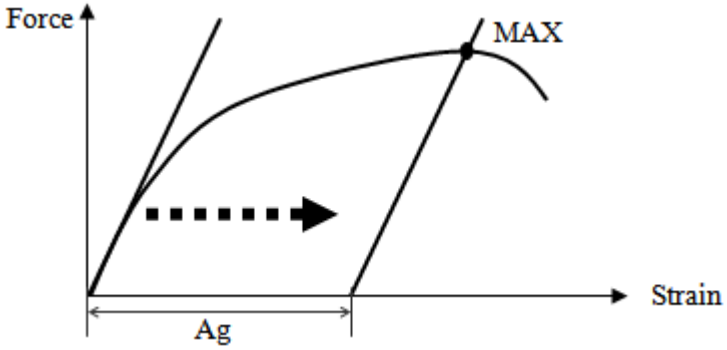
Single **Tension** **Metal**

Indicates process can be specified only for tensile testing metal specimens in the single mode.

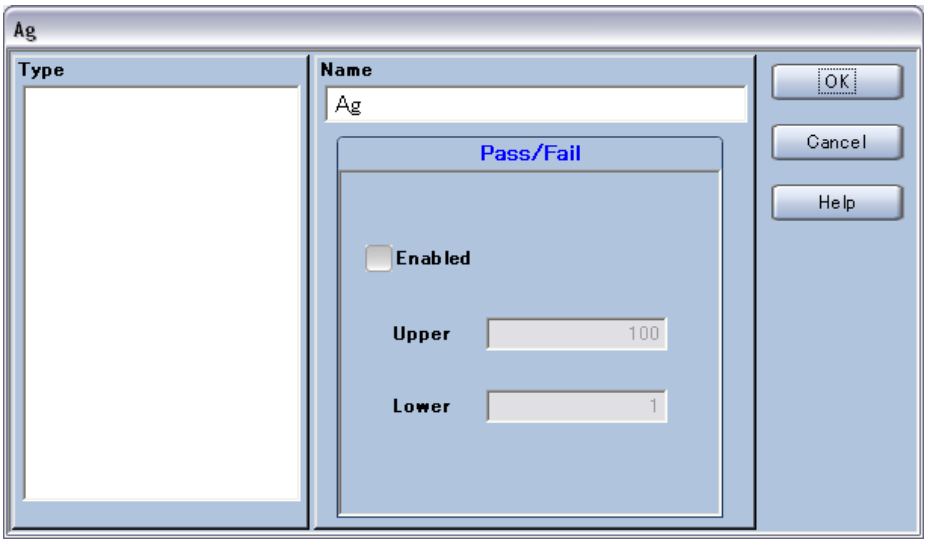
Ag

Single **Tension** **Metal**

This process determines the Ag value, which is the distance to the point on the X-axis (strain) where a straight line extending through the point of maximum [Force] and parallel to the elastic, intersects the X-axis. This requires specifying that maximum force and elastic be determined beforehand.



Parameters



- None

Supplemental Information

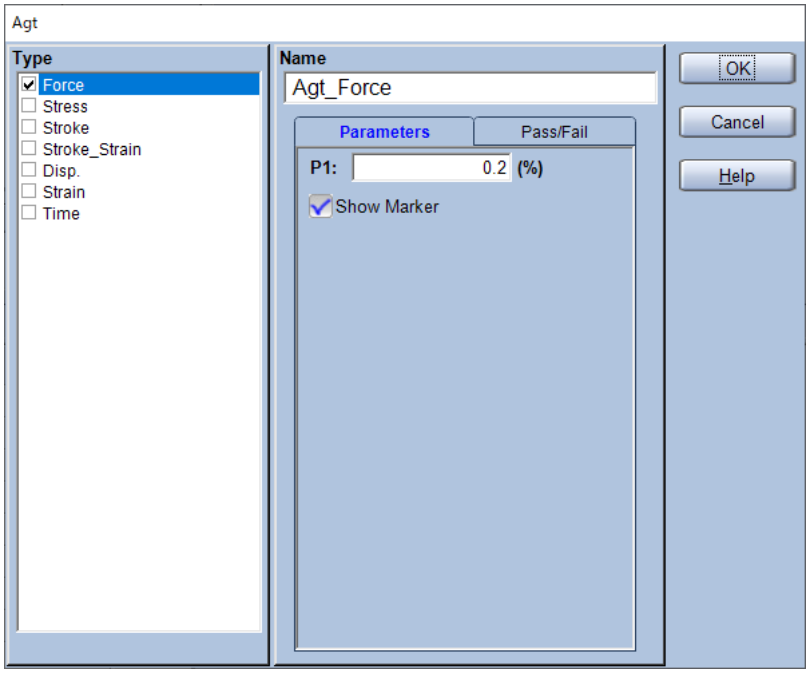
This value can be edited only if [Metal, etc.] is selected in the [Material] list box in the [Specimen] dialog box.

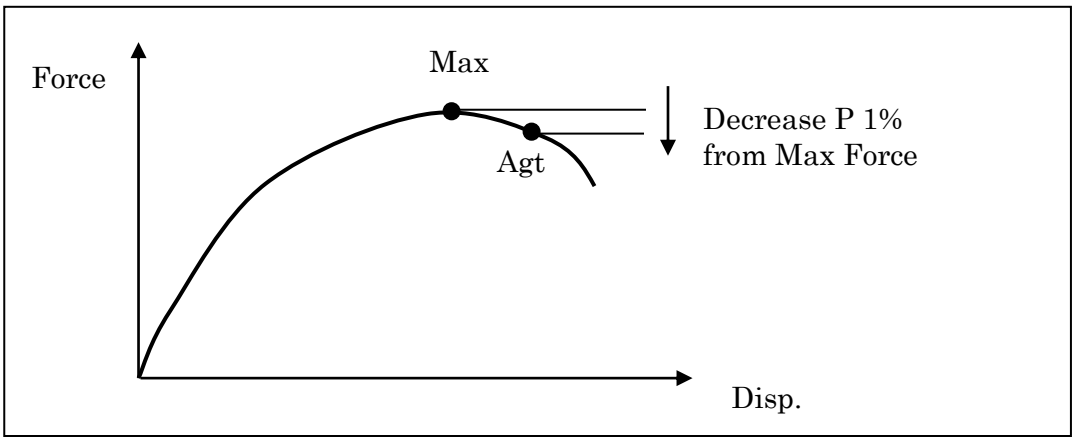
Agt

Single **Control** **Mark**

Determine the Agt value for the point after the Max point and down by the specified percentage from the Max point. The Max point is the Max point of the data processing result when the data processing item "Max" is registered. If not registered, the test point is the Max point in the raw data.

Parameters





-Percent of Max (P1) (Initial value 0.2%)

Type

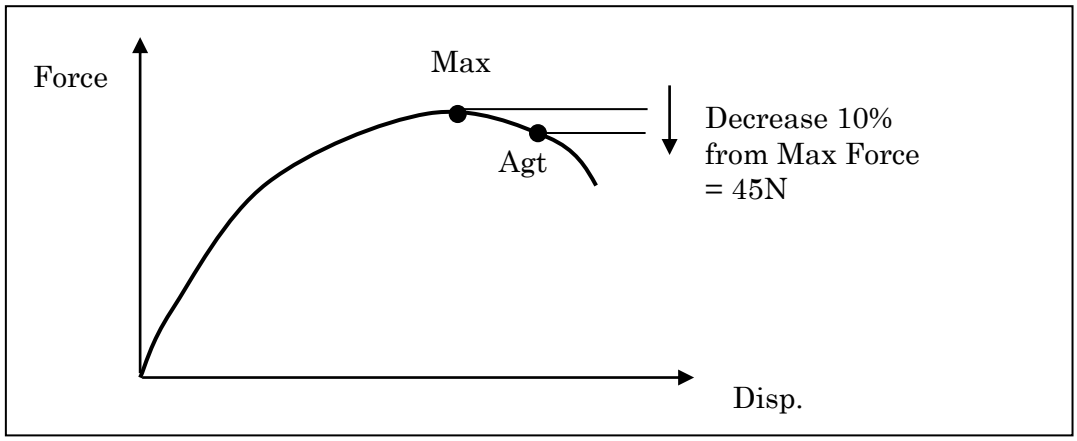
Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Example

The detection value P1 is 10% and the Max is 50 N.

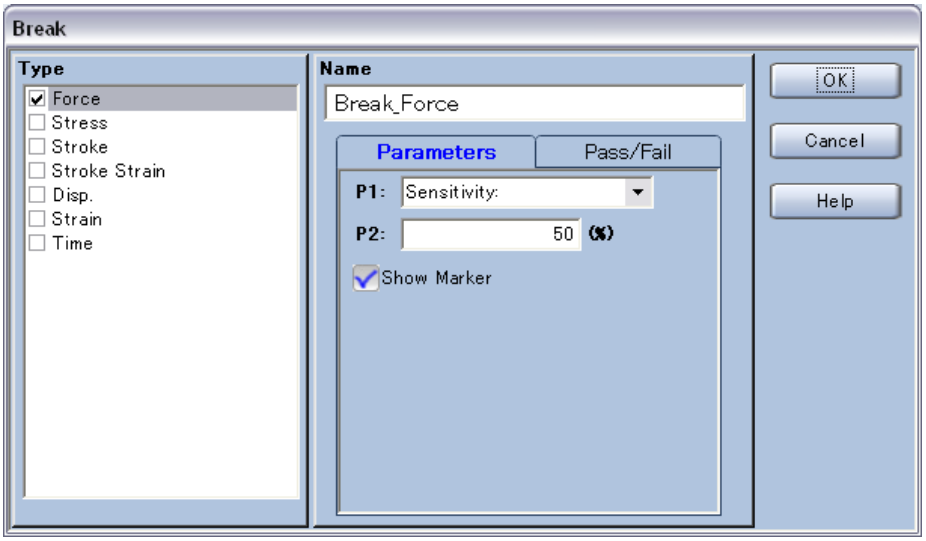


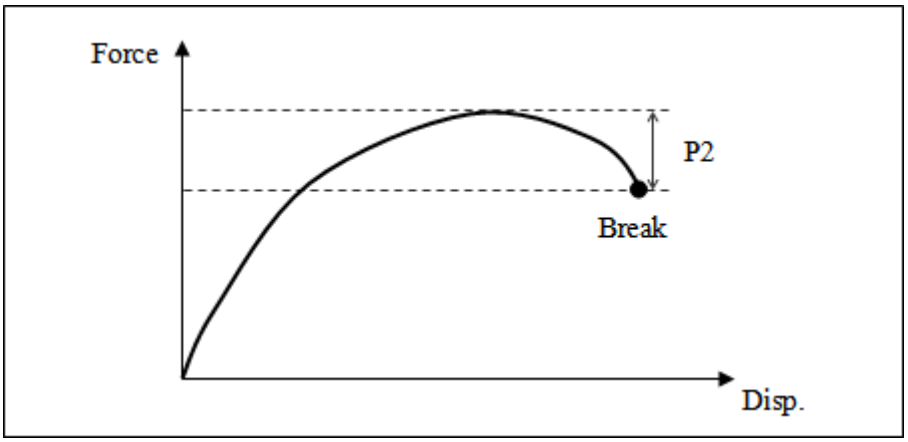
Break

Single **Control** **Mark**

Determines the break point as the point among the most recent sampling data that meets the specified parameter.

Parameters





- Detection Parameter (P1)

Sensitivity	Level
% Level of Max.	Last 100 msec

- Detection Value (P2)

Type

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

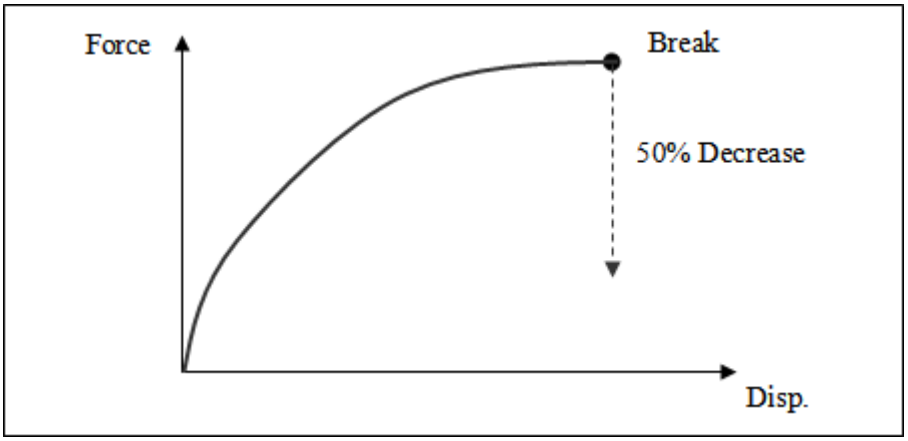
Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Example

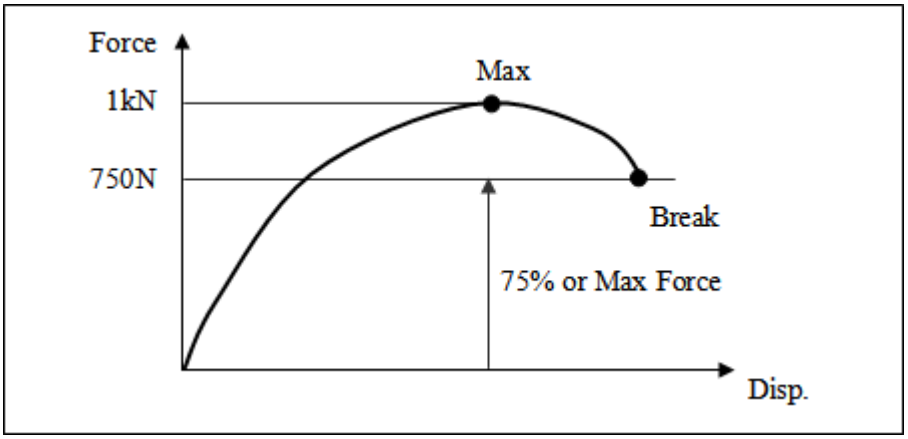
Assume the detection parameter P1 is [Sensitivity], the detection value P2 is 50%, and the load cell capacity (full scale) is 5 kN.

The [Break] is determined as the point which the decrease of force exceeds 2.5kN (= 5kN × 50%) in 1 second.



Assume the detection parameter P1 is [% Level of Max.], the detection value P2 is 25%, and the [Max] is 1 kN.

The [Break] is determined as the point when [Force] drops (by 25%) to 75% of [Max].



Elastic(Standard)

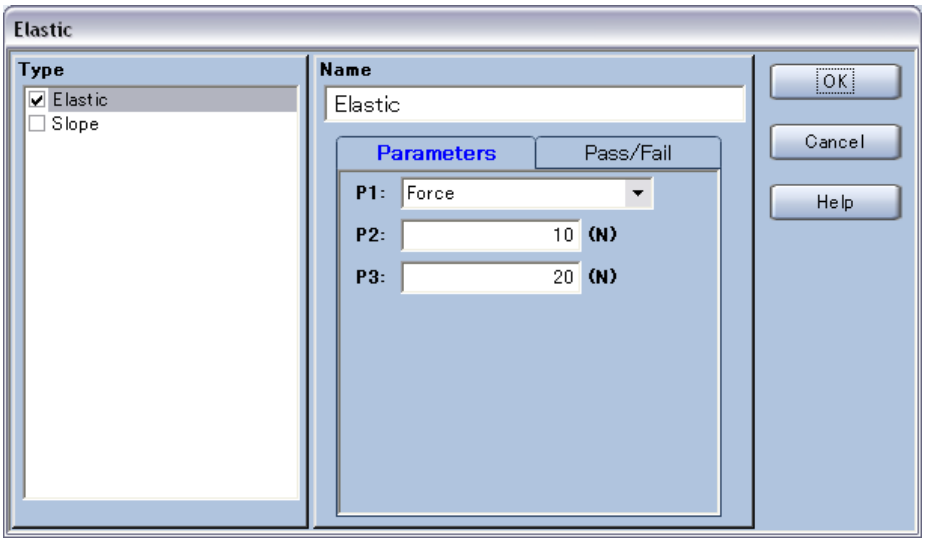
Single **Control**

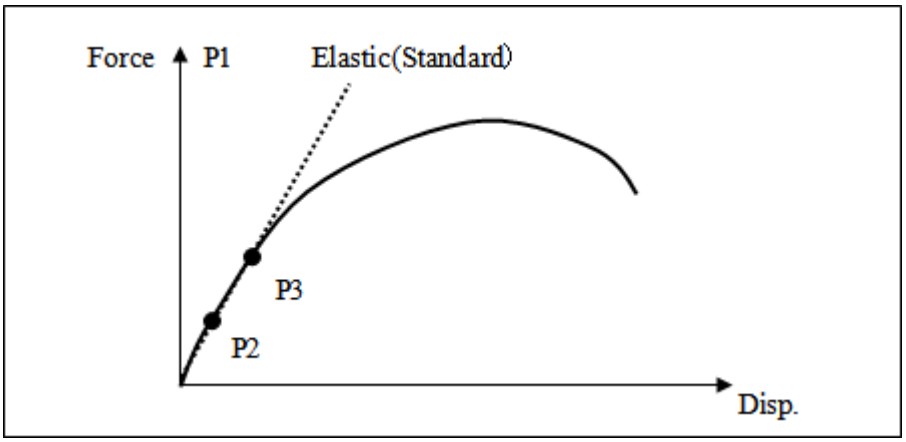
Determines elastic using the least squares method from all data between two pre-specified points.

$$\text{Elastic} = \text{Slope} \times \frac{\text{Gauge Length}}{\text{Cross-Sectional Area of Specimen}}$$

Slope	Calculated from all data between two points using least squares method
Gauge Length	Value specified via the [Sensor] or [Specimen] dialog box.
Cross-Sectional Area of Specimen	Calculated from dimensions specified in the [Specimen] dialog box.

Parameters





- Sensor Type (P1)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Starting Point (P2)

- Ending Point (P3)

* Parameter values are in terms of units specified in the [System] window.

Type

- Elastic

Calculated from [Slope], [Gauge Length], and [Specimen Cross-Sectional Area].

- Slope

Calculates only slope of elastic.

Example

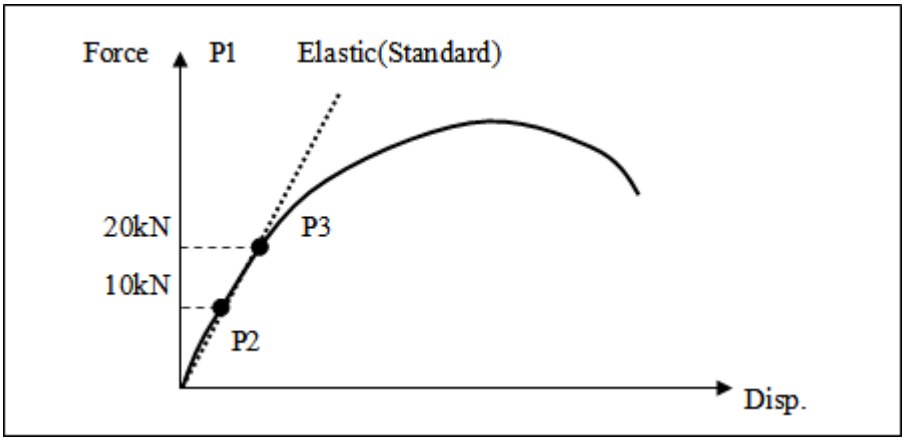
Assume P1 is set to [Force], P2 to 10 kN, P3 to 20 kN, gauge length to 50 mm, and the specimen is a flat plate 2 mm thick and 5 mm wide.

$$\text{Elastic} = \text{Slope} \times \frac{50\text{mm}}{2\text{mm} \times 5\text{mm}}$$

Note

Slope is calculated using least squares method.

The formula used to calculate cross-sectional area depends on specimen shape. For more details, refer to Methods for "Calculating Stress and Strain".



Related keyword: "Calculating Stress and Strain"

Elastic(Chord)

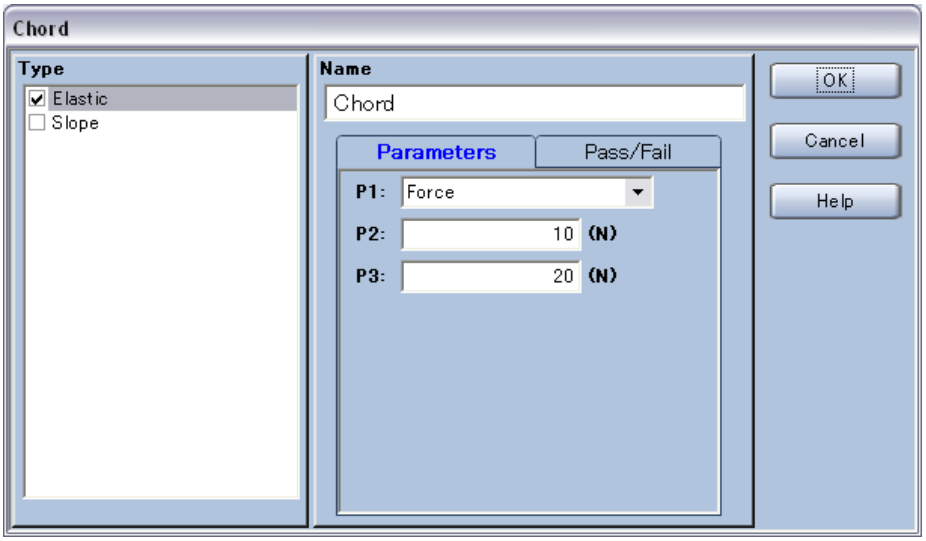
Single **Control**

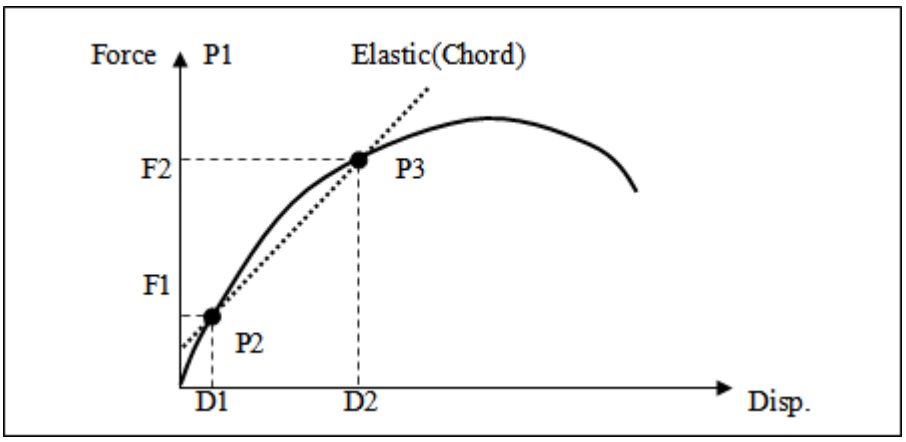
Determines elastic from the slope of a straight line between two pre-specified force or [Disp.] points.

$$\text{Elastic} = \text{Slope} \times \frac{\text{Gauge Length}}{\text{Cross - Sectional Area of Specimen}}$$

Slope	Calculated from difference between two points.
Gauge Length	Value specified via the [Sensor] or [Specimen] dialog box.
Cross-Sectional Area of Specimen	Calculated from dimensions specified in the [Specimen] dialog box.

Parameters





- Sensor Type (P1)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Starting Point (P2)

- Ending Point (P3)

* Parameter values are in terms of units specified in the [System] window.

Type

- Elastic

Calculated from [Slope], [Gauge Length], and [Specimen Cross-Sectional Area].

- Slope

Calculates only slope of elastic.

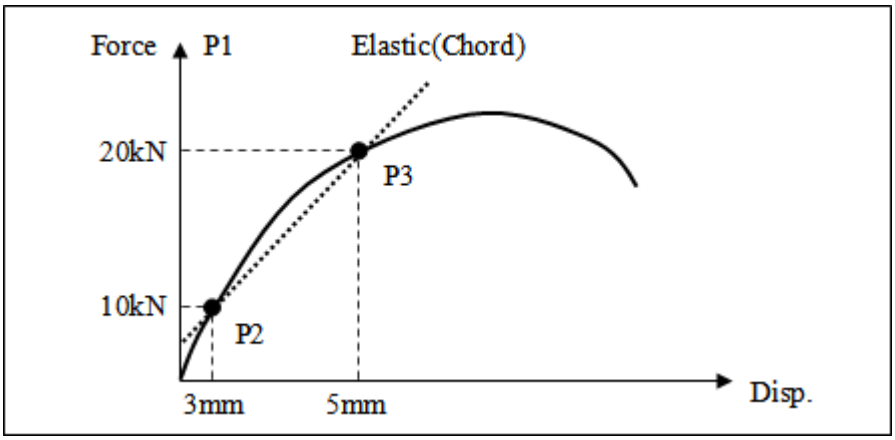
Example

Assume [Force] F1 is set to 10 kN, F2 to 20 kN, Disp. D1 to 3 mm, D2 to 5 mm, and gauge length to 50 mm, for a plastic flat plate specimen 2 mm thick and 5 mm wide.

$$\text{Elastic} = \frac{20\text{kN} - 10\text{kN}}{5\text{mm} - 3\text{mm}} \times \frac{50\text{mm}}{2\text{mm} \times 5\text{mm}}$$

Note

The formula used to calculate cross-sectional area depends on specimen shape. For more details, refer to Methods for Calculating Cross-Sectional Area.



Related keyword: “Calculating Stress and Strain”

Elastic(Secant)

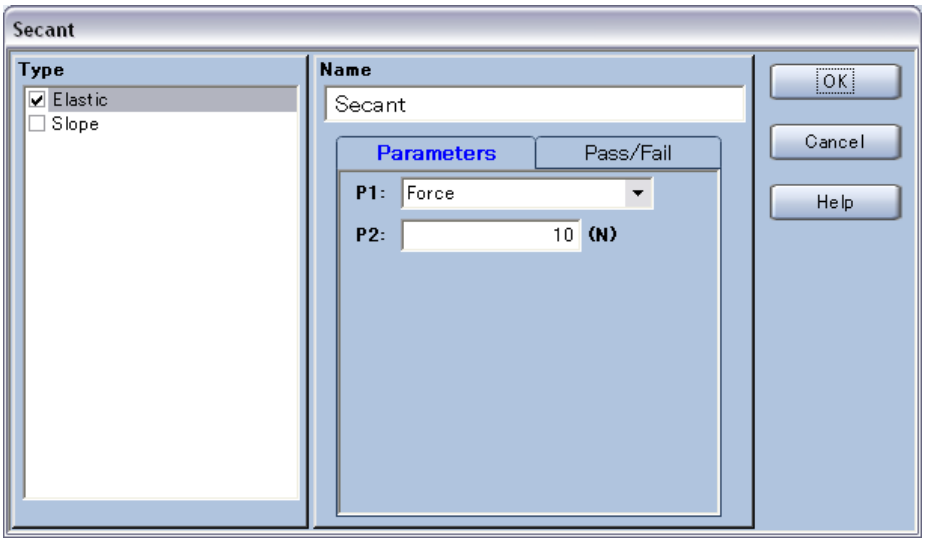
Single **Control**

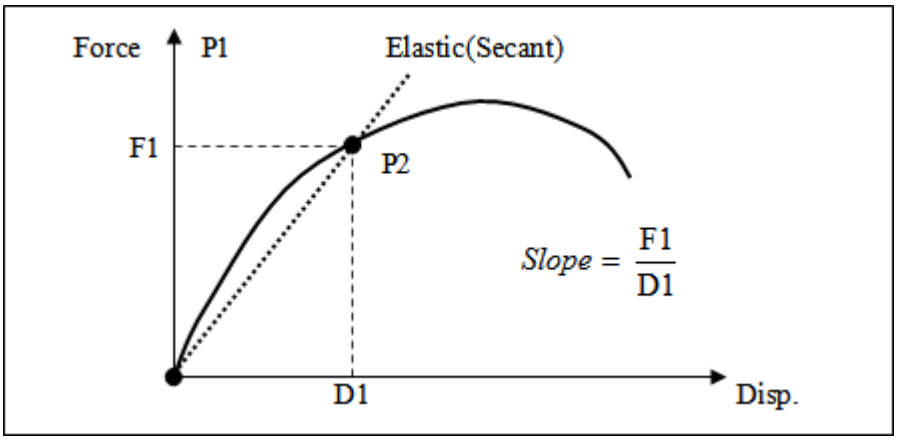
Determines elastic from the test starting point and another pre-specified point.

$$\text{Elastic} = \text{Slope} \times \frac{\text{Gauge Length}}{\text{Cross-Sectional Area of Specimen}}$$

Slope	Calculated from difference between starting point of test and another specified point.
Gauge Length	Value specified via the [Sensor] or [Specimen] dialog box.
Cross-Sectional Area of Specimen	Calculated from dimensions specified in the [Specimen] dialog box.

Parameters





- Sensor Type (P1)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Value for Sensor Type (P2)

* Parameter values are in terms of units specified in the [System] window.

Type

- Elastic

Calculated from [Slope], [Gauge Length], and [Specimen Cross-Sectional Area].

- Slope

Calculates only slope of elastic.

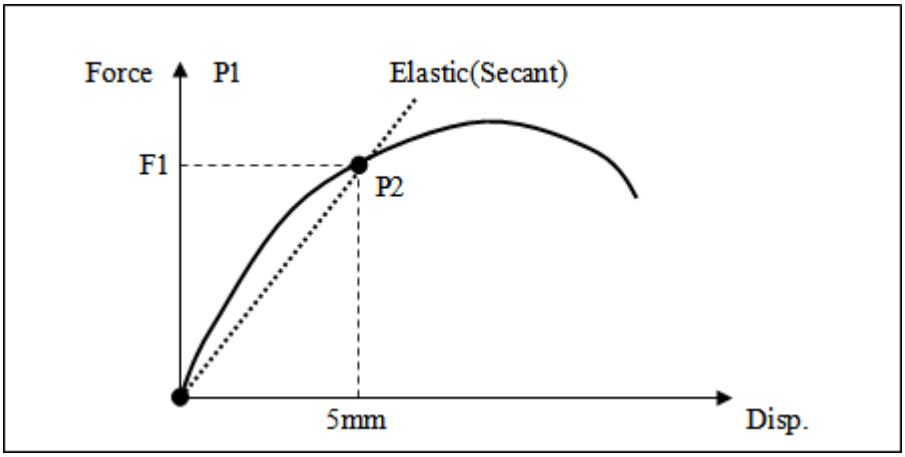
Example

Assume [Force] F1 is set to 20 N, Disp. D1 to 5 mm, and gauge length to 50 mm, for a plastic flat plate specimen 2 mm thick and 5 mm wide.

$$\text{Elastic} = \frac{20\text{kN}}{5\text{mm}} \times \frac{50\text{mm}}{2\text{mm} \times 5\text{mm}}$$

Note

The formula used to calculate cross-sectional area depends on specimen shape. For more details, refer to Methods for Calculating Cross-Sectional Area.



Related keyword: “Calculating Stress and Strain”

Elastic(Tangent)

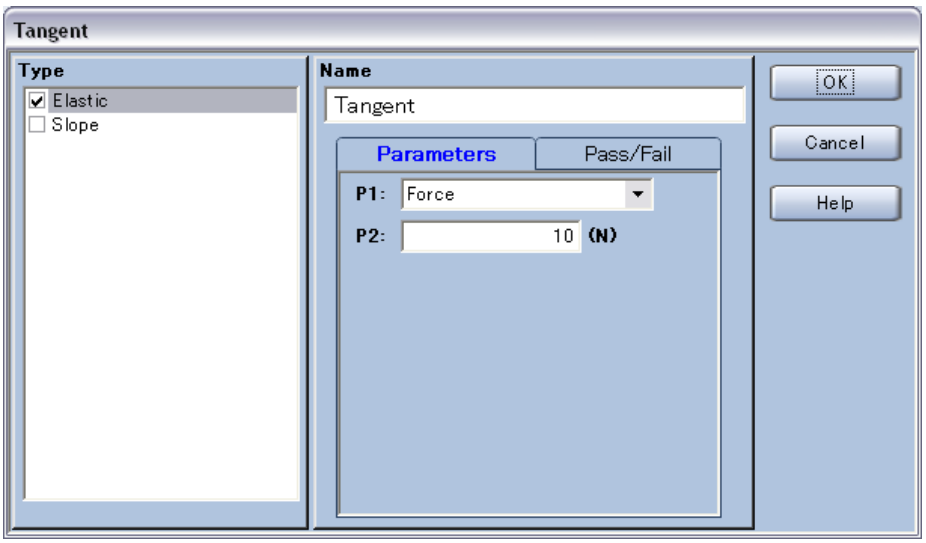
Single **Control**

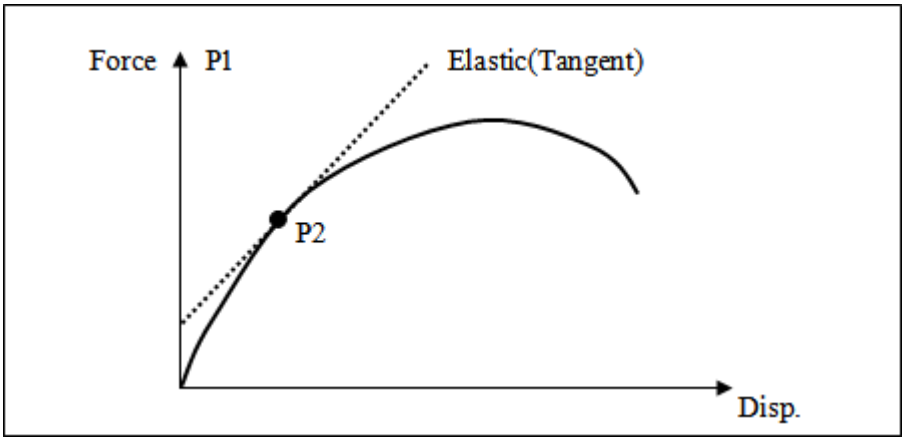
Determines elastic from the slope of a straight line between a point just before and after a pre-specified point.

$$\text{Elastic} = \text{Slope} \times \frac{\text{Gauge Length}}{\text{Cross-Sectional Area of Specimen}}$$

Slope	Slope of straight line through a point on either side of a specified point.
Gauge Length	Value specified via the [Sensor] or [Specimen] dialog box.
Cross-Sectional Area of Specimen	Calculated from dimensions specified in the [Specimen] dialog box.

Parameters





- Sensor Type (P1)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Value for Sensor Type (P2)

* Parameter values are in terms of units specified in the [System] window.

Parameters

- Elastic

Calculated from [Slope], [Gauge Length], and [Specimen Cross-Sectional Area].

- Slope

Calculates only slope of elastic.

Example

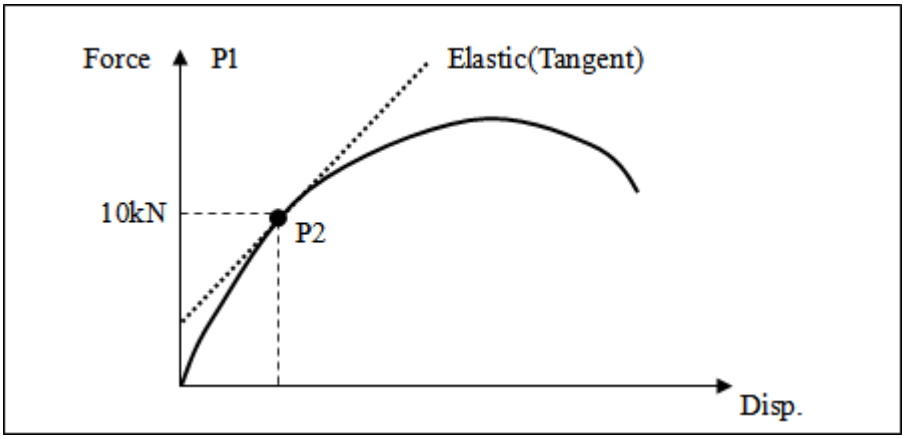
Assume P1 is set to [Force], P2 to 10 N, gauge length to 50 mm, and the specimen is a flat plastic plate 2 mm thick and 5 mm wide.

$$\text{Elastic} = \text{Slope} \times \frac{50\text{mm}}{2\text{mm} \times 5\text{mm}}$$

Note

Slope is slope of straight line through a point on either side of a specified point.

The formula used to calculate cross-sectional area depends on specimen shape. For more details, refer to Methods for Calculating Cross-Sectional Area.



Related keyword: “Calculating Stress and Strain”

Elastic(Max. Slope)

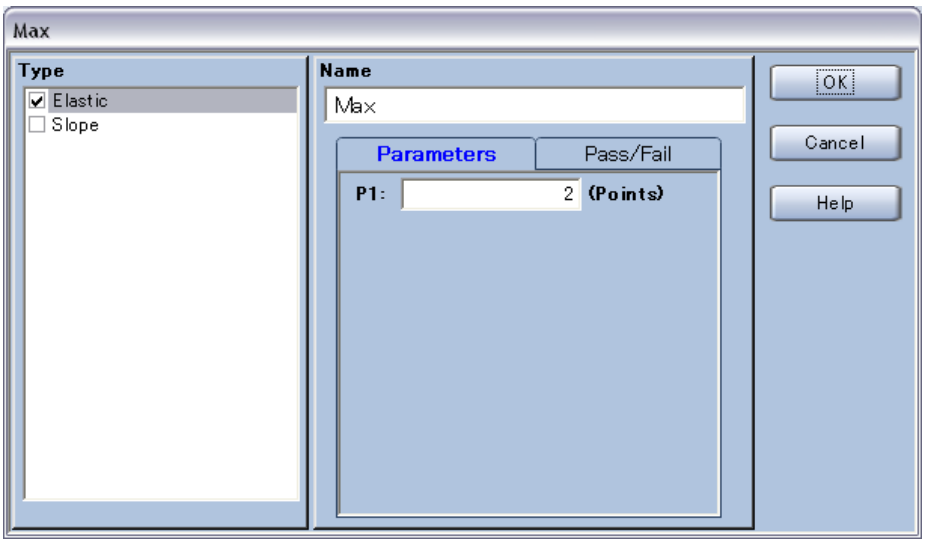
Single Control

Calculates elastic based on the maximum slope found from calculating the slope at specified intervals for all sampling data. The black nodes shown below indicate sampling data points.

$$\text{Elastic} = \text{Slope} \times \frac{\text{Gauge Length}}{\text{Cross-Sectional Area of Specimen}}$$

Slope	Calculated as maximum slope from all data points.
Gauge Length	Value specified via the [Sensor] or [Specimen] dialog box.
Cross-Sectional Area of Specimen	Calculated from dimensions specified in the [Specimen] dialog box.

Parameters



- Data Intervals for Calculating Slope (P1)

Type

- Elastic

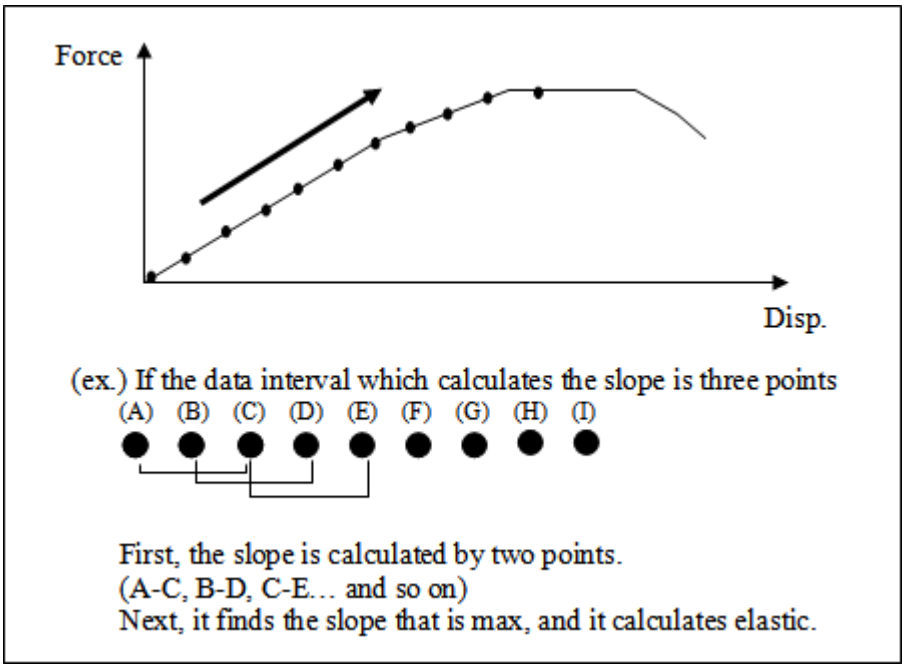
Calculated from [Slope], [Gauge Length], and [Specimen Cross-Sectional Area].

- Slope

Calculates only slope of elastic.

Example

Assume P1 is set to 3 points for calculating slope.

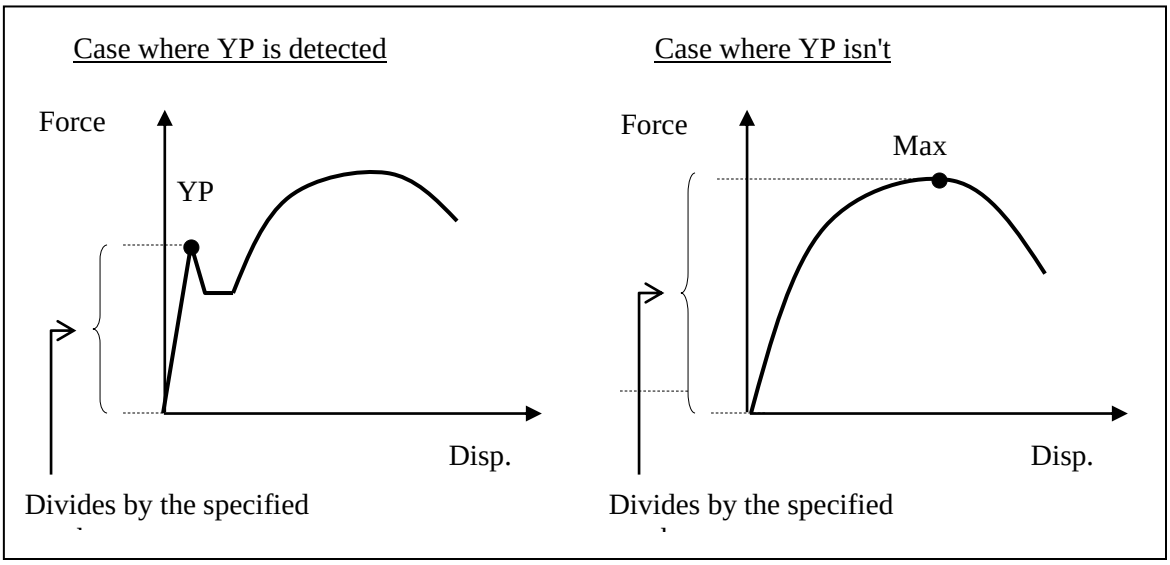


Elastic(Devided)

Single **Control**

When there is the YP, determines elastic by using data until reaching [YP]. When there is no YP, determines elastic by using data until reaching [Max].

Data from [Disp. Origin] to [YP or Max] is divided, and the inclination of "line of tangency" of each area is calculated by using the least squares. This data processing item is determined by using the value that is the maximum of these inclinations.



Note

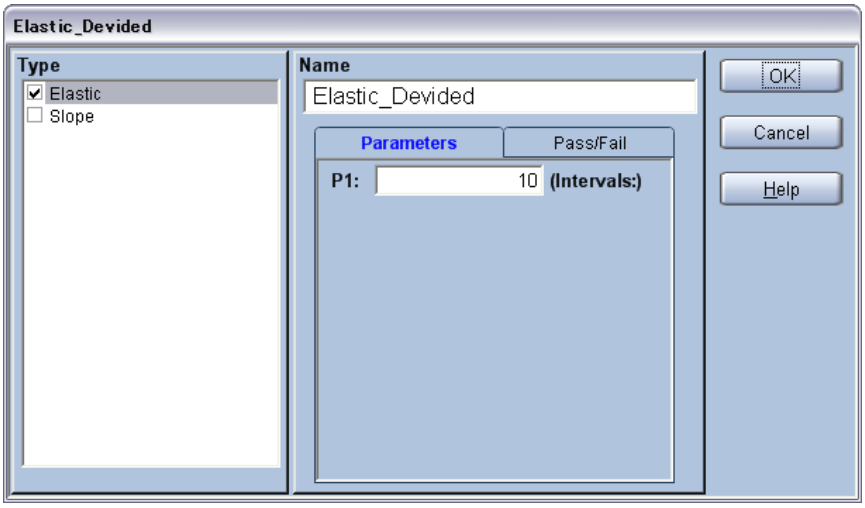
Please set the [YP] to the data processing item when you obtain [Elastic(Devided)].

When you don't set [YP], this data processing item is determined by using data until reaching [Max].

$$\text{Elastic} = \text{Slope} \times \frac{\text{Gauge Length}}{\text{Cross - Sectional Area of Specimen}}$$

Slope	Calculated by using the above-mentioned logic.
Gauge Length	Value specified via the [Sensor] or [Specimen] dialog box.
Cross-Sectional Area of Specimen	Calculated from dimensions specified in the [Specimen] dialog box.

Parameters



- Number of Intervals (P1)

Type

- Elastic

Calculated from [Slope], [Gauge Length], and [Specimen Cross-Sectional Area].

- Slope

Calculates only slope of elastic.

Elastic Loop

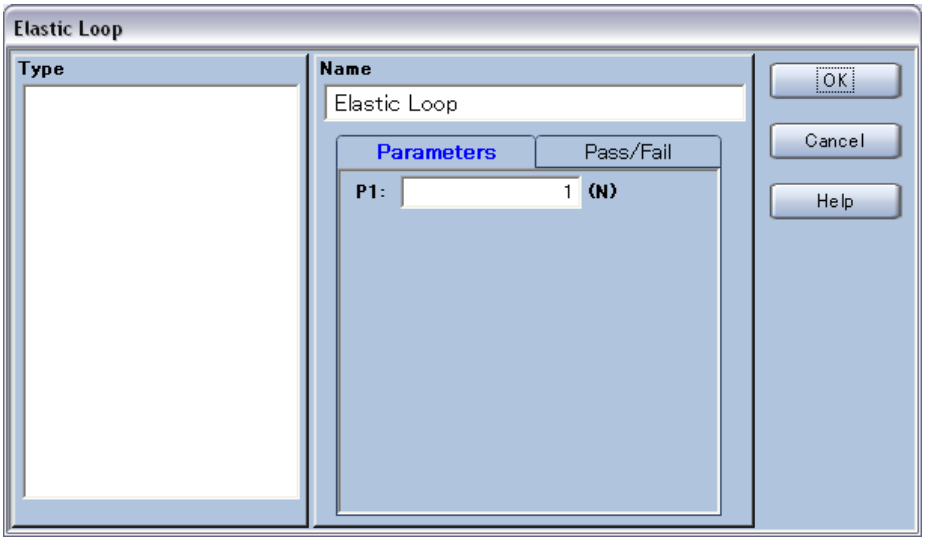
Single Tension Metal

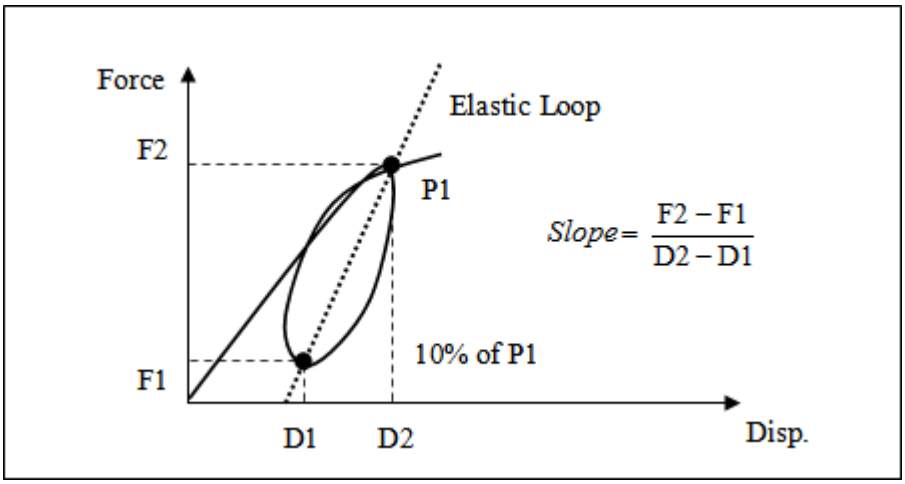
This data processing item is calculated between [Disp.] at specified [Force] and 10% of that force.

$$\text{Elastic} = \text{Slope} \times \frac{\text{Gauge Length}}{\text{Cross - Sectional Area of Specimen}}$$

Slope	Slope is calculated between [Disp.] at specified [Force] and 10% of that force.
Gauge Length	Value specified via the [Sensor] or [Specimen] dialog box.
Cross-Sectional Area of Specimen	Calculated from dimensions specified in the [Specimen] dialog box.

Parameters





- Force (P1)

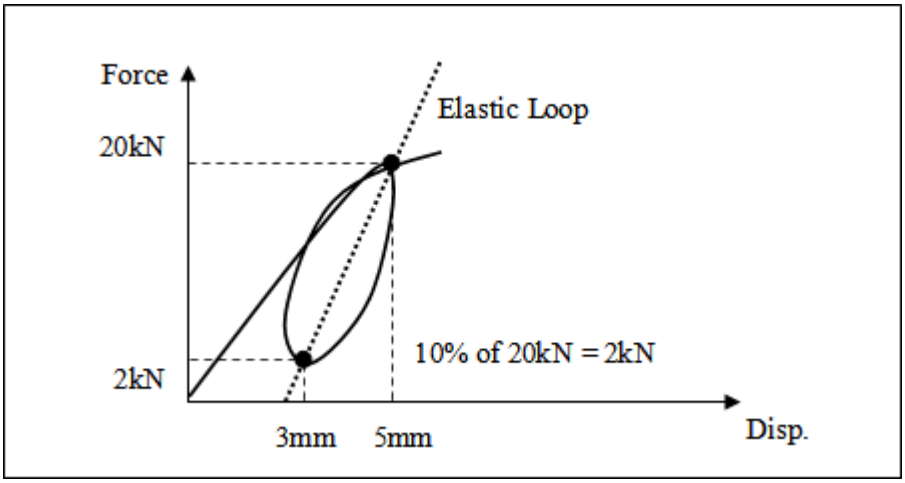
Example

Force F2: 20 kN, Disp. D1: 3 mm, D2: 5 mm, Gauge Length: 50 mm, Specimen: Plastic Flat Plate, Thickness: 2 mm, and Width: 5 mm.

$$\text{Elastic} = \frac{20\text{kN} - 2\text{kN}}{5\text{mm} - 3\text{mm}} \times \frac{50\text{mm}}{2\text{mm} \times 5\text{mm}}$$

Note

The formula used to calculate cross-sectional area depends on specimen shape. For more details, refer to Methods for Calculating Cross-Sectional Area.



Related keyword: “Calculating Stress and Strain”

Supplemental Information

This process can be selected only if [Metal, etc.] is selected in the [Material] list box in the [Specimen] dialog box.

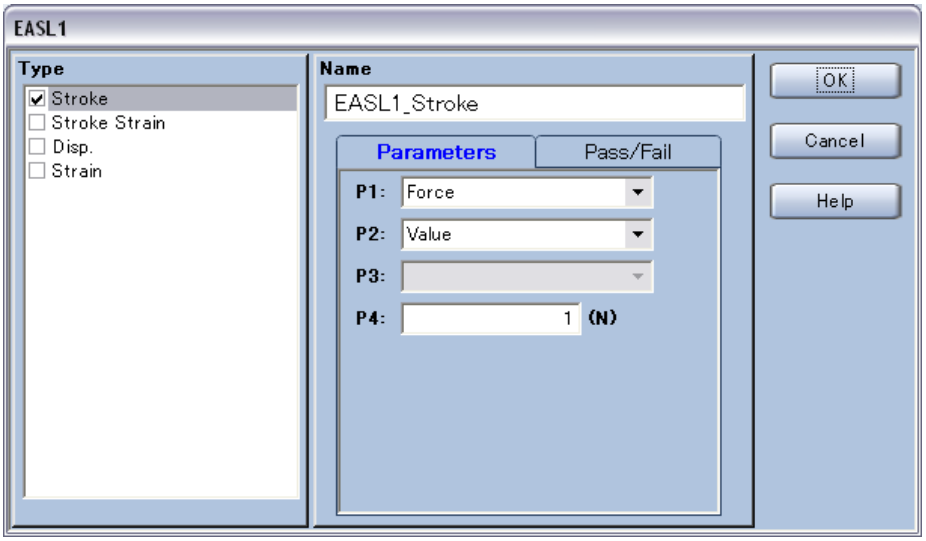
In addition, testing machine movements will be different than for other [Single] testing modes.

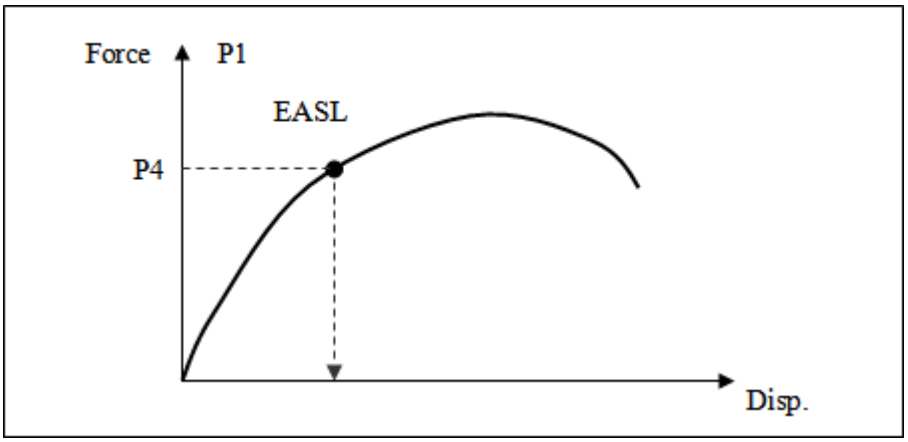
Elongation at Specified Load (EASL)

Single

Determines EASL as the [Disp.] that satisfies the specified parameter. Up to ten EASL parameters can be specified.

Parameters





- Method for Specifying Parameter (P1): Select either [Force] or [Stress].

- Method for Specifying Parameter (P2): Select either [Formula] or [Value].

- Formula (P3): Enabled to specify only if [Formula] is specified in the [P2] box.

- Value (P4): Enabled to specify only if [Value] is specified in the [P2] box.

* Parameter values are in terms of units specified in the [System] window.

Type

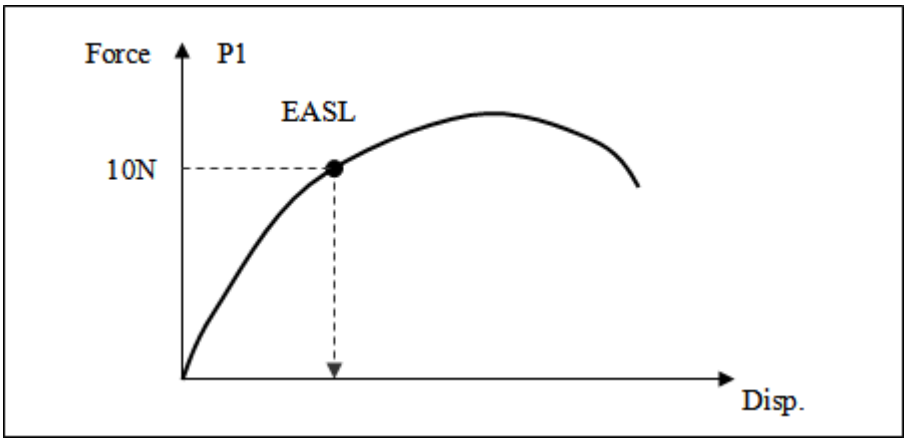
Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Stroke	Stroke (strain)
Disp.	Disp. (strain)

Example

Assume the sensor type P1 is set to [Force], the specified parameter type P2 is [Value], and the value P4 is 10N.



Energy

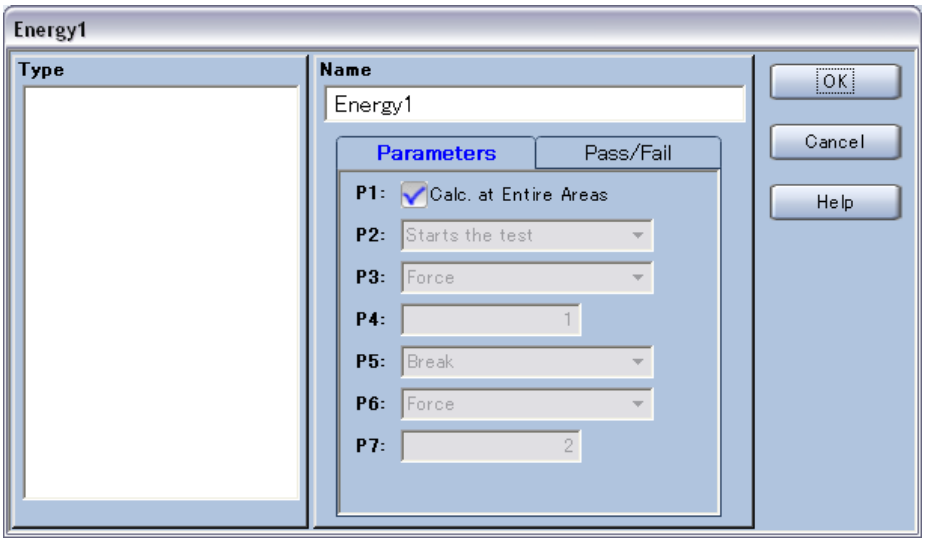
Single **Control**

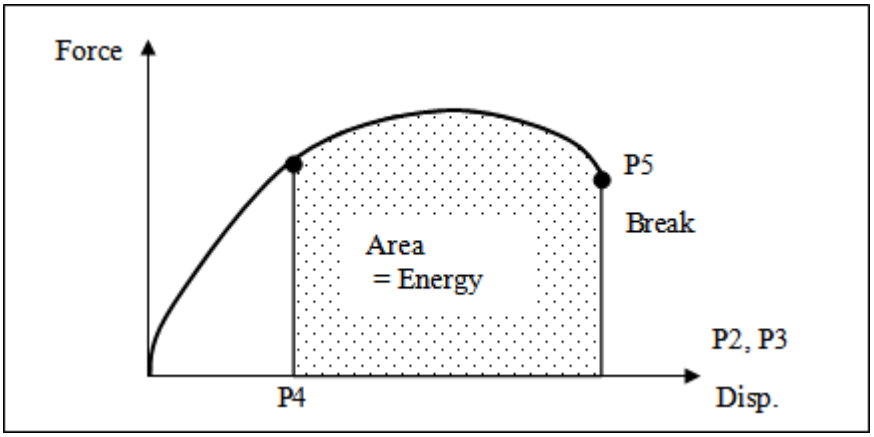
Determines the energy for the entire area under the curve or for a specified range. Up to five energy regions can be specified.

Note

Energy is considered the sum of the [Force] measured during a test, integrated over [Disp.].

Parameters





- Specifies whether to calculate total area or not (P1)

- Type of Starting Point (P2)

Start of Test	Channel	Control Area
Upper Yield Point	Max. Force	Break
% of Max.	-	-

- Type of Sensor at Starting Point (P3)

Selected from sensor types specified under [Channel] in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Starting Value (P4)

P3 and P4 can be specified only if a [Channel] was specified for P2.

- Type of Ending Point (P5)

Start of Test	Channel	Control Area
Upper Yield Point	Max. Force	Break
% of Max.	-	-

- Type of Sensor at Ending Point (P6)

same as P3

- Ending Value (P7)

P6 and P7 can be specified only if [Channel] was specified for P5.

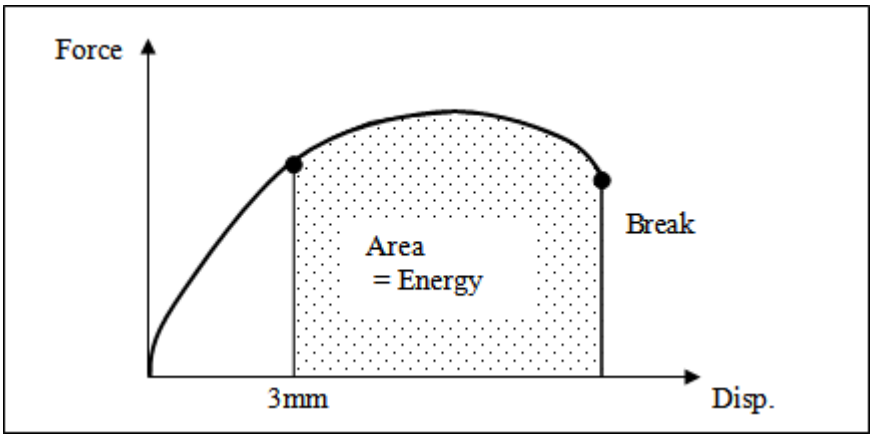
Note

P2 to P7 can only be specified if the total area is not being calculated.

Parameter values are in terms of units specified in the [System] window.

Example

Assume P1: Total area not calculated, P2: Channel, P3: Disp., P4: 3 mm, and P5: Break.



Fitted

Single Tension Metal

Determines the percent elongation at fracture from the pre-test gauge length (Lo) and post-test length (Lu), after the fractured specimen is fitted back together.

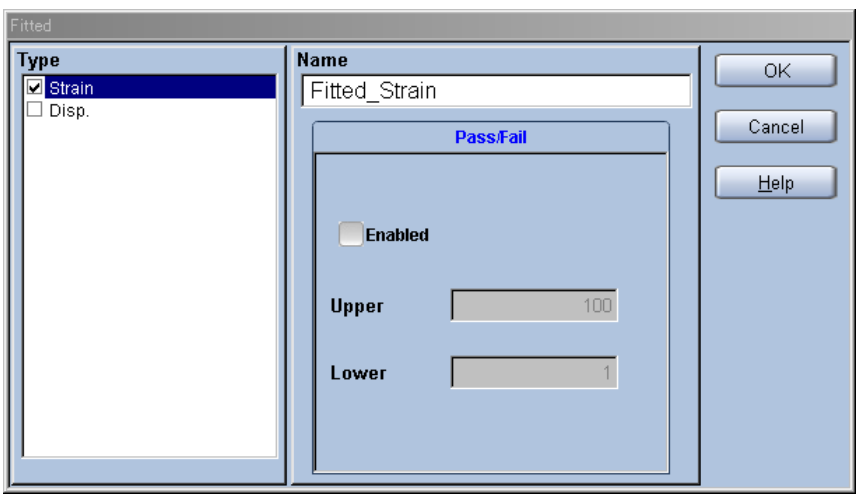
The formula of "Fitted_Strain" is as follows

$$\text{Fitted}\ (\%) = \frac{\text{Lu} - \text{Lo}}{\text{Lo}} \times 100(\%)$$

Gauge Length Lo	Pre-test gauge length
Gauge Length Lu	Post-test gauge length

"Fitted_Disp" = Lu (Post-test gauge length)

Parameters



- Strain
- Disp.

Example

Assume pre-test gauge length Lo is 50 mm and post-test gauge length Lu is 70 mm.

$$\text{Fitted}\ (\%) = \frac{70\text{mm} - 50\text{mm}}{50\text{mm}} \times 100(\%) = 40\%$$

Supplemental Information

This value can be edited only if [Metal, etc.] is specified as the [Material] in the [Specimen] dialog box.

Flexural Compliance

Single

Determine the flexural compliance decribed in the appendix JA of JIS Z7171:2016.

When not using an extensometer:

$$C_M = \frac{S_{RE}}{F} - \frac{L_{RE}^3}{4E_{RE}b_{RE}h_{RE}^3}$$

Where	C_M	:	Machine compliance between 2 set points (mm/N)
	S_{RE}	:	Displacemnet between 2 set points (mm)
	F	:	Force (N)
	E_{RE}	:	Tensile elastic of reference specimen (MPa)
	L_{RE}	:	Support length (mm)
	b_{RE}	:	Width of reference specimen (mm)
	h_{RE}	:	Thickness of reference specimen (mm)

When using an extensomter:

$$C_M = \frac{1}{F}(S^* - \Delta S_{RE})$$

Where	S^*	:	Displacement of stroke (mm)
	ΔS_{RE}	:	Dispalcement of extesometer (mm)

■ Parameters

・Sensor Type (P1)

Select sensor type for staring and ending points to calculate the compliance.

・Staring Point (P2)

・Ending Point (P3)

Set 2 points to calculate the compliance.

・Tensile elastic of reference specimen (P4)

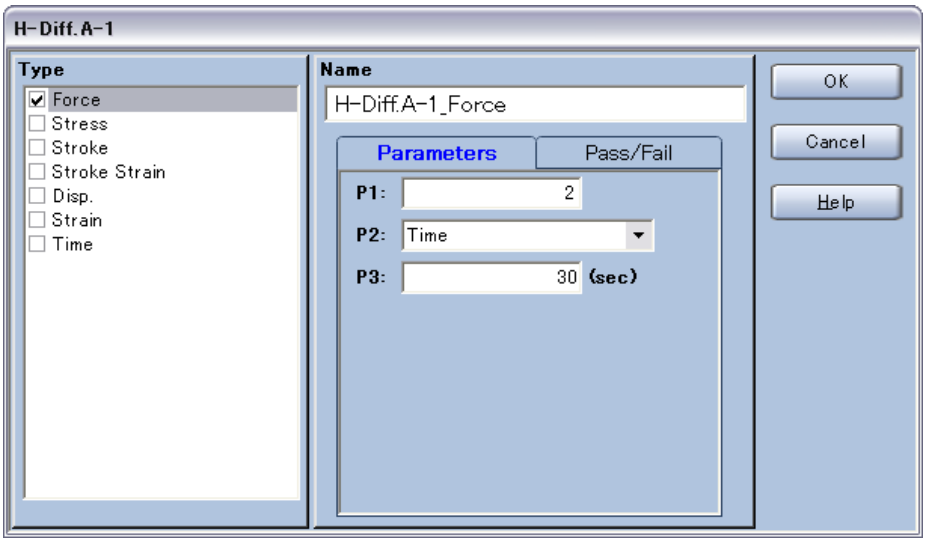
When not using an extensometer, enter the tensile elasic of reference spmenen (MPa).

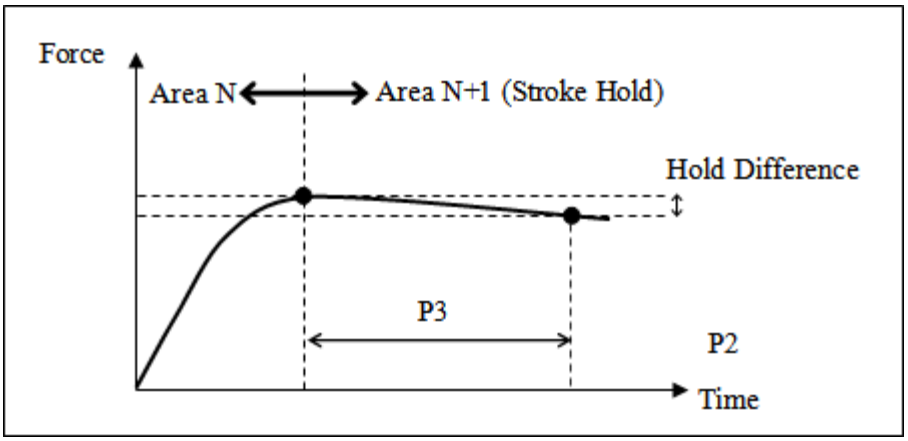
Hold Difference

Single

Determines the difference between “hold starting point” and the specified point. Up to 10 hold differences can be specified.

Parameters





- Area Number (P1)

Specifies the area number for applying the [Hold] setting as control action.

- Sensor Type (P2)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Value for Sensor Type (P3)

* Parameter values are in terms of units specified in the [System] window.

Type

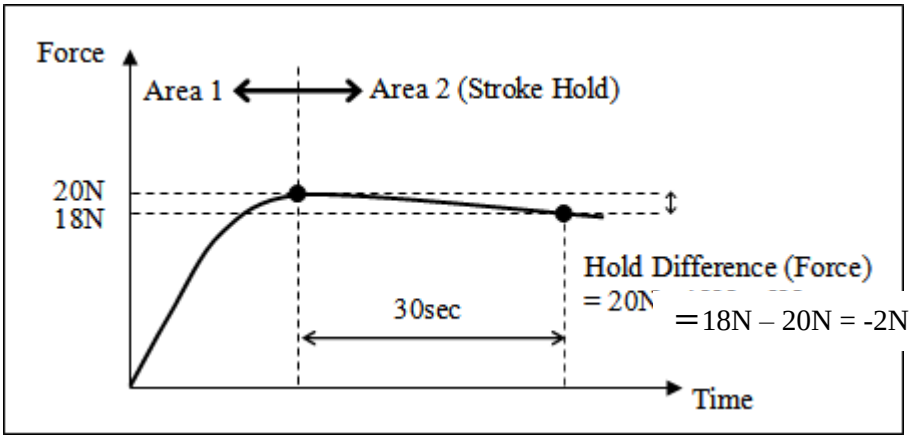
Select from types specified via the [Channel] settings, in the same manner as for sensor types (P2).

Specify default settings in the same way as well.

Example

Assume area1 is “Stroke up”, and area2 is “Stroke hold”.

Area number P1 is “2”, sensor type P2 is [Time], and the time value P3 is “30” seconds.

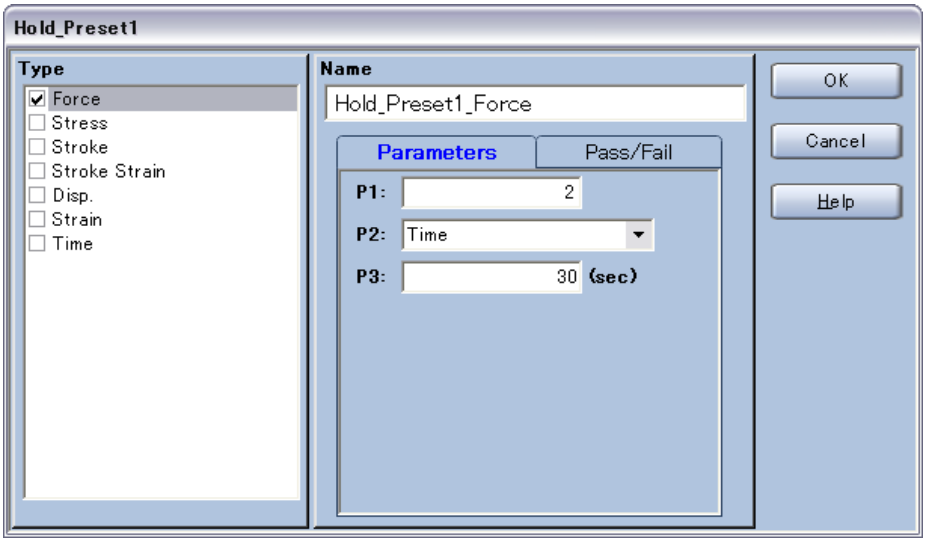


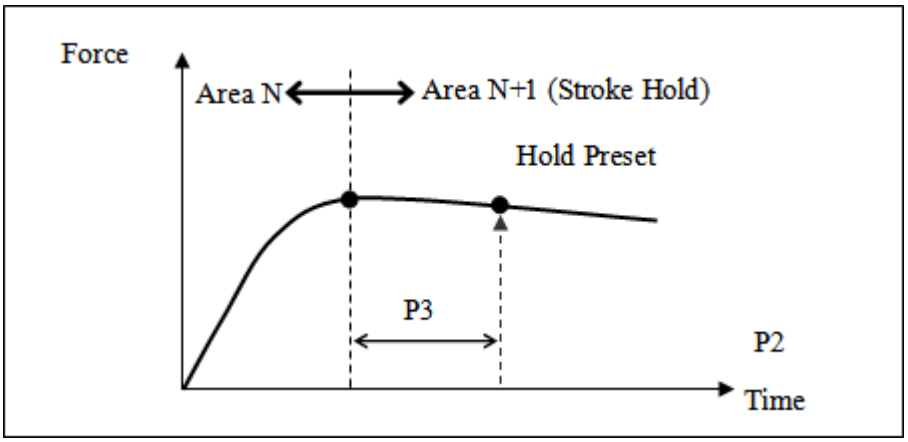
Hold Preset

Single

Determines presets in the hold. Up to 10 hold presets can be specified.

Parameters





- Area Number (P1)

Specifies the area number for applying the [Hold] setting as control action.

- Sensor Type (P2)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Value for Sensor Type (P3)

* Parameter values are in terms of units specified in the [System] window.

Type

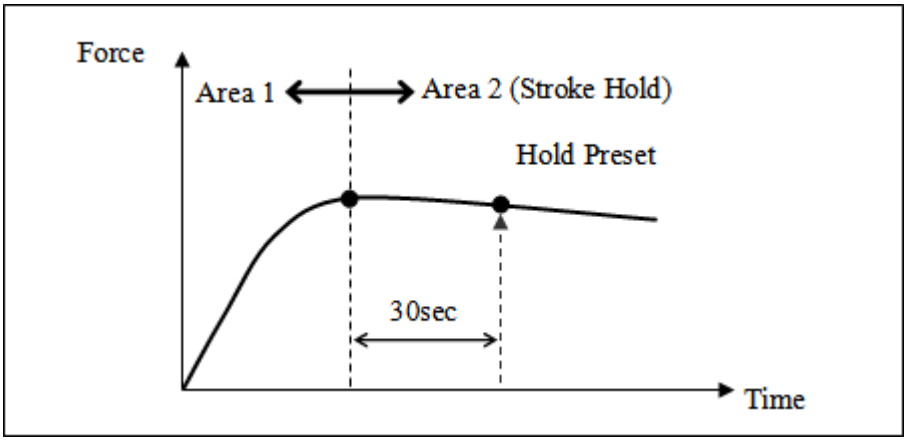
Select from types specified via the [Channel] settings, in the same manner as for sensor types (P2).

Specify default settings in the same way as well.

Example

Assume area1 is “Stroke up”, and area2 is “Stroke hold”.

Area number P1 is “2”, sensor type P2 is [Time], and the time value P3 is “30” seconds.

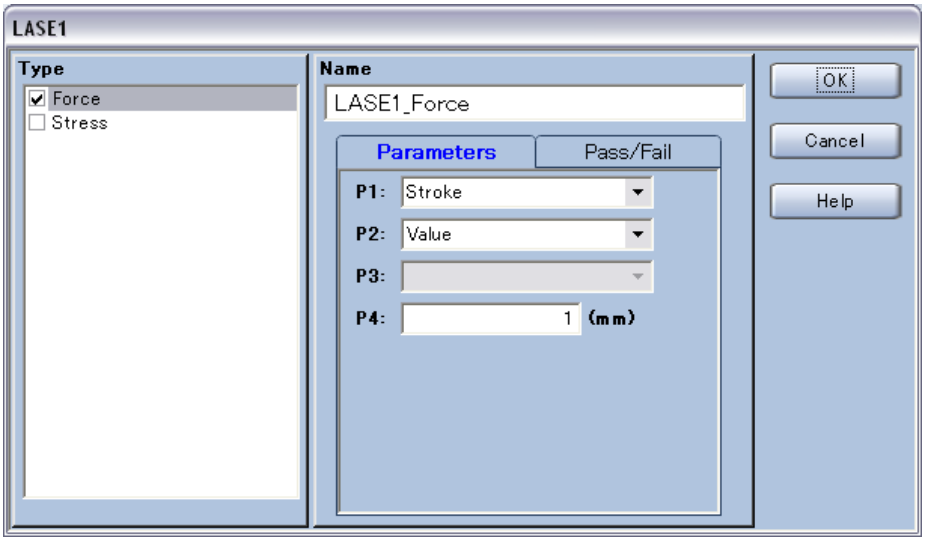


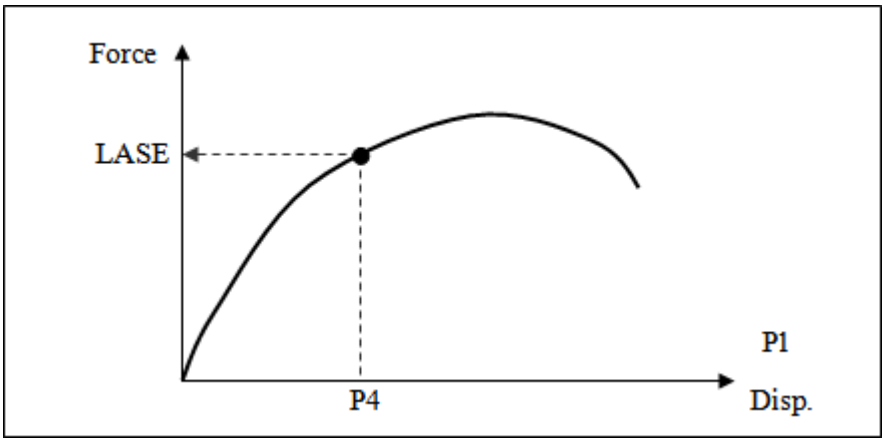
Load at Specified Elongation (LASE)

Single

Determines LASE as the [Force] that satisfies the specified parameter. Up to ten LASE parameters can be specified.

Parameters





- Sensor Type (P1)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Stroke	Stroke (strain)
Disp.	Disp. (strain)

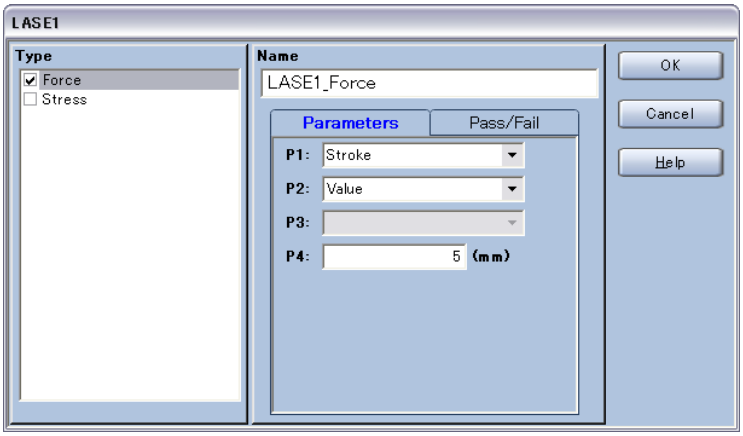
- Method for Specifying Parameter (P2): Select either [Formula] or [Value].

- Formula (P3): Enabled to specify only if [Formula] is specified in the [P2] box.

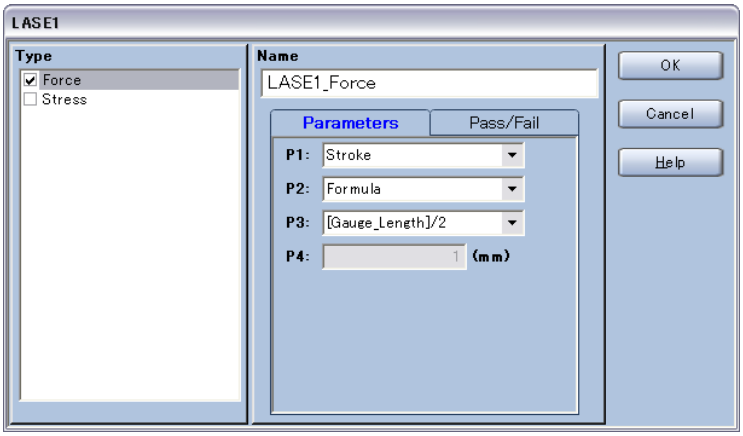
- Value (P4): Enabled to specify only if [Value] is specified in the [P2] box.

* Parameter values are in terms of units specified in the [System] window.

[P3] is disabled if [Value] is selected in [P2].



[P4] is disabled if [Formula] is selected in [P2].



Type

Force

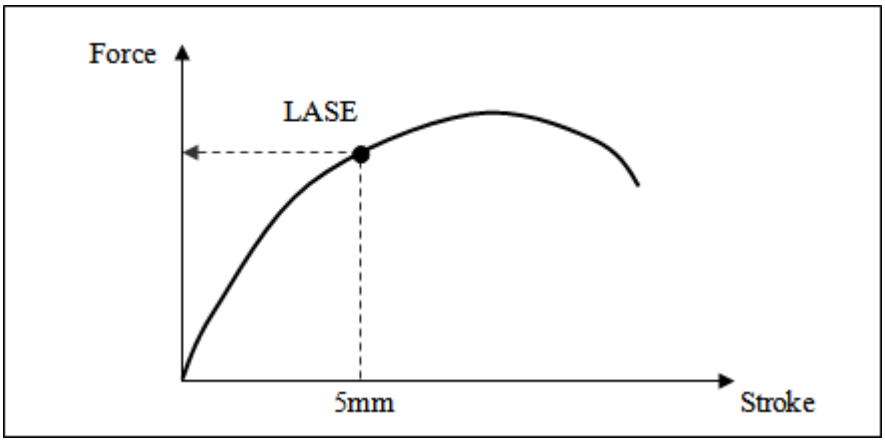
Force calculated from specified parameter.

Stress

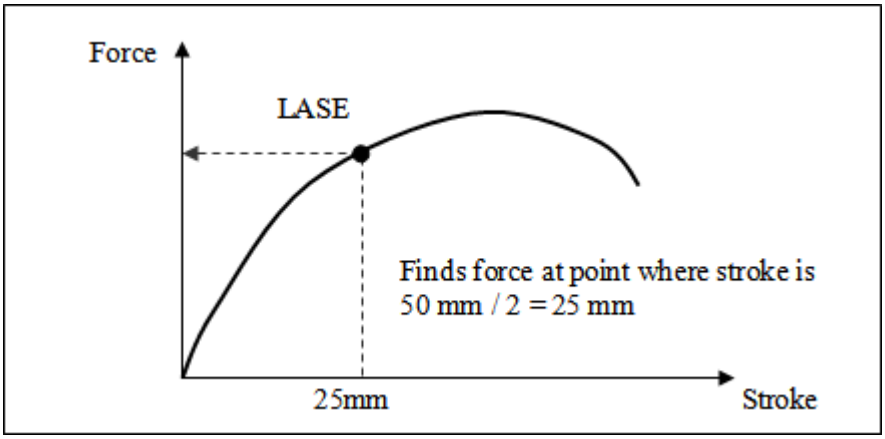
Calculated from the [Force] for the specified parameter.

Example

Assume the sensor type P1 is set to [Stroke], the specified parameter type P2 is [Value], and the value P4 is 5 mm.



Assume the sensor type P1 is set to [Stroke], the specified parameter type P2 is [Formula], and the formula P3 is gauge length divided by 2, where gauge length is 50 mm.

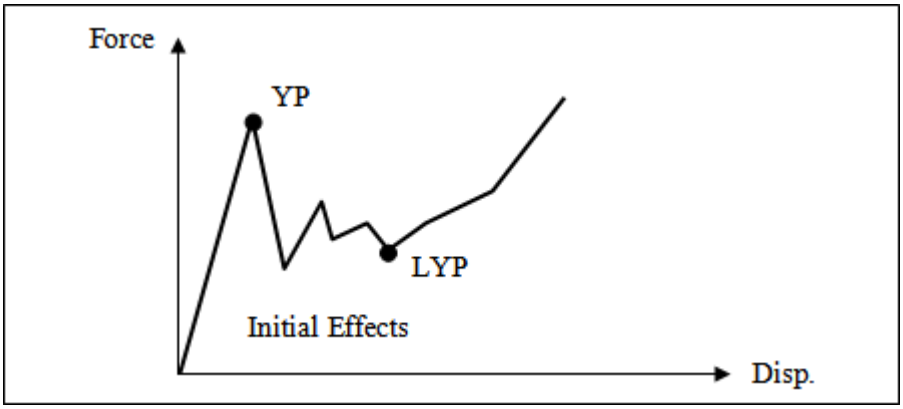


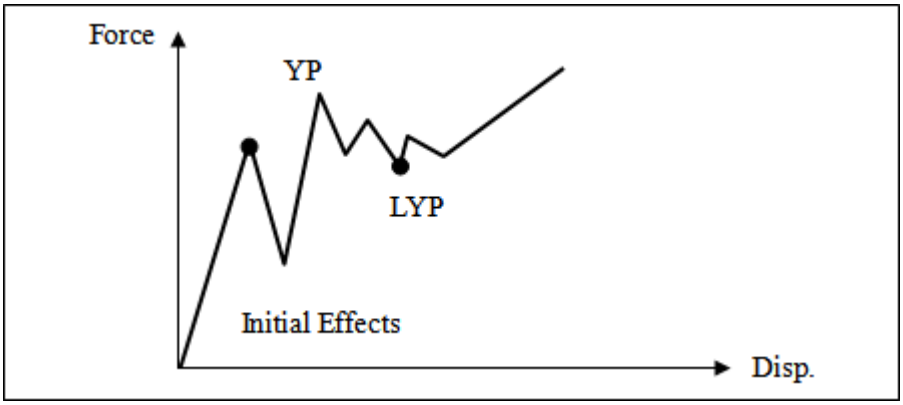
Lower Yield Point

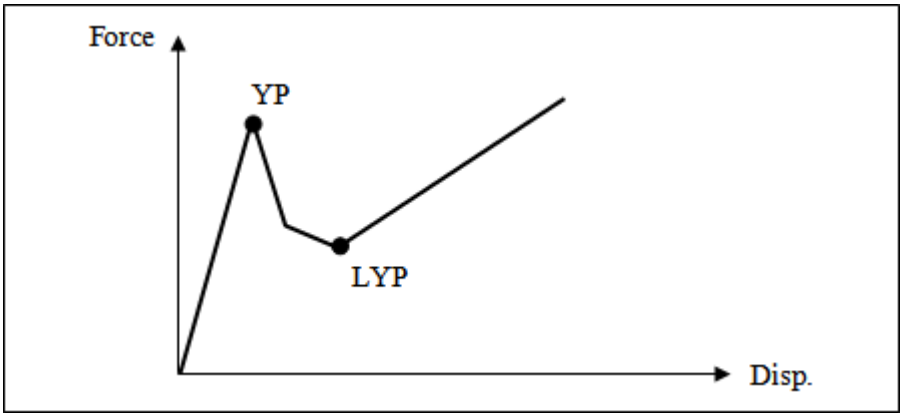
Single **Control** **Mark**

Determines the lower yield point as the point where the [Force] is minimum, excluding any inertial effects, provided the upper yield point is determined as well.

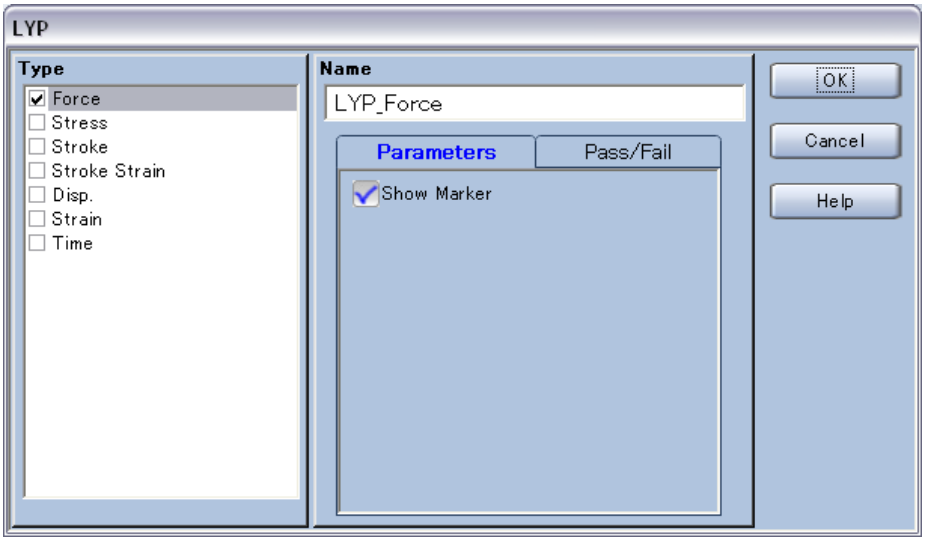
Allows selecting from three detection algorithms, depending on the graph pattern.







Parameters



- None. No need to enter any calculation points.

Type

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

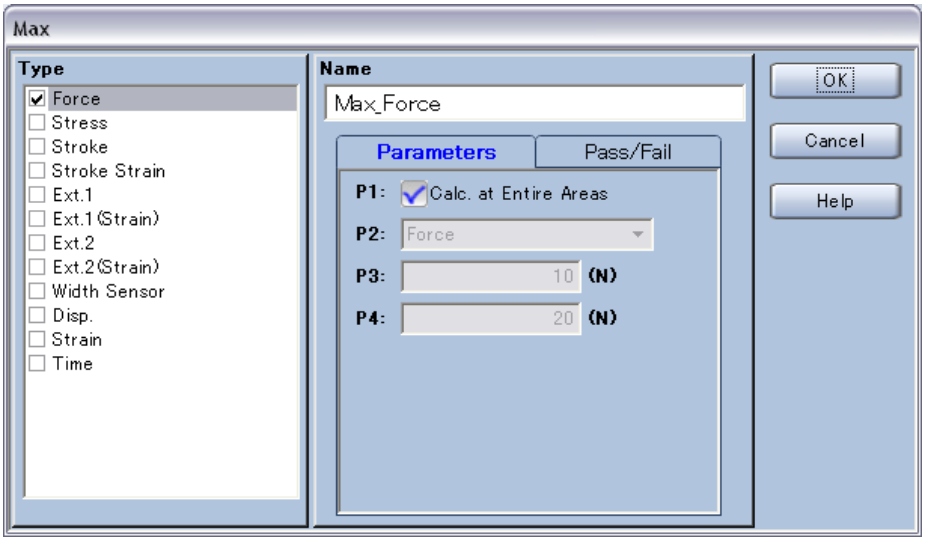
Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

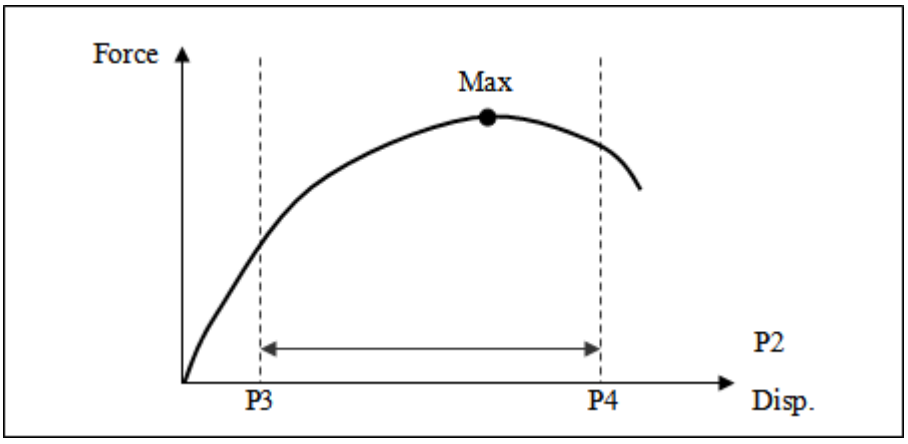
Max

Single **Control** **Mark**

Determines the maximum [Force] over a specified range or the entire area.

Parameters





- Specifies whether or not to search entire area (P1)

- Sensor Type (P2)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Starting Value (P3)

- Ending Value (P4)

* Parameter values are in terms of units specified in the [System] window.

Note

P2 to P4 can be specified only if [Calc. at Entire Areas] is not selected.

Type

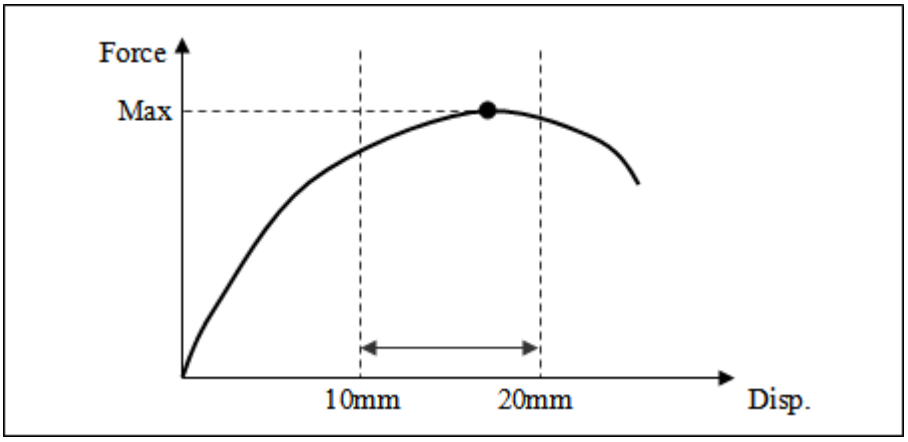
Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Example

Assume [Calc. at Entire Areas] is not selected and P2 is set to [Disp.], P3 to 10 mm, and P4 to 20 mm.

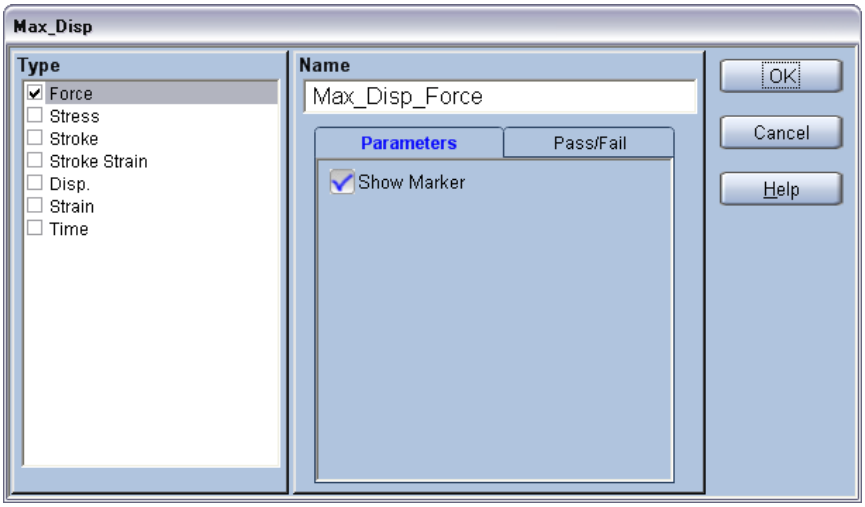


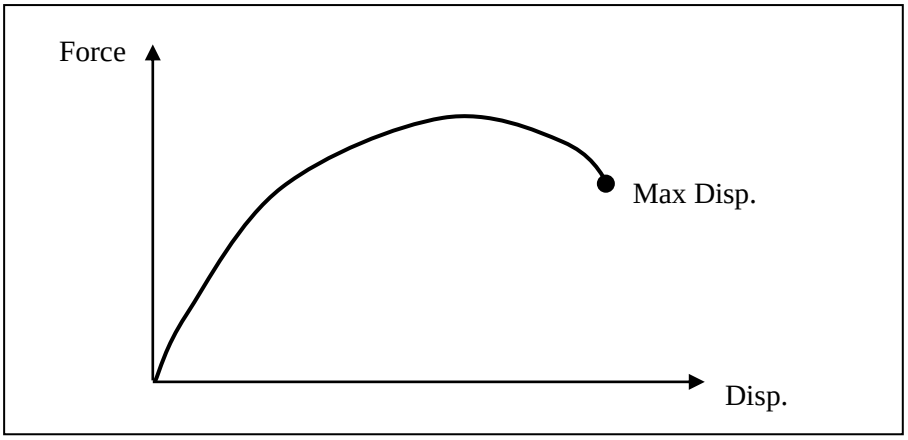
Max Disp.

Single Control Mark

Determines the maximum [Disp.] over the entire area.

Parameters





Type

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

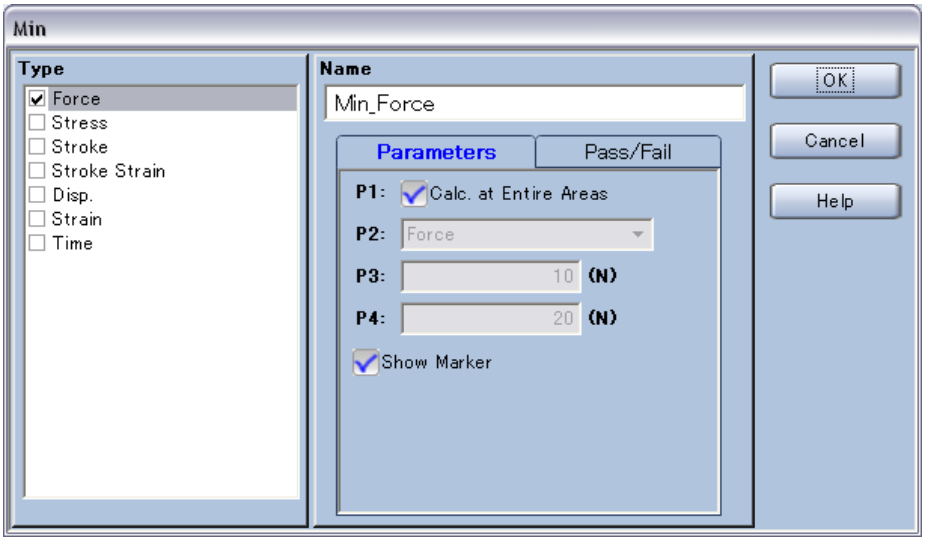
Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

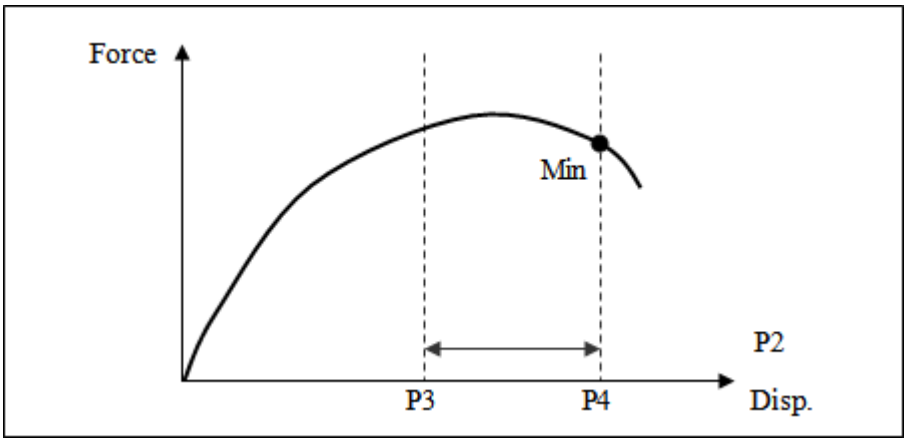
Min

Single Control Mark

Determines the minimum [Force] over a specified range or the entire area.

Parameters





- Specifies whether or not to search entire area (P1)

- Sensor Type (P2)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Starting Point (P3)

- Ending Point (P4)

* Parameter values are in terms of units specified in the [System] window.

Note

P2 to P4 can be specified only if [Calc. at Entire Areas] is not selected.

Type

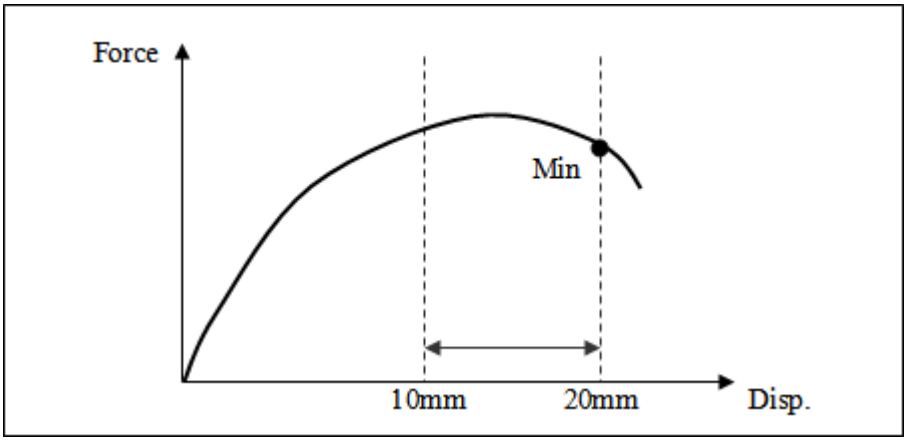
Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Example

Assume [Calc. at Entire Areas] is not selected and P2 is set to [Disp.], P3 to 10 mm, and P4 to 20 mm.

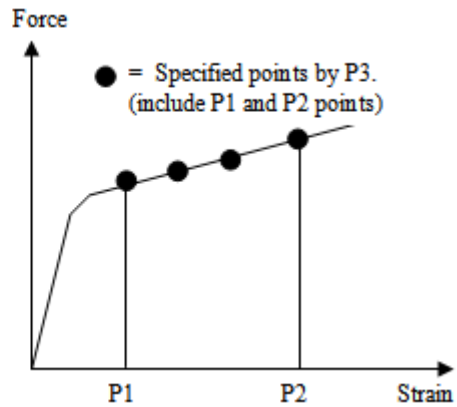


n-Value

Single Tension Metal

Uses the least squares method to determine the n-value for two specified points and specified number of intervals.

$$n = \frac{N\sum_{i=1}^{N}(\log \varepsilon_i \log \sigma_i) - (\sum_{i=1}^{N}\log \varepsilon_i \sum_{i=1}^{N}\log \sigma_i)}{N\sum_{i=1}^{N}(\log \varepsilon_i)^2 - (\sum_{i=1}^{N}\log \varepsilon_i)^2}$$



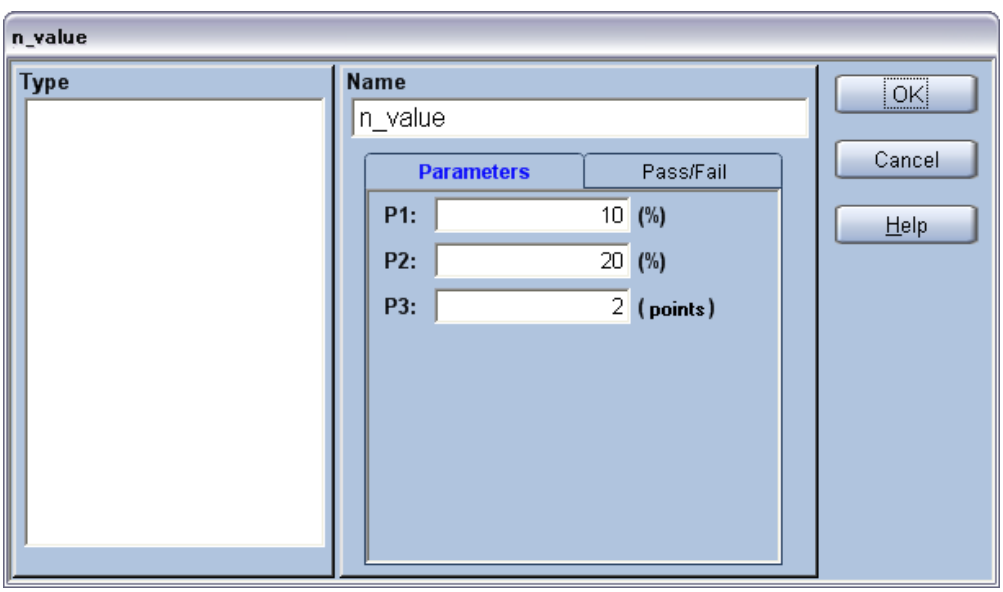
where;

N = the number of data pairs (P3)

ε = True Strain

σ = True Stress

Parameters



- Strain Starting Point (P1)
- Strain Ending Point (P2)
- Number of points for calculating (P3)

Supplemental Information

This value can be edited only if [Metal, etc.] is selected in the [Material] list box in the [Specimen] dialog box.

new-n-Value

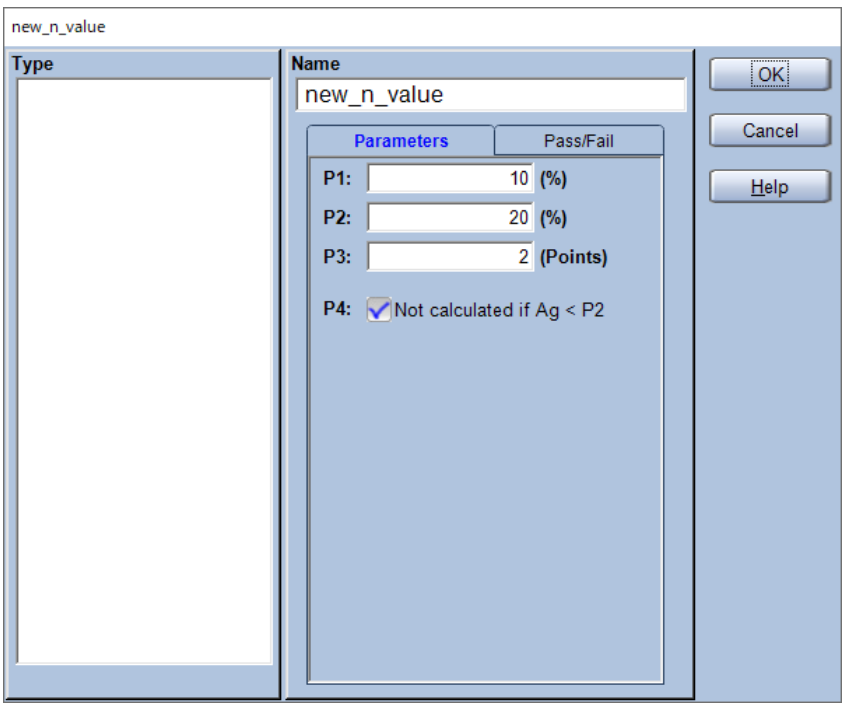
Single Tension Metal

Determines the n-value with the method described in this section. The least squares method is used to determine the n-value for two specified points and specified number of intervals.

To calculate the “new-n-value”, “Elastic”, “Non-propotional elongation”, and “Ag” have to be selected.

The calculation formula is almost same as “n-value”, but the calculation for “True Strain” is different.

Parameters



- Plastic Strain Starting Point (P1)

- Plastic Strain Ending Point (P2)

- Number of points for calculating (P3)

- If Ag<P2, new-n-value is not calculated (P4)

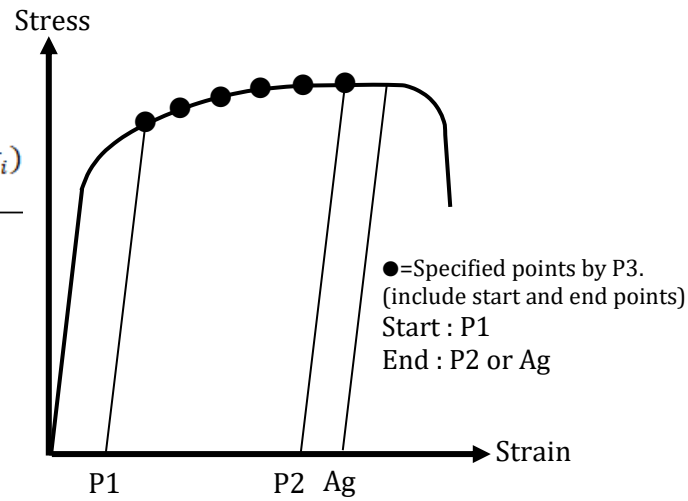
$$n = \frac{N\sum_{i=1}^{N}(\log\varepsilon_i \log\sigma_i) - (\sum_{i=1}^{N}\log\varepsilon_i \sum_{i=1}^{N}\log\sigma_i)}{N\sum_{i=1}^{N}(\log\varepsilon_i)^2 - (\sum_{i=1}^{N}\log\varepsilon_i)^2}$$

where:

N = the number of data pairs

ε = true strain

σ = True stress



Type

- n Hardening coefficient

- C Intensity coefficient

$\sigma = C\varepsilon^n$

Supplemental Information

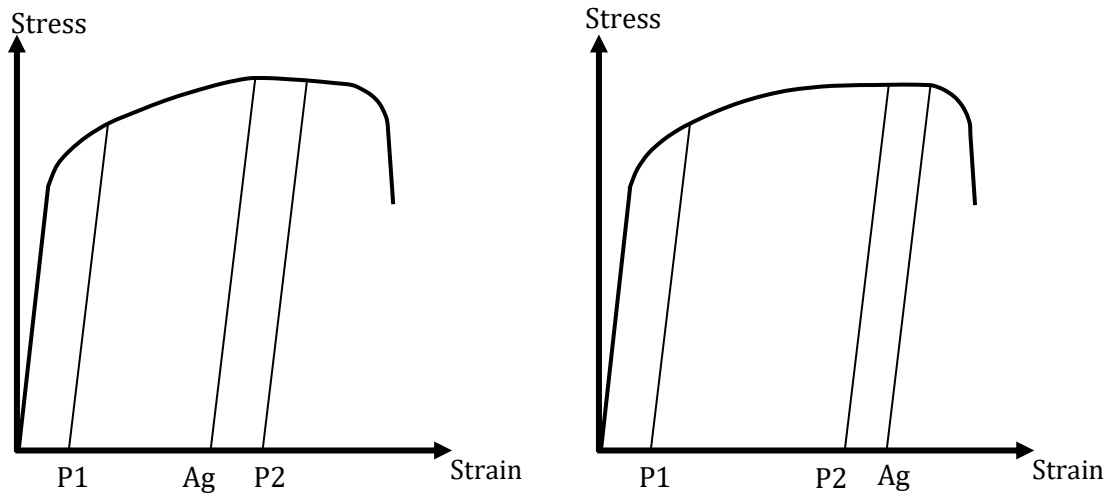
This value can be edited only if [Metal] or [Other] is selected in the [Material] list box in the [Specimen] dialog box.

Calculation range of new-n-Value

The calculation range of new-n-value depends on the relationship between Ag and P2.

P1：start point of calculation range of new-n-value

P2：end point of calculation range of new-n-value



(1)**Ag < P2** (2)**P2 < Ag**

	(1) Ag<P2	(2) P2<Ag
P4:☑ "If Ag<P2, new-n-value is not calculated"	New-n-value is not calculated (「--.--」)	New-n-value is calculated between P1 and P2
P4:☐ "If Ag<P2, new-n-value is not calculated"	New-n-value is calculated between P1 and Ag	New-n-value is calculated between P1 and P2

True Strain Calculation

The calculation formula differs from the value of "Elastic Strain". It is described as follows:

Elastic Strain $e_{elastic}$ = (deltaL – L_{nonL}) / Le

Where deltaL : elongation from test starts., L_{nonL} : non-propotional elongation,

Le: Initial Gauge Length

When "Elastic Strain" is less than 10% of the total elongation, true strain is described as follows:

True Strain e = ln (Le+deltaL) / Le

When "Elastic Strain" is equal to or more than 10% of the total elongation, true strain is described as follows:

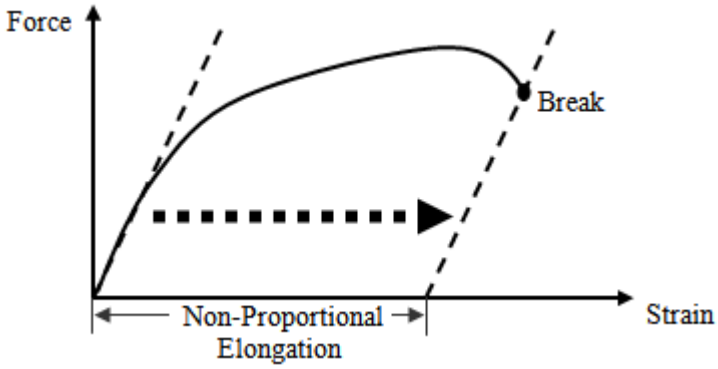
True Strain e = ln [(Le+deltaL)/Le – F/ S*me]]

where F: Force, S: Cross sectional area of specimen, me: elastic

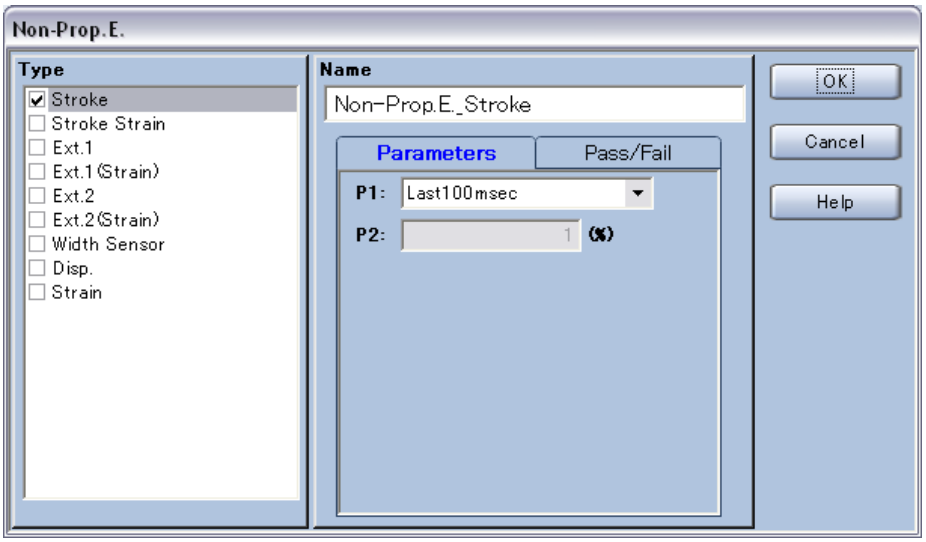
Non-Proportional Elongation

Single Tension Metal

Determines the non-proportional elongation at the distance to the point on the X-axis (strain) where a straight line extending through the break point and parallel to the elastic, intersects the X-axis. This requires specifying that elastic be determined beforehand.



Parameters



- Detection Parameter (P1)

Sensitivity	Level
% Level of Max.	Last 100 msec

- Detection Value (P2)

Type

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Stroke	Stroke (strain)
Disp.	Disp. (strain)

Supplemental Information

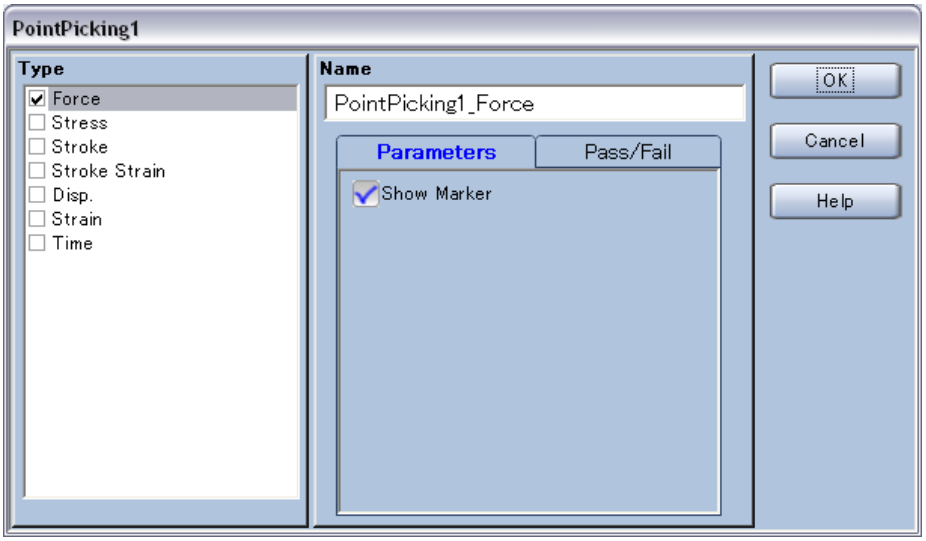
This value can be edited only if [Metal, etc.] is specified as the [Material] in the [Specimen] dialog box.

Point Picking

Single Control

Determines post-test information for any point on the graph specified with the mouse pointer. Up to 20 points can be specified.

Parameters



- None

Type

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Poisson's

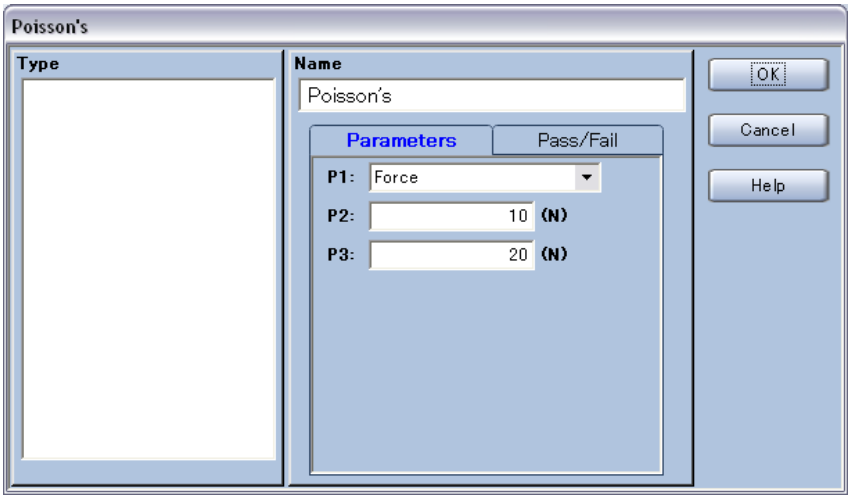
Single Tension Compression

Determines [Poisson's] from longitudinal elastic in the direction of testing and transverse elastic in the perpendicular direction.

Poisson's = Longitudinal Elastic ÷ Transverse Elastic

This is the same as the ratio of transverse strain to longitudinal strain.

Type



- Sensor Type (P1)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Starting Point (P2)

- Ending Point (P3)

* Parameter values are in terms of units specified in the [System] window.

Supplemental Information

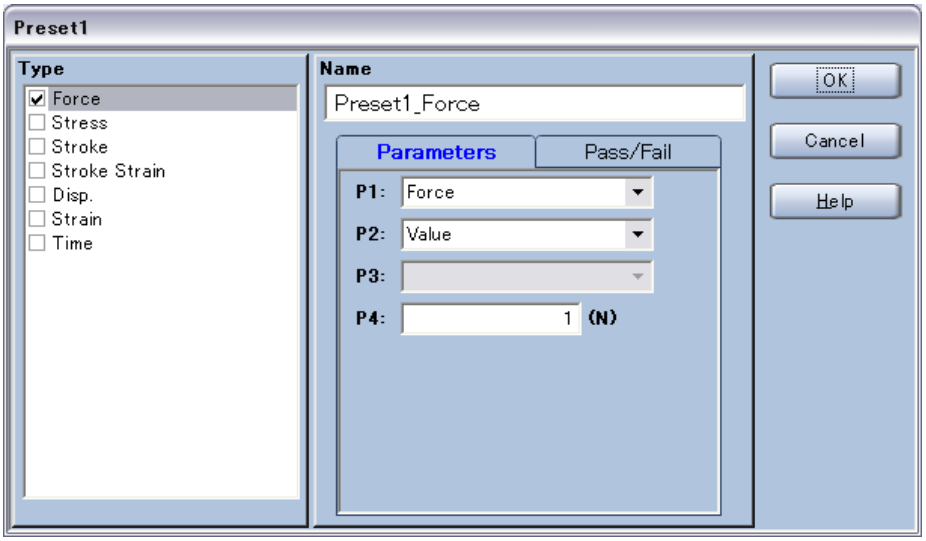
This parameter can be edited only if [Width] is specified in the [Sensor] dialog box.

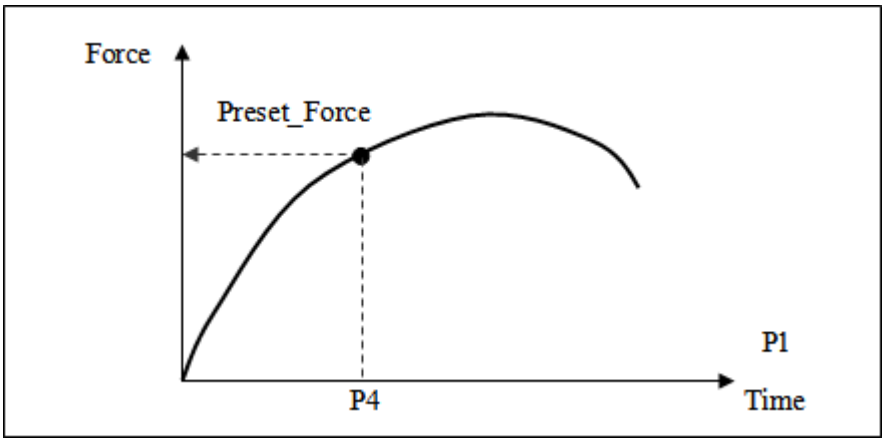
Preset

Single **Control**

Finds the specific points corresponding to a specified parameter. Up to 10 preset points can be specified.

Parameters





- Sensor Type (P1)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Method for Specifying Parameter (P2): Select either [Formula] or [Value].

- Formula (P3): Enabled to specify only if [Formula] is specified in the [P2] box.

- Value (P4): Enabled to specify only if [Value] is specified in the [P2] box.

* Parameter values are in terms of units specified in the [System] window.

Type

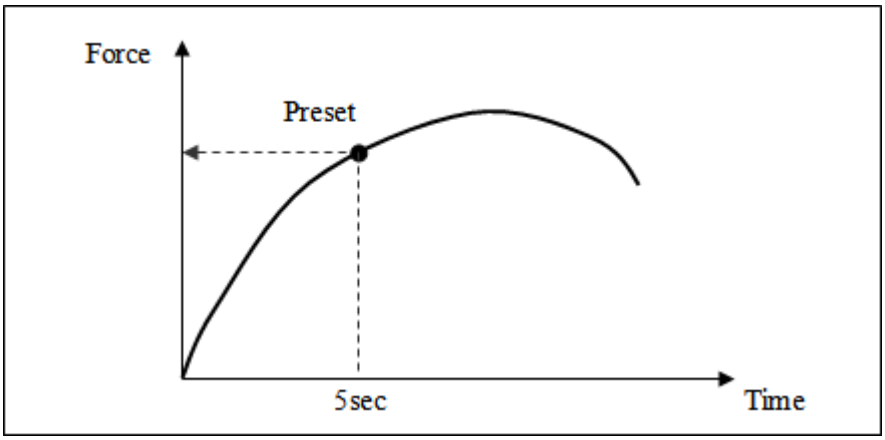
Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Example

Assume the sensor type P1 is set to [Time], the specified parameter type P2 is [Value], and the value P4 is 5 sec.



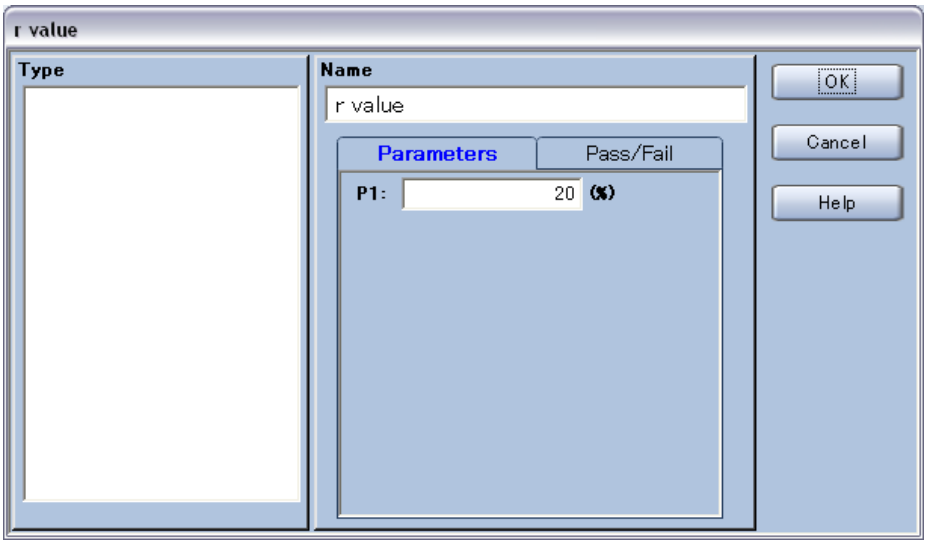
R-Value

Single **Tension** **Metal**

Determines the r-value from the specimen width when elongation reaches a specified value.

$$r = \frac{\ln W_0 / W}{\ln L / L_0 - \ln W_0 / W}$$

Parameters



- Strain (P1)

Supplemental Information

The above can be specified only if [Width Sensor] is specified in the [Sensor] window and [Metal, etc] is selected in the [Material] list box in the [Specimen] dialog box.

Reduction

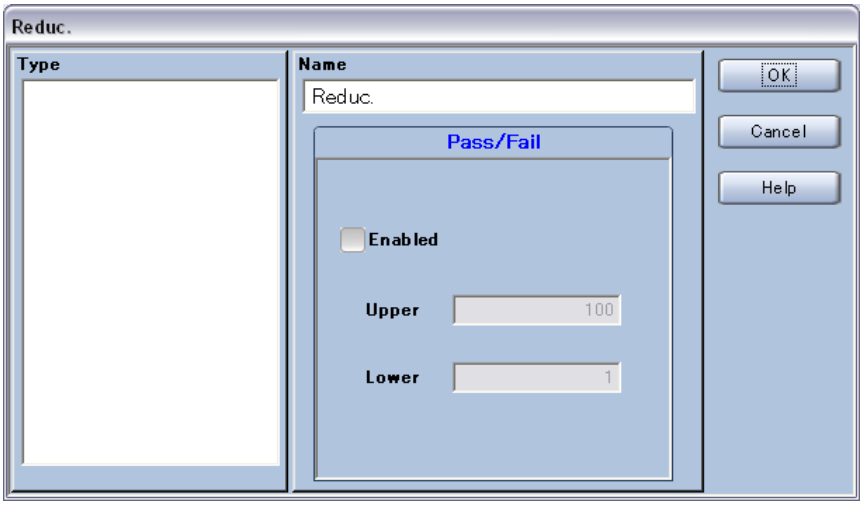
Single **Tension** **Metal**

Calculates percent reducting from initial cross-sectional area (So) and the minimum cross-sectional area (Su) after fracture, determined by carefully realigning the fracture surfaces of fractured parts.

$$\text{Reduction} = \frac{\text{So} - \text{Su}}{\text{So}} \times 100(\%)$$

Initial Cross Section So	Cross-sectional area of specimen before test.
Final Cross Section Su	Cross-sectional area of specimen after test.

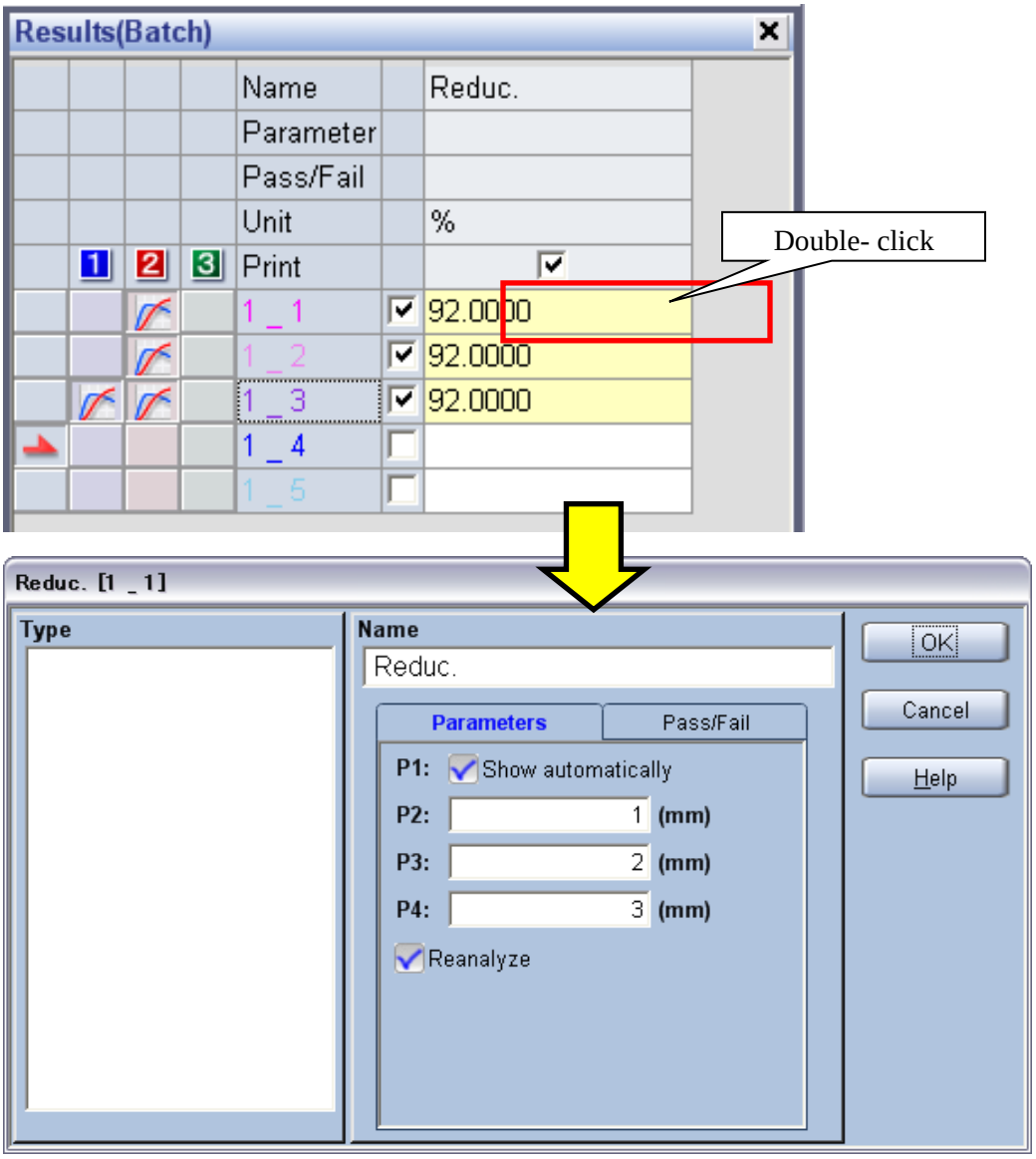
Parameters (in [Create New Method] or [Open Method])



- None

Parameters (in [Recalculate])

Double click the cell which displays the value of Reducation.



- Setting of the size dialog (P1)

Selecting this checkbox, the dialog like following is appears automatically. Enter size values of specimen to the dialog.

Not selecting this checkbox, dialog is not appeared, and P2~P4 is used for calculating of the [Reduction].

- Size of specimen (P2)

Already entered value is displayed in this item.

If P1 and [Reanalize] is selected, the value of cross section area is calculated with this parameter.

The type of value depends on the specimen shape.

shape	Type of P2
Plate	Enter Thickness of the specimen after test.
Rod	Enter Diameter of the specimen after test.
Tube 1	Enter Outer_Diameter of the specimen after test.
Tube 2	
Tube 3	
Balance	Enter Length of the specimen after test.
Ring	Enter Thickness of the specimen after test.
ORing	
Yarn	Enter Linear_Density of the specimen after test.
Hexagonal	Enter a_side of the specimen after test.
Area	Enter Area of the specimen after test.

- Size of specimen (P3)

Already entered value is displayed in this item.

If P1 and [Reanalize] is selected, the value of cross section area is calculated with this parameter.

The type of value depends on the specimen shape.

shape	Type of P3
Plate	Enter Widthof the specimen after test.
Rod	Nonuse
Tube 1	Enter Inner_Diameter of the specimen after test.
Tube 2	Enter Thickness of the specimen after test.
Tube 3	
Balance	Enter Weight of the specimen after test.
Ring	Enter Thickness of the specimen after test.
ORing	Enter Ring_In.Diameter of the specimen after test.
Yarn	Nonuse
Hexagonal	
Area	

- Size of specimen (P4)

Already entered value is displayed in this item.

If P1 and [Reanalize] is selected, the value of cross section area is calculated with this parameter.

The type of value depends on the specimen shape.

shape	Type of P4
Plate	Nonuse
Rod	
Tube 1	
Tube 2	
Tube 3	Enter Widthof the specimen after test.
Balance	Enter Density of the specimen after test.
Ring	Nonuse
ORing	
Yarn	
Hexagonal	
Area	

Example

Assume pre-test specimen cross section So is 6 mm2 and post-test specimen cross section Su is 3 mm2.

$$\text{Reduction} = \frac{6\text{mm}^2 - 3\text{mm}^2}{6\text{mm}^2} \times 100(\%) = 50\%$$

Supplemental Information

This value can be edited only if [Metal, etc.] is specified as the [Material] in the [Specimen] dialog box.

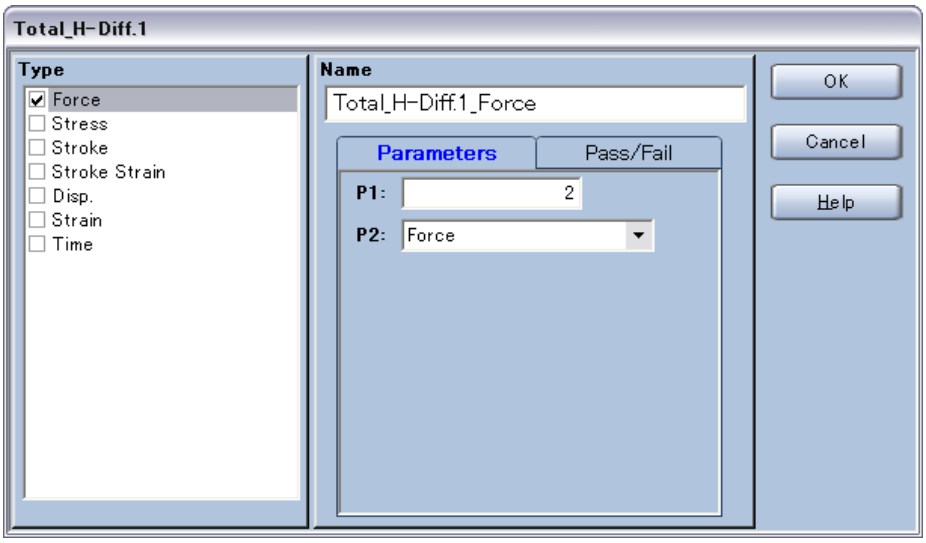
Total Hold Difference

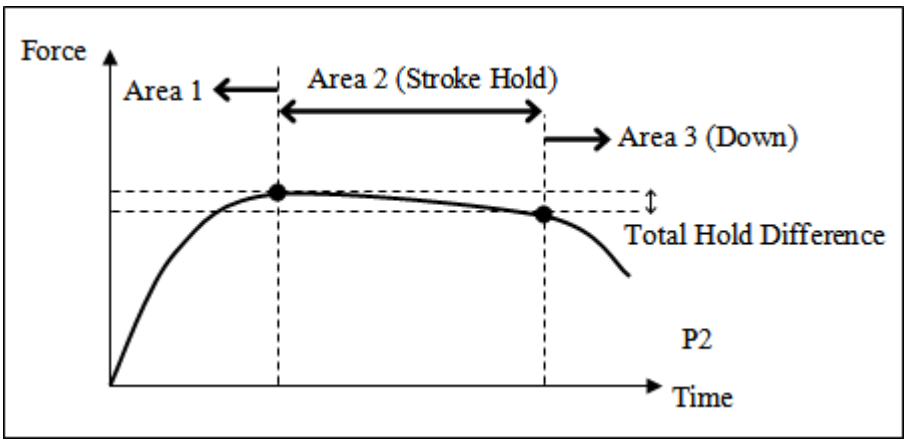
Single

This data processing item is calculated by the point that is the max difference value between all holding points of the specified sensor and the hold starting point.

Up to ten points can be specified for determining total change.

Parameters





- Area Number (P1)

Specifies the area number for applying the [Hold] setting as control action.

- Sensor Type (P2)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Type

Select from types specified via the [Channel] settings, in the same manner as for sensor types (P2).

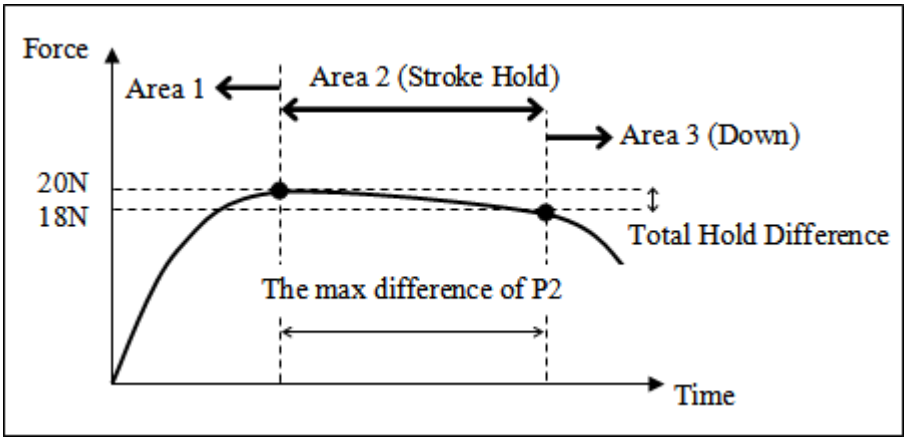
Specify default settings in the same way as well.

Example

Assume area1 is “Stroke up”, area2 is “Stroke hold”, and area3 is “Stroke down”.

Area number P1 is “2”, sensor type P2 is [Time].

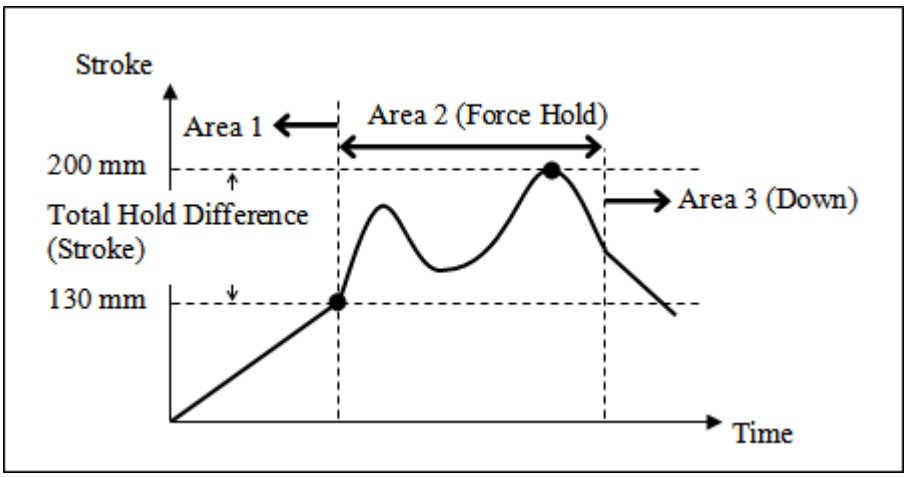
Total Hold Difference (Force) = 20N – 18N = 2N



Assume area1 is “Stroke up”, area2 is “Force hold”, and area3 is “Stroke down”.

Area number P1 is “2”, sensor type P2 is [Stroke].

Total Hold Difference (Stroke) = 20mm – 130mm = 70mm



Upper Yield Point

Single Control Mark

The upper yield point refers to the point of maximum [Force], when the [Force] exceeds a given elastic limit and only Disp. increases, while the yield force begins to decrease.

One of the following three calculation methods can be selected depending on the shape pattern of this decrease in yield force.

- Upper Yield Point (%FS), UpperYield Point(%YP)

Determines the upper yield point by detecting how much the [Force] changes.

- Upper Yield Point (Disp.)

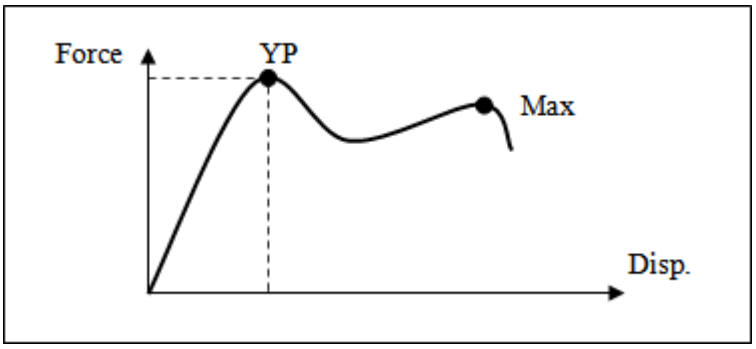
Determines the upper yield point by finding the maximum [Force] up to a given [Disp.].

- Upper Yield Point (Points)

Determines the upper yield point by finding the first point where the [Force] decreases or stays the same.

Supplemental Information about Maximum Force

If the upper yield point has the highest [Force] of all sampling data, as shown below, then the next [Max] following the upper yield point is considered the maximum force.



Upper Yield Point (%FS)

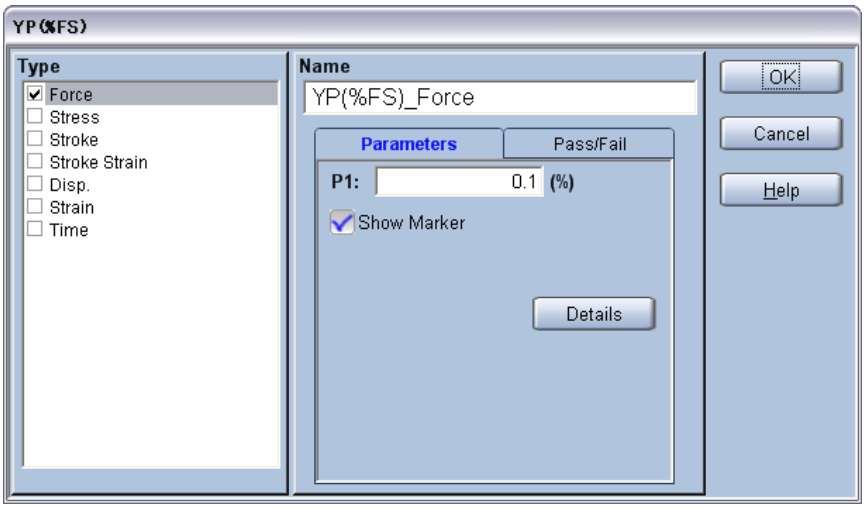
Single Control Mark

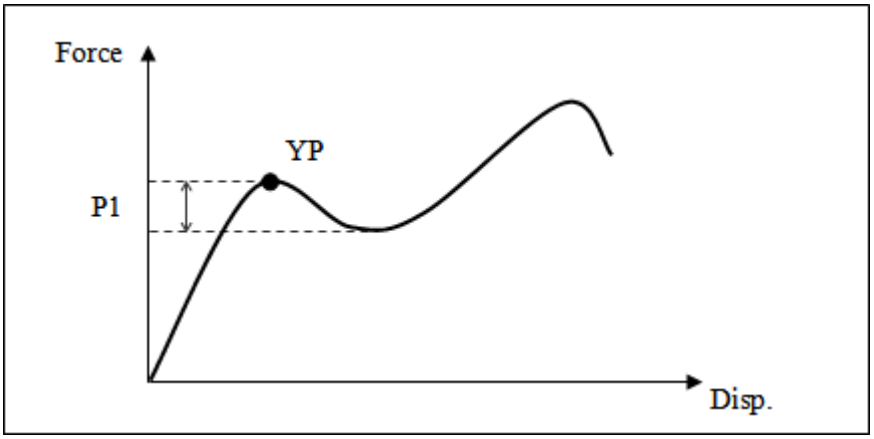
Determines the upper yield point by detecting how much the [Force] changes.

The upper yield point is detected as the first point within a given amount of change, where the [Force] first decreases after starting the test. However, any decreases that occur within the first 1% of [Fullscale Force] will not be considered the upper yield point.

If the “Force” become larger than “YP” within 0.05%(Strain) after detecting “YP”. This point is not registered to the “YP”.

Parameters





- Sensitivity of Detecting Decreased Force: Proportion of [Fullscale Force] (P1).

* For more about [Fullscale Force], see the section below.

Fullscale Force

For the four models indicated below, full scale is given as full scale range.

For all other models, full scale is given as the load cell capacity.

Models with Range as Full Scale	AG-IS
	EZGRAPH
	MST-I
	AG-I
Models with Load Cell Capacity as Full Scale	All models other than the four above.

Type

Selected in the [Full Scale] list box in the [Sensor] dialog box.

Default settings allow selecting from the following.

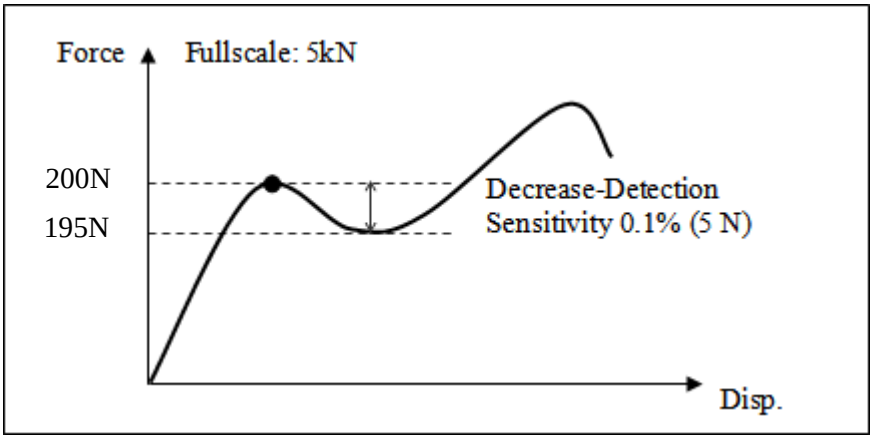
Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Example

Assume a load cell capacity (full scale) of 5 kN and decrease-detection sensitivity P1 of 0.1% (5 N).

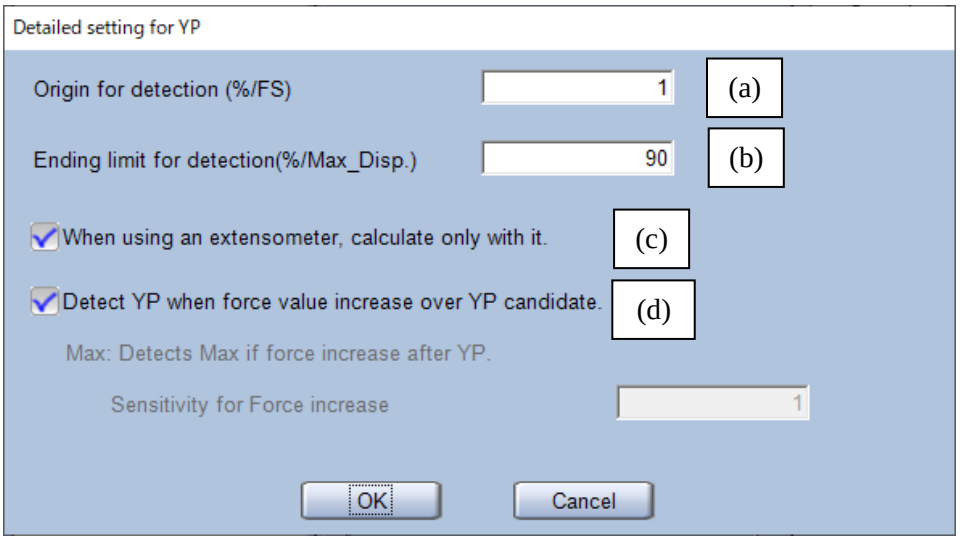
Force Decrease = 200 N – 195 N = 5 N

Force Decrease ≥ Decrease-Detection Sensitivity 0.1% (5 N)



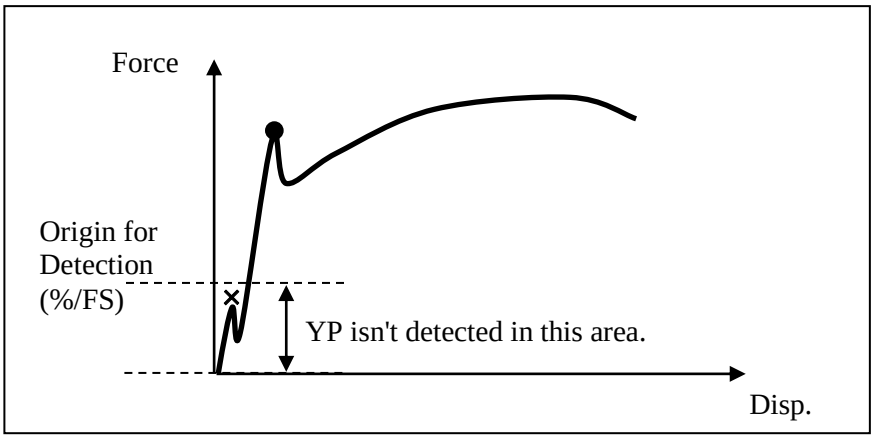
Details

When [Detail] button is clicked, this dialog box appears.



(a) Origin for detection (%/FS)

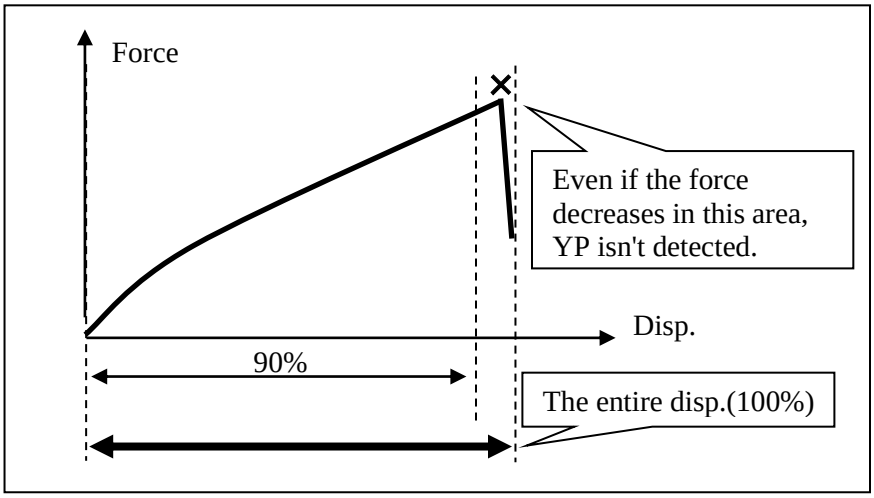
Specify the sensitivity to use for detecting YP, where the decrease in force is specified in terms of a percent of full scale range.



(b) Ending limit for detection (%/Max_Disp.)

Specify the area to use for detecting YP, where the decrease in Disp. is specified in terms of a percent of [Max_Disp.].

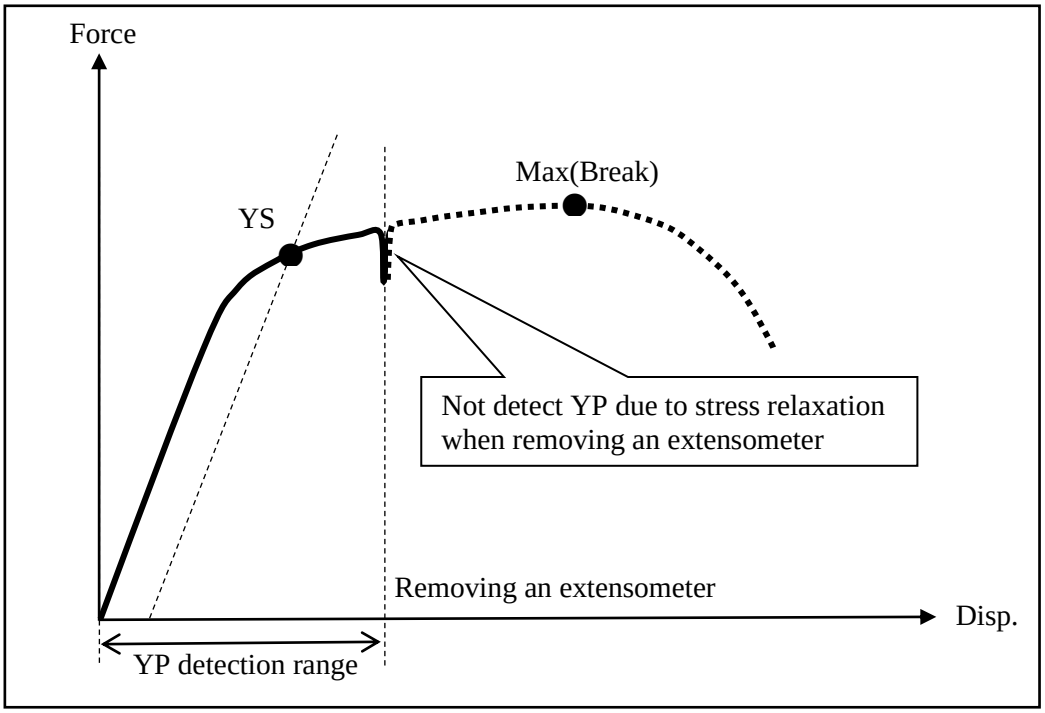
For example, if this parameter is 90%, YP isn't detected in the following area.



(c)When using an extensometer, calculate only with an extensometer

Check ON

To detect YP, only the extensometer data is used. For specimens not exhibiting yield point phenomena, not detect a decrease in test force as “YP” due to stress relaxation when removing the extensometer.



Check OFF

YP is detected when the specified amount of test force drop is detected regardless of whether the extensometer is removed or not.

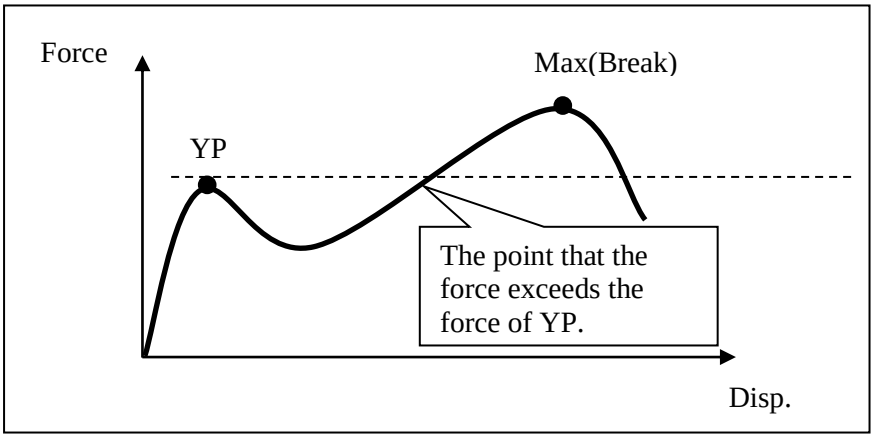
When not using an extensometer

Whether this setting is ON or OFF, YP detection is always calculated over the entire range.

(d) Detect YP when force value increase over YP candidate.

Check ON

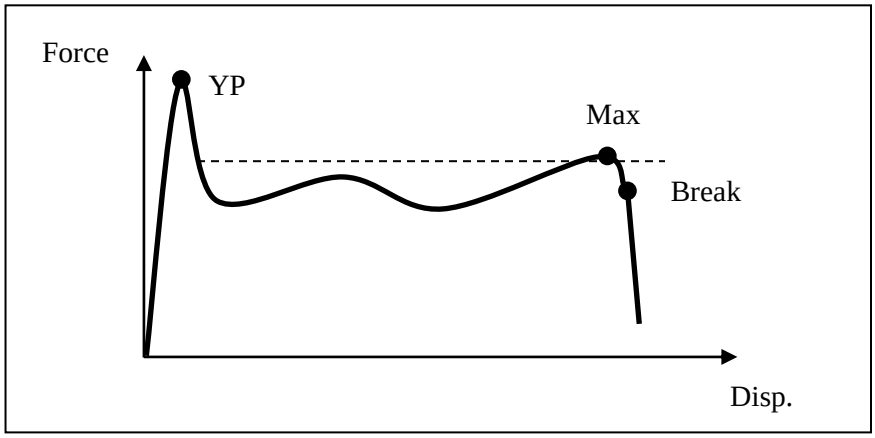
Only when the force exceeds the force of YP, the point is detected as YP.



Check OFF

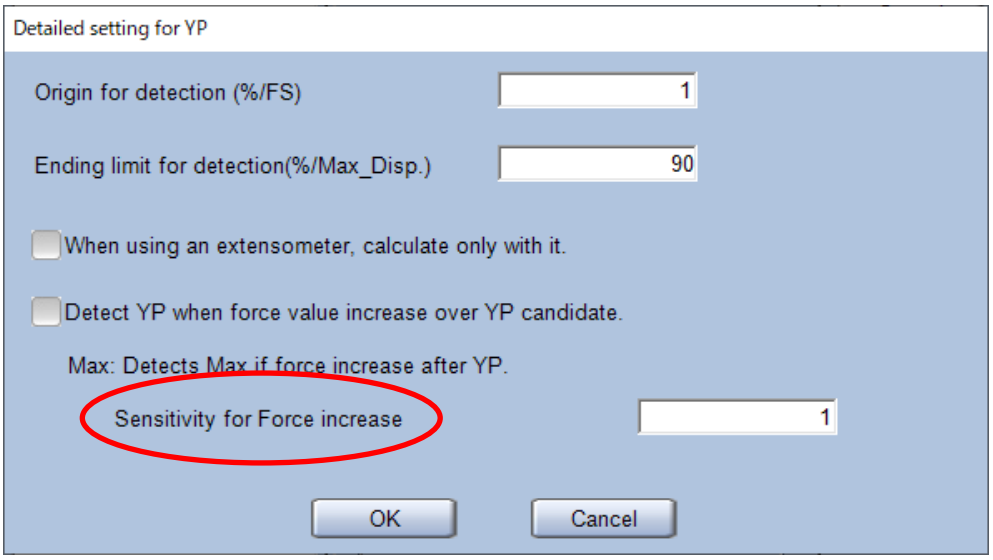
Even if the force doesn't exceed the force of YP, YP is detected.

[Max] is detected in the area where Disp. is larger than YP.

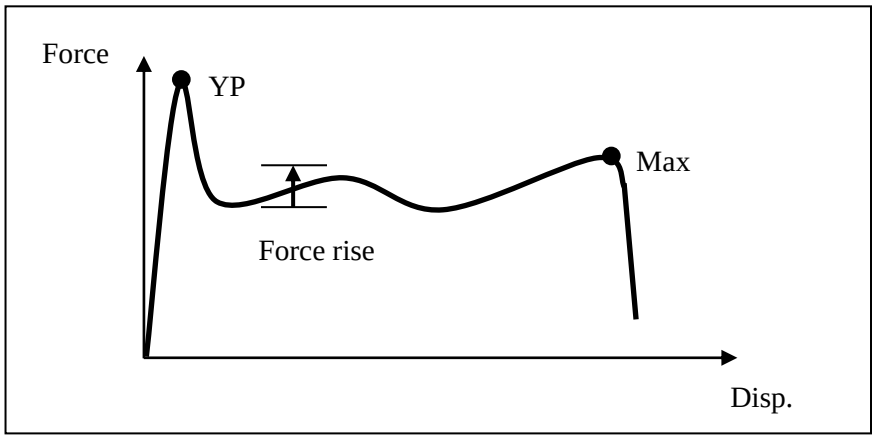


Sensitivity for Force increase

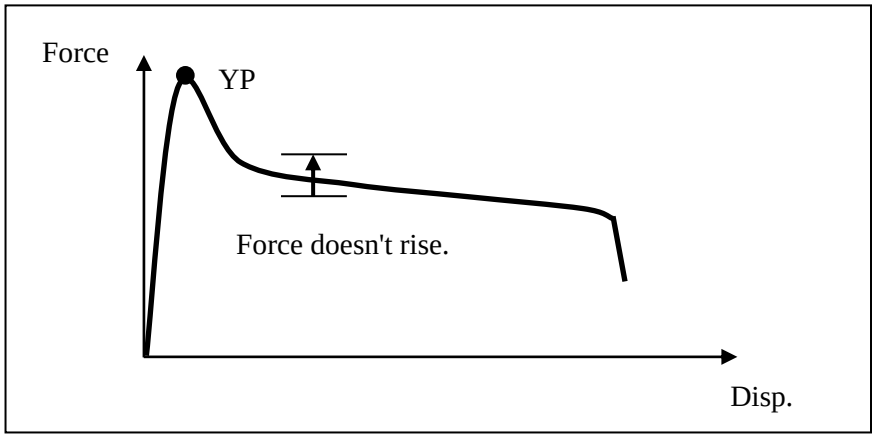
When there is no increase in the force after YP, YP is detected by the [Sensitivity for Force increase] setting.



Whether force rises is checked in the area where Disp. is larger than YP.



When the force doesn't increase, [MAX] is not detected.

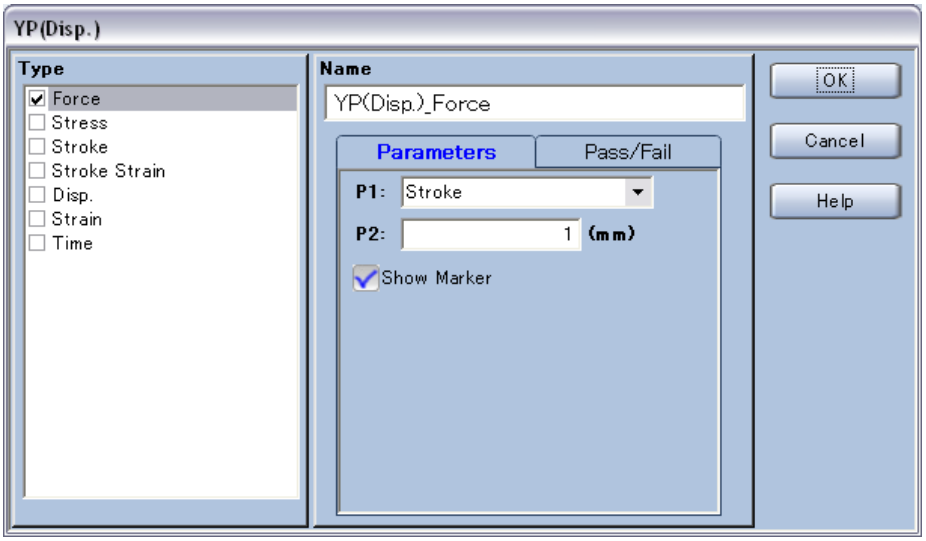


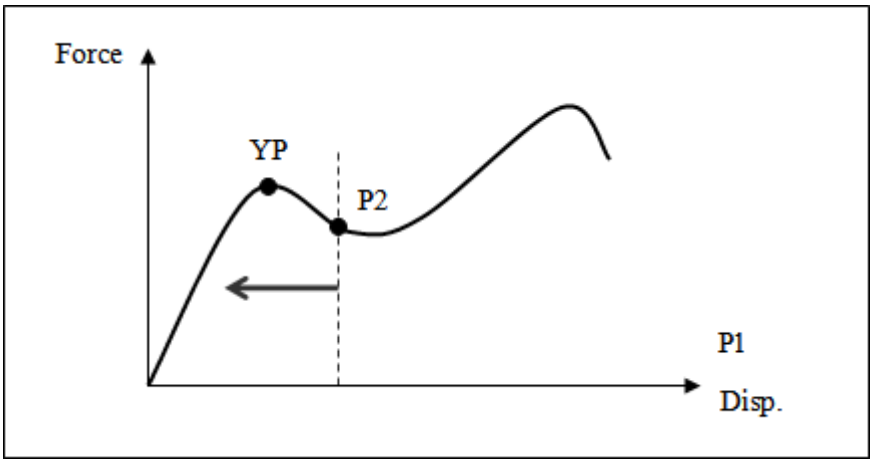
Upper Yield Point (Disp.)

Single Control Mark

Determines the upper yield point as the maximum [Force] occurring up to the specified calculation point.

Parameters





- Sensor Type Used to Indicate Calculation Point (P1)

Stroke	Stroke (strain)
Disp.	Disp. (strain)

- Calculation Point (P2)

* Parameter values are in terms of units specified in the [System] window.

Type

Select from types specified via the [Channel] settings in the [Sensor] dialog box.

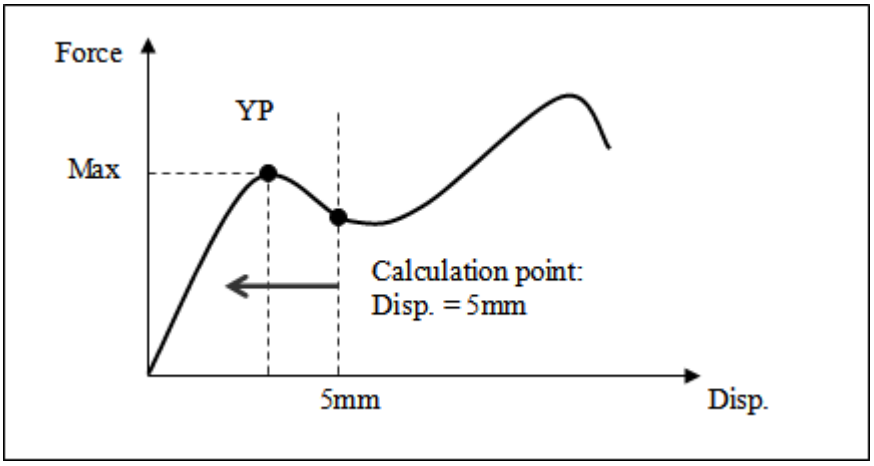
Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Example

Assume a P1 Sensor Type of [Stroke] and P2 Calculation Point of 5 mm.

The process finds the maximum [Force] occurring within the first 5 mm of [Disp.], as shown below.

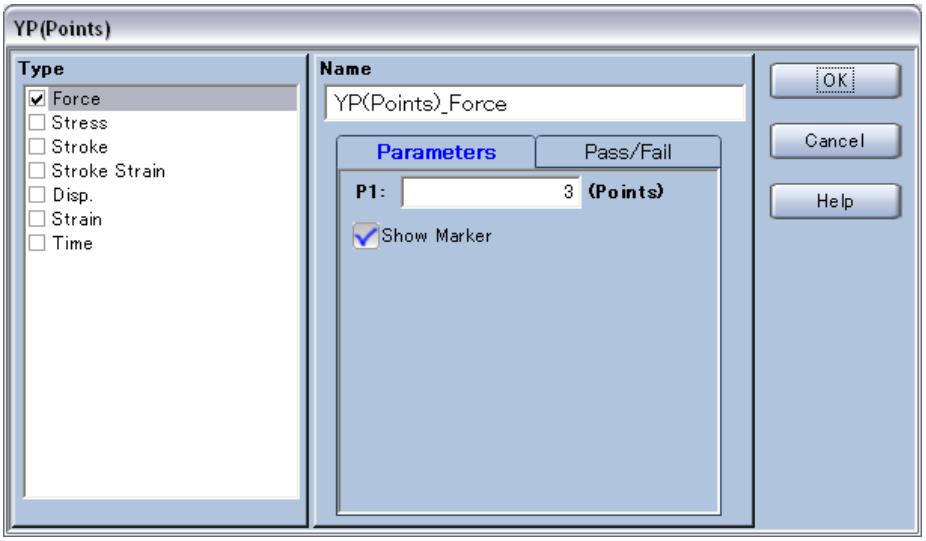


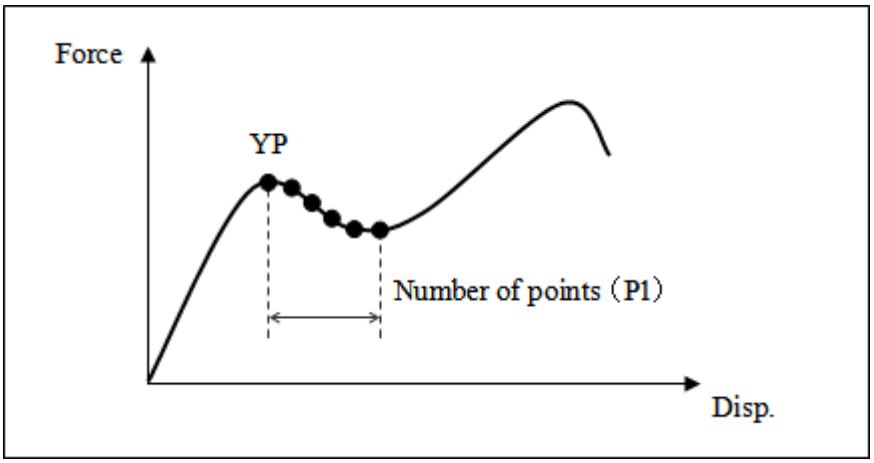
Upper Yield Point (Points)

Single Control Mark

Determines the upper yield point as the first of the points where the [Force] decreases or stays the same for the specified number of points.

Parameters





- Number of consecutive points where the [Force] decreases or stays the same (P1)

Type

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

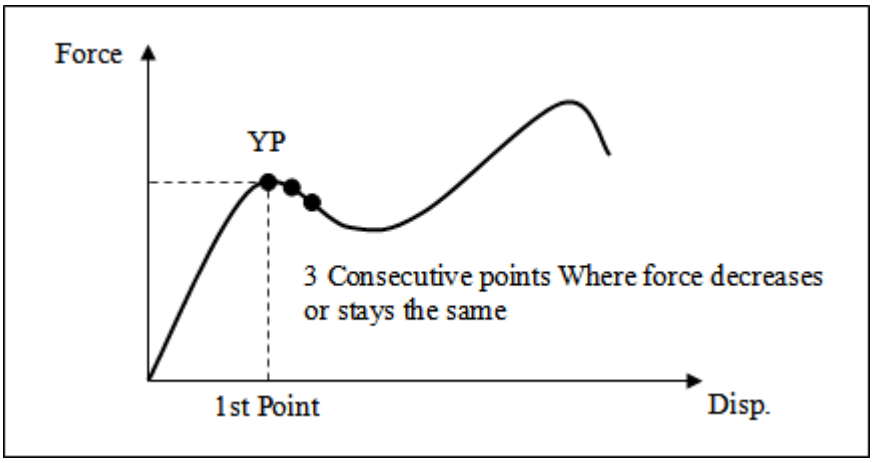
Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Example

Assume P1, the number of consecutive points, is set to 3.

This process finds the first of three consecutive points where the [Force] decreases or stays the same.



Upper Yield Point (%YP)

Single Control Mark

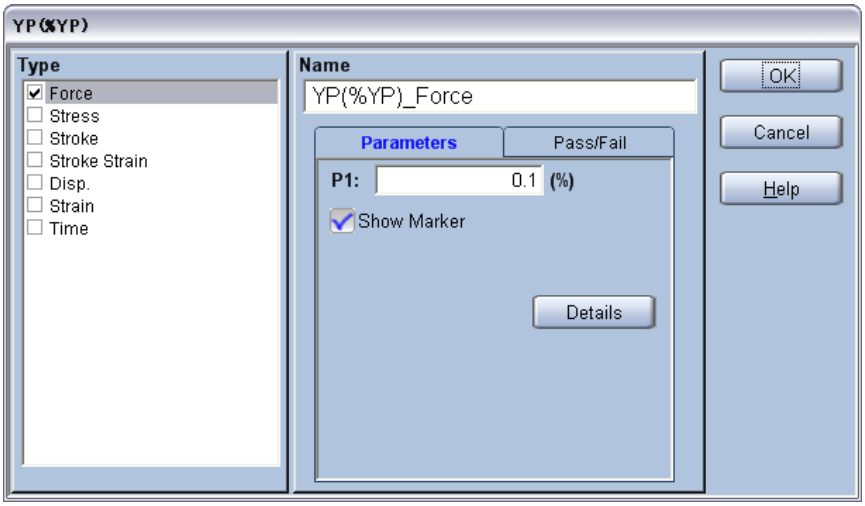
Determines the upper yield point by detecting how much the [Force] changes.

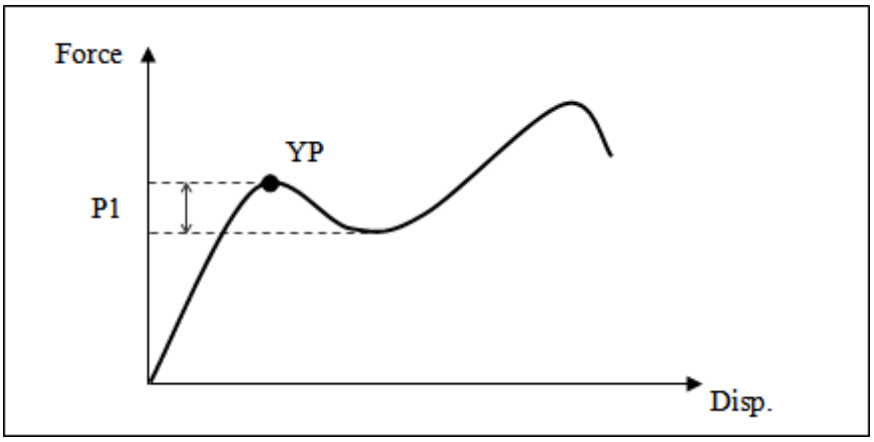
The upper yield point is detected as the first point within a given amount of change, where the [Force] first decreases after starting the test. However, any decreases that occur within the first 1% of [Fullscale Force] will not be considered the upper yield point.

* For more about [Fullscale Force], see the section below.

If the “Force” become larger than “YP” within 0.05%(Strain) after detecting “YP”. This point is not registered to the “YP”.

Parameters





Sensitivity of Detecting Decreased Force: Proportion of the candidate of YP

which is peak force from the start point to the current position (P1).

Fullscale Force

For the four models indicated below, full scale is given as full scale range.

For all other models, full scale is given as the load cell capacity.

Models with Range as Full Scale	AG-IS
	EZGRAPH
	MST-I
	AG-I
Models with Load Cell Capacity as Full Scale	All models other than the four above.

Type

Select from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

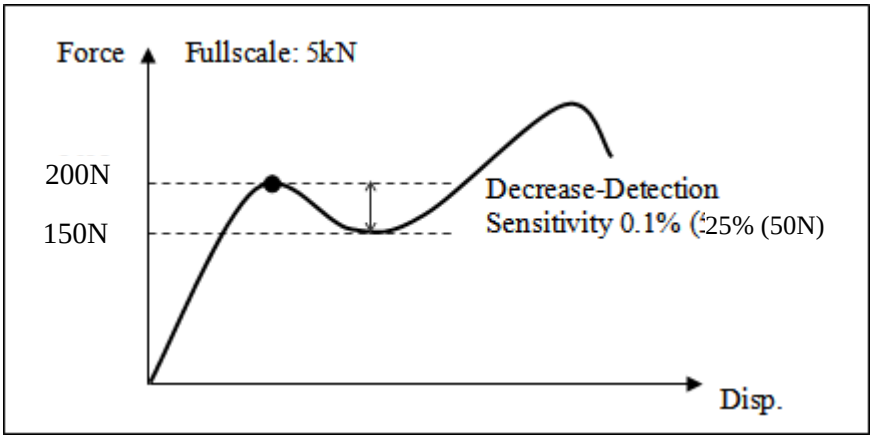
Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Example

Assume a load cell capacity (full scale) of 5 kN, The candidate of YP: 200N, and decrease-detection sensitivity P1 of 25% (50 N).

Force Decrease = 200 N – 150 N = 50 N

Force Decrease ≥ Decrease-Detection Sensitivity 25% (50 N)



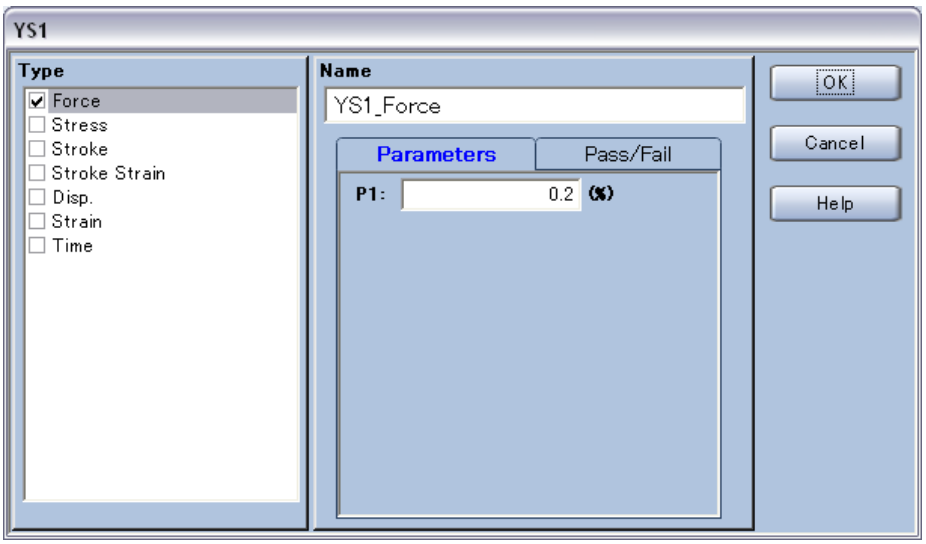
※For the details, see the description of “Upper Yield Point (%FS)”

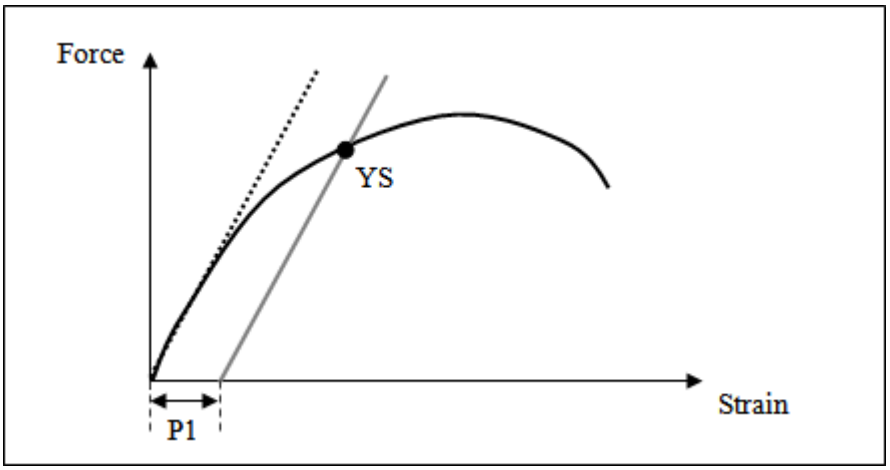
Yield Strength (1 and 2)

Single Control

Determines the strength as the point where a straight line, with the average slope calculated for the elastic and shifted horizontally by a given offset, intersects the graph. Up to two yield strength points can be specified. If the type of elastic is “Elastic Loop”, the horizontal offset is shifted from the displacement origin.

Parameters





- Proportion of Specimen Gauge Length (P1): Offset

Type

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

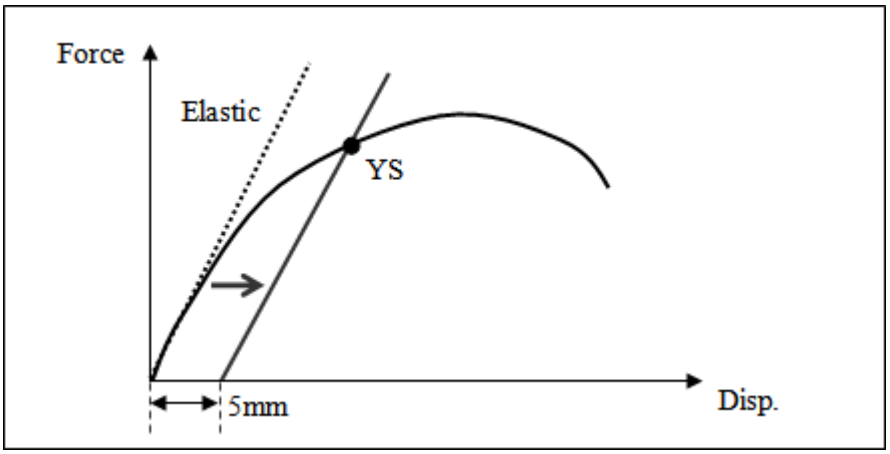
Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Example

Assume offset P1 is set to 10% and gauge length to 50 mm.

Disp.: 50 mm x 10% = 5 mm.



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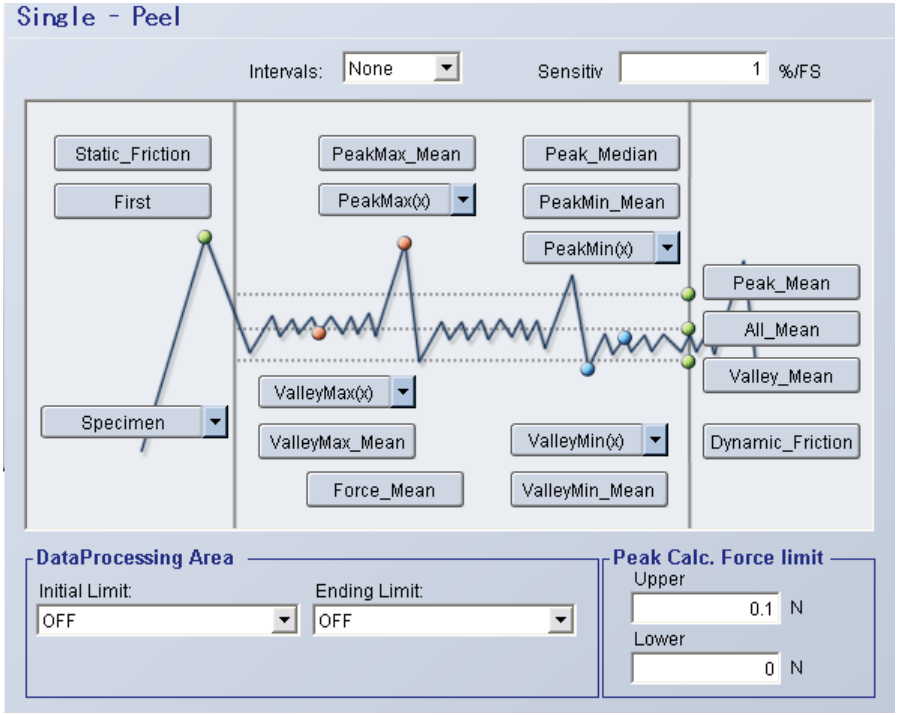
Chapter 2
DataProcessing Items in Single Mode Peel Test

Data Processing Range for Peel Tests

For peel tests, multiple data processing areas can be divided into multiple intervals. Data processing processes will differ depending on the number of intervals.

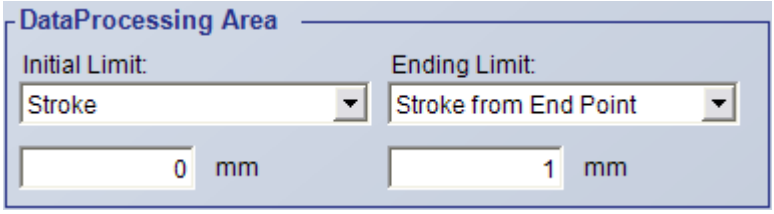


Sensitivity

Specify the sensitivity to use for detecting peaks, where the decrease in force is specified in terms of a percent of full scale range.

Data Processing Area

The area for calculating peaks is specified in terms of a starting and ending point.



To specify the starting point (= Initial Limit), select from the following.

- OFF (starting point is first data point)

- First (Peak)

- Stroke

To specify the ending point (= Ending Limit), select from the following.

- OFF (ending point is last data point)

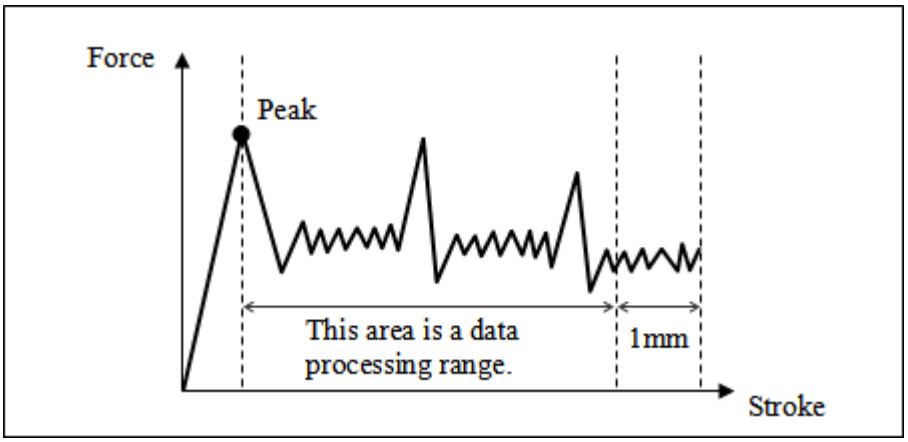
- Stroke from End point

- Stroke from Start point

- Last Peak

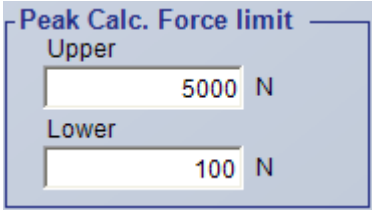
Example

Assume the starting point (= Initial Limit) is a peak (= [First]) and the ending point (= Ending Limit) is a 1 mm stroke from the end point.



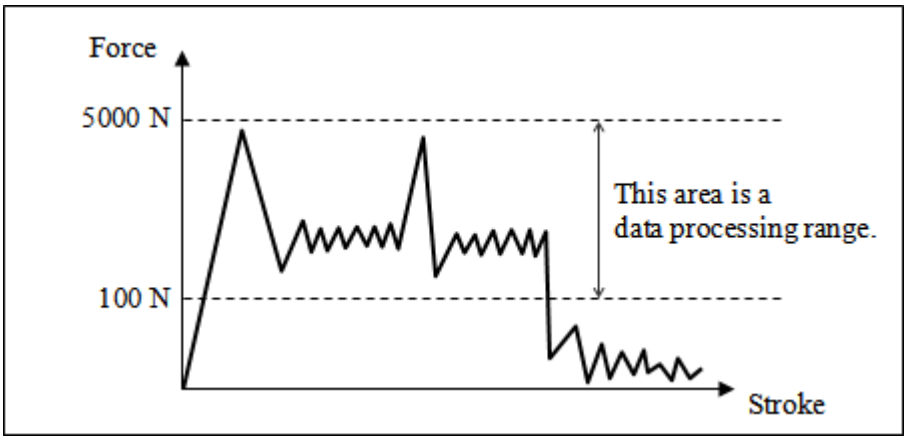
Excluding Force Range for Peak Calculation

A range of [Force] can be excluded from peak calculations by specifying the upper and lower limits.



Example

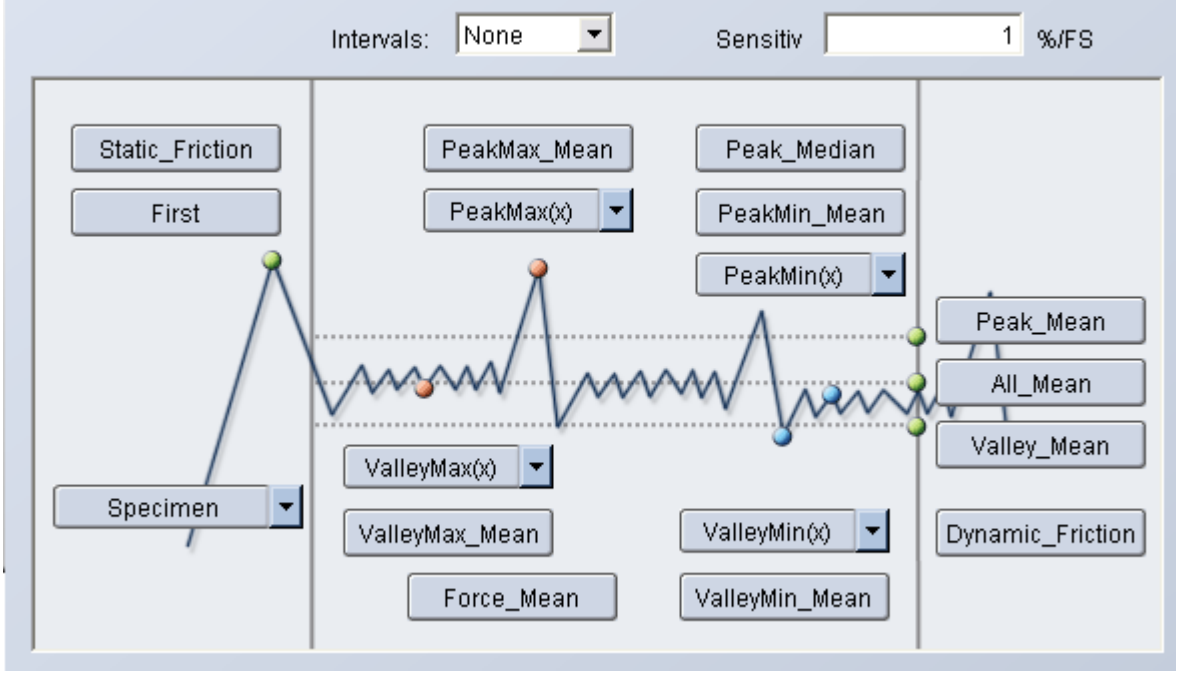
Assume the upper limit is 5000 N and the lower limit is 100 N.



No Intervals

If data processing areas are not divided into intervals, the following processes are available for selection.

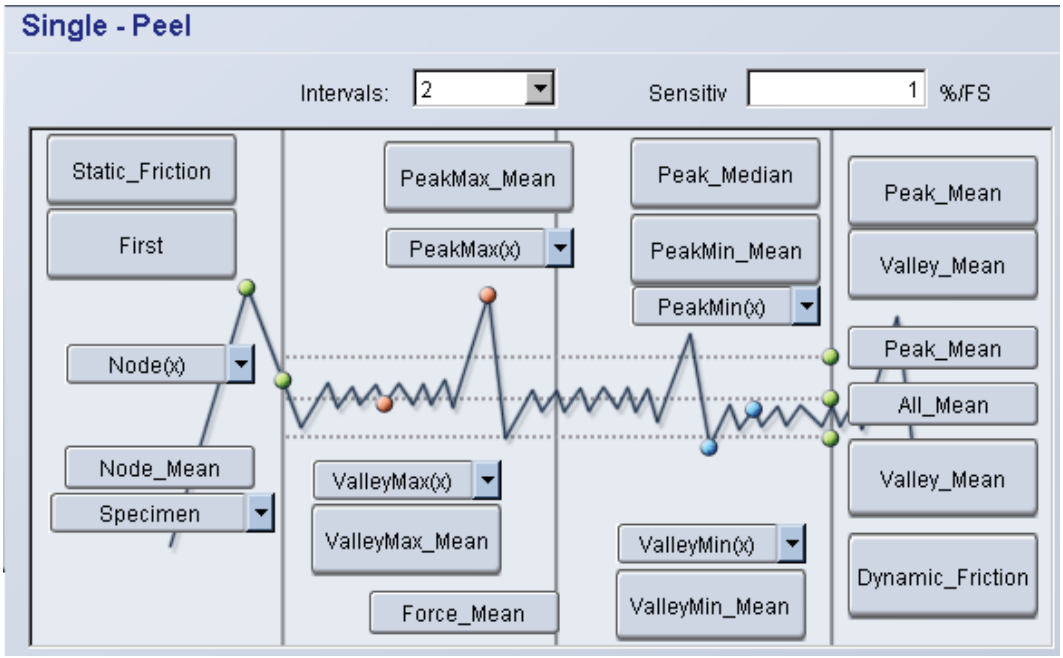
- Force_Mean
- Static_Friction
- All_Mean
- Dynamic_Friction
- PeakMax 1 to 10
- PeakMax_Mean
- PeakMin 1 to 10
- PeakMin_Mean
- Peak_Median
- Peak_Mean
- First
- ValleyMax 1 to 10
- ValleyMax_Mean
- ValleyMin 1 to 10
- ValleyMin_Mean
- Valley_Mean



With Intervals

If data processing areas are divided into 2 or more intervals, the following processes are available for selection.

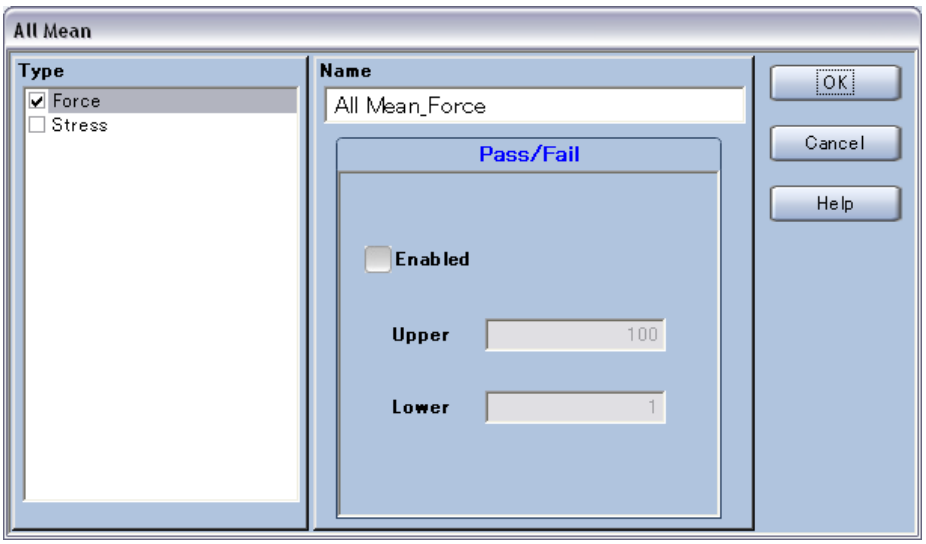
- Node (x)
- Node_Mean
- Force_Mean
- Static_Friction
- All_Mean
- Dynamic_Friction
- PeakMax 1 to 10
- PeakMax_Mean
- PeakMin 1 to 10
- PeakMin_Mean
- Peak_Median
- Peak_Mean
- Peak_Mean (All)
- First
- ValleyMax 1 to 10
- ValleyMax_Mean
- ValleyMin 1 to 10
- ValleyMin_Mean
- Valley_Mean
- Valley_Mean (All)

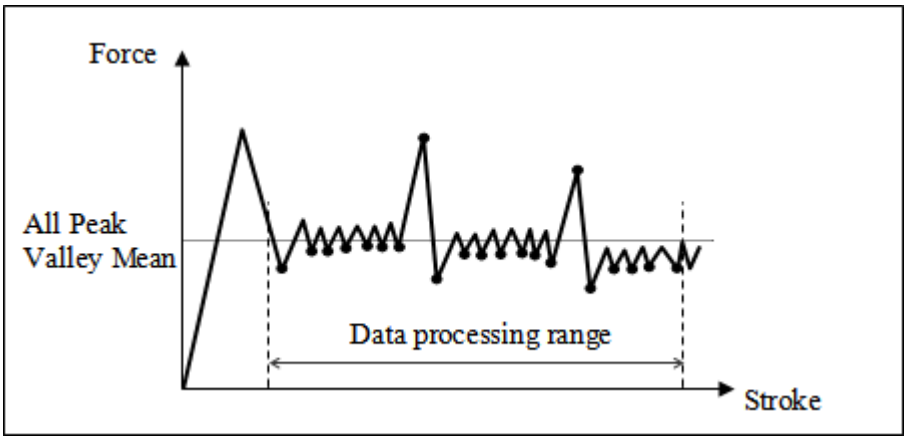


All Mean

Determines the mean for all peaks within the data processing area.

Parameters





- None

Type

- Force

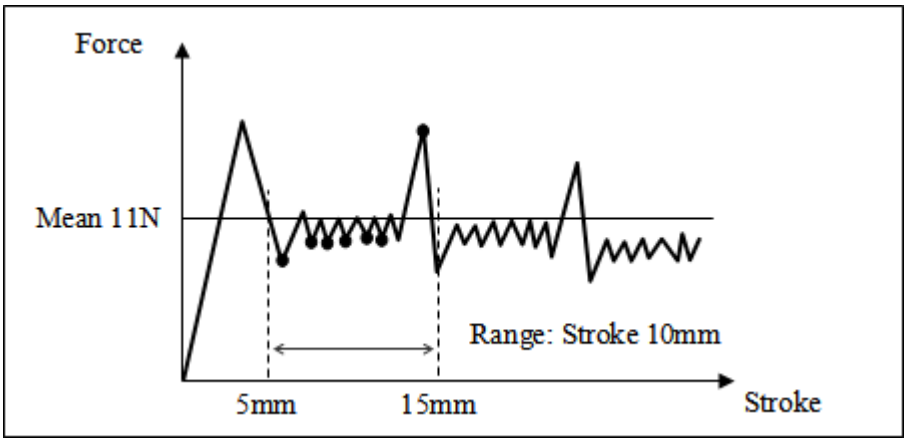
Determines the mean [Force] of all peaks.

- Stress

Determines the mean stress of all peaks.

Example

Assume the starting point of the data processing area is after a 5 mm stroke, the ending point is a 10 mm stroke from the starting point, and the area include one peak with a [Force] of 21 N and six valleys with forces 7 N, 9 N, 9 N, 10 N, 11 N, and 10N.

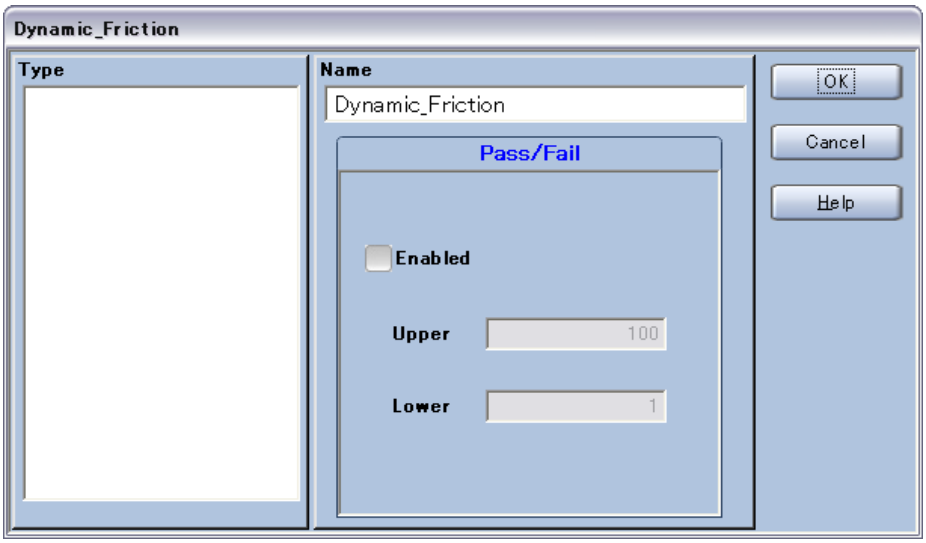


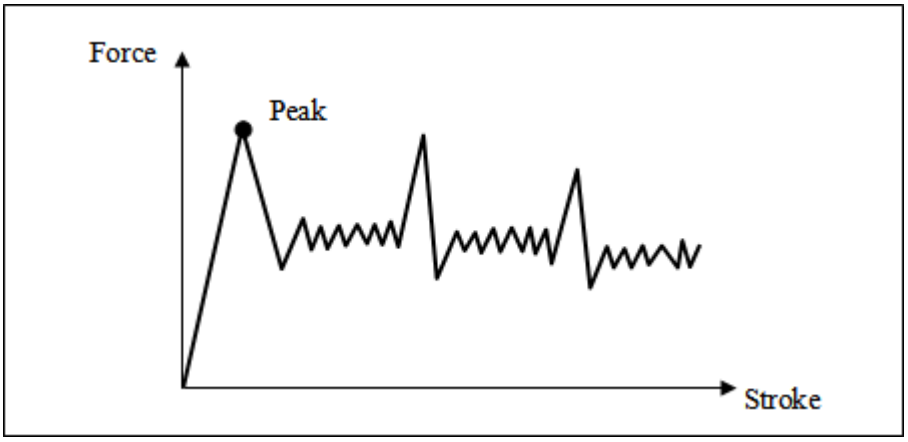
Dynamic Friction

Determines the coefficient of dynamic friction from the mean [Force] after the first peak and the mass of the moving weight.

$$\text{Dynamic Friction} = \frac{\text{Mean Force After First Peak}}{\text{Mass of Moving Weight}}$$

Parameters





- None

Type

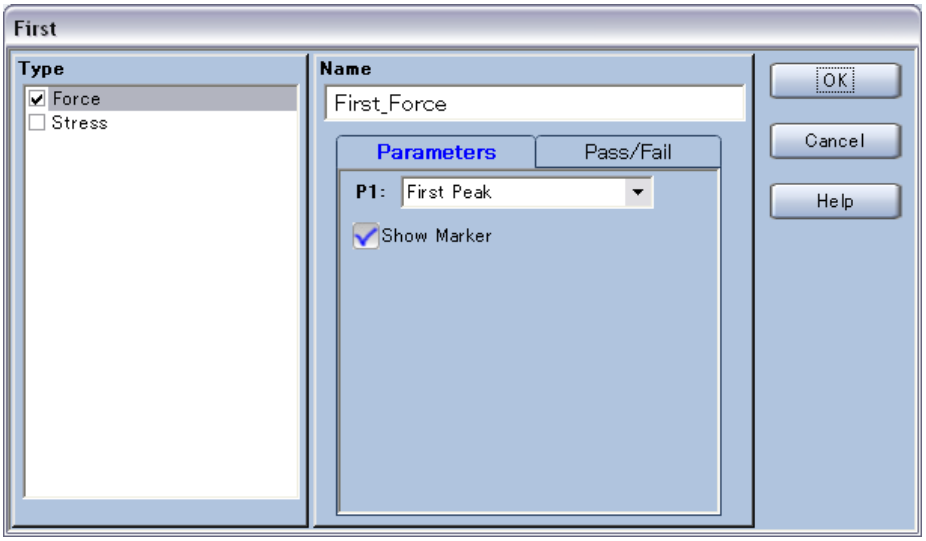
- None

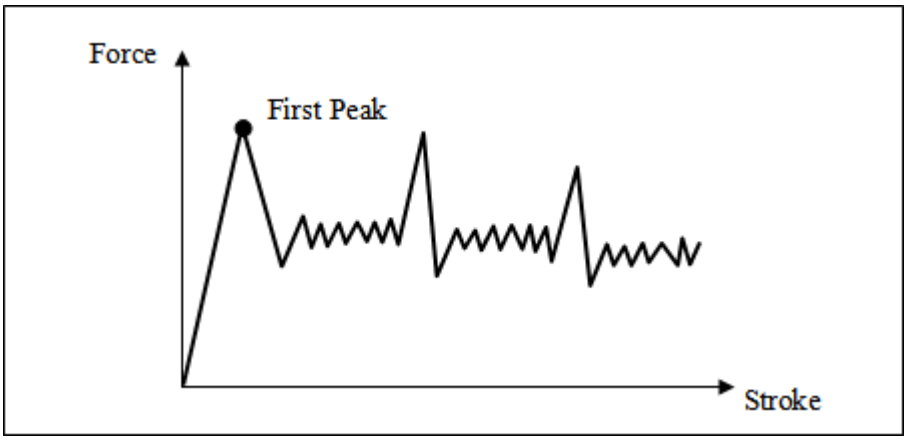
First

Mark

Determines the FirstPeak as the first peak to satisfy the specified parameter.

Parameters





- Parameters for First Peak (P1): Select either [First Peak] or [Max. Peak].

Type

- Force

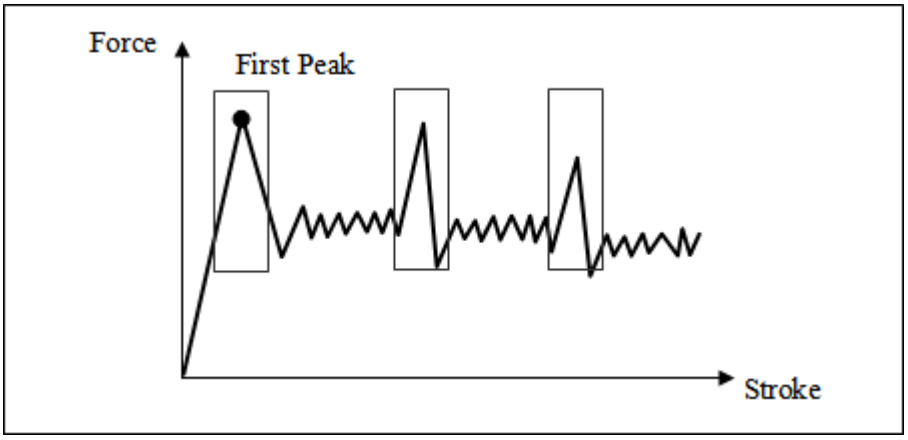
Determines the [Force] at the [First].

- Stress

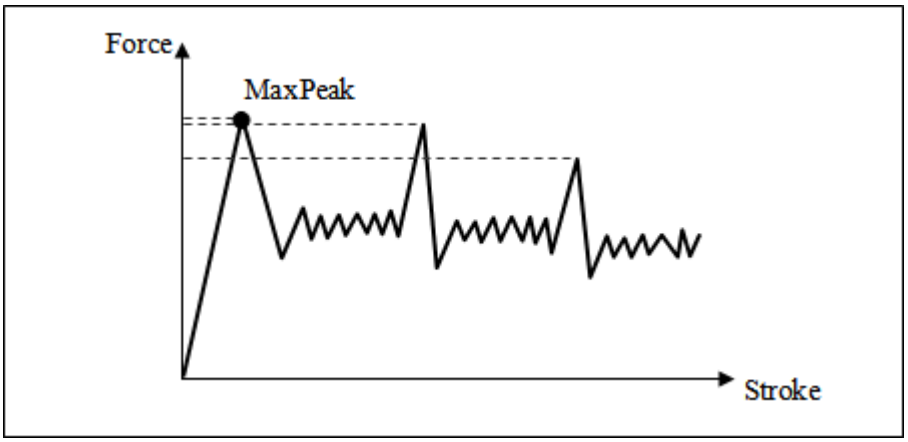
Determines the stress at the [First].

Example

Assume the parameter P1 is [First Peak].



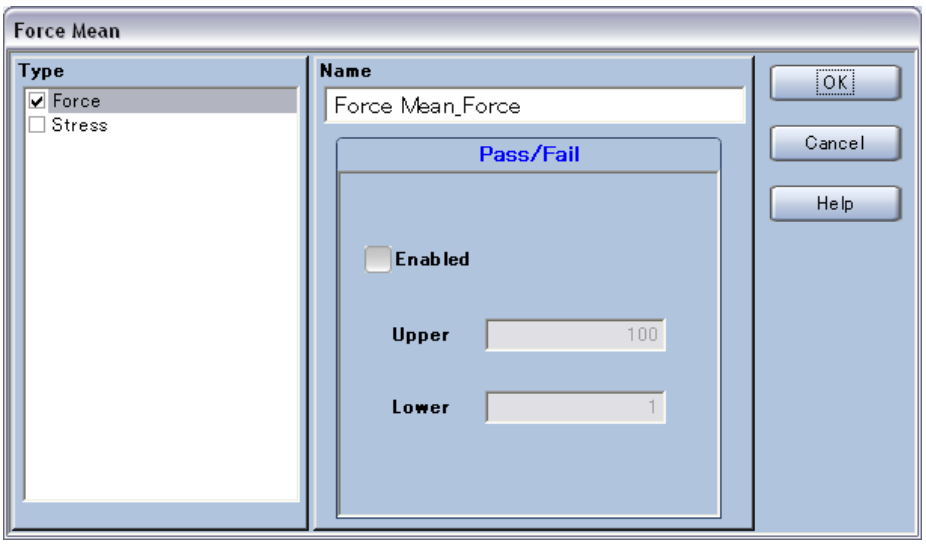
Assume the parameter P1 is [Max. Peak].

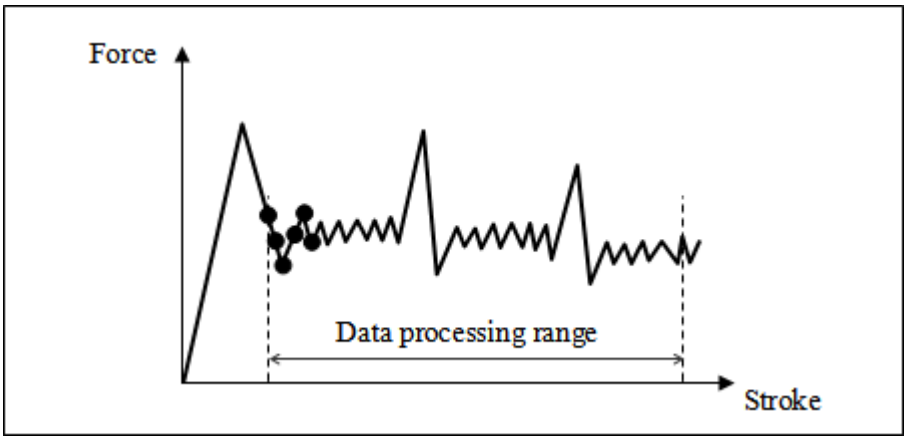


Force Mean

Determines the mean [Force] of all sampling data within the data processing area.

Parameters





- None

Type

- Force

Determines the mean [Force].

- Stress

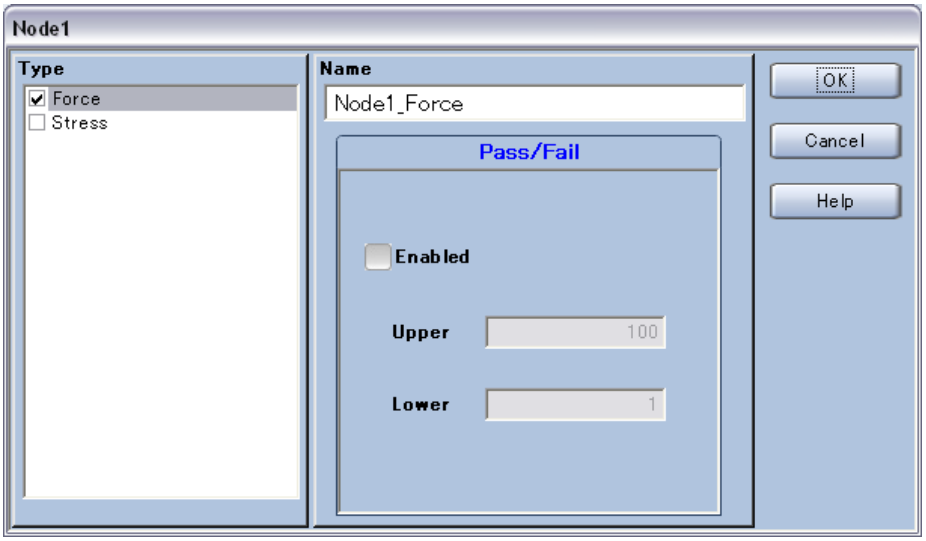
Determines the mean stress.

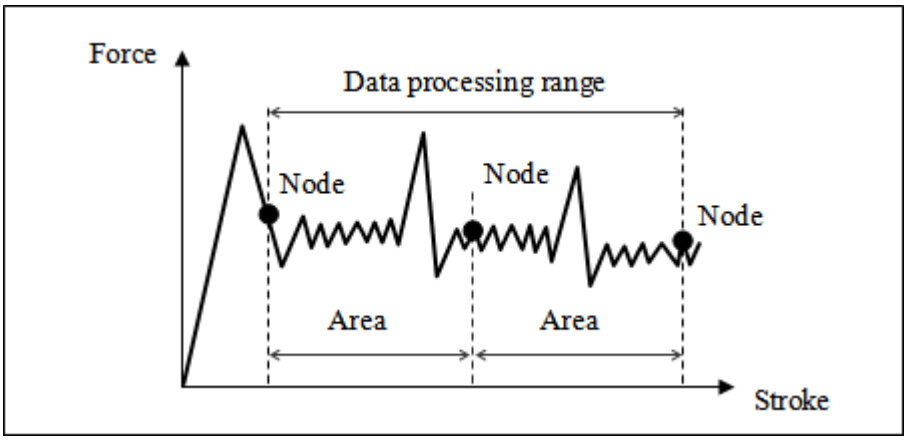
Node (x)

Mark

Determines the points where lines separating areas, when divided into intervals, intersect the graph.

Parameters





- None

Type

- Force

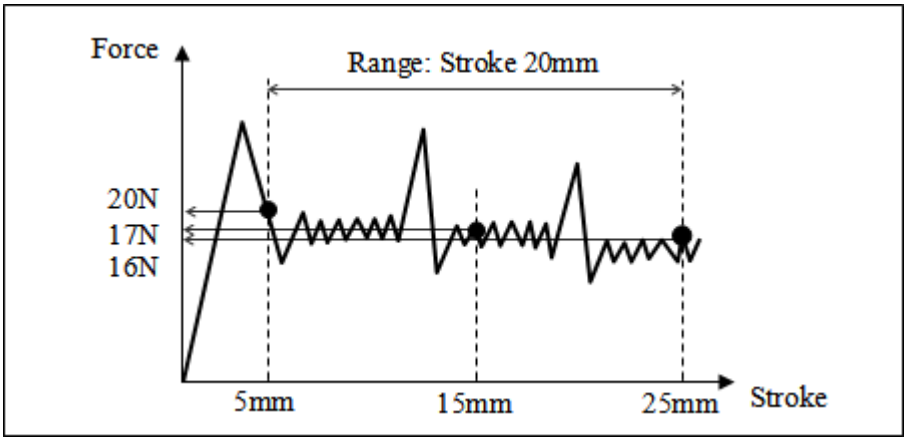
Determines the [Force] at nodes.

- Stress

Determines the stress at nodes.

Example

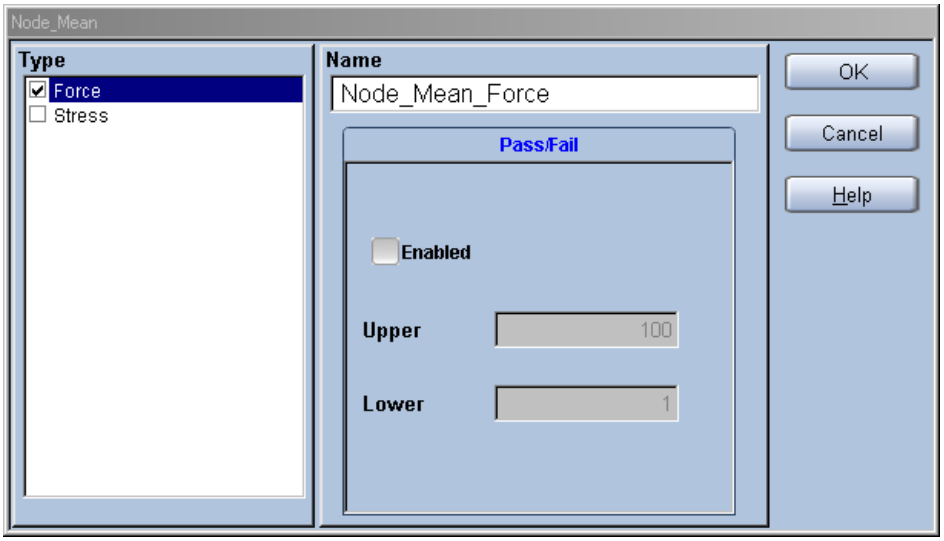
Assume the starting point of a data processing area is after a stroke of 5 mm and the ending point is a 20 mm stroke after the starting point.

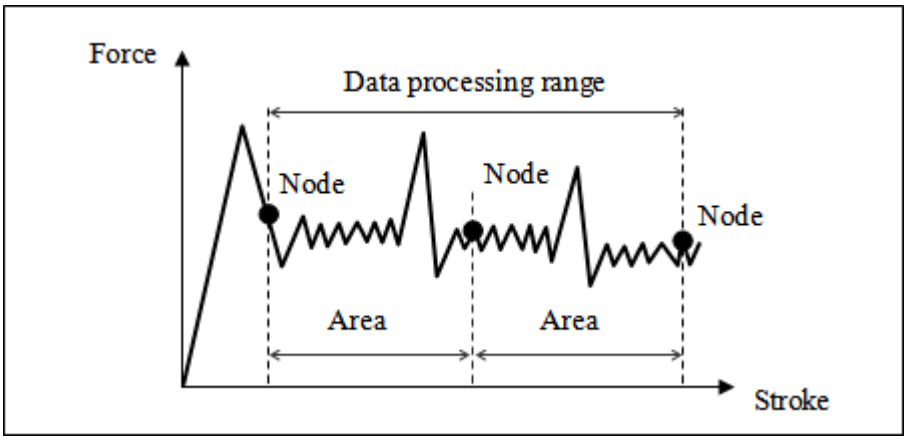


Node_Mean

Determines the average values of all “Node(X)”.

Parameters





- None

Type

- Force

Determines the [Force] .

- Stress

Determines the stress .

Notes:

The statistics value calculates as follows, that is different form standard algorithm

- Maximum : Get the maximum value of all “Node(x)”, it’s not the maxmum of “Node_mean”.
- Minimum : Get the minimum value of all “Node(x)”, it’s not the minimum of “Node_mean”.
- Average: Get the average value of all “Node(x)”, it’s not the average of “Node_mean”.

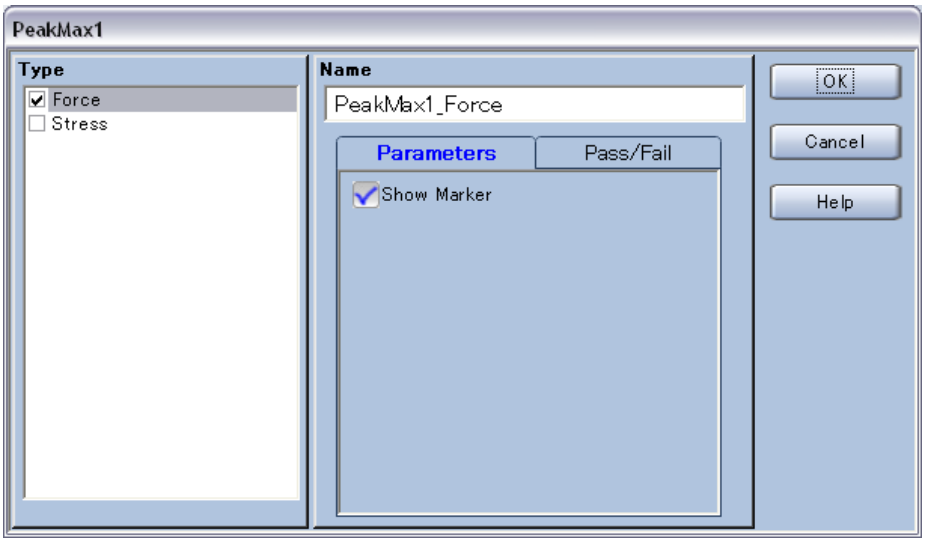
PeakMax (x)

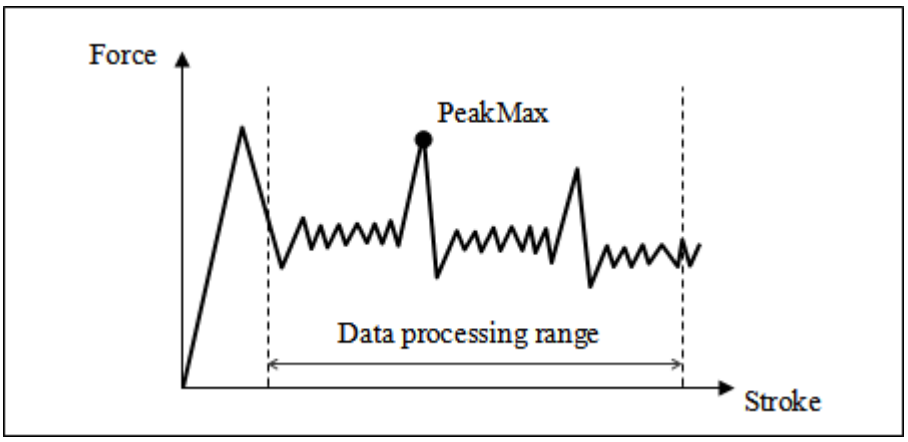
Mark

Determines the maximum peaks within the data processing area.

Up to 9 of [PeakMax] can be ranked, where the maximum peak is PeakMax1, the next maximum peak is PeakMax2, the next maximum peak is PeakMax3, and so on.

Parameters





- None

Type

- Force

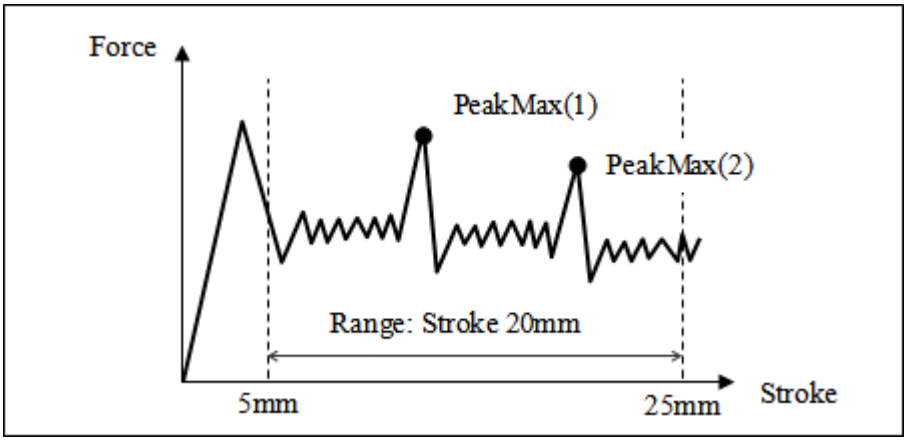
Determines the [Force] at the [PeakMax].

- Stress

Determines the stress at the [PeakMax].

Example

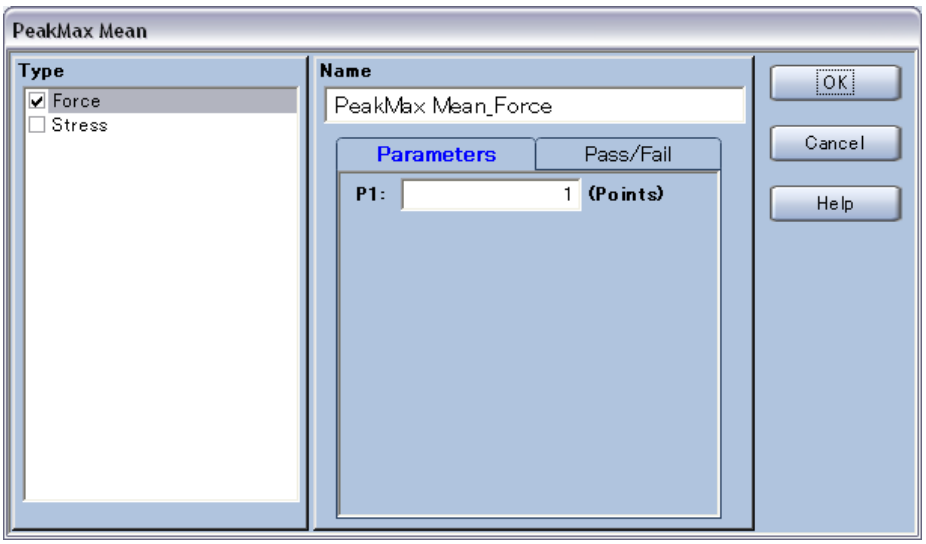
Assume the starting point of a data processing area is after a stroke of 5 mm and the ending point is a 20 mm stroke after the starting point.

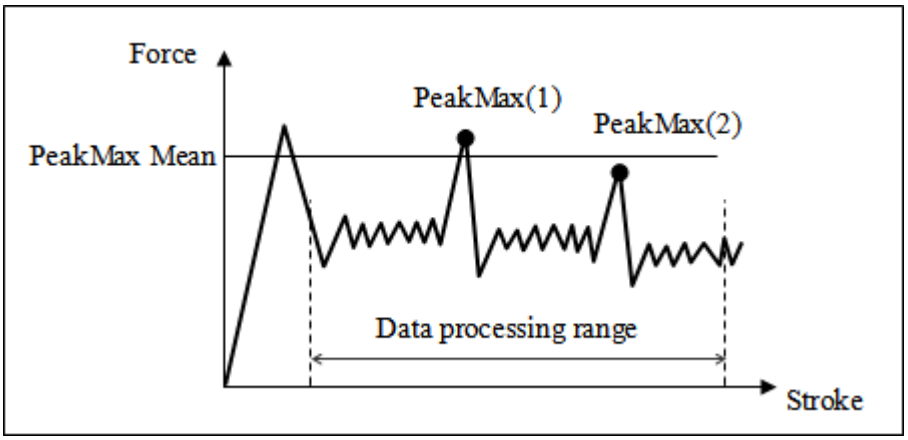


PeakMax Mean

Determines the mean of the specified number of the [PeakMax].

Parameters





- Number of PeakMax (P1)

Type

- Force

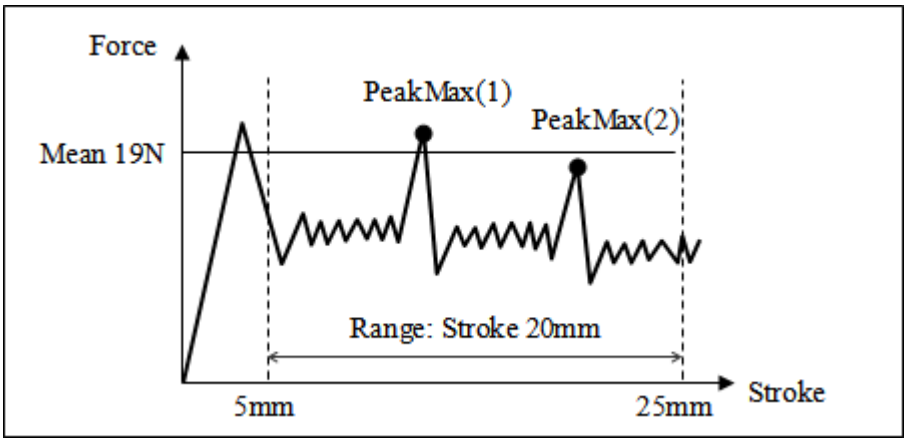
Determines the mean [Force] of the specified [PeakMax].

- Stress

Determines the mean stress of the specified [PeakMax].

Example

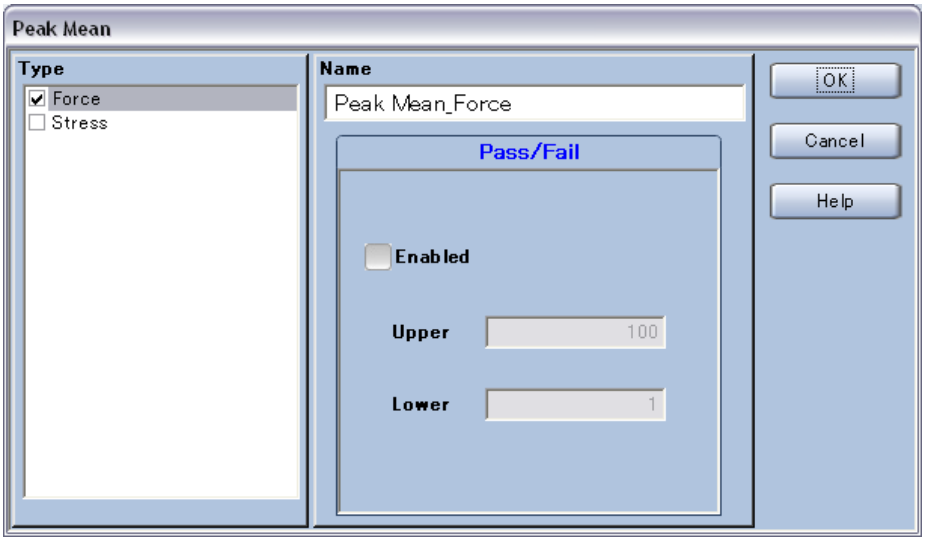
Assume the data processing area starts after a 5 mm stroke, ends a 20 mm stroke after the starting point, and the two [PeakMax] are to be averaged, where the [Force] at PeakMax1 is 20 N and PeakMax2 is 18 N.

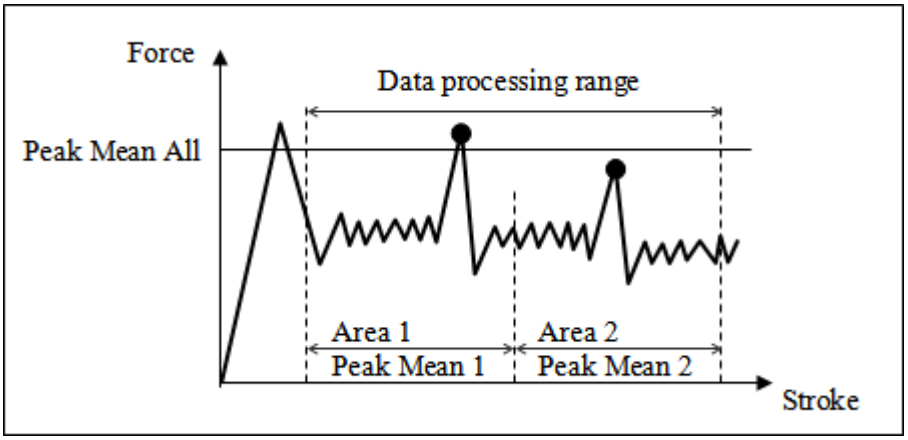


Peak Mean / Peak Mean All

Peak Mean finds the mean of all peaks within each area. Peak Mean All finds the mean of all peaks in all areas, when divided into intervals.

Parameters





- None

Type

- Force

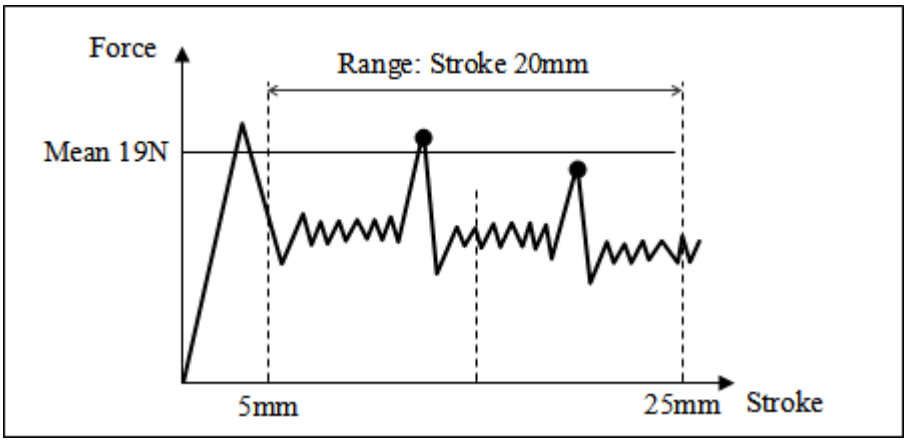
Determines the ratio of PeakMean to PeakMean All for [Force].

- Stress

Determines the ratio of PeakMean to PeakMean All for stress.

Example

Assume the data processing area starts after a 5 mm stroke, ends a 20 mm stroke after the starting point, and there is a 20 N peak in Area1 and an 18 N peak in Area2, where PeakMean All is to be determined.

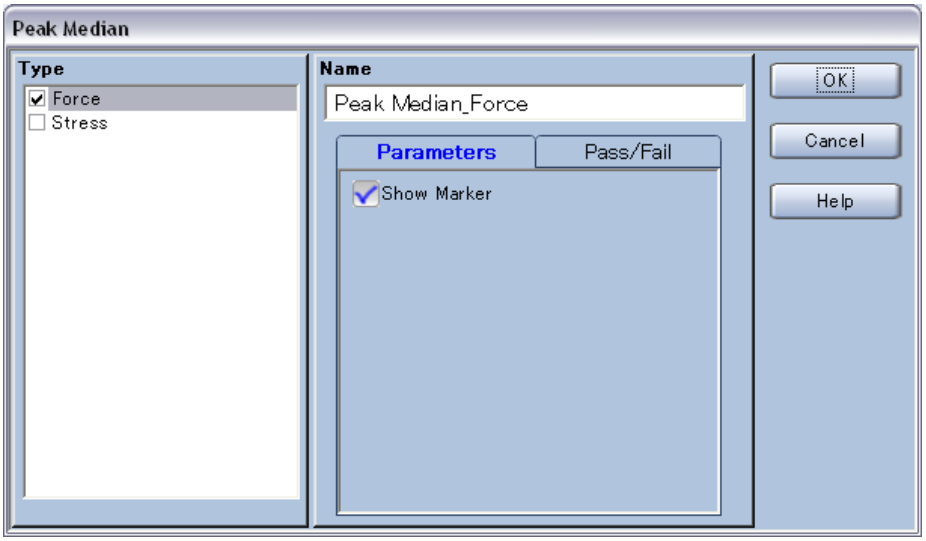


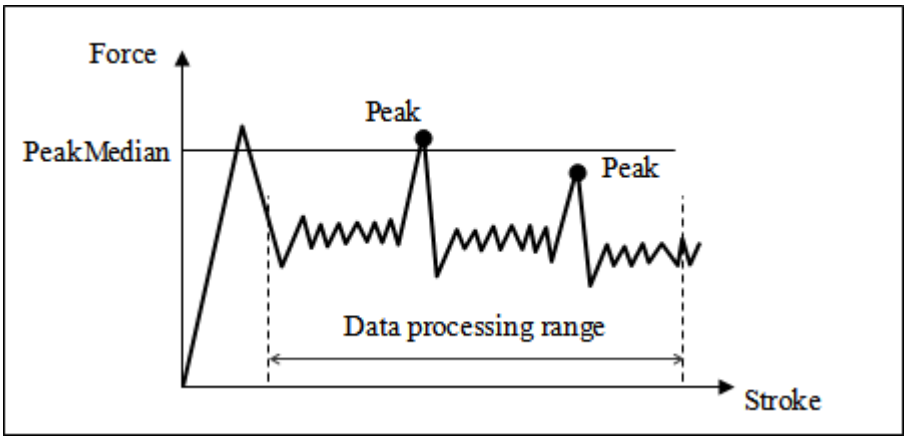
Peak Median

Mark

Determines the median peak from all peaks within the data processing area. If there are an even number of peaks, the middle two are averaged.

Parameters





- None

Type

- Force

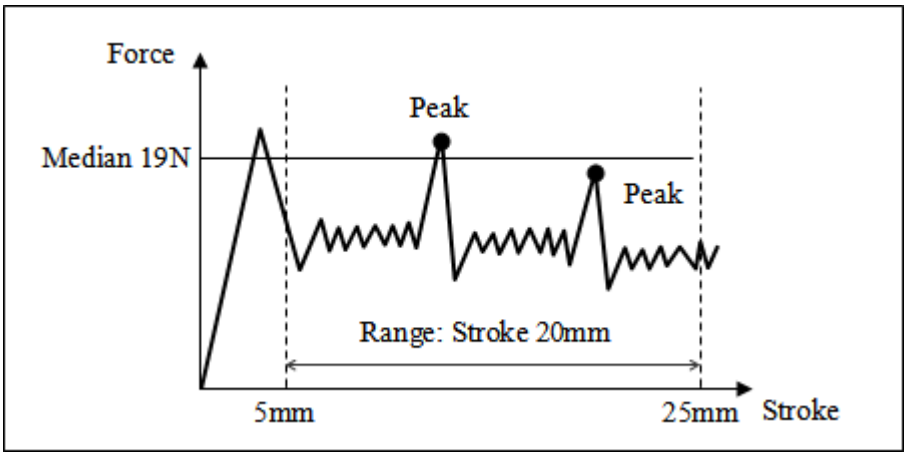
Determines the [Force] at the [Peak Median].

- Stress

Determines the stress at the [Peak Median].

Example

Assume the data processing area starts after a 5 mm stroke, ends a 20 mm stroke after the starting point, and there are two peaks (P1) within the processing area, where the [Force] of each peak is 18 N and 20 N, respectively.



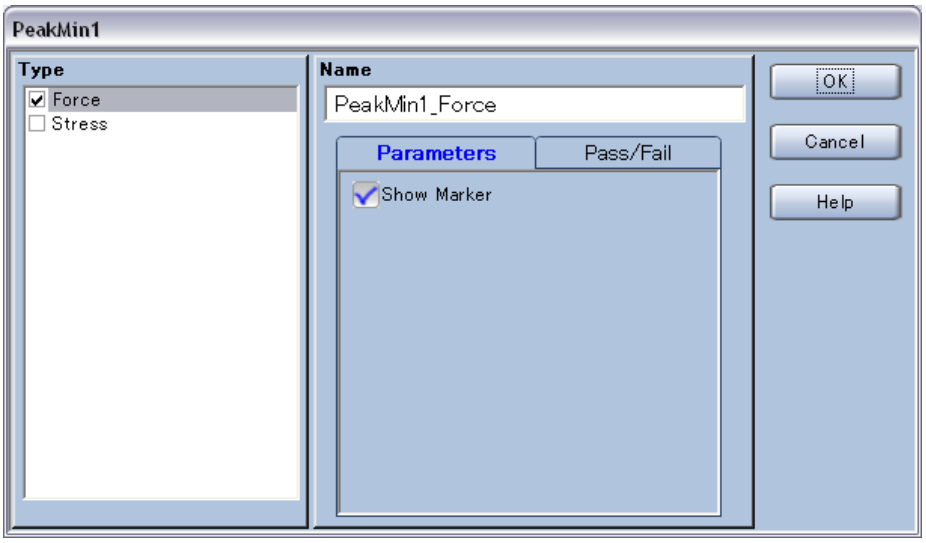
PeakMin (x)

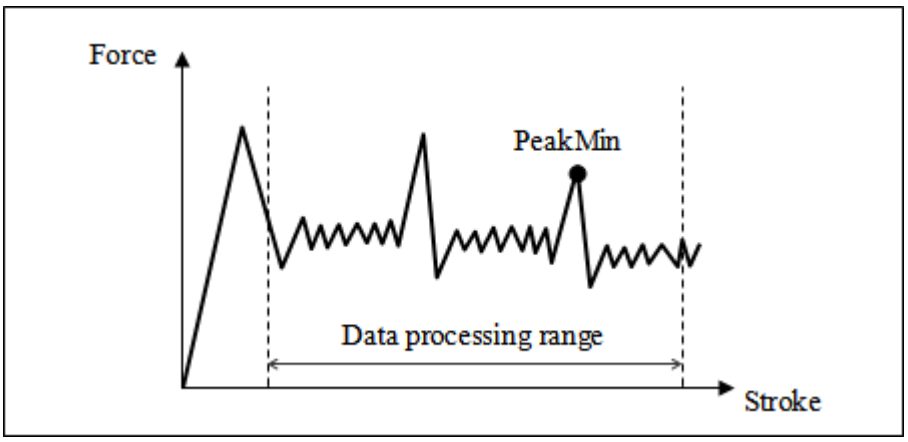
Mark

Determines the minimum peaks within the data processing area.

Up to 9 of the [PeakMin] can be ranked, where the minimum peak is PeakMin1, the next minimum peak is PeakMin2, the next minimum peak is PeakMin3, and so on.

Parameters





- None

Type

- Force

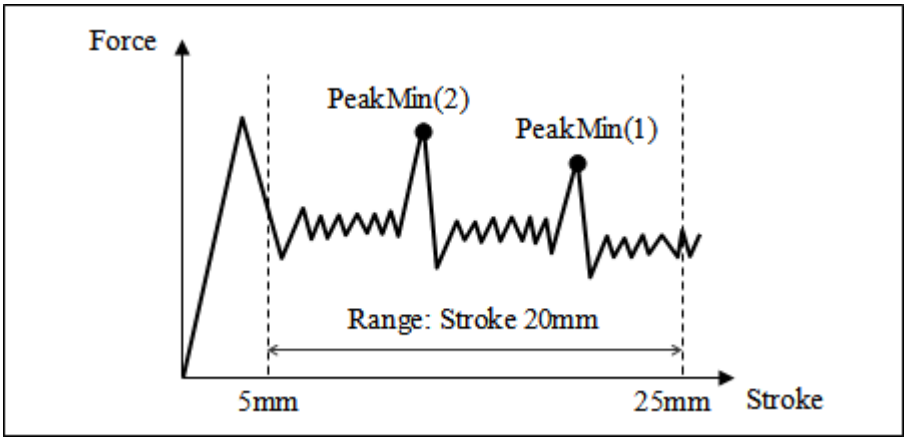
Determines the [Force] at the [PeakMin].

- Stress

Determines the stress at the [PeakMin].

Example

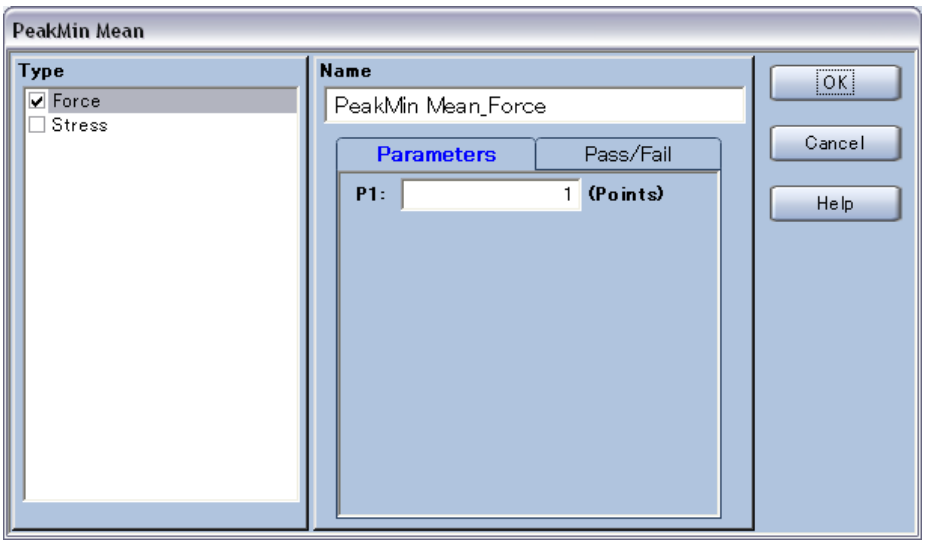
Assume the starting point of a data processing area is after a stroke of 5 mm and the ending point is a 20 mm stroke after the starting point.

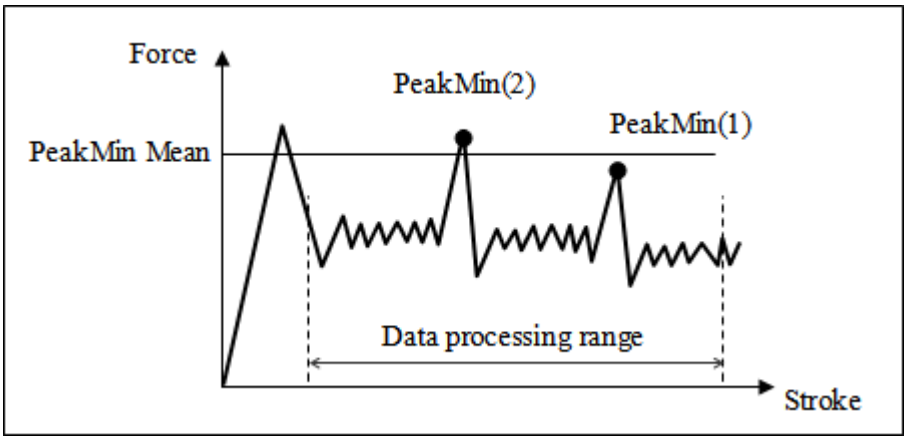


PeakMin Mean

Determines the mean of the specified number of the [PeakMin].

Parameters





- Number of PeakMin (P1)

Type

- Force

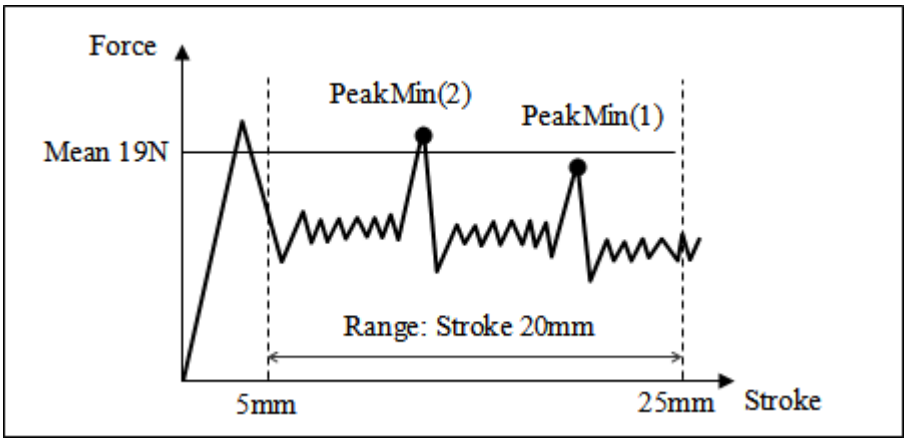
Determines the [Force] at the [PeakMin].

- Stress

Determines the stress at the [PeakMin].

Example

Assume the data processing area starts after a 5 mm stroke, ends a 20 mm stroke after the starting point, and the two [PeakMin] (P1) are to be averaged, where the [Force] at PeakMin1 is 18 N and PeakMin2 is 20 N.

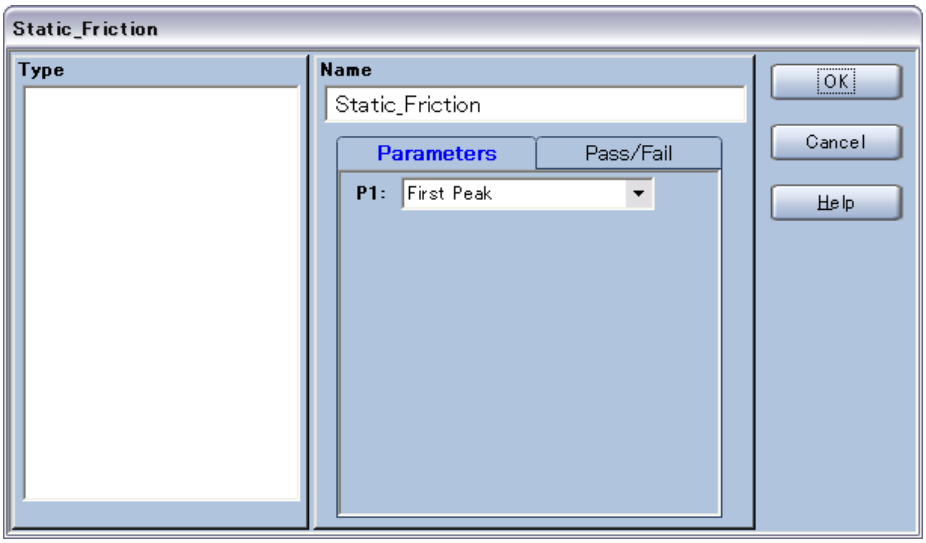


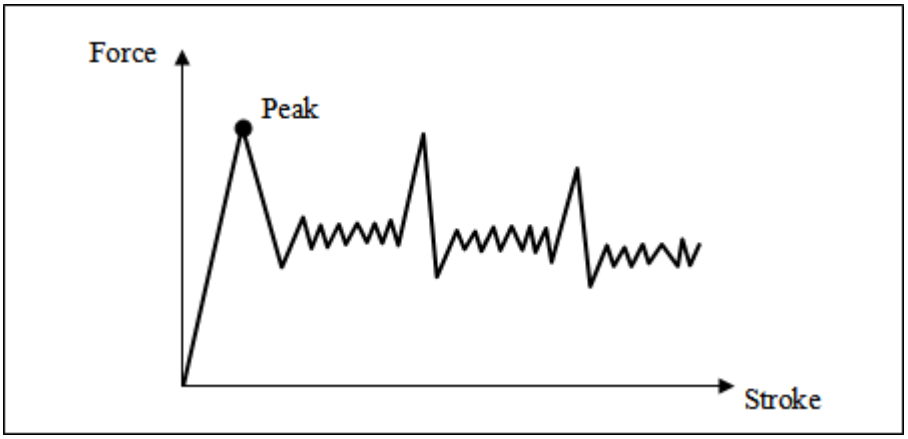
Static Friction

Determines the coefficient of static friction from the [Force] up to the first peak and the mass of the moving weight.

$$\text{Static Friction} = \frac{\text{Test Force at First Peak}}{\text{Mass of Moving Weight}}$$

Parameters





- Parameters for First Peak (P1): Select either [First Peak] or [Max. Peak].

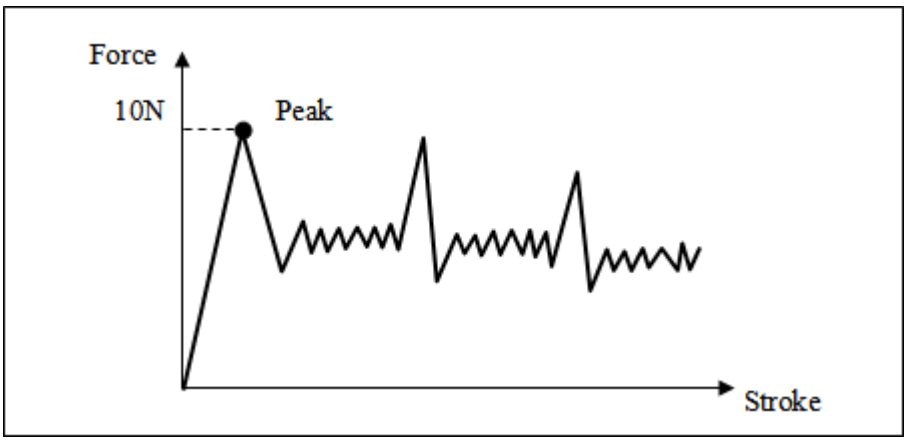
Type

- None

Example

Assume the moving weight has a mass of 200 gf and the [Force] at the first peak is 10 N.

$$\text{Static Friction} = \frac{10\text{N}}{200\text{gf}} = \frac{10\text{N}}{1.96133\text{N}} = 5.0985$$



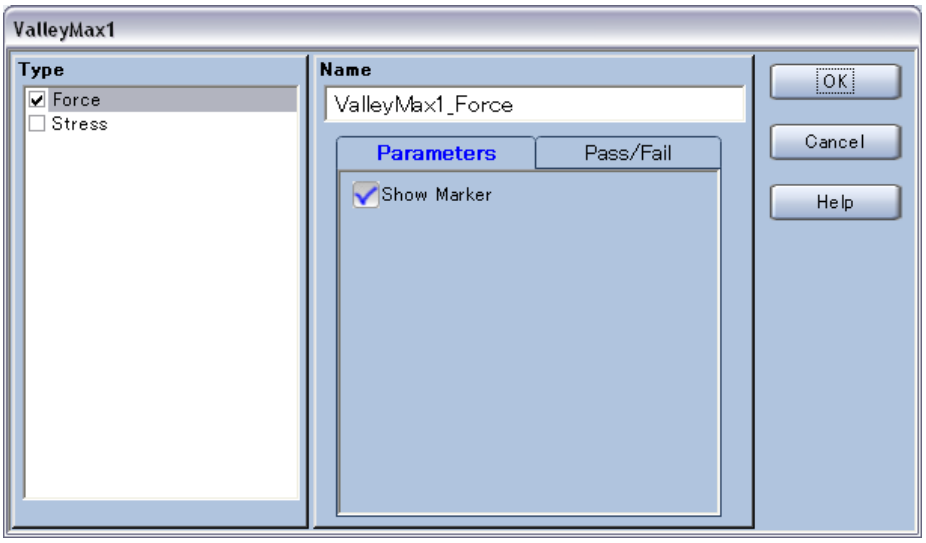
ValleyMax

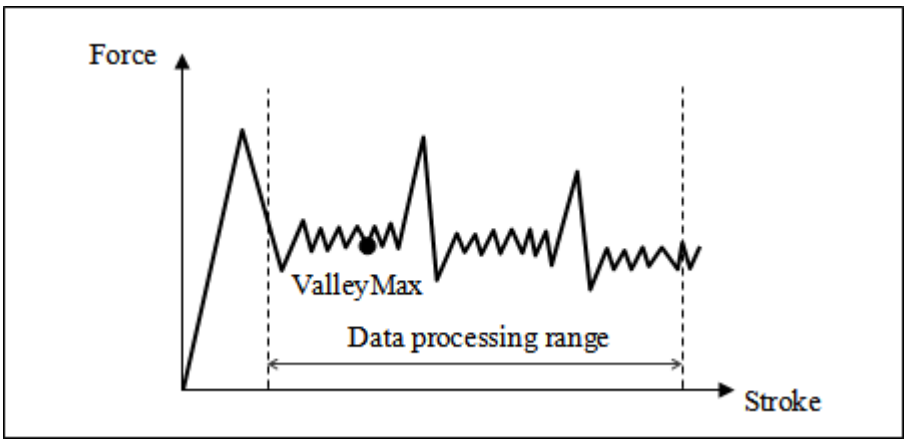
Mark

Determines the maximum valleys within the data processing area.

Up to 9 of the maximum valleys can be ranked, where the maximum valley is ValleyMax1, the next maximum valley is ValleyMax2, the next maximum valley is ValleyMax3, and so on.

Parameters





- None

Type

- Force

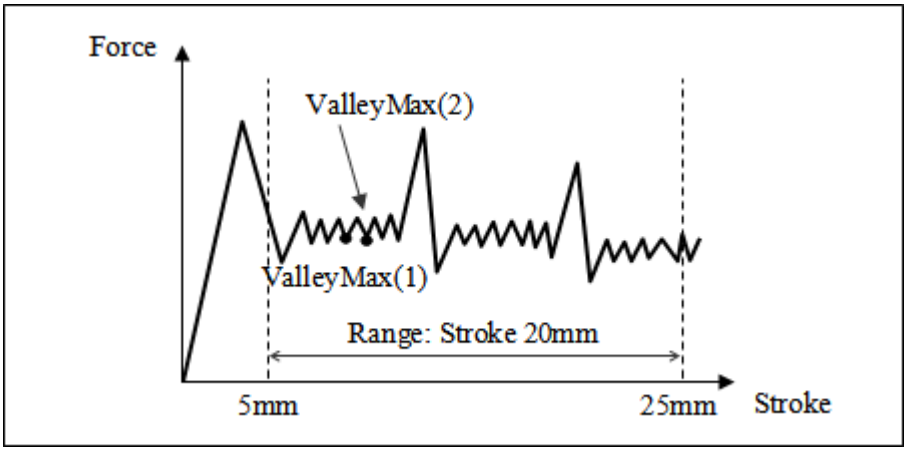
Determines the [Force] at the maximum valley.

- Stress

Determines the stress at the maximum valley.

Example

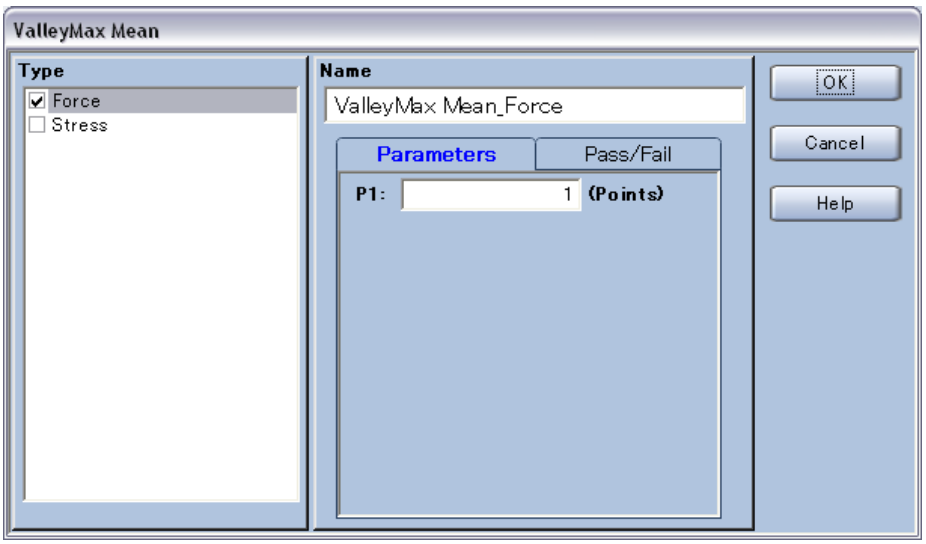
Assume the starting point of a data processing area is after a stroke of 5 mm and the ending point is a 20 mm stroke after the starting point.

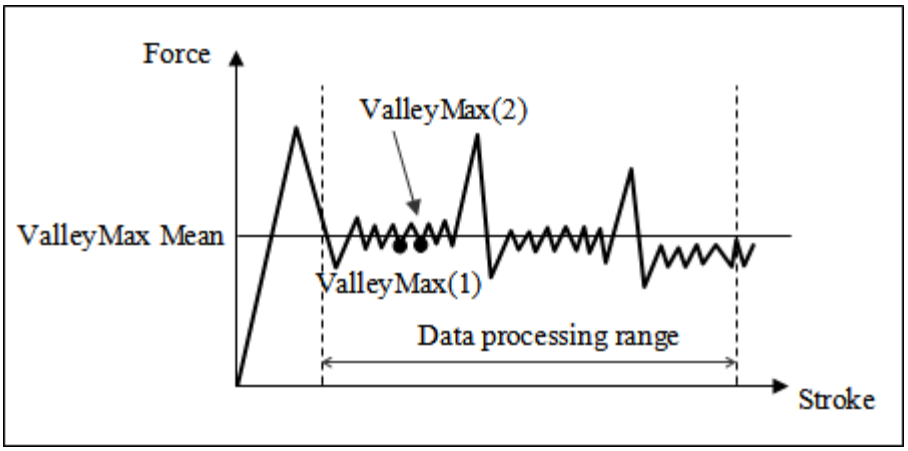


ValleyMax Mean

Determines the mean of the specified number of the maximum valleys.

Parameters





- Number of Maximum Valley Points (P1)

Type

- Force

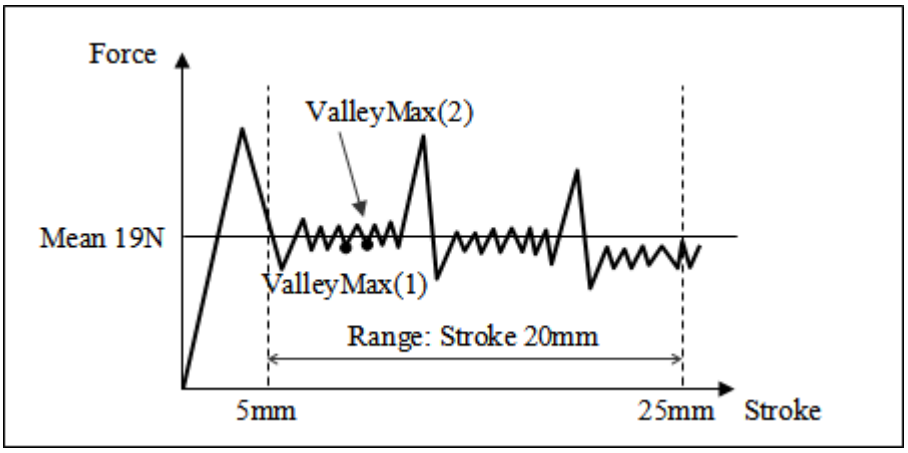
Determines the mean [Force] of the specified maximum valleys.

- Stress

Determines the mean stress of the specified maximum valleys.

Example

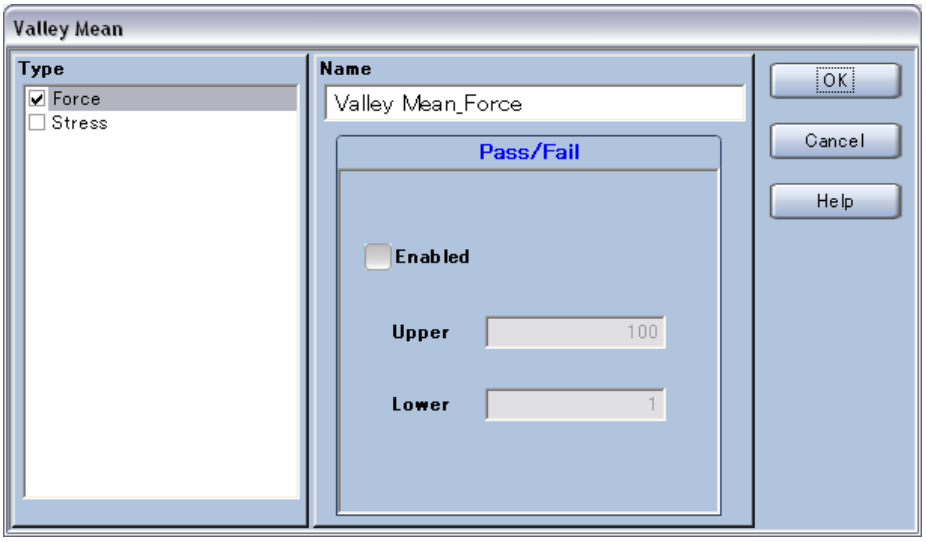
Assume the data processing area starts after a 5 mm stroke, ends a 20 mm stroke after the starting point, and the two maximum valleys (P1) are to be averaged, where the [Force] at Valley Max1 is 20 N and ValleyMax2 is 18 N.

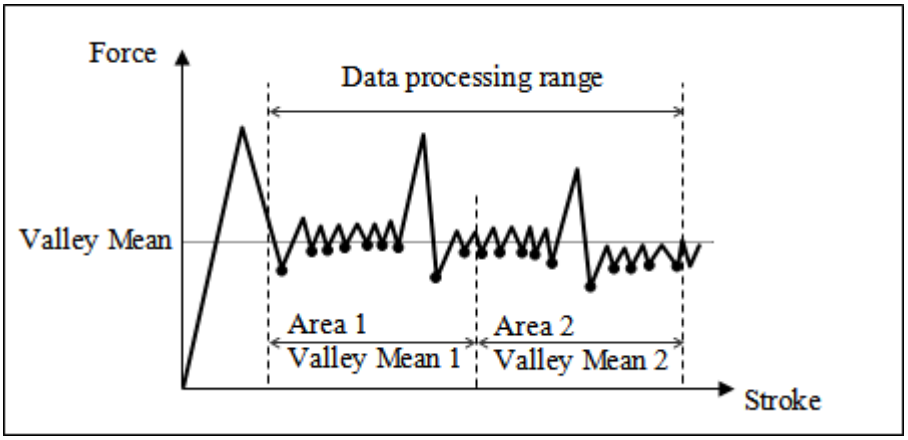


Valley Mean / Valley Mean All

Valley Mean finds the mean of all valleys within each area. Valley Mean All finds the mean of all valleys in all areas, when divided into intervals.

Parameters





- None

Type

- Force

Determines the ratio of ValleyMean to ValleyMean All for [Force].

- Stress

Determines the ratio of ValleyMean to ValleyMean All for stress.

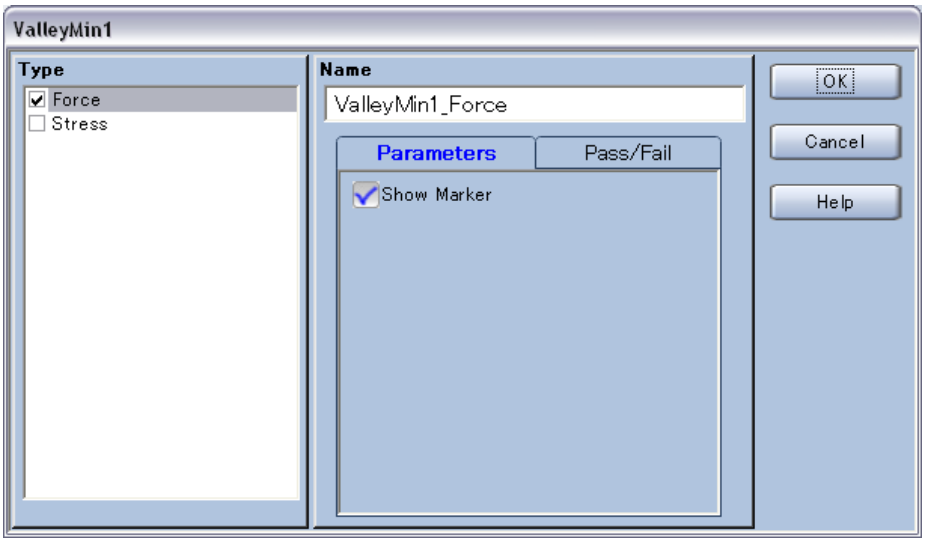
ValleyMin (x)

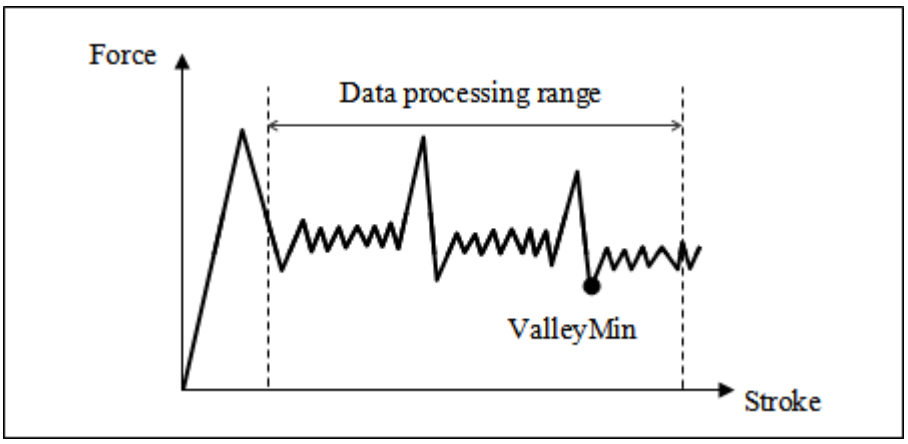
Mark

Determines the minimum valleys within the data processing area.

Up to 9 of the [ValleyMin] can be ranked, where the minimum valley is ValleyMin1, the next minimum valley is ValleyMin2, and the next minimum valley is ValleyMin3, and so on.

Parameters





- None

Type

- Force

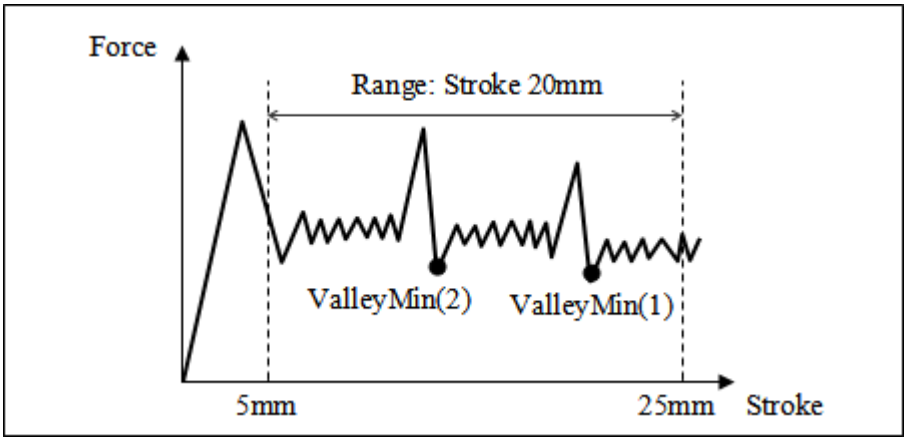
Determines the [Force] at the [ValleyMin].

- Stress

Determines the stress at the [ValleyMin].

Example

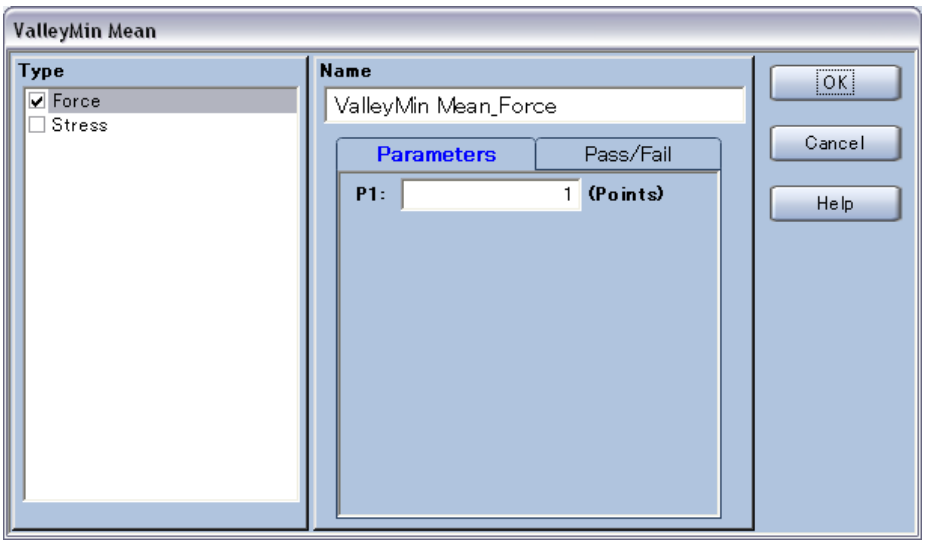
Assume the starting point of a data processing area is after a stroke of 5 mm and the ending point is a 20 mm stroke after the starting point.

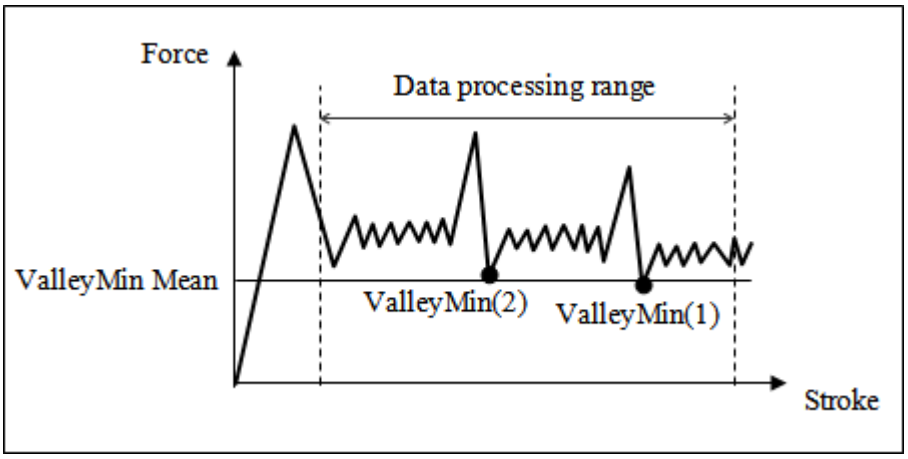


ValleyMin Mean

Determines the mean of the specified number of the [ValleyMin].

Parameters





- Number of ValleyMin (P1)

Type

- Force

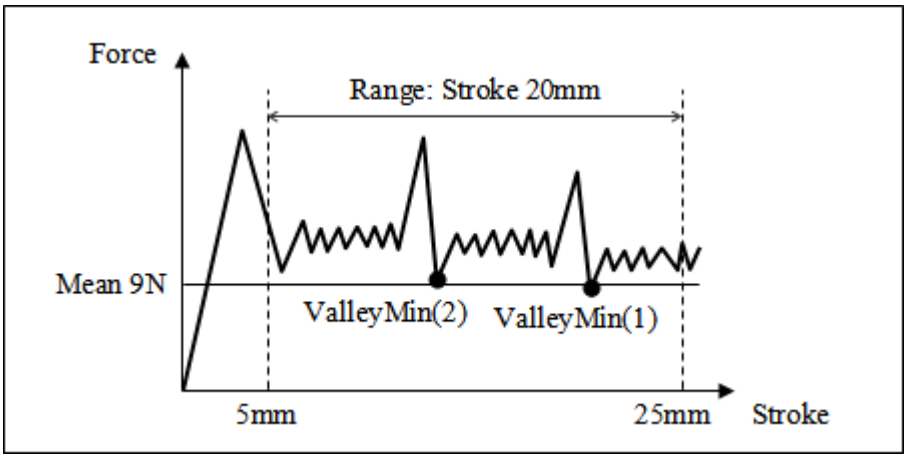
Determines the mean [Force] of the specified [ValleyMin].

- Stress

Determines the mean stress of the specified [ValleyMin].

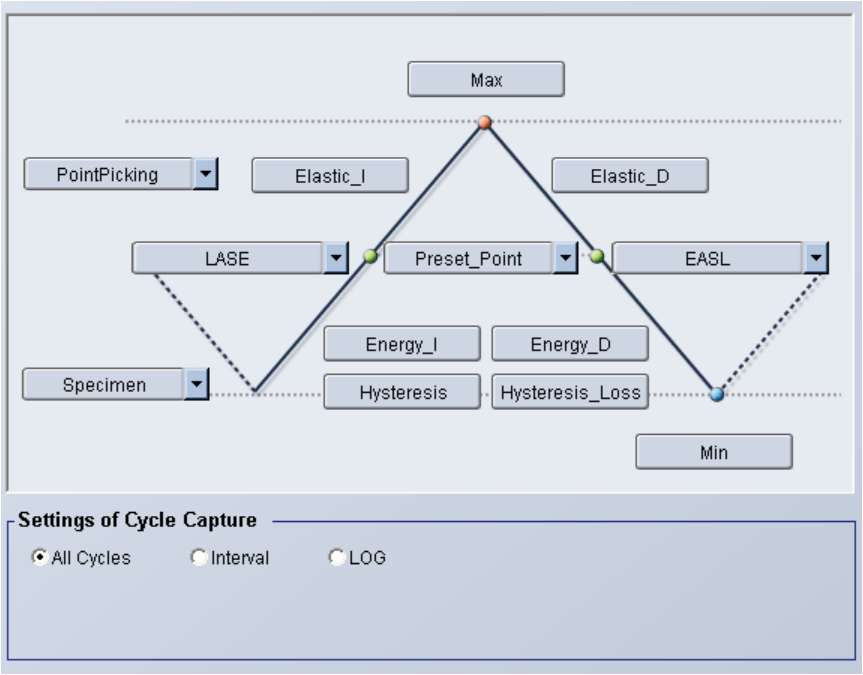
Example

Assume the data processing area starts after a 5 mm stroke, ends a 20 mm stroke after the starting point, and the two [ValleyMin] (P1) are to be averaged, where the [Force] at ValleyMin1 is 8 N and ValleyMin2 is 10 N.



Chapter 3
Data Processing Items in the Cycle Mode

This chapter describes processing data in the cycle mode. In the cycle mode, the procedure for processing tensile, compression, and bending data is the same.



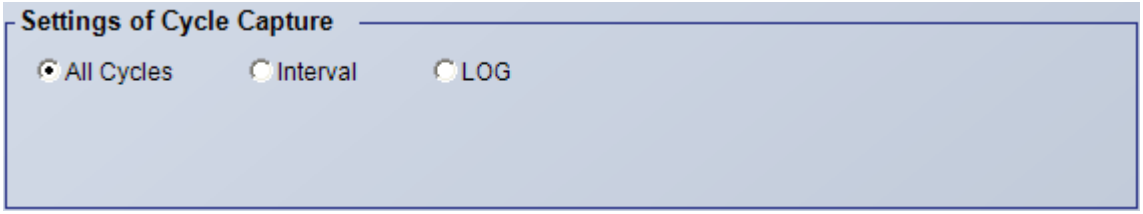
Data Acquisition Cycle Settings

Select from the three sampling data acquisition cycles.

This setting cannot be changed during re-analysis.

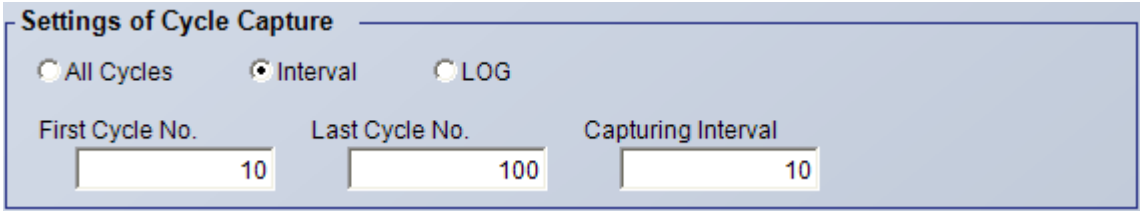
- All Cycles

Collects data from all cycles.



- Interval

Collects data at equal intervals specified via [First Cycle No.], [Last Cycle No.], and [Capturing Interval] settings.



Example

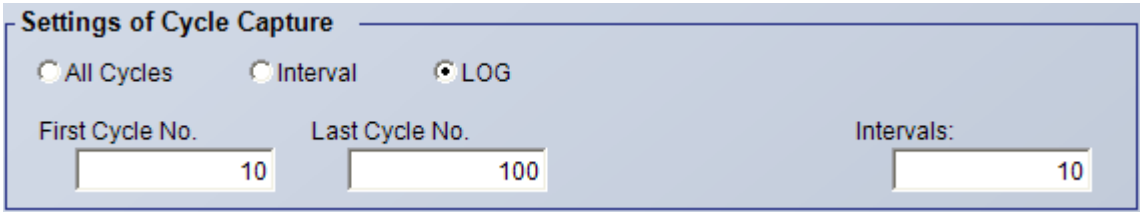
Assume the starting cycle number is 10, the last is 100, and the number of intervals is 10.

Data Acquisition Cycle: 1, 10, 20, 30, 40, 50, 60, 70, 80, 90, and 100

* Data is always acquired on the first cycle.

- Log

Collects data at logarithmic intervals specified via [First Cycle No.], [Last Cycle No.], and [Intervals] settings.



Example

Assume the starting cycle number is 10, the last is 100, and the number of intervals is 10.

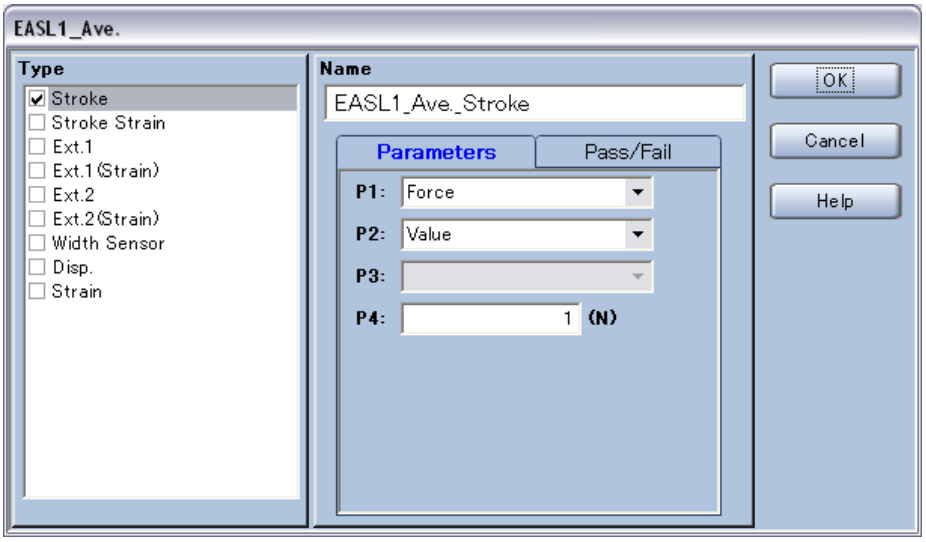
Data Acquisition Cycle: 1, 10, 13, 16, 20, 25, 32, 40, 50, 63, 79, and 100

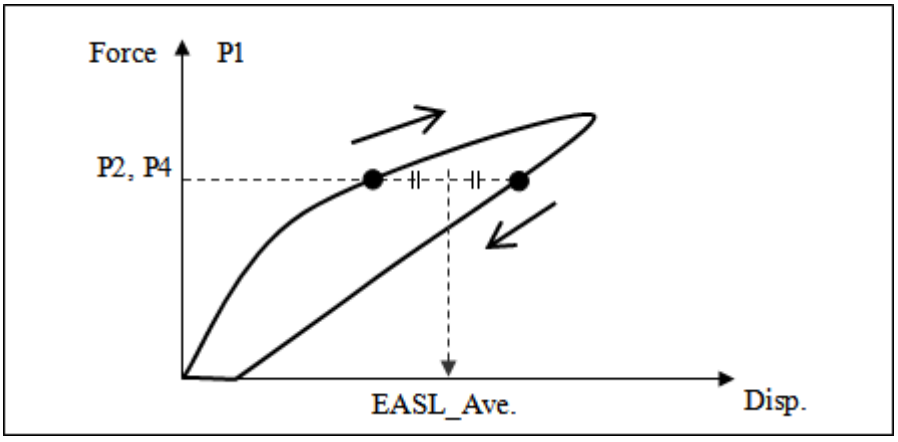
* Data is always acquired on the first cycle.

EASL_Ave - Cycle Mode

Determines EASL_Ave as the mean [Force] value when the [Force] is increasing or decreasing, which satisfies the specified parameter . Up to ten EASL_Ave parameters can be specified.

Parameters





- Method for Specifying Parameter (P1): Select either [Force] or [Stress].

- Method for Specifying Parameter (P2): Select either [Formula] or [Value].

- Formula (P3): Enabled to specify only if [Formula] is specified in the [P2] box.

- Value (P4): Enabled to specify only if [Value] is specified in the [P2] box.

* Parameter values are in terms of units specified in the [System] window.

Type

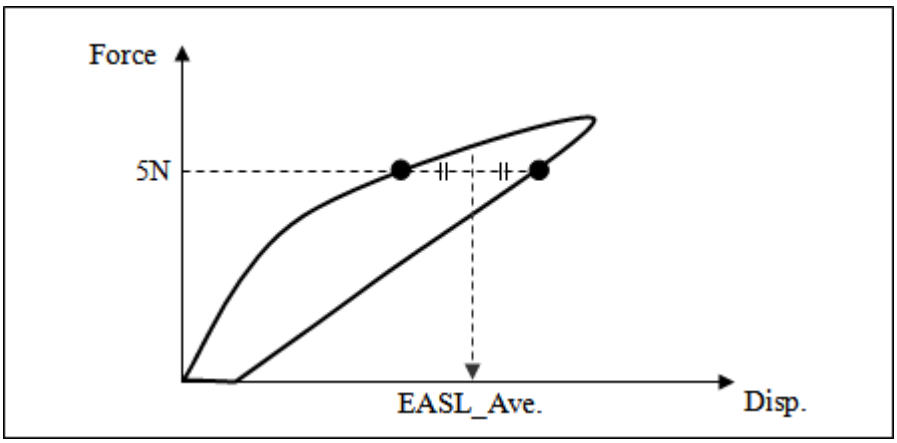
Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Stroke	Stroke (strain)
Disp.	Disp. (strain)

Example

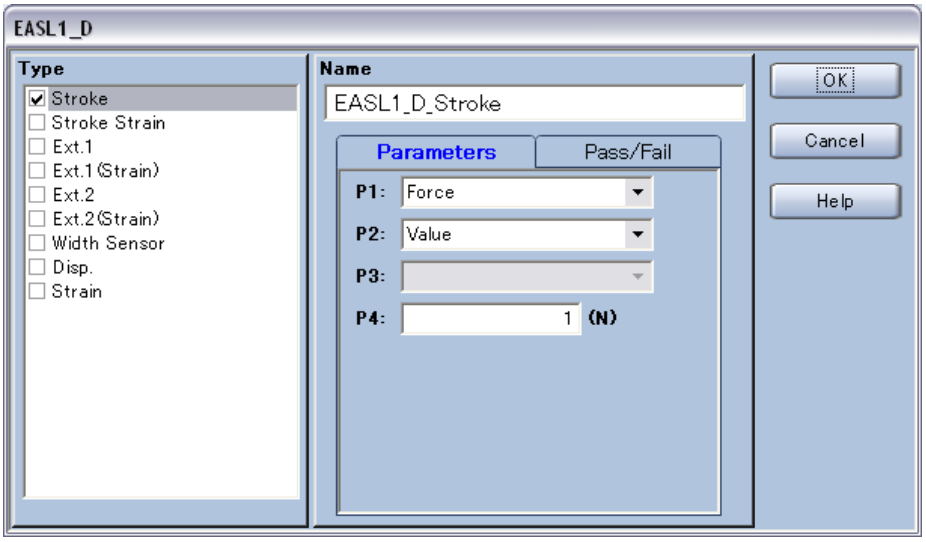
Assume the sensor type P1 is set to [force], the specified parameter type P2 is [value], and the value P4 is 5N.

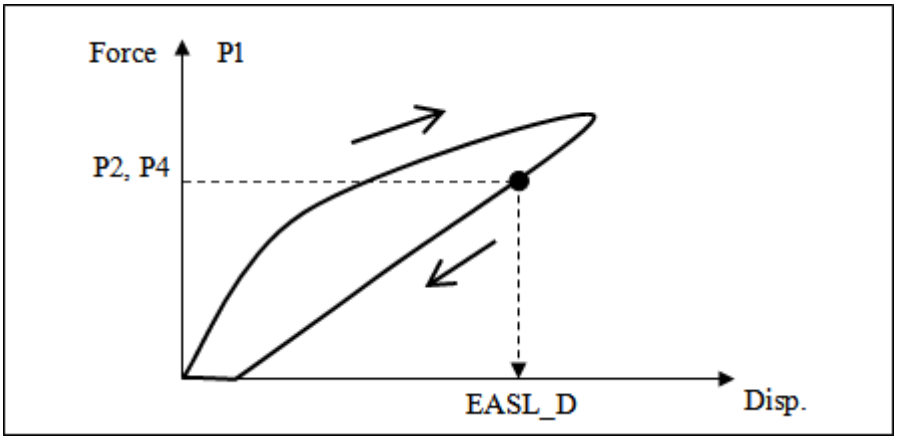


EASL_D - Cycle Mode

Determines EASL_D as the [Force] value when the [Force] is decreasing that satisfies the specified parameter . Up to ten EASL_D parameters can be specified.

Parameters





- Method for Specifying Parameter (P1): Select either [Force] or [Stress].

- Method for Specifying Parameter (P2): Select either [Formula] or [Value].

- Formula (P3): Enabled to specify only if [Formula] is specified in the [P2] box.

- Value (P4): Enabled to specify only if [Value] is specified in the [P2] box.

* Parameter values are in terms of units specified in the [System] window.

Type

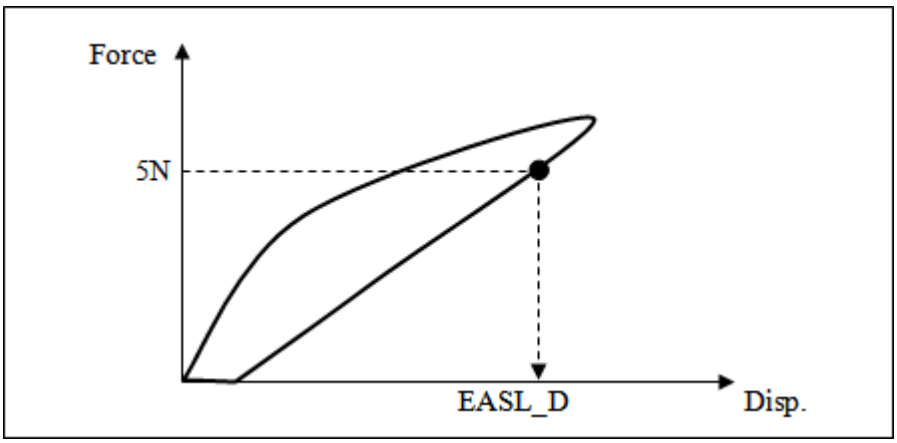
Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Stroke	Stroke (strain)
Disp.	Disp. (strain)

Example

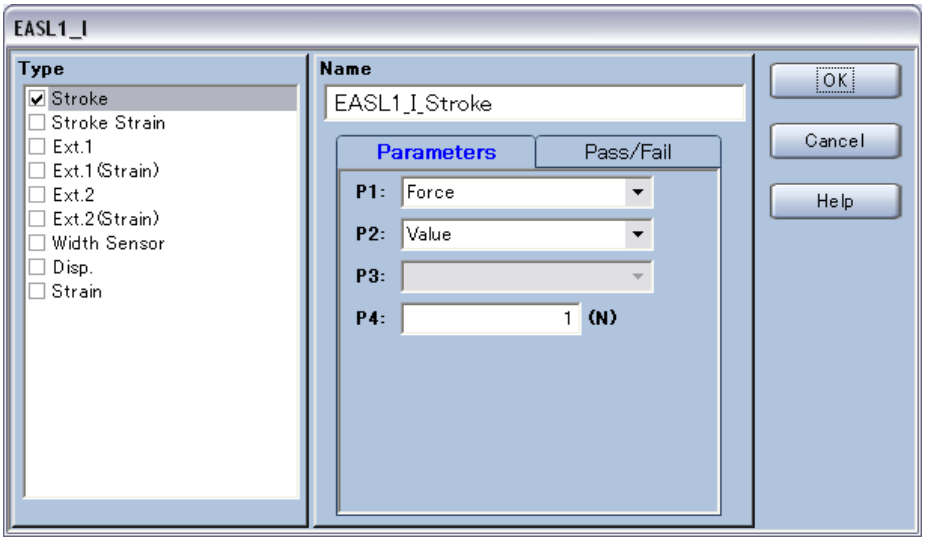
Assume the sensor type P1 is set to [force], the specified parameter type P2 is [value], and the value P4 is 5N.

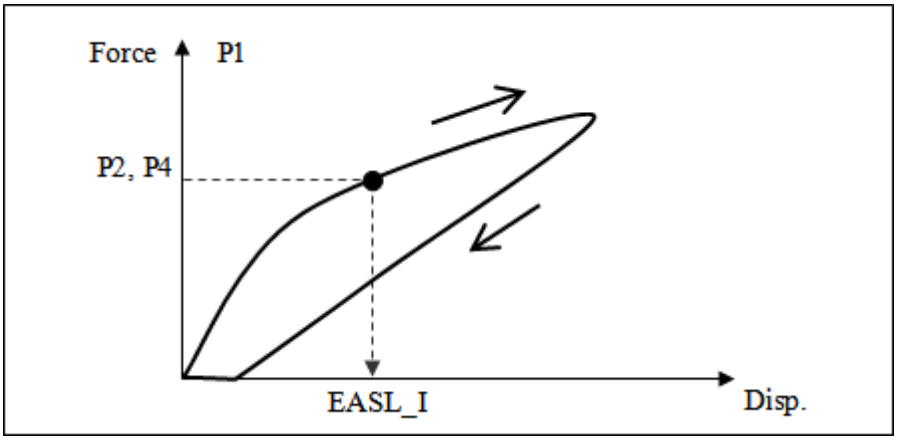


EASL_I - Cycle Mode

Determines EASL_I as the [Force] value when the [Force] is increasing that satisfies the specified parameter . Up to ten EASL_I parameters can be specified.

Parameters





- Method for Specifying Parameter (P1): Select either [Force] or [Stress].

- Method for Specifying Parameter (P2): Select either [Formula] or [Value].

- Formula (P3): Enabled to specify only if [Formula] is specified in the [P2] box.

- Value (P4): Enabled to specify only if [Value] is specified in the [P2] box.

* Parameter values are in terms of units specified in the [System] window.

Type

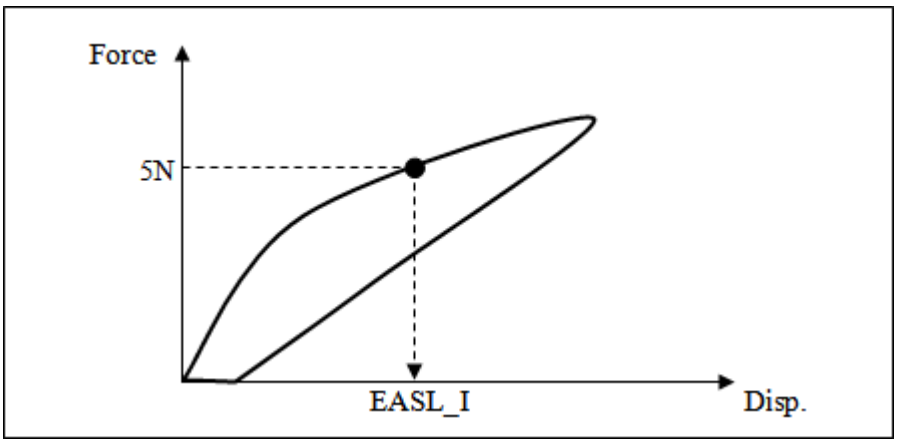
Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Stroke	Stroke (strain)
Disp.	Disp. (strain)

Example

Assume the sensor type P1 is set to [Force], the specified parameter type P2 is [Value], and the value P4 is 5N.



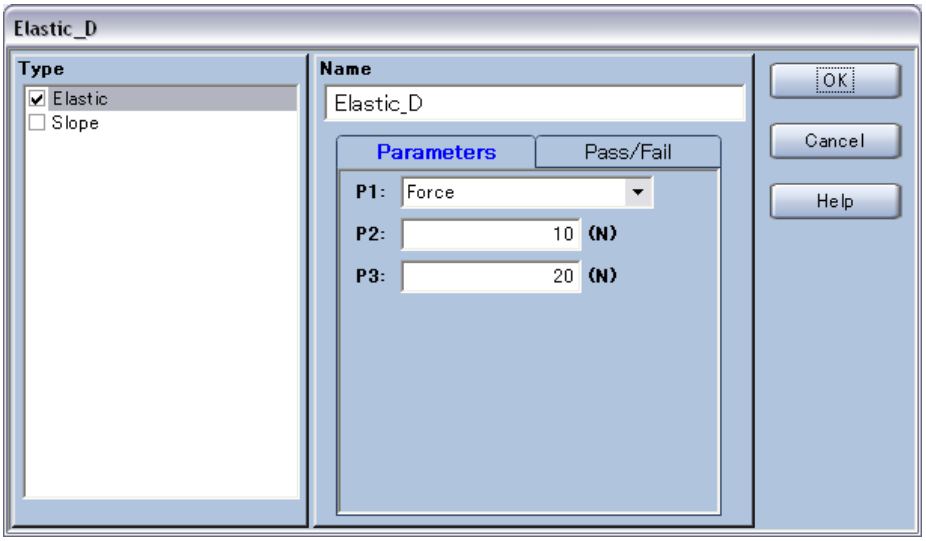
Elastic_D - Cycle Mode

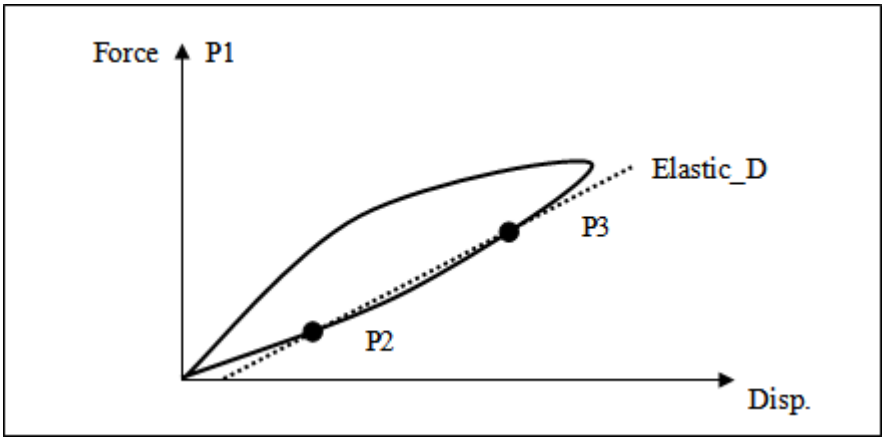
Determines elastic using the least squares method from all data between two pre-specified points.

$$\text{Elastic} = \text{Slope} \times \frac{\text{Gauge Length}}{\text{Cross - Sectional Area of Specimen}}$$

Slope	Calculated from all data between two points using least squares method.
Gauge Length	Value specified via the [Sensor] or [Specimen] dialog box.
Cross-Sectional Area of Specimen	Calculated from dimensions specified in the [Specimen] dialog box.

Parameters





- Sensor Type (P1)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Starting Point (P2)

- Ending Point (P3)

* Parameter values are in terms of units specified in the [System] window.

Type

- Elastic

Calculated from [Slope], [Gauge Length], and [Specimen Cross-Sectional Area].

- Slope

Calculates only slope of elastic.

Example

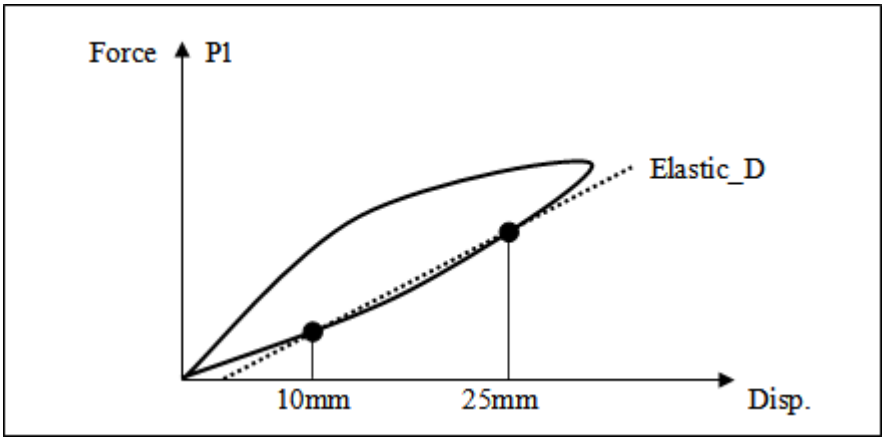
Assume P1 is set to [Disp.], P2 to 10 mm, P3 to 25 mm, gauge length to 50 mm, and the specimen is a flat plastic plate 2 mm thick and 5 mm wide.

$$\text{Elastic} = \text{Slope} \times \frac{50\text{mm}}{2\text{mm} \times 5\text{mm}}$$

Note

Slope is calculated using least squares method.

The formula used to calculate cross-sectional area depends on specimen shape. For more details, refer to Methods for Calculating Cross-Sectional Area.



Related keyword: “Calculating Stress and Strain”

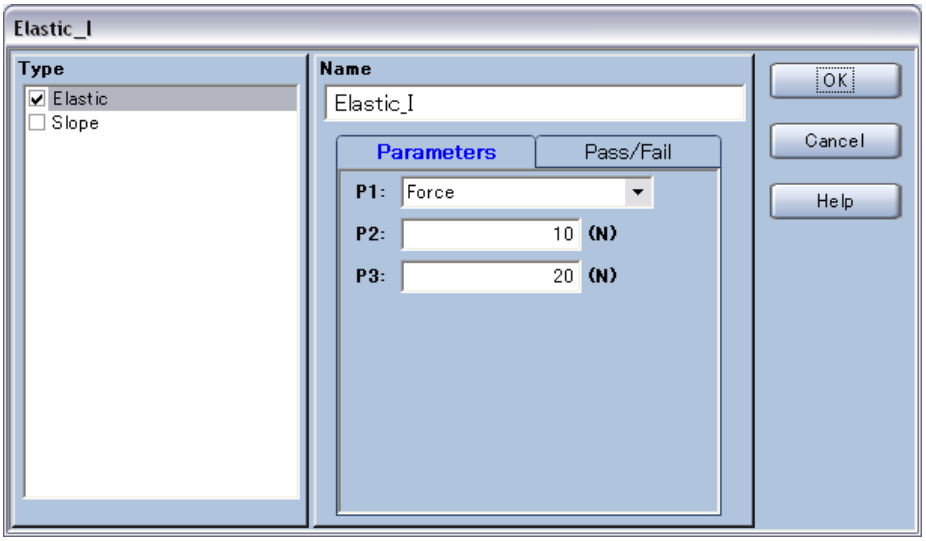
Elastic_I - Cycle Mode

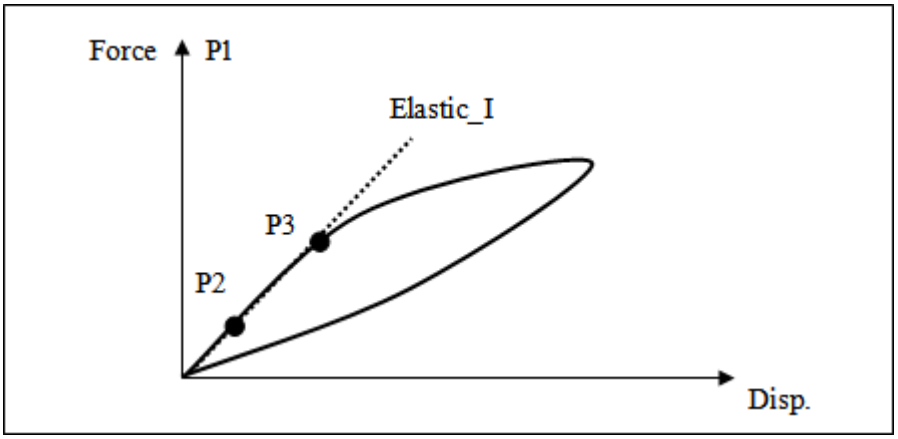
Determines elastic using the least squares method from all data between two pre-specified points.

$$\text{Elastic} = \text{Slope} \times \frac{\text{Gauge Length}}{\text{Cross - Sectional Area of Specimen}}$$

Slope	Calculated from all data between two points using least squares method
Gauge Length	Value specified via the [Sensor] or [Specimen] dialog box.
Cross-Sectional Area of Specimen	Calculated from dimensions specified in the [Specimen] dialog box.

Parameters





- Sensor Type (P1)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Starting Point (P2)

- Ending Point (P3)

* Parameter values are in terms of units specified in the [System] window.

Type

- Elastic

Calculated from [Slope], [Gauge Length], and [Specimen Cross-Sectional Area].

- Slope

Calculates only slope of elastic.

Example

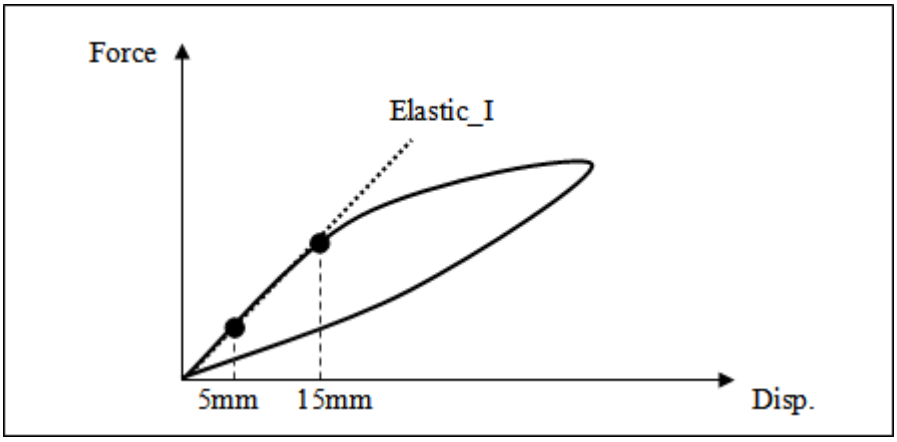
Assume P1 is set to [Disp.], P2 to 5 mm, P3 to 15 mm, gauge length to 50 mm, and the specimen is a flat plastic plate 2 mm thick and 5 mm wide.

$$\text{Elastic} = \text{Slope} \times \frac{50\text{mm}}{2\text{mm} \times 5\text{mm}}$$

Note

Slope is calculated using least squares method.

The formula used to calculate cross-sectional area depends on specimen shape. For more details, refer to Methods for Calculating Cross-Sectional Area.

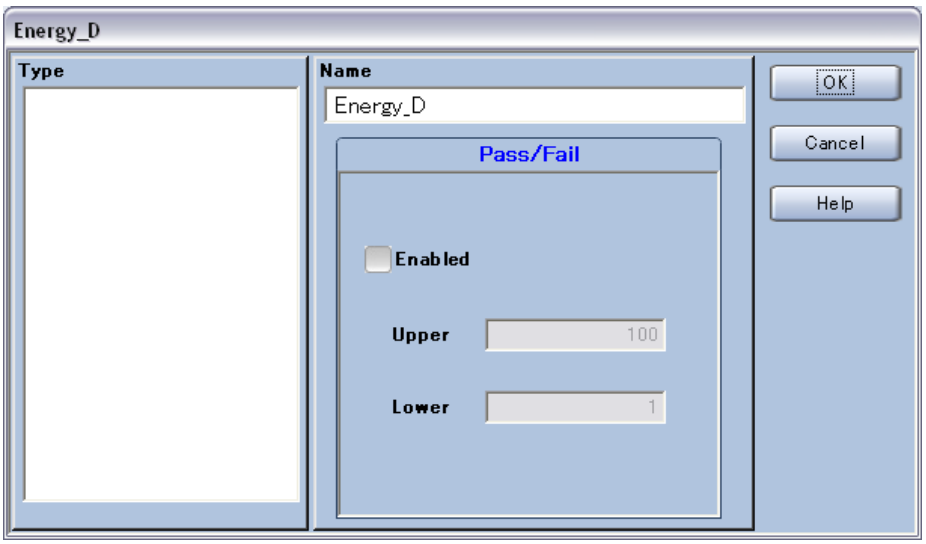


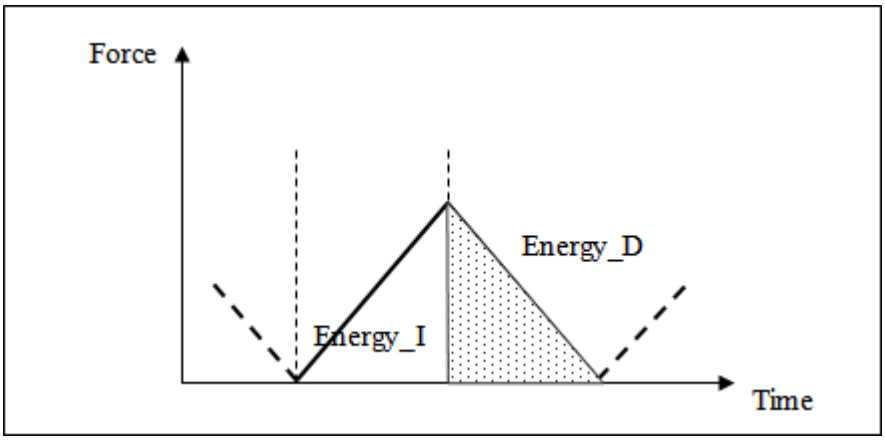
Related keyword: “Calculating Stress and Strain”

Energy_D - Cycle Mode

Determines the energy when the [Force] is decreasing.

Parameters





- None

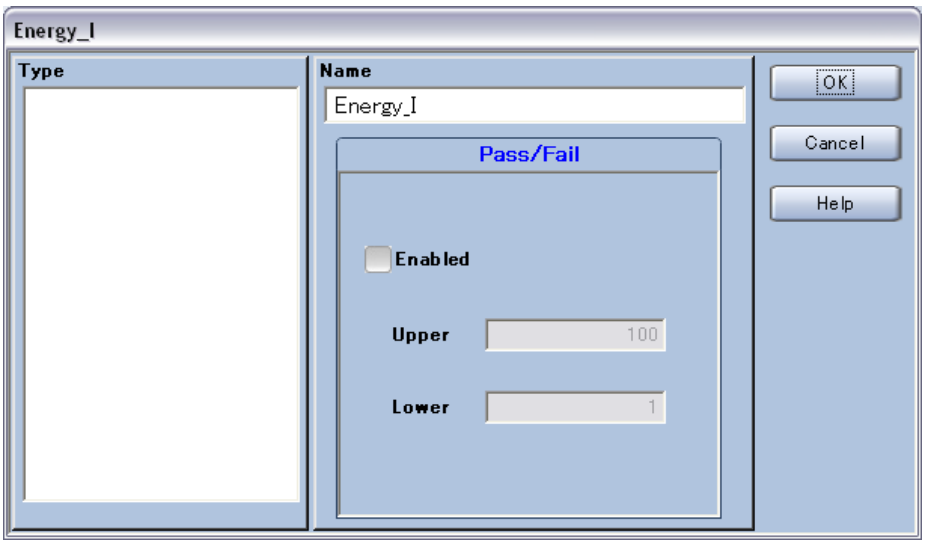
Type

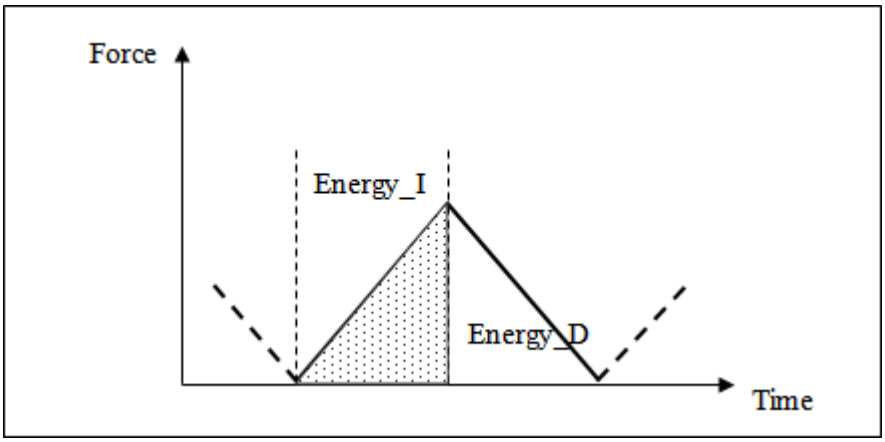
None

Energy_I - Cycle Mode

Determines the energy when the [Force] is increasing.

Parameters





- None

Type

None

Hysteresis

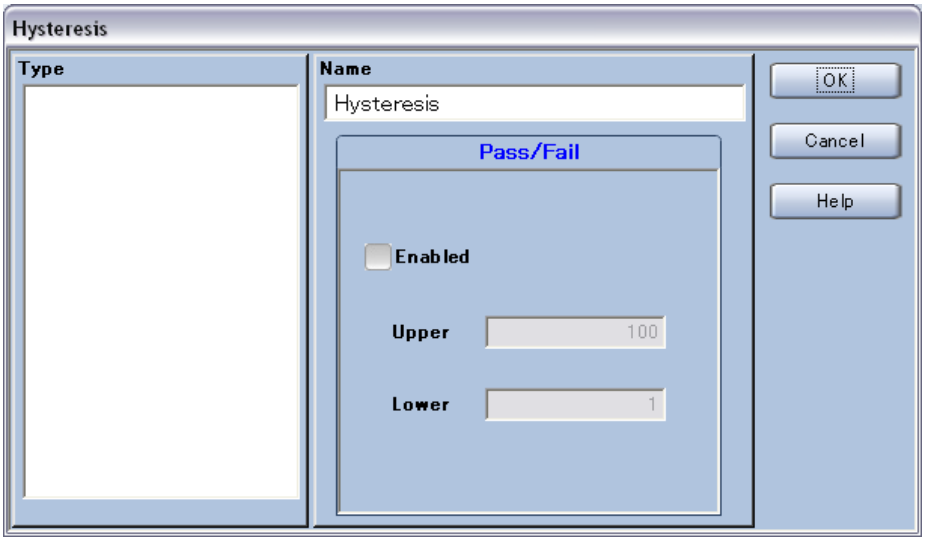
Determines hysteresis as the difference between the energy required to produce a fixed increase in [Disp.] and the energy released when decreasing that [Disp.].

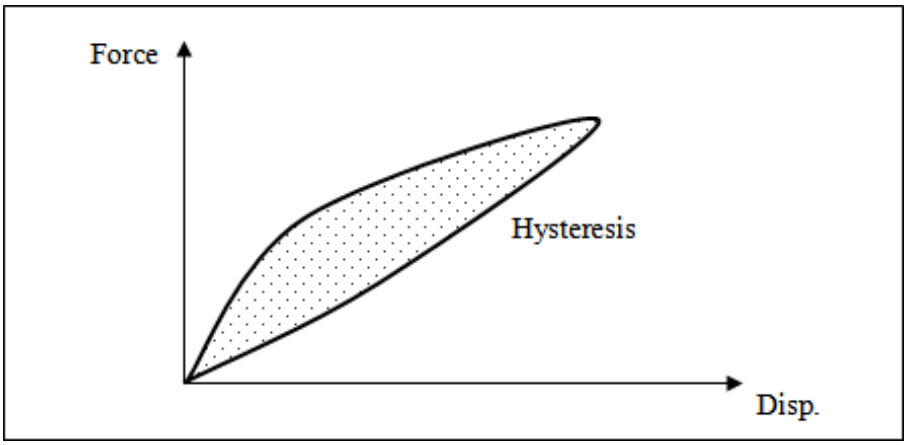
Hysteresis = Energy (when increasing) - Energy (when decreasing)

Note

If [Disp.] is on the horizontal axis, hysteresis is equivalent to the area between the increasing and decreasing portions of the energy graph.

Parameters





- None

Type

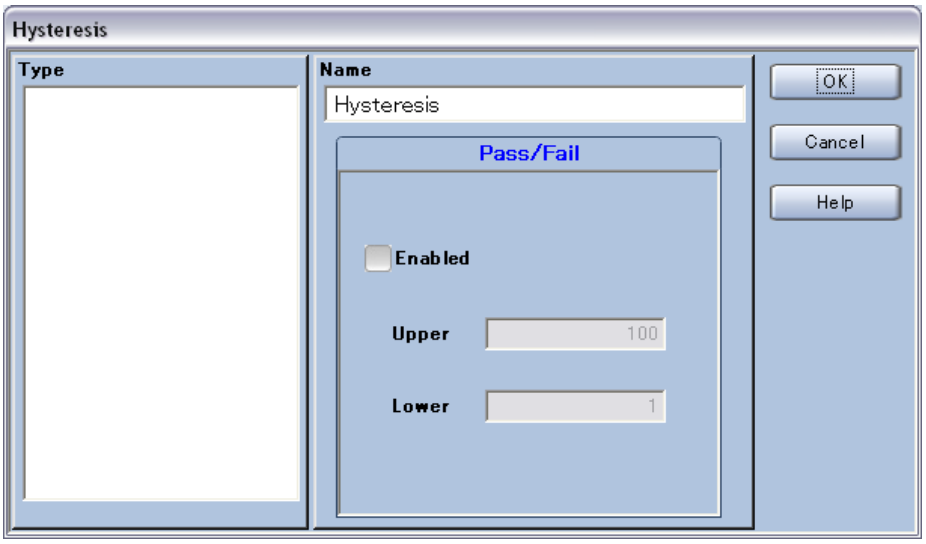
None

Hysteresis Loss

Determines hysteresis loss as the ratio of hysteresis to increase in energy.

$$\text{Hysteresis Loss} = \frac{\text{Hysteresis}}{\text{Energy (when increasing)}}$$

Parameters



- None

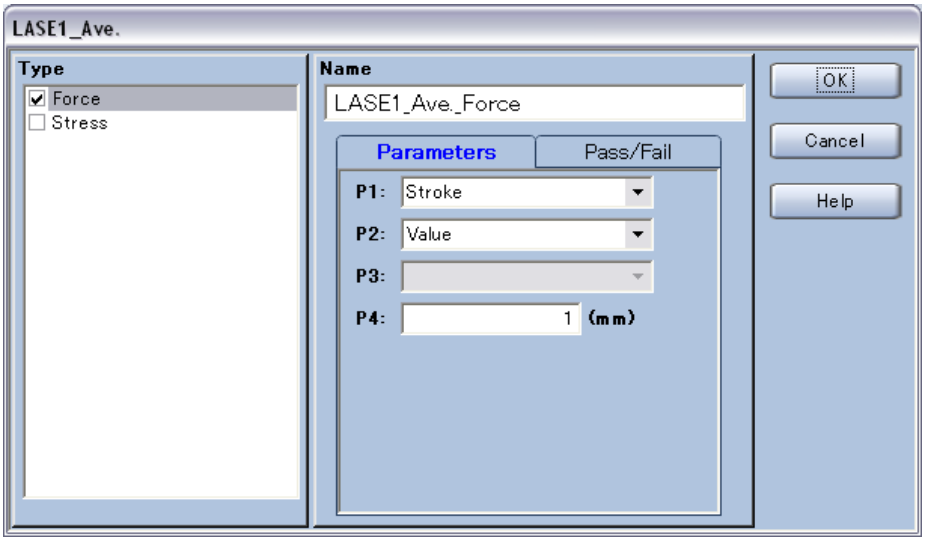
Type

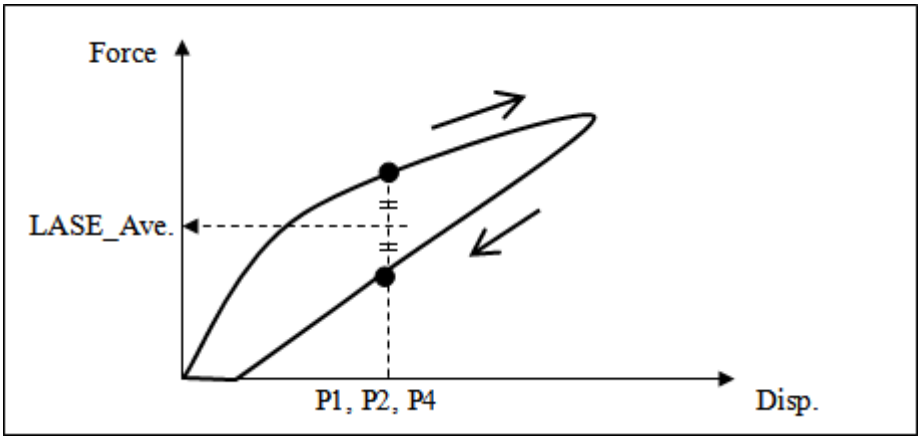
None

LASE_Ave. - Cycle Mode

Determines LASE_Ave as the mean [Force] value when the [Force] is increasing or decreasing, which satisfies the specified parameter . Up to ten LASE_Ave parameters can be specified.

Parameters





- Sensor Type (P1)

Stroke	Stroke (strain)
Disp.	Disp. (strain)

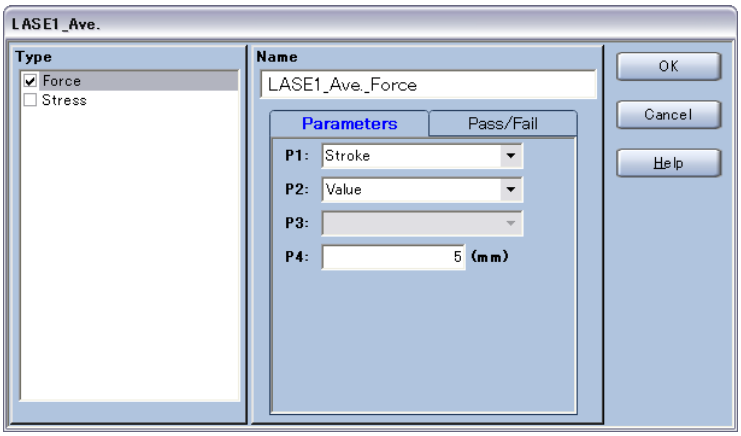
- Method for Specifying Parameter (P2): Select either [Formula] or [Value].

- Formula (P3): Enabled to specify only if [Formula] is specified in the [P2] box.

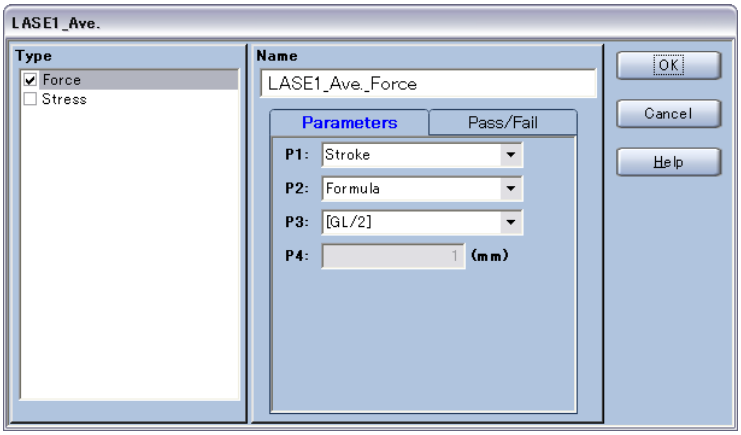
- Value (P4): Enabled to specify only if [Value] is specified in the [P2] box.

* Parameter values are in terms of units specified in the [System] window.

[P3] is disabled if [Value] is selected in [P2].



[P4] is disabled if [Formula] is selected in [P2].



Type

- Force

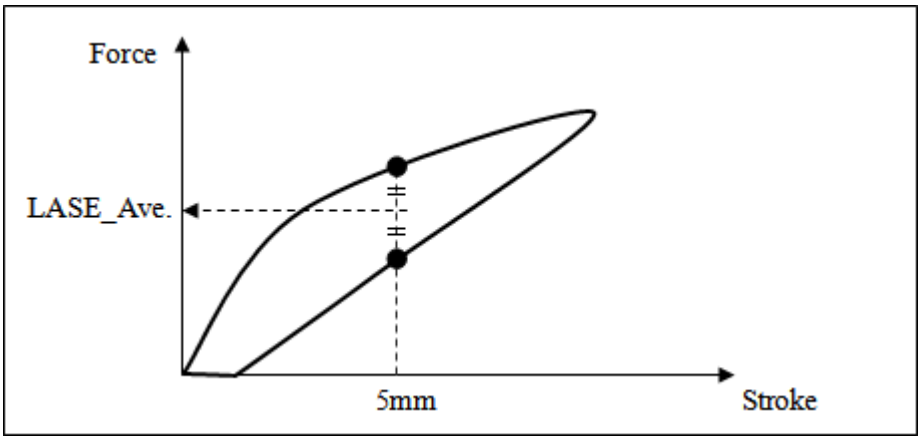
Calculated from specified parameter.

- Stress

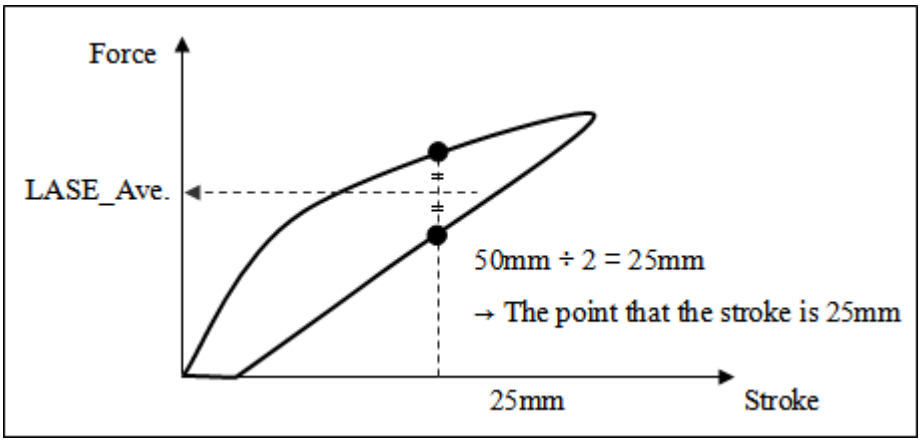
Stress calculated from the specified [Force].

Example

Assume the sensor type P1 is set to [Stroke], the specified parameter type P2 is [Value], and the value P4 is 5 mm.



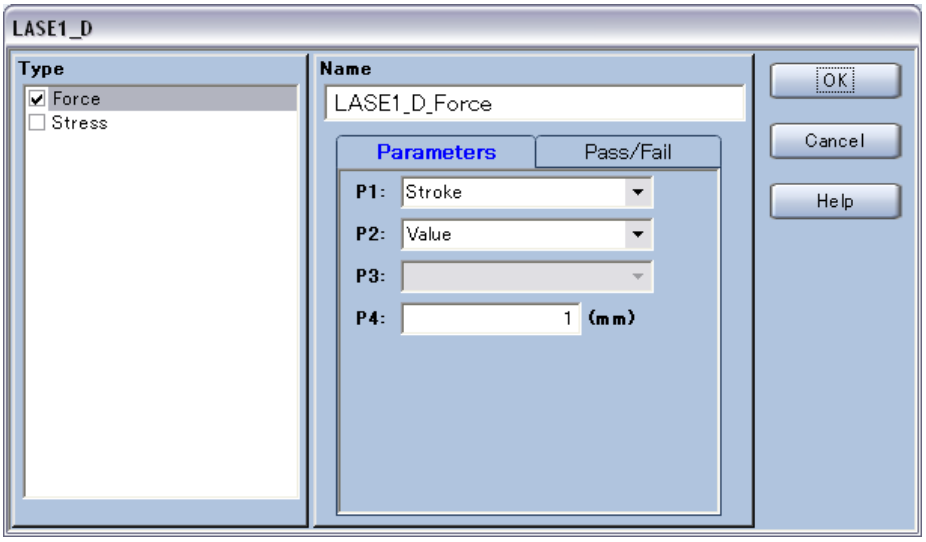
Assume the sensor type P1 is set to [Stroke], the specified parameter type P2 is [Value], and the value P4 is 5 mm.

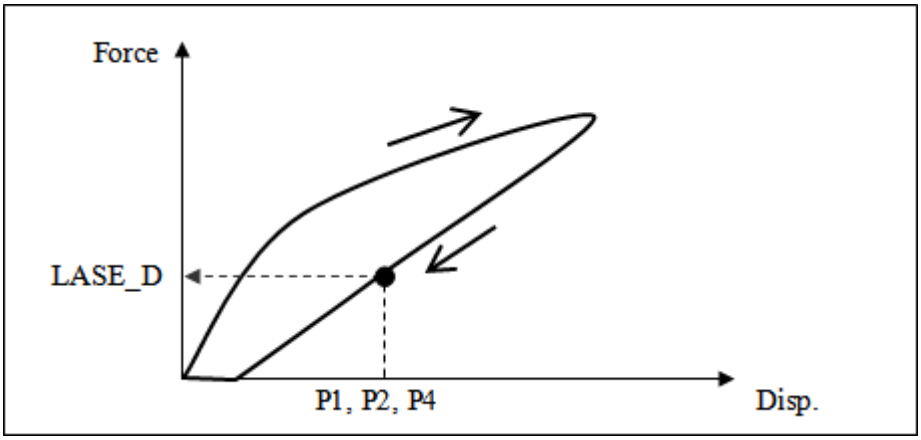


LASE_D - Cycle Mode

Determines LASE_D as the [Force] value when the [Force] is decreasing that satisfies the specified parameter. Up to ten LASE_D parameters can be specified.

Parameters





- Sensor Type (P1)

Stroke	Stroke (strain)
Disp.	Disp. (strain)

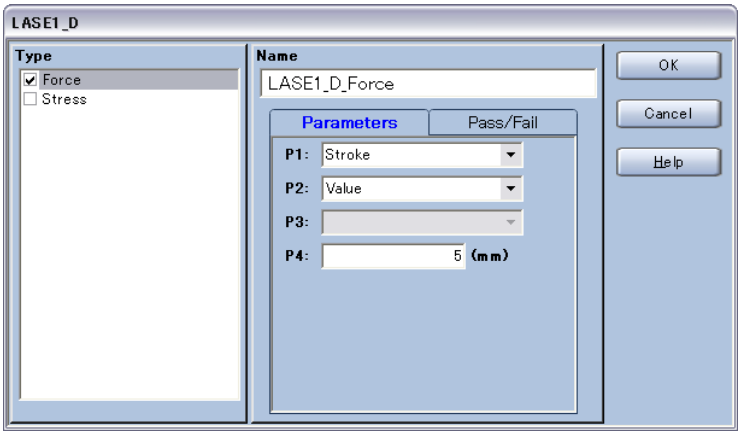
- Method for Specifying Parameter (P2): Select either [Formula] or [Value].

- Formula (P3): Enabled to specify only if [Formula] is specified in the [P2] box.

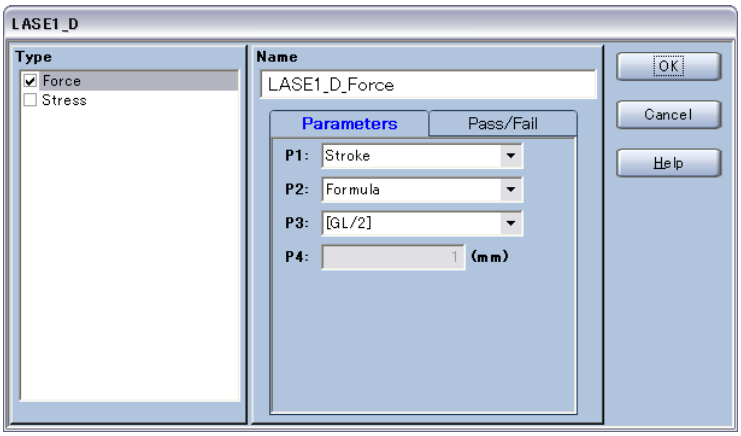
- Value (P4): Enabled to specify only if [Value] is specified in the [P2] box.

* Parameter values are in terms of units specified in the [System] window.

[P3] is disabled if [Value] is selected in [P2].



[P4] is disabled if [Formula] is selected in [P2].



Type

- Force

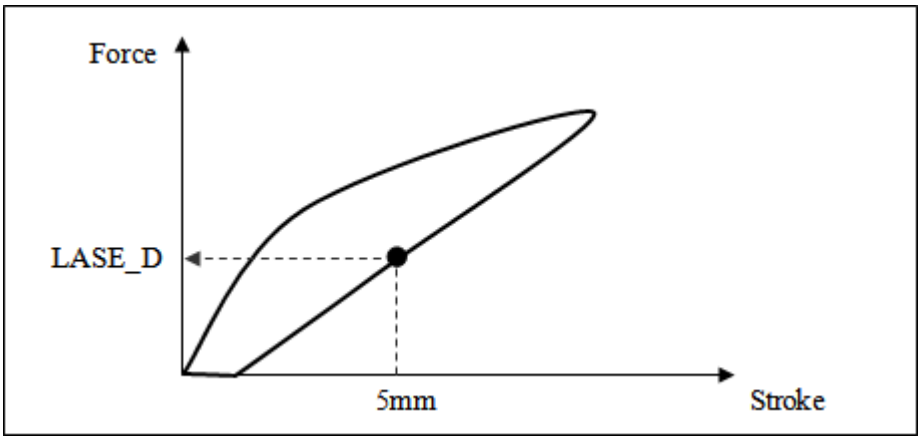
Calculated from specified parameter.

- Stress

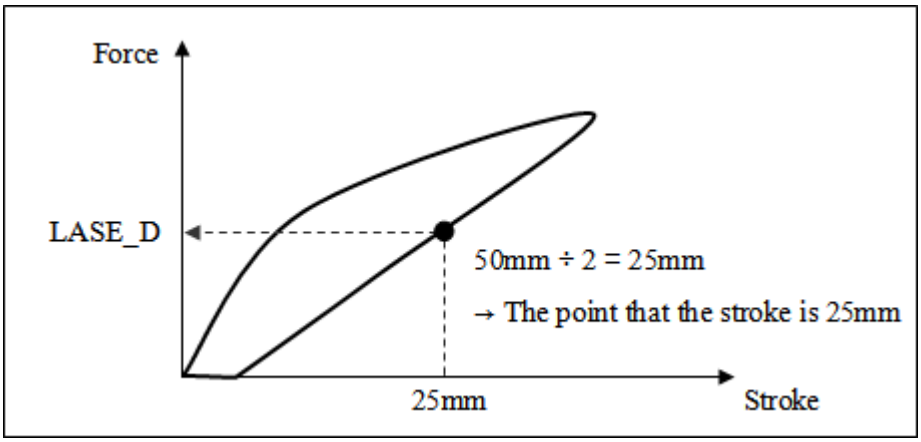
Stress calculated from the specified [Force].

Example

Assume the sensor type P1 is set to [Stroke], the specified parameter type P2 is [Value], and the value P4 is 5 mm.



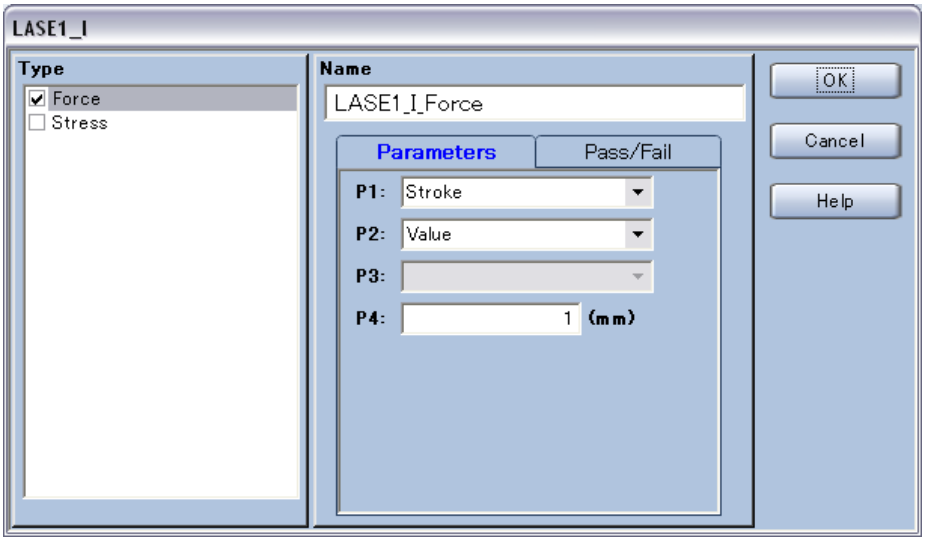
Assume the sensor type P1 is set to [Stroke], the specified parameter type P2 is [Value], and the value P4 is 5 mm.

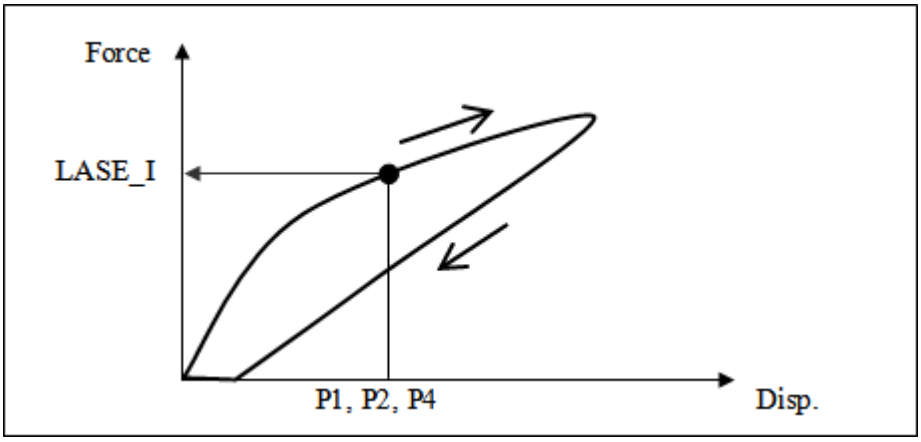


LASE_I - Cycle Mode

Determines LASE_I as the [Force] value that satisfies the specified parameter when the [Force] is increasing. Up to ten LASE_I parameters can be specified.

Parameters





- Sensor Type (P1)

Stroke	Stroke (strain)
Disp.	Disp. (strain)

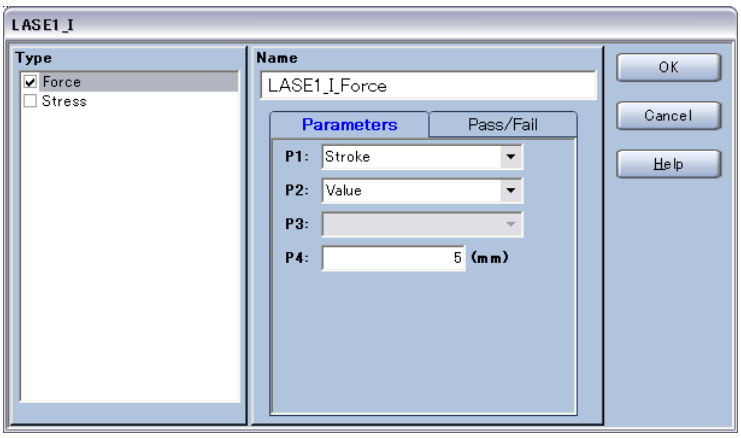
- Method for Specifying Parameter (P2): Select either [Formula] or [Value].

- Formula (P3): Enabled to specify only if [Formula] is specified in the [P2] box.

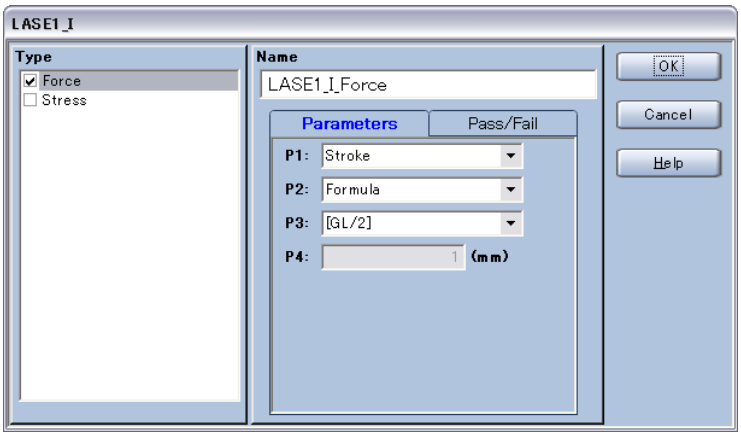
- Value (P4): Enabled to specify only if [Value] is specified in the [P2] box.

* Parameter values are in terms of units specified in the [System] window.

[P3] is disabled if [Value] is selected in [P2].



[P4] is disabled if [Formula] is selected in [P2].



Type

- Force

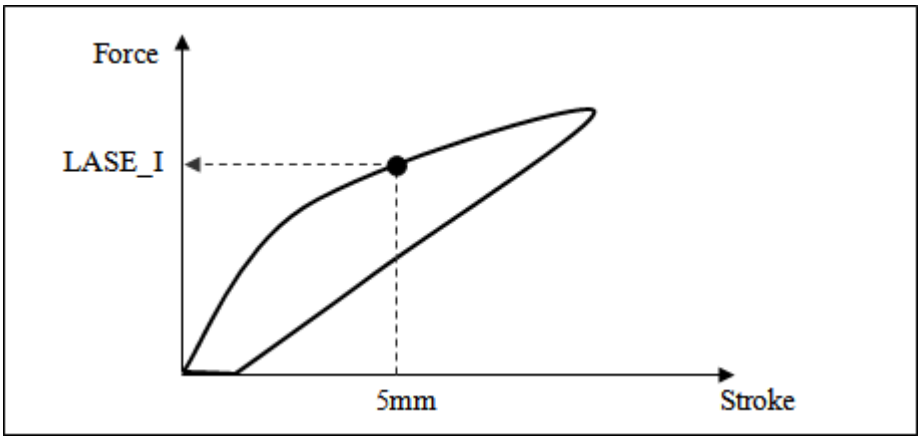
Calculated from specified parameter.

- Stress

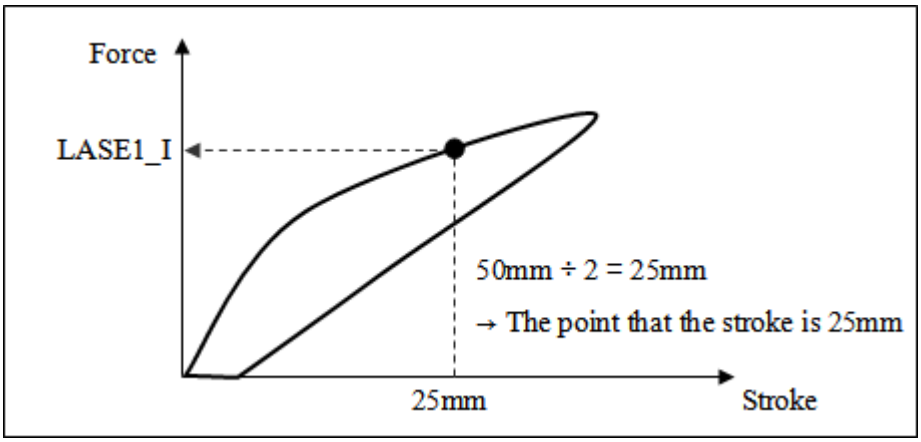
Stress calculated from the specified [Force].

Example

Assume the sensor type P1 is set to [Stroke], the specified parameter type P2 is [Value], and the value P4 is 5 mm.



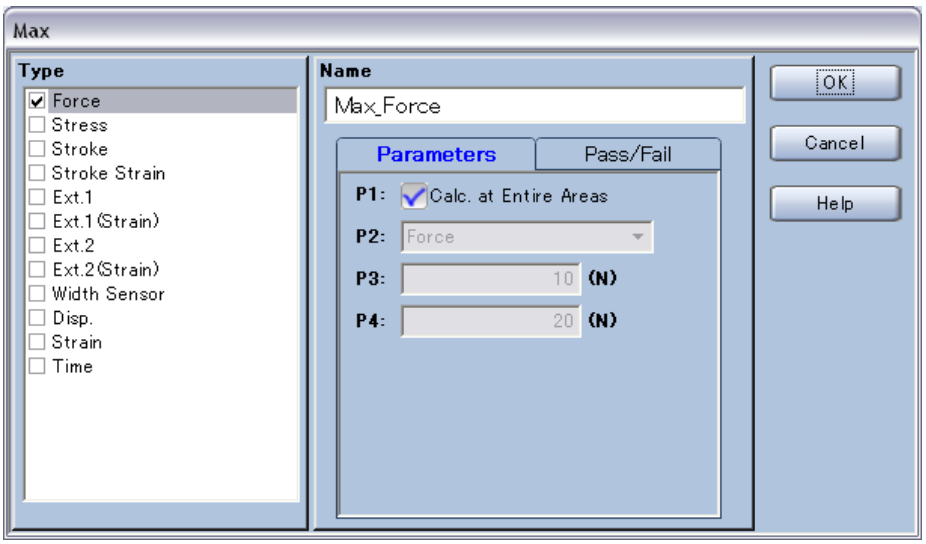
Assume the sensor type P1 is set to [Stroke], the specified parameter type P2 is [Formula], and the formula P3 is gauge length divided by 2, where gauge length is 50 mm.

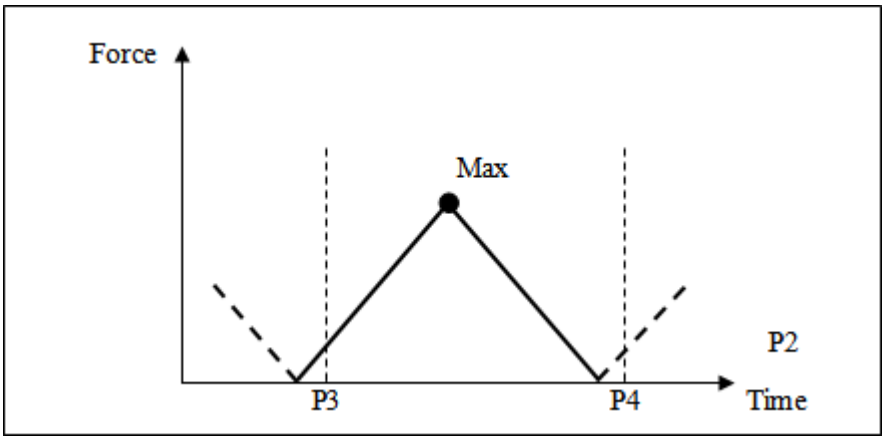


Max - Cycle Mode

Determines the maximum [Force] over a specified range or the entire area.

Parameters





- Specifies whether or not to search entire area (P1)

- Sensor Type (P2)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Starting Point (P3)

- Ending Point (P4)

* Parameter values are in terms of units specified in the [System] window.

* P2 to P4 can be specified only if [Calc. at Entire Areas] is not selected.

Type

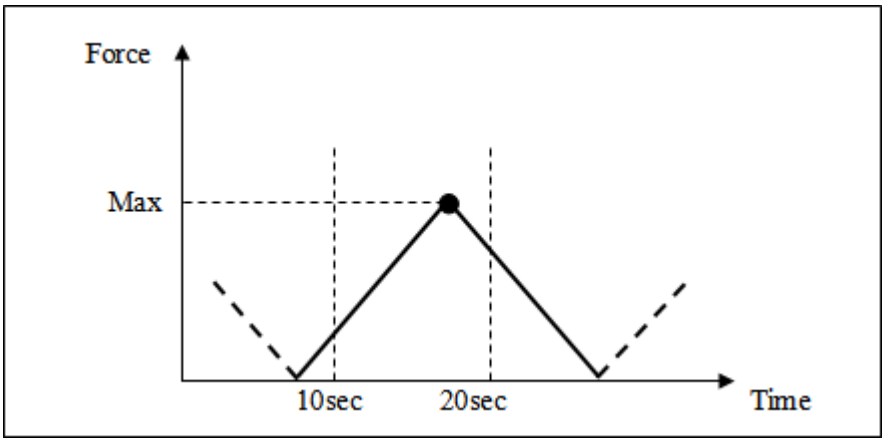
Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Example

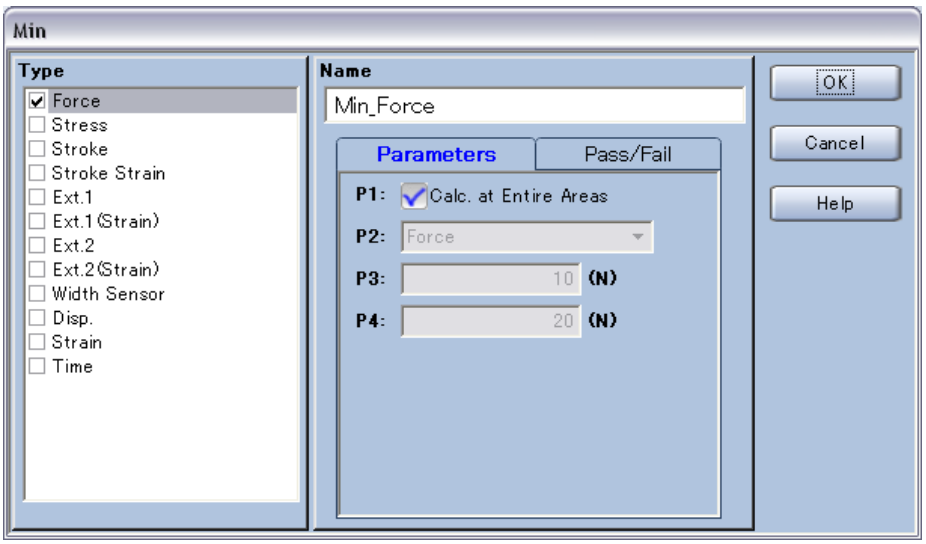
Assume [Calc. at Entire Areas] is not selected and P2 is set to [Time], P3 to 10 sec, and P4 to 20 sec.

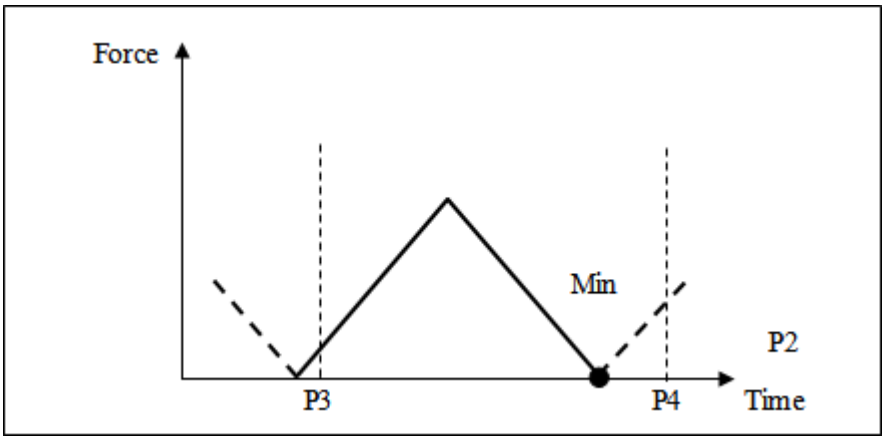


Min - Cycle Mode

Determines the minimum [Force] over a specified range or the entire area.

Parameters





- Specifies whether or not to search entire area (P1)

- Sensor Type (P2)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Starting Point (P3)

- Ending Point (P4)

* Parameter values are in terms of units specified in the [System] window.

* P2 to P4 can be specified only if [Calc. at Entire Areas] is not selected.

Type

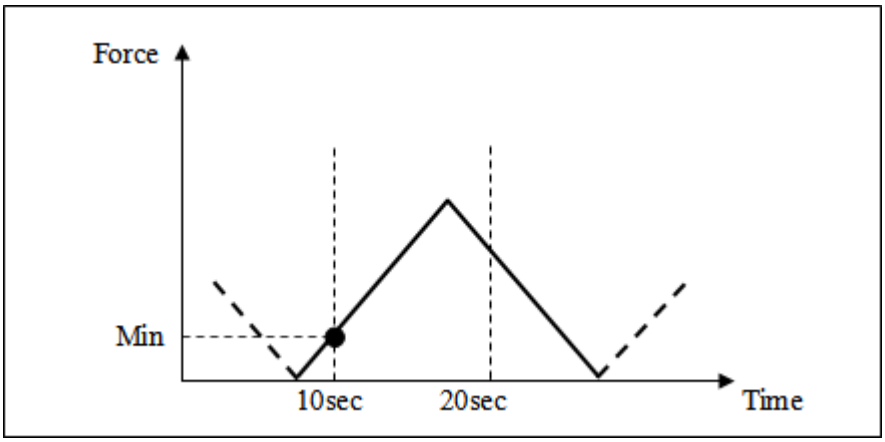
Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Example

Assume [Calc. at Entire Areas] is not selected and P2 is set to [Time], P3 to 10 sec, and P4 to 20 sec.

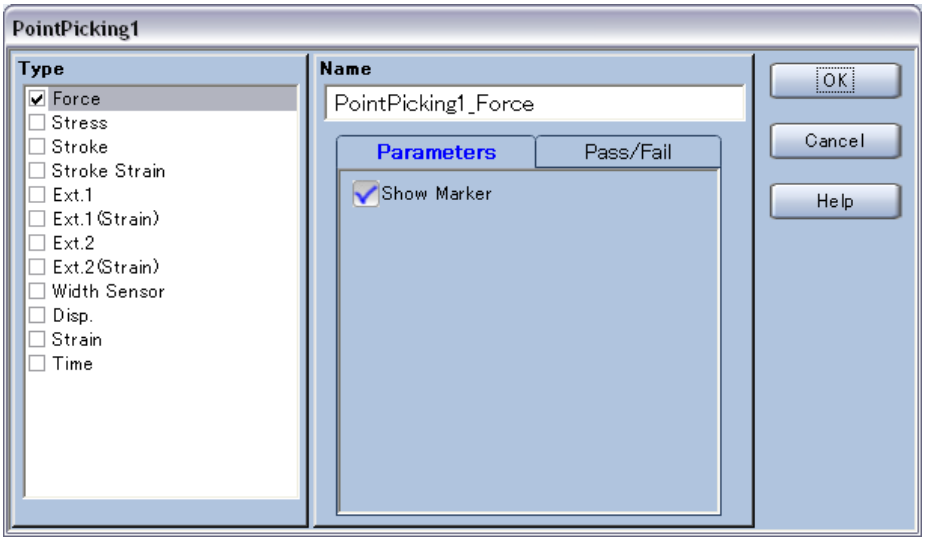


Point Picking - Cycle Mode

Mark

Determines post-test information for any point on the graph specified with the mouse pointer. Up to 20 points can be specified.

Parameters



- None

Type

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

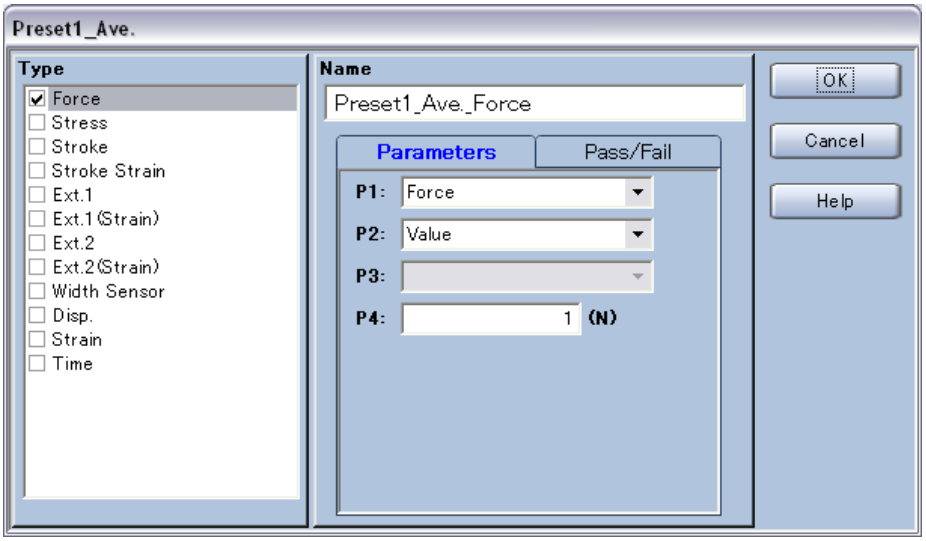
Default settings allow selecting from the following.

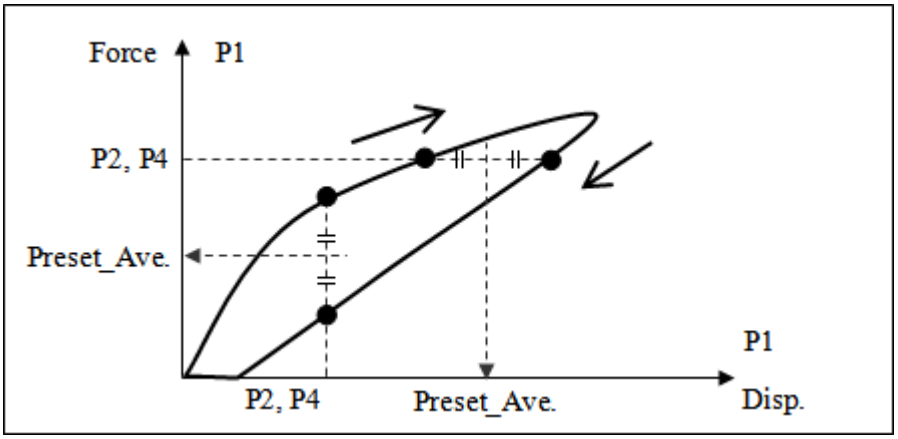
Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Preset_Ave - Cycle Mode

Determines the Preset_Ave value as the mean value of specific points when [Force] is increasing and decreasing that correspond to a specified parameter. Up to ten Preset_Ave points can be specified.

Parameters





- Sensor Type (P1)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Method for Specifying Parameter (P2): Select either [Formula] or [Value].

- Formula (P3): Enabled to specify only if [Formula] is specified in the [P2] box.

- Value (P4): Enabled to specify only if [Value] is specified in the [P2] box.

* Parameter values are in terms of units specified in the [System] window.

Type

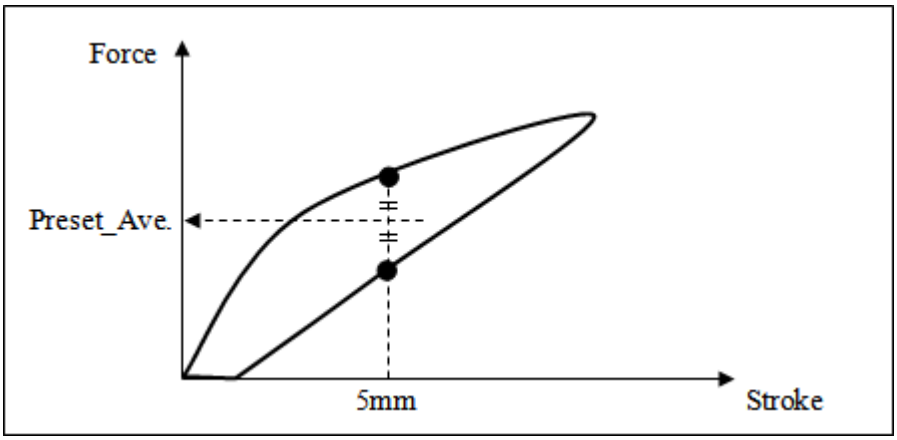
Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Example

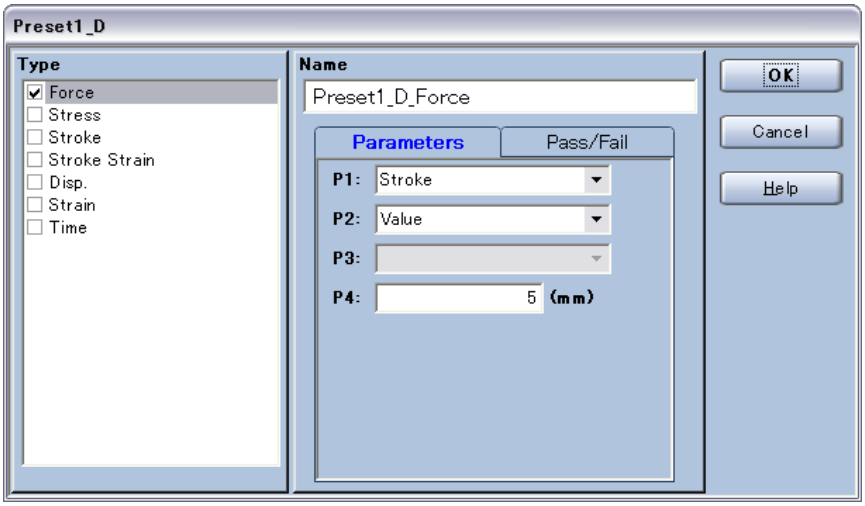
Assume the sensor type P1 is set to [Stroke], the specified parameter type P2 is [Value], and the value P4 is 5 mm.

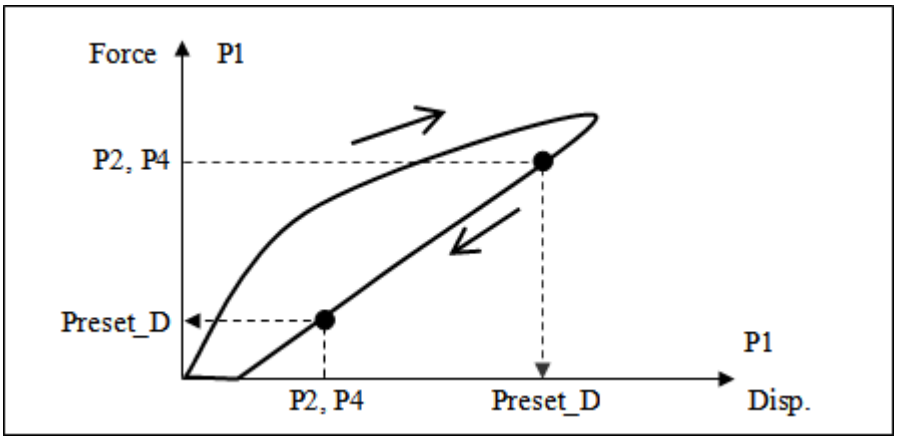


Preset_D - Cycle Mode

Determines the Preset_D value as the specific point when [Force] is decreasing that corresponds to a specified parameter. Up to ten Preset_D points can be specified.

Parameters





- Sensor Type (P1)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Method for Specifying Parameter (P2): Select either [Formula] or [Value].

- Formula (P3): Enabled to specify only if [Formula] is specified in the [P2] box.

- Value (P4): Enabled to specify only if [Value] is specified in the [P2] box.

* Parameter values are in terms of units specified in the [System] window.

Type

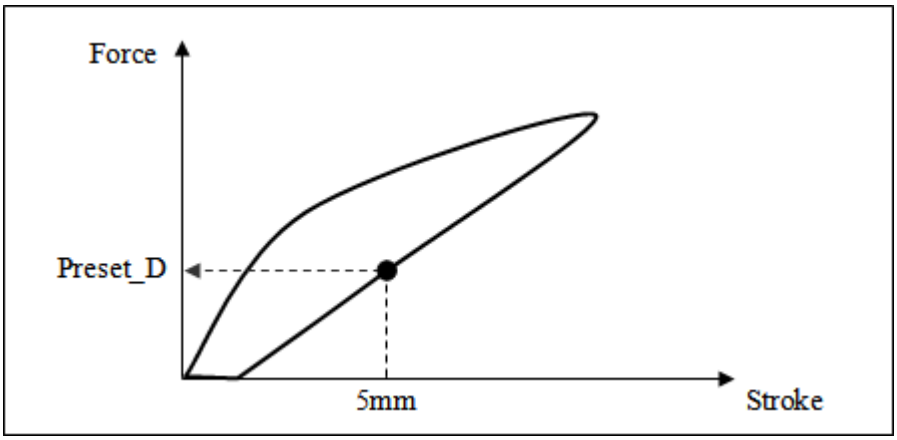
Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Example

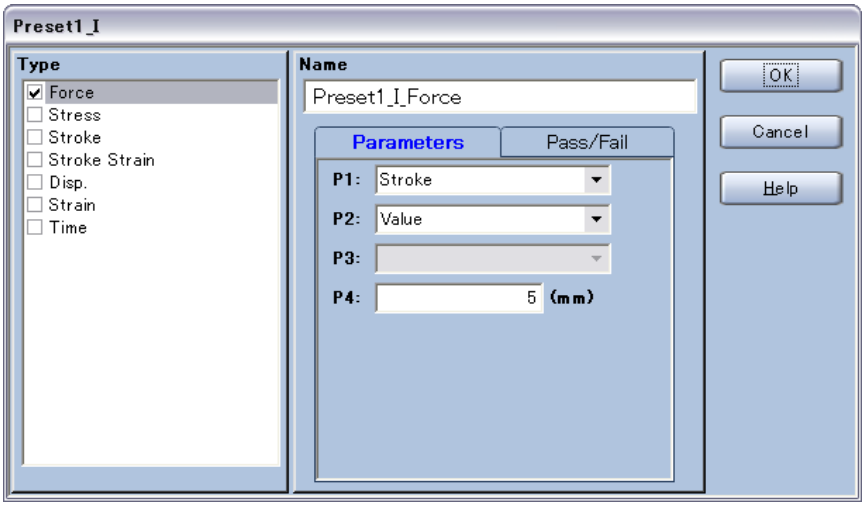
Assume the sensor type P1 is set to [Stroke], the specified parameter type P2 is [Value], and the value P4 is 5 mm.

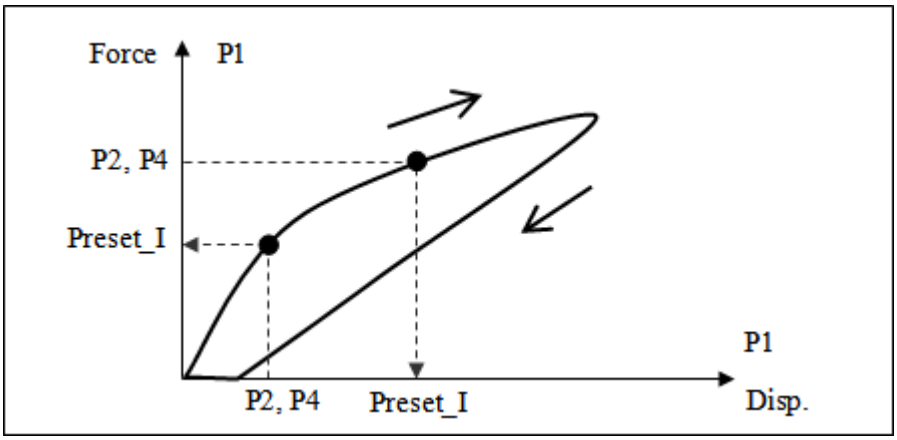


Preset_I - Cycle Mode

Determines the Preset_I value as the specific point when [Force] is increasing that corresponds to a specified parameter. Up to ten Preset_I points can be specified.

Parameters





- Sensor Type (P1)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Method for Specifying Parameter (P2): Select either [Formula] or [Value].

- Formula (P3): Enabled to specify only if [Formula] is specified in the [P2] box.

- Value (P4): Enabled to specify only if [Value] is specified in the [P2] box.

* Parameter values are in terms of units specified in the [System] window.

Type

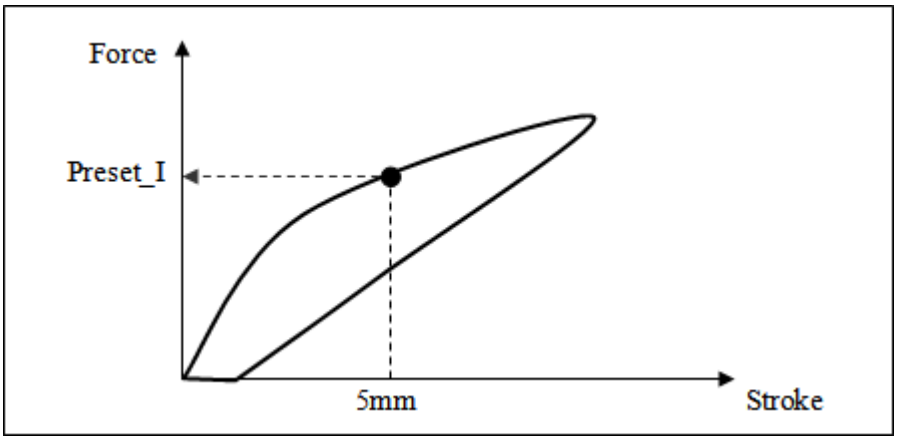
Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Example

Assume the sensor type P1 is set to [Stroke], the specified parameter type P2 is [Value], and the value P4 is 5 mm.



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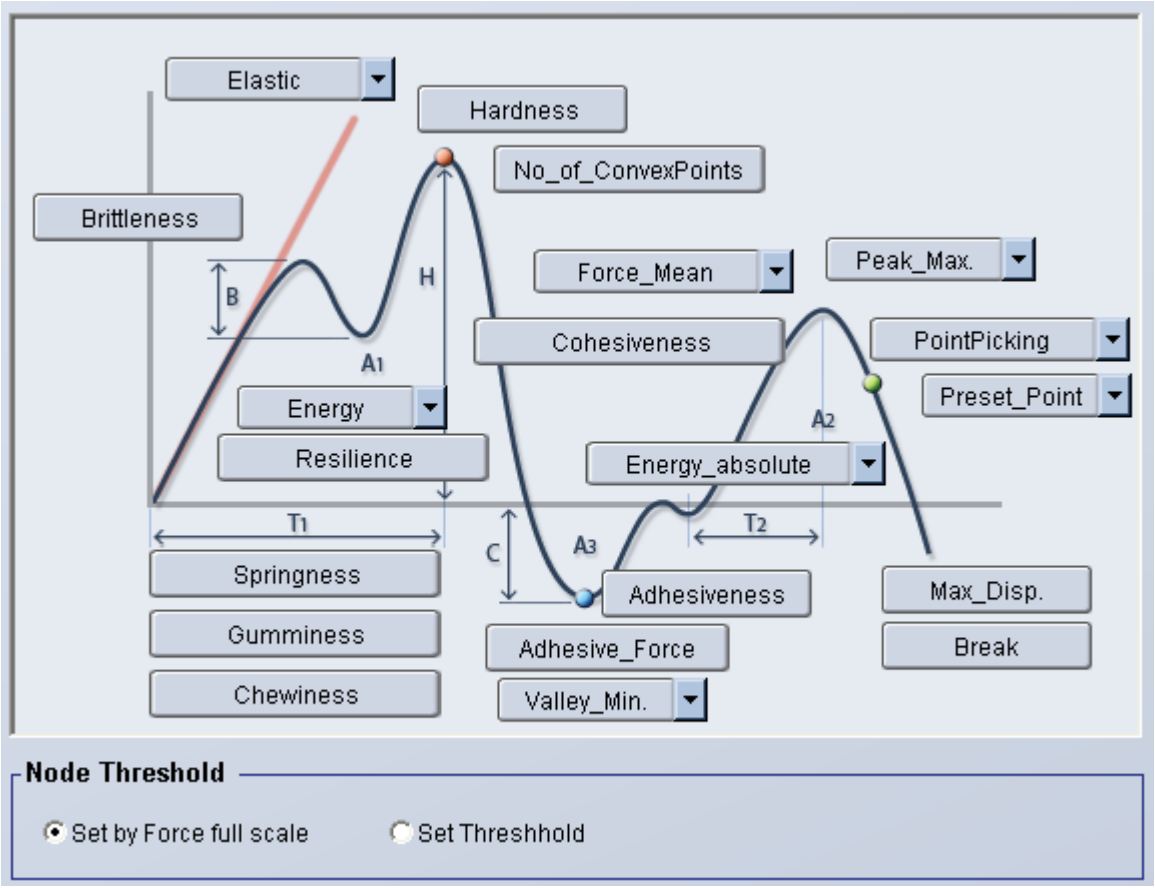
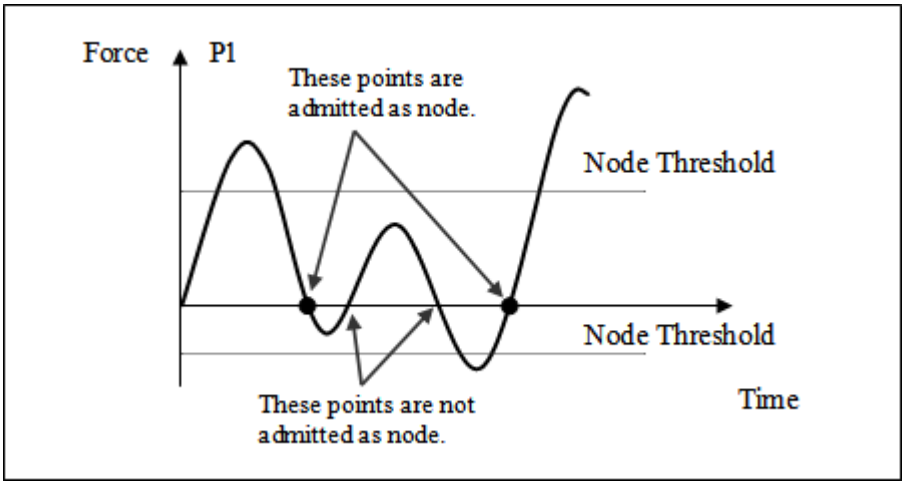
Chapter 4
Data Processing Items in Texture Mode

This chapter describes processing data in the texture mode. In the texture mode, the procedure for processing tensile, compression, and bending data is the same.



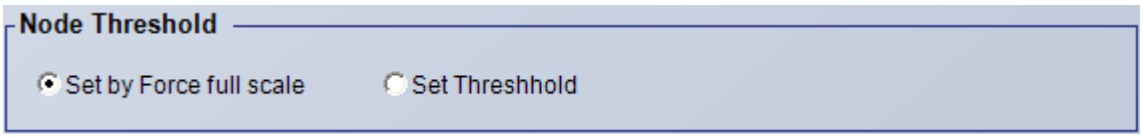
Node Threshold

When [Force] moves more than a specified threshold value away from the zero force axis, then returns to the zero axis, the point where the graph crosses the zero axis is considered a node.



Select either of the following node threshold parameters.

- Set by Force Full Scale



Node thresholds are determined by the following equation.

$$\text{Node Threshold} = \text{Test Force Range} \times \frac{0.01}{500}$$

For the four models indicated below, the [Force] range is given as the full scale range.

For all other models, the [Force] range is given as the load cell capacity.

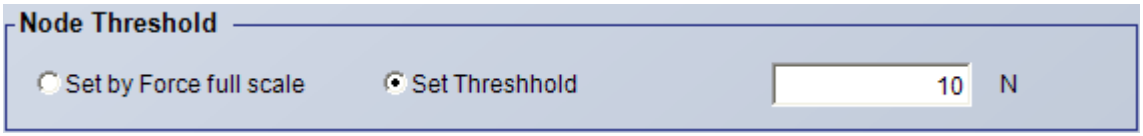
Models With Full-Scale Range as the Force Range	AG-IS
	EZGRAPH
	MST-I
	AG-I
Models With Load Cell Capacity as the Force Range	All models other than the four above.

Example

Assume the model is AG-X and the load cell capacity is 100 kN.

$$\text{NodeThreshold} = 100 \times 0.01 \div 500 = 0.002\text{kN} = 2\text{N}$$

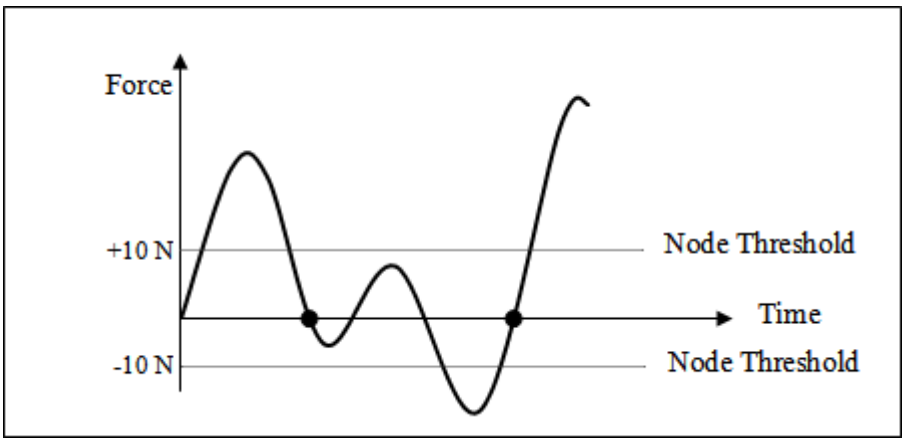
- Set Threshhold



A user specified value is used as the node threshold (does not depend on model).

Example

Assume node threshold is specified as 10 N.

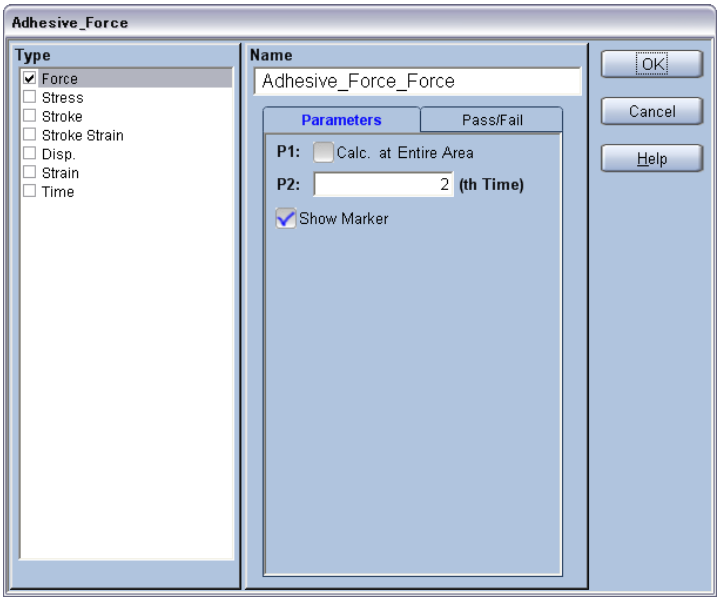


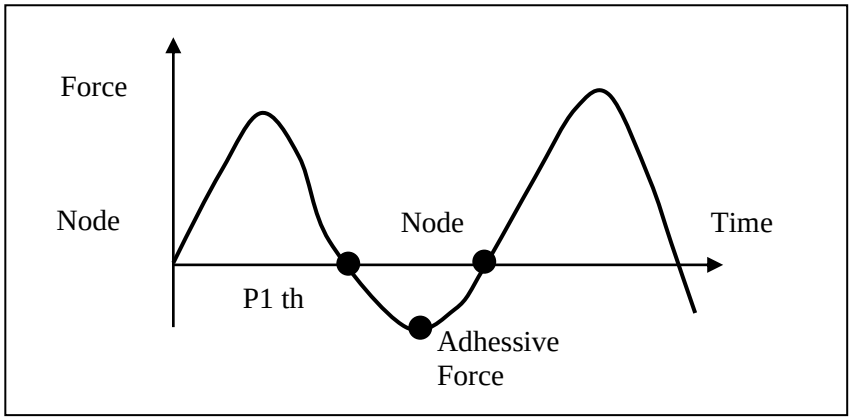
Adhessive Force

Mark

Determines the minimum [Force] over a specified range or the entire area.

Parameters





- Specifies whether or not to search entire area (P1)

- Number of Intersections (P2):

  Data is process between the specified starting node and the next node.

Type

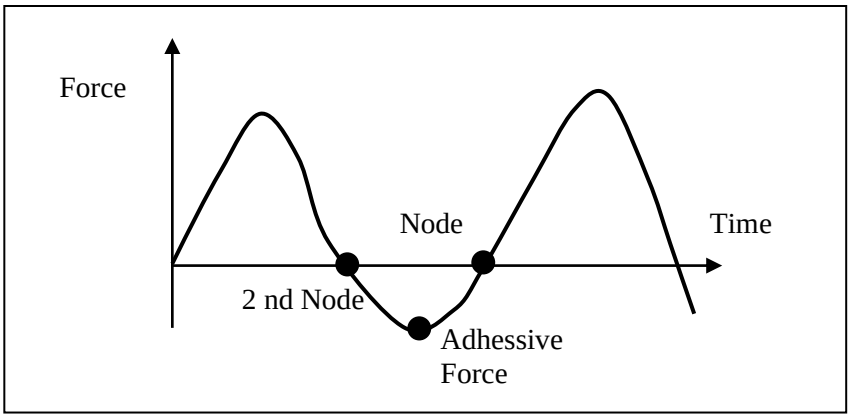
Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

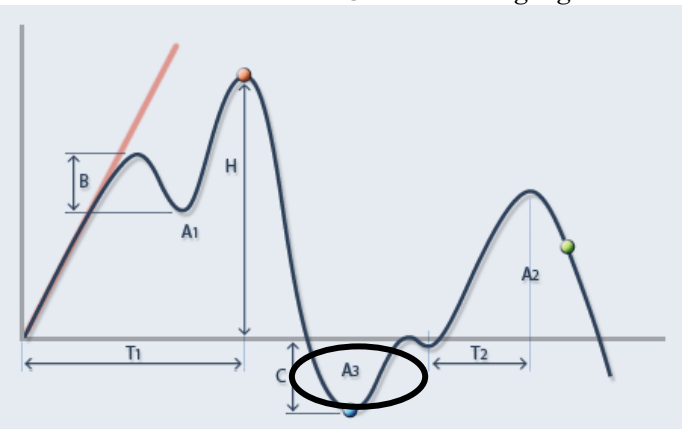
Example

Assume [Calc. at Entire Areas] is not selected and P2 is set to 1 (first intersection).

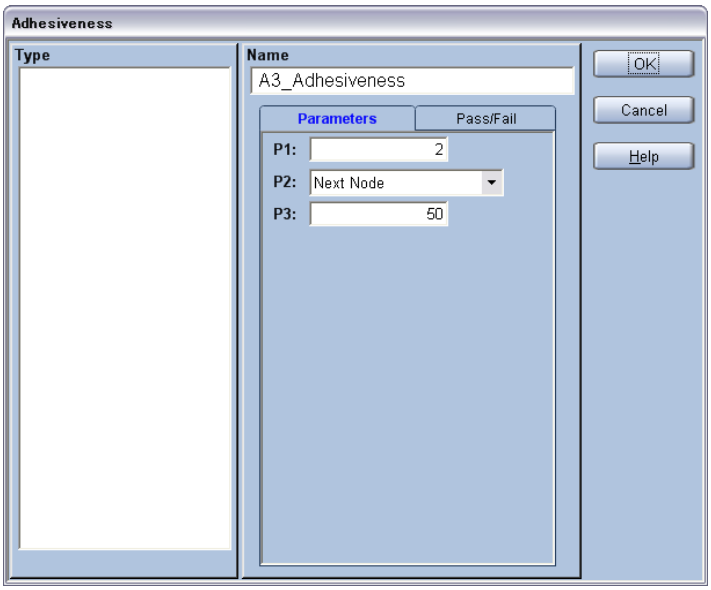


Adhessiveness

Adhesiveness is area of A3 in following figure.



Parameters



- Start Point Parameter (P1): Select [Node] or [Auto].

  For P2 - P4 in [Auto] mode, refer to the later pages.

- Number of Intersections (P2): Starting node.

- End Point Parameter (P3): Ending point parameter is selected from the following.

Next Node	Next Peak	Next Valley
Next Max.	Next Min.	-

- Peak and Valley Detection Sensitivity (P4)

  Specifies sensitivity to use for detecting [Next Peak] and [Next Valley]. Sensitivity is specified as a percent of full-scale range specified for decrease of [Force].

Type

None

Auto Mode

An area in which a compression plate does not attach to the sample can be removed by using [Auto] mode.

When you use [Auto] mode, register data processing items “Energy absolute 1” and “Energy absolute 2” previously.

In case of P1 [Auto], other parameters are following:

- Point Number (P2): Enter number of points.

- End Point Parameter (P3): Not use.

- End Point (P4): Enter the ratio (%) to load cell capcity (or range full scale).

Determine the 2 points (calculation start point and calculation end point) as following process. Determines energy as the area formed by the graph between these two specified points and the zero [Force] line

1. Find the minimum force point between “Energy absolute 1” and “Energy absolute 2”.

2. Search forward from minimum force point and find the point which satisfiy at least one of the following 3 rules. This point is calculation start point of “Adhesiveness”.

   a) Point just before crossing line segment with test force 0.

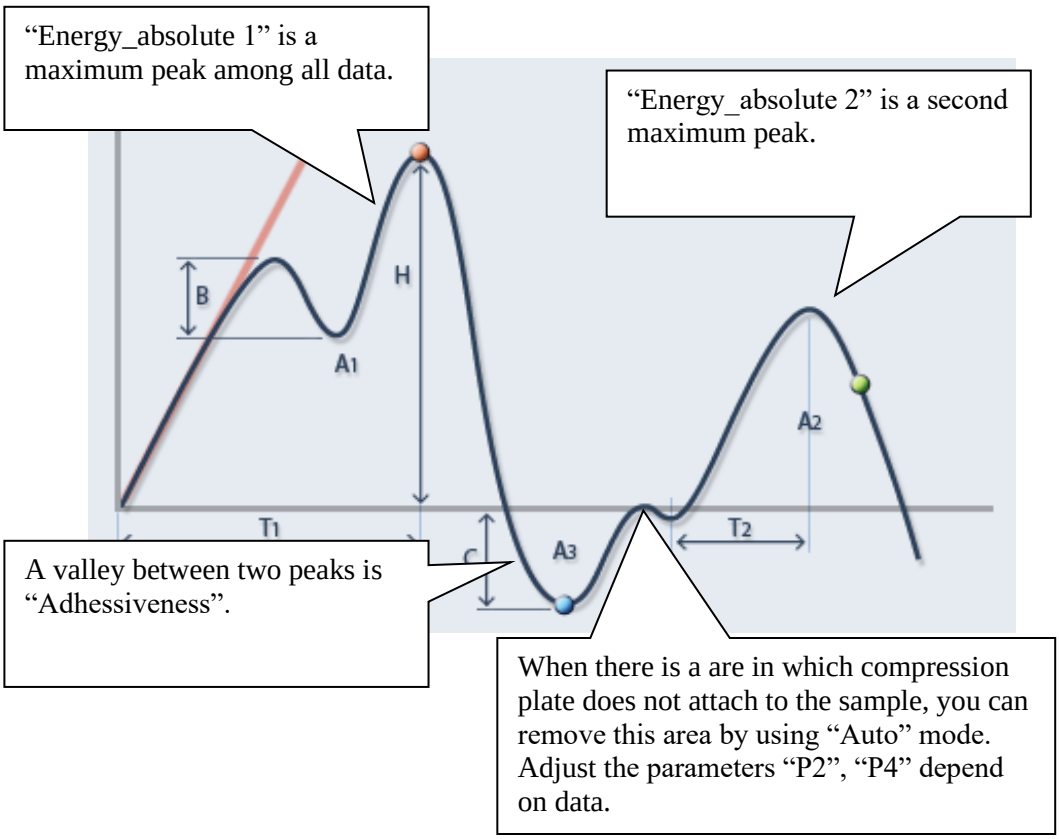
   b) Point just before the calculation end point of “Energy absolute 1”.

3. Search backward from minimum force point and find the point which satisfiy at least one of the following 3 rules. This point is calculation end point of “Adhesiveness”.

   a) Point just before crossing line segment with test force 0.

   b) First point in the continueous (more than P2) of the lower force changing than P4 value.

   c) Point just before the calculation start point of “Energy absolute 2”.



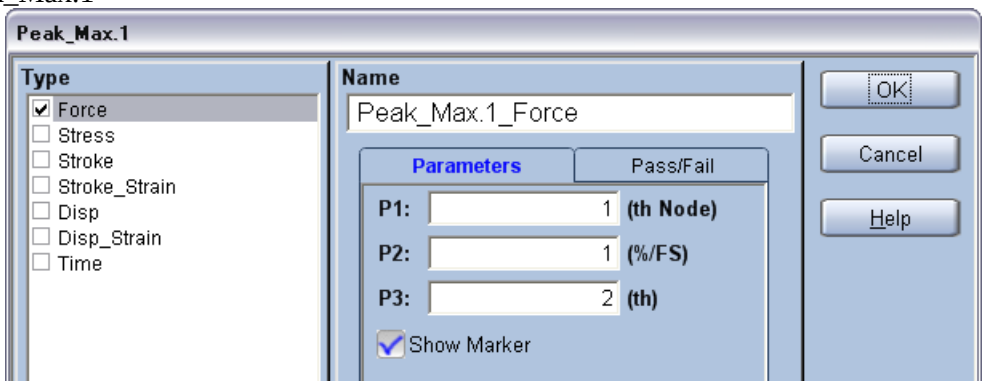
Brittleness

Brittleness is expressed by the following calculating fomula.

B = Peak_Max.Force – Valley_min.Force

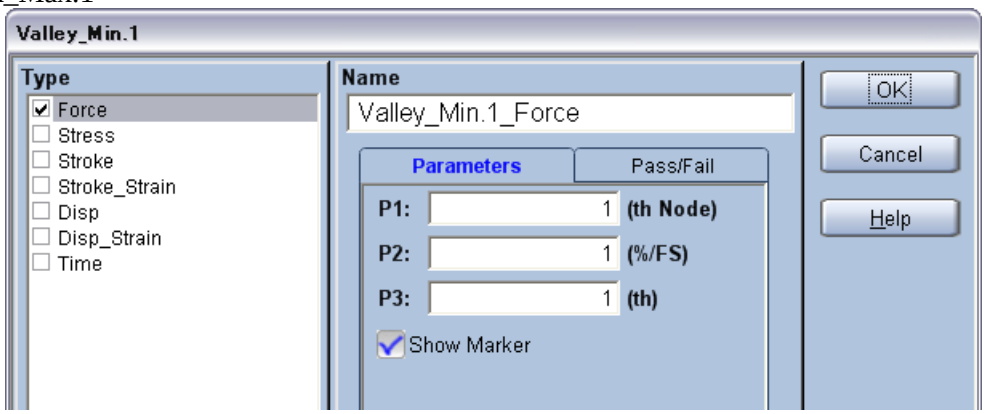
When you use this item, register data processing items “Peak_Max.1_Force” and “Valley_Min.1_Force”. These items parameter is following.

Peak_Max.1



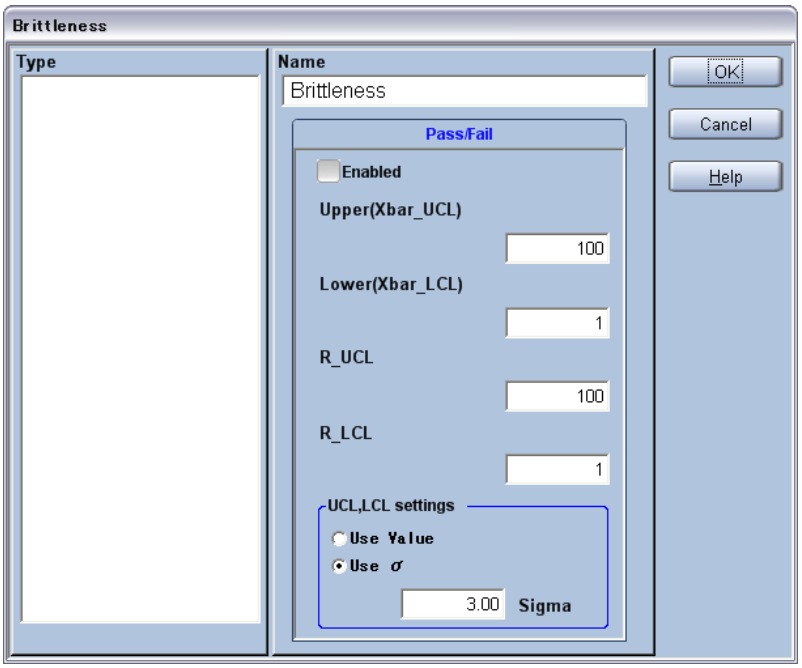
P1 : 1 (th Node)
P2 : 1 (%/FS)
P3 : 2 (th)

Peak_Max.1



P1 : 1 (th Node)
P2 : 1 (%/FS)
P3 : 1 (th)

Parameters



Type

None

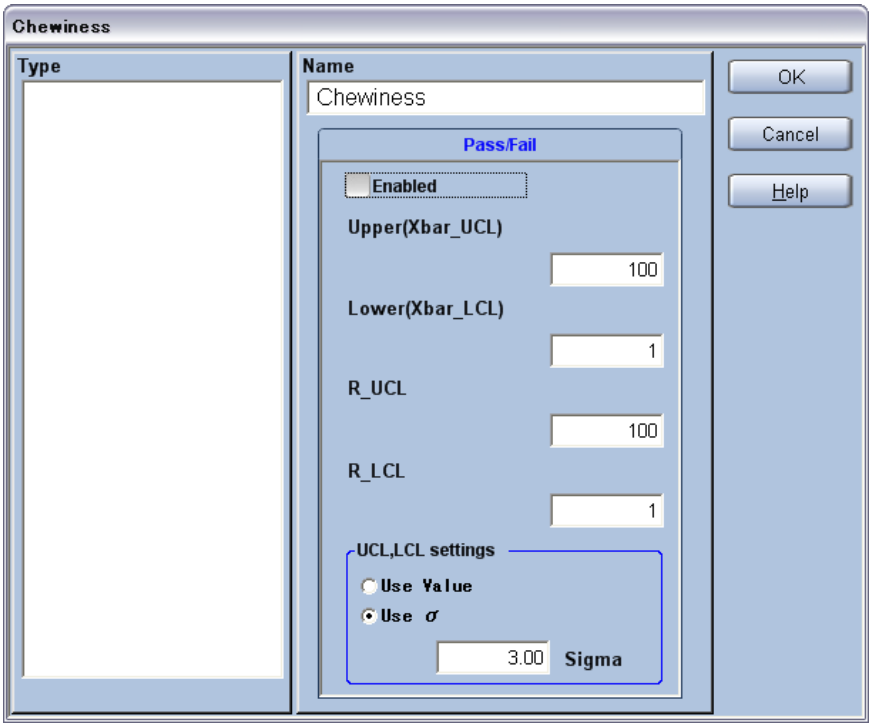
Chewiness

Chewiness is expressed by following equation.

Chewiness = Guminess * Springness

When you use this item, register data processing items “Guminess” and “Springness” previously.

Parameters



None

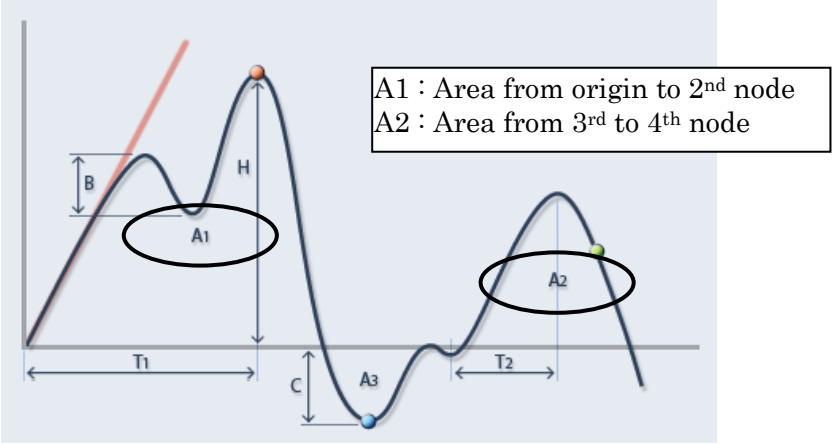
Type

None

Cohessiveness

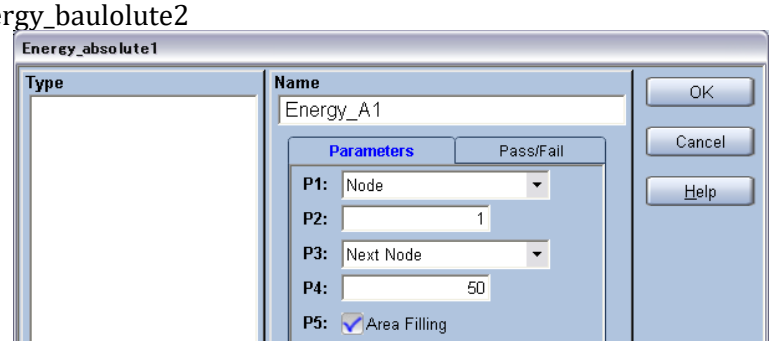
Cohesiveness is expressed by the following calculating fomula.

Cohesiveness = A2 / A1



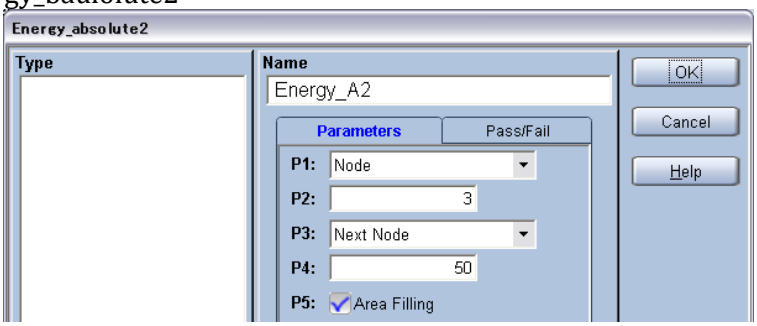
When you use this item, register data processing items “Energy_abusolute1” and “Energy_abusolute2”. These items parameter is following.

Energy_baulolute2



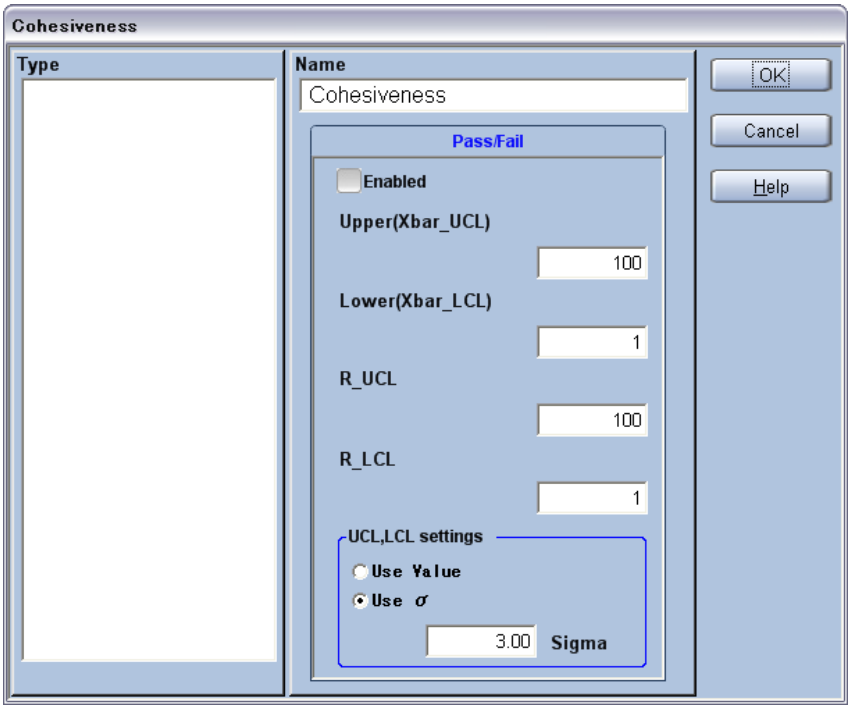
P1 : Node
P2 : 1 (th node)
P3 : Next Node
P4 : 50 (%/FS)

Energy_baulolute2



P1 : Node
P2 : 3 (th node)
P3 : Next Node
P4 : 50 (%/FS)

Parameters



None

Type

None

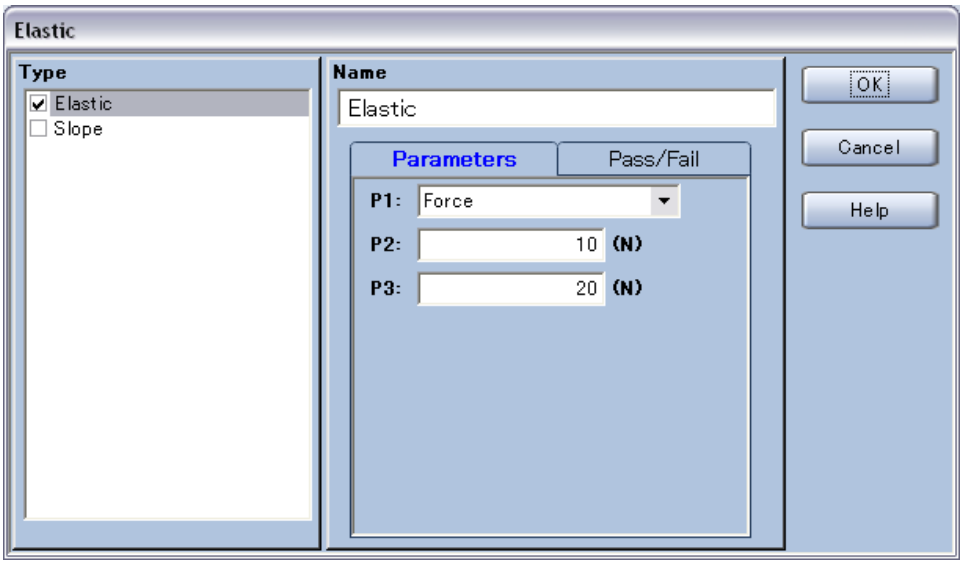
Elastic(Standard) - Texture Mode

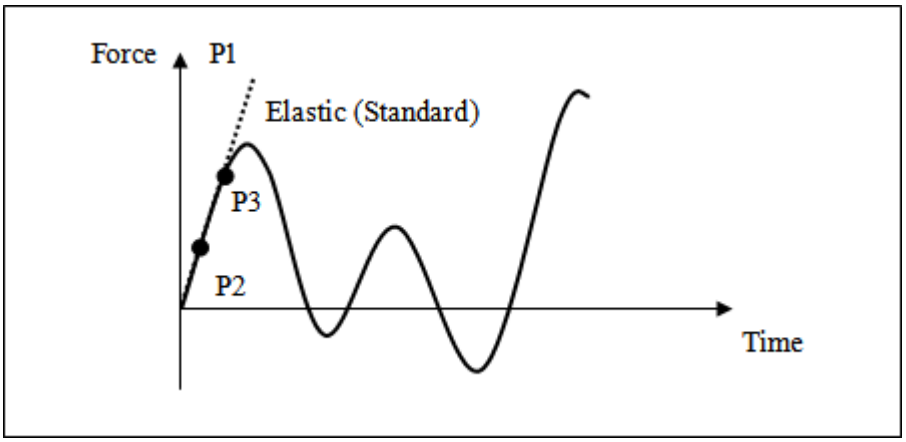
Determines elastic using the least squares method from all data between two pre-specified points. If multiple parameters are specified, elastic is determined from the first applicable point.

$$\text{Elastic} = \text{Slope} \times \frac{\text{Gauge Length}}{\text{Cross - Sectional Area of Specimen}}$$

Slope	Calculated from all data between two points using least squares method.
Gauge Length	Value specified via the [Sensor] or [Specimen] dialog box.
Cross-Sectional Area of Specimen	Calculated from dimensions specified in the [Specimen] dialog box.

Parameters





- Sensor Type (P1)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Starting Point (P2)

- Ending Point (P3)

* Parameter values are in terms of units specified in the [System] window.

Type

- Elastic

Calculated from [Slope], [Gauge Length], and [Specimen Cross-Sectional Area].

- Slope

Calculates only slope of elastic.

Example

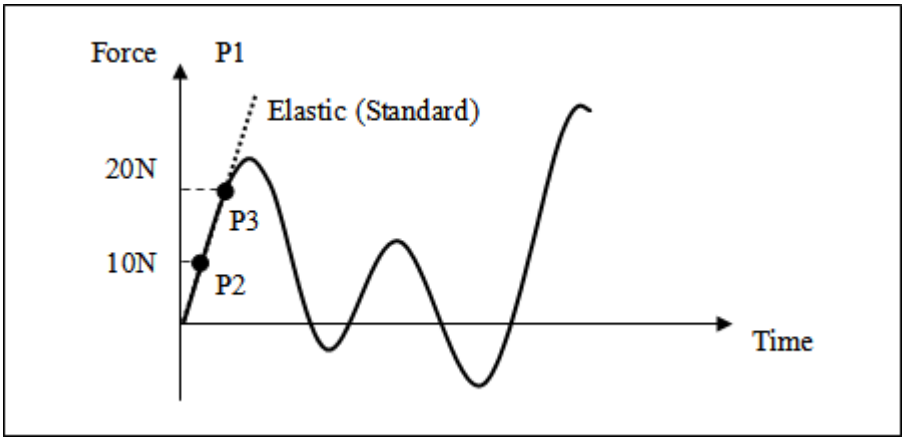
Assume P1 is set to [Force], P2 to 10 N, P3 to 20 N, gauge length to 50 mm, and the specimen is a flat plastic plate 2 mm thick and 5 mm wide.

$$\text{Elastic} = \text{Slope} \times \frac{50\text{mm}}{2\text{mm} \times 5\text{mm}}$$

Note

Slope is calculated using least squares method.

The formula used to calculate cross-sectional area depends on specimen shape. For more details, refer to Methods for Calculating Cross-Sectional Area.



Related keyword: “Calculating Stress and Strain”

Elastic(Chord) - Texture Mode

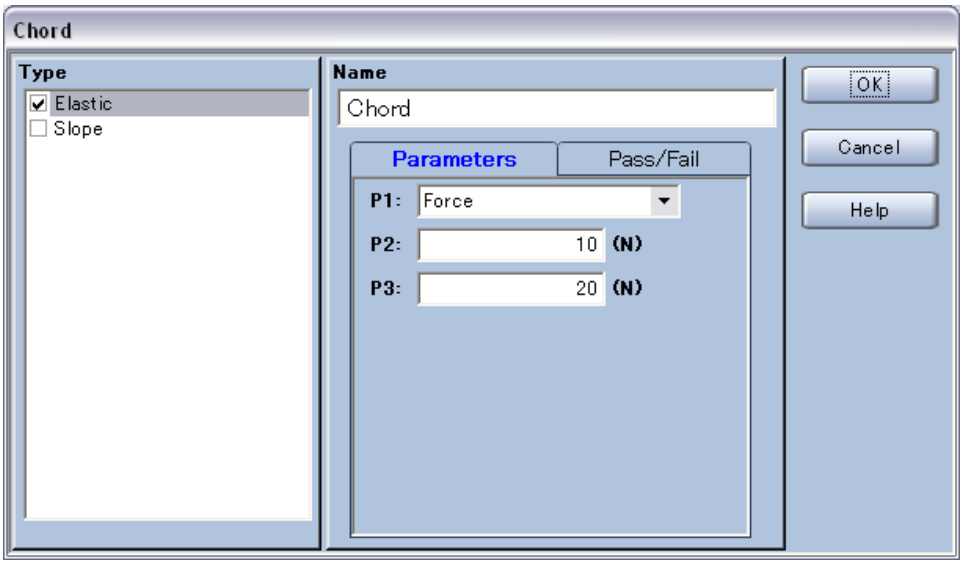
Determines elastic from the slope of a straight line between two pre-specified force or [Disp.] points.

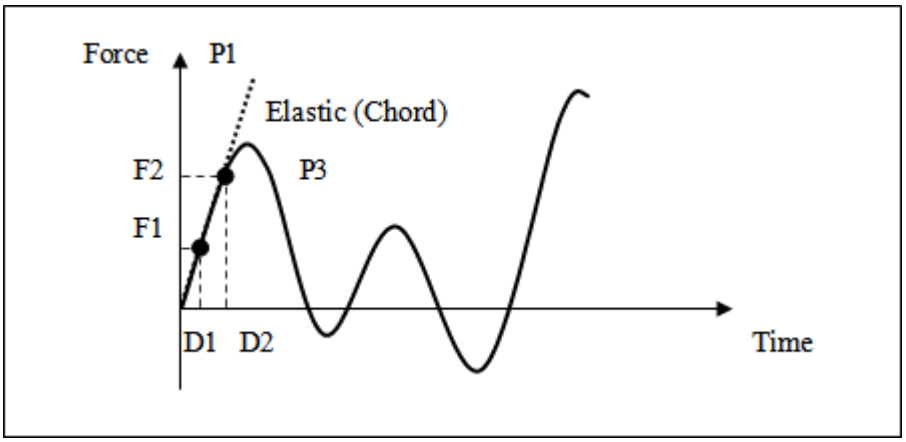
If multiple parameters are specified, elastic is determined from the first applicable point.

$$\text{Elastic} = \text{Slope} \times \frac{\text{Gauge Length}}{\text{Cross - Sectional Area of Specimen}}$$

Slope	Calculated from difference between two points.
Gauge Length	Value specified via the [Sensor] or [Specimen] dialog box.
Cross-Sectional Area of Specimen	Calculated from dimensions specified in the [Specimen] dialog box.

Parameters





- Sensor Type (P1)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Starting Point (P2)

- Ending Point (P3)

* Parameter values are in terms of units specified in the [System] window.

Type

- Elastic

Calculated from [Slope], [Gauge Length], and [Specimen Cross-Sectional Area].

- Slope

Calculates only slope of elastic.

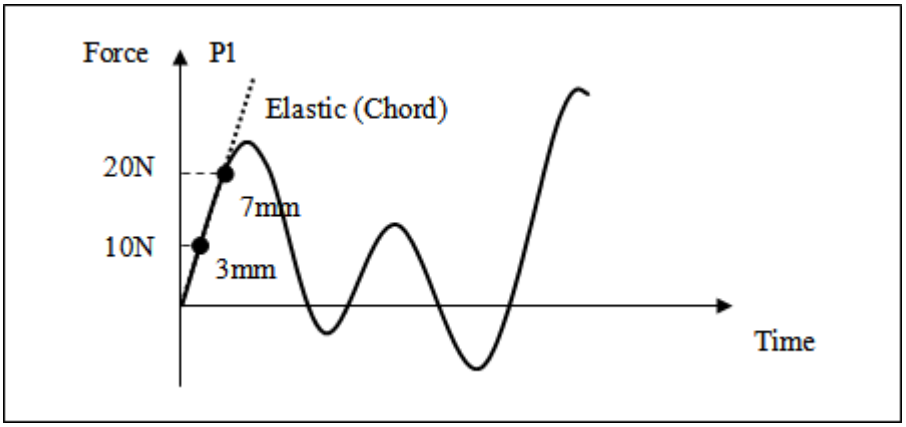
Example

Assume [Force] F1 is set to 10 N, F2 to 20 N, Disp. D1 to 3 mm, D2 to 7 mm, and gauge length to 50 mm, for a plastic flat plate specimen 2 mm thick and 5 mm wide.

$$\text{Elastic} = \frac{20\text{N} - 10\text{N}}{7\text{mm} - 3\text{mm}} \times \frac{50\text{mm}}{2\text{mm} \times 5\text{mm}}$$

Note

The formula used to calculate cross-sectional area depends on specimen shape. For more details, refer to Methods for Calculating Cross-Sectional Area.



Related keyword: “Calculating Stress and Strain”

Elastic(Secant) - Texture Mode

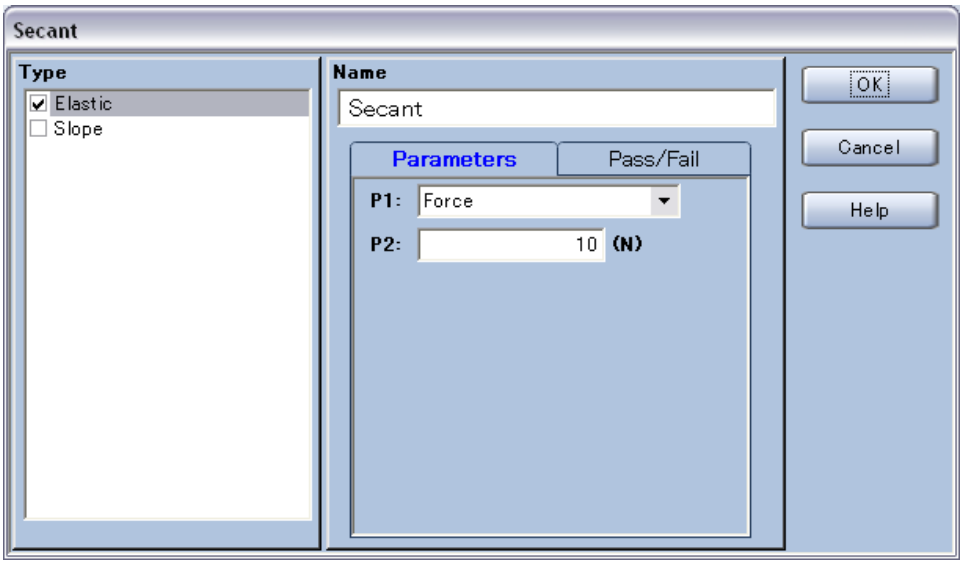
Determines elastic from the test starting point and another pre-specified point.

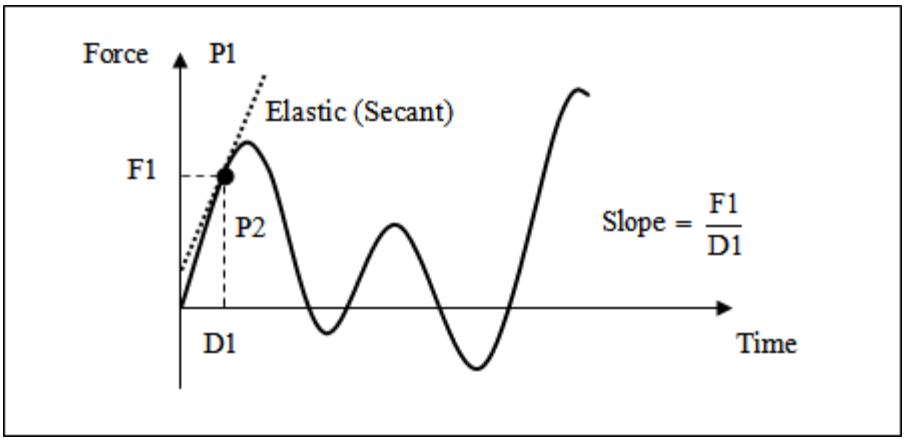
If multiple parameters are specified, elastic is determined from the first applicable point.

$$\text{Elastic} = \text{Slope} \times \frac{\text{Gauge Length}}{\text{Cross - Sectional Area of Specimen}}$$

Slope	Calculated from difference between starting point of test and another specified point.
Gauge Length	Value specified via the [Sensor] or [Specimen] dialog box.
Cross-Sectional Area of Specimen	Calculated from dimensions specified in the [Specimen] dialog box.

Parameters





- Sensor Type (P1)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Value for Sensor Type (P2)

* Parameter values are in terms of units specified in the [System] window.

Type

- Elastic

Calculated from [Slope], [Gauge Length], and [Specimen Cross-Sectional Area].

- Slope

Calculates only slope of elastic.

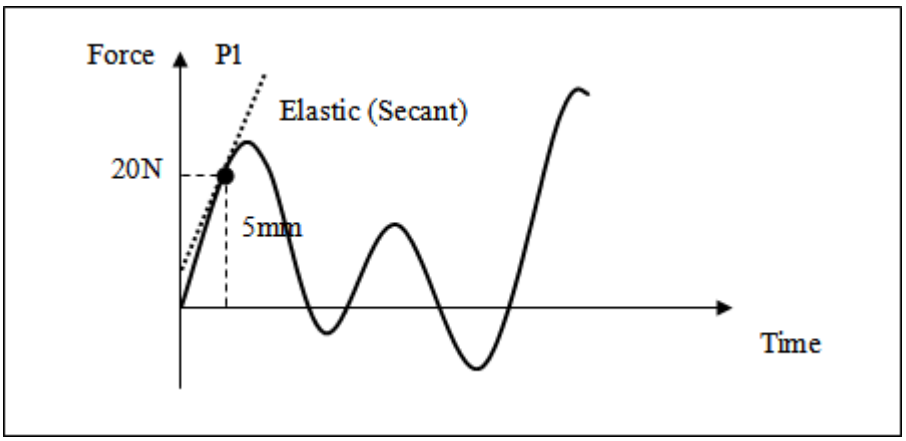
Example

Assume [Force] F1 is set to 20 N, Disp. D1 to 5 mm, and gauge length to 50 mm, for a plastic flat plate specimen 2 mm thick and 5 mm wide.

$$\text{Elastic} = \frac{20\text{N}}{5\text{mm}} \times \frac{50\text{mm}}{2\text{mm} \times 5\text{mm}}$$

Note

The formula used to calculate cross-sectional area depends on specimen shape. For more details, refer to Methods for Calculating Cross-Sectional Area.



Related keyword: “Calculating Stress and Strain”

Elastic(Tangent) - Texture Mode

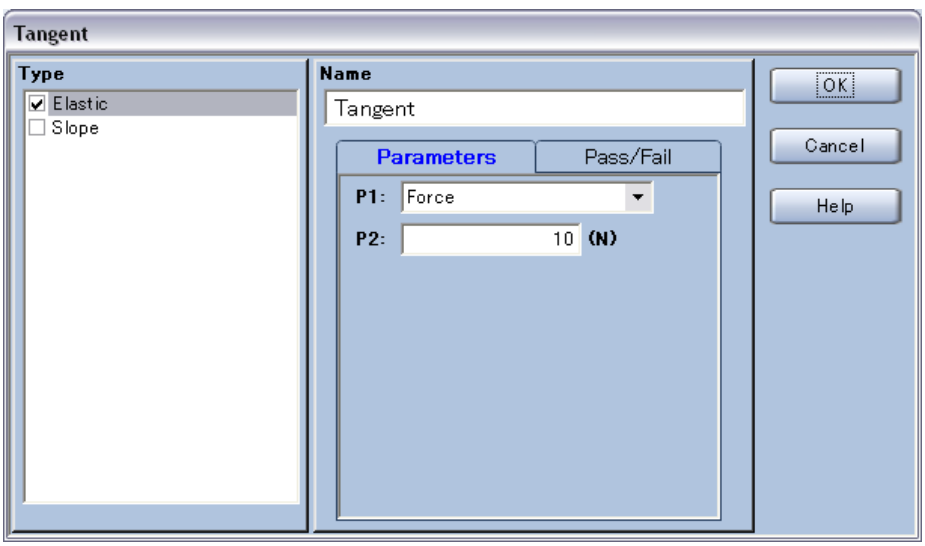
Determines elastic from the slope of a straight line between a point just before and after a pre-specified point.

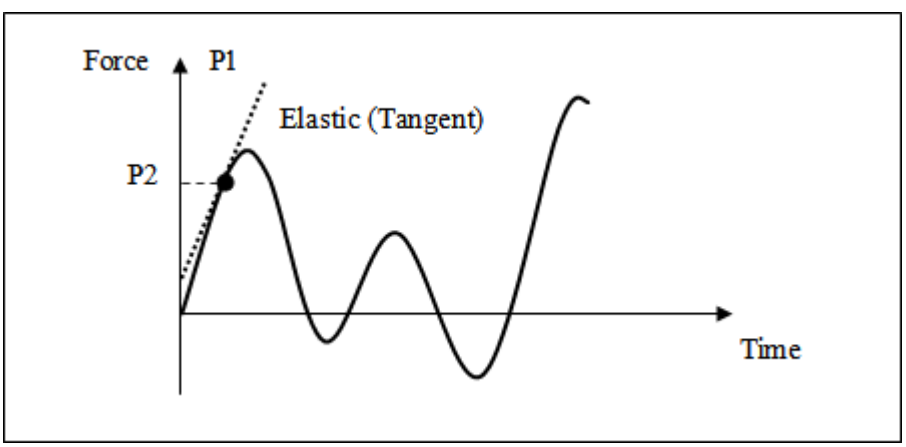
If multiple parameters are specified, elastic is determined from the first applicable point.

$$\text{Elastic} = \text{Slope} \times \frac{\text{Gauge Length}}{\text{Cross - Sectional Area of Specimen}}$$

Slope	Slope of straight line through a point on either side of a specified point.
Gauge Length	Value specified via the [Sensor] or [Specimen] dialog box.
Cross-Sectional Area of Specimen	Calculated from dimensions specified in the [Specimen] dialog box.

Parameters





- Sensor Type (P1)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Value for Sensor Type (P2)

* Parameter values are in terms of units specified in the [System] window.

Type

- Elastic

Calculated from [Slope], [Gauge Length], and [Specimen Cross-Sectional Area].

- Slope

Calculates only slope of elastic.

Example

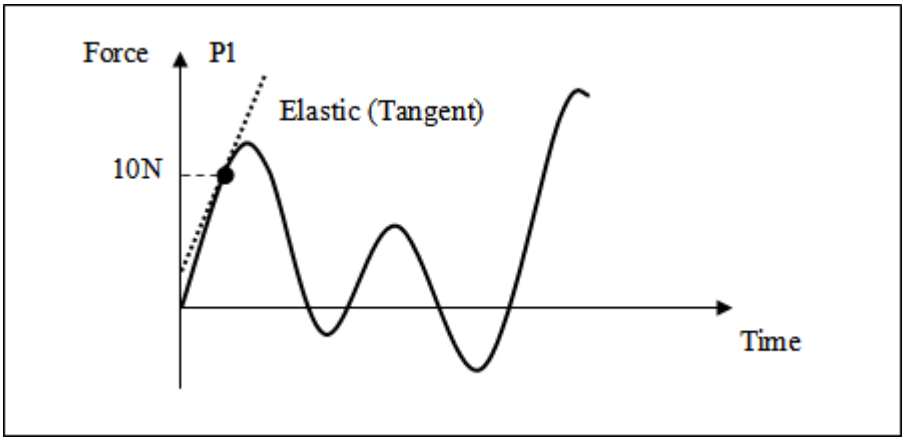
Assume P1 is set to [Force], P2 to 10 N, gauge length to 50 mm, and the specimen is a flat plastic plate 2 mm thick and 5 mm wide.

$$\text{Elastic} = \text{Slope} \times \frac{50\text{mm}}{2\text{mm} \times 5\text{mm}}$$

Note

Slope is slope of straight line through a point on either side of a specified point.

The formula used to calculate cross-sectional area depends on specimen shape. For more details, refer to Methods for Calculating Cross-Sectional Area.



Related keyword: “Calculating Stress and Strain”

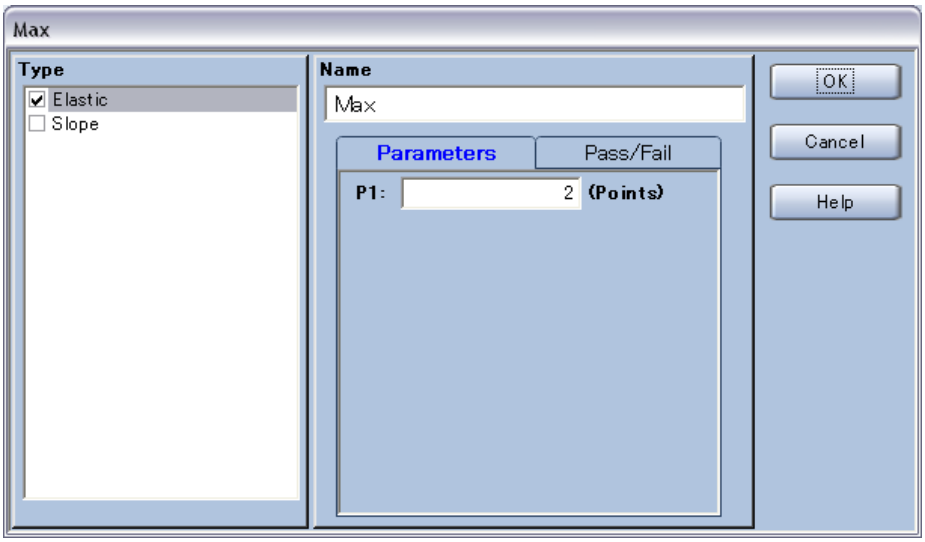
Elastic(Max. Slope) - Texture Mode

Calculates modulus of elastic based on the maximum slope found from calculating the slope at specified intervals for all sampling data. The black nodes shown below indicate sampling data points.

$$\text{Elastic} = \text{Slope} \times \frac{\text{Gauge Length}}{\text{Cross - Sectional Area of Specimen}}$$

Slope	Calculated as maximum slope from all data points.
Gauge Length	Value specified via the [Sensor] or [Specimen] dialog box.
Cross-Sectional Area of Specimen	Calculated from dimensions specified in the [Specimen] dialog box.

Parameters



- Data Intervals for Calculating Slope (P1)

Type

- Elastic

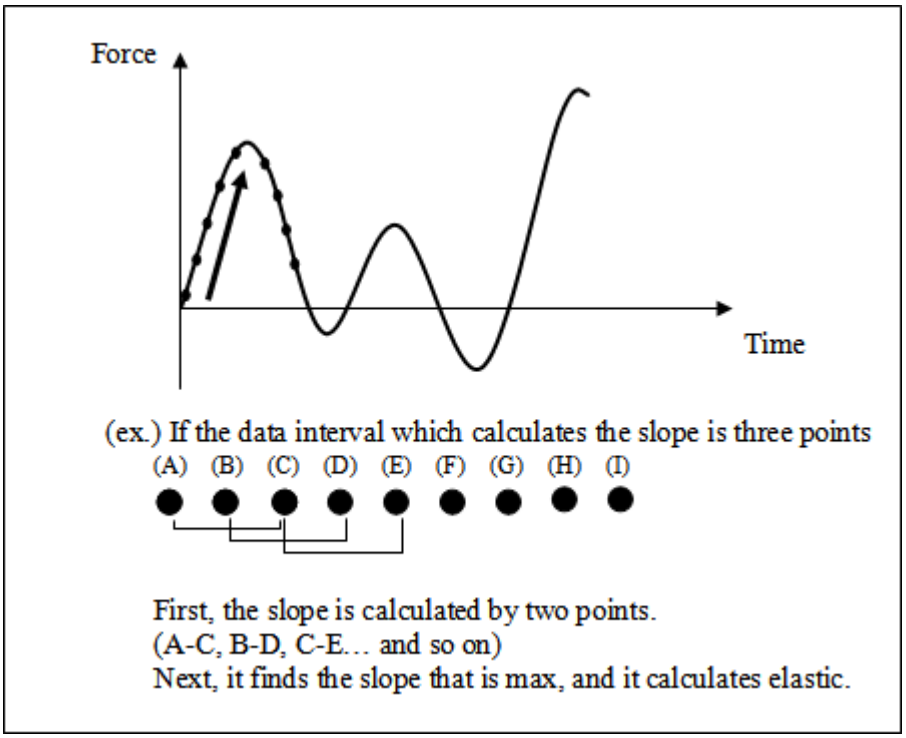
  Calculated from [Slope], [Gauge Length], and [Specimen Cross-Sectional Area].

- Slope

  Calculates only slope of elastic.

Example

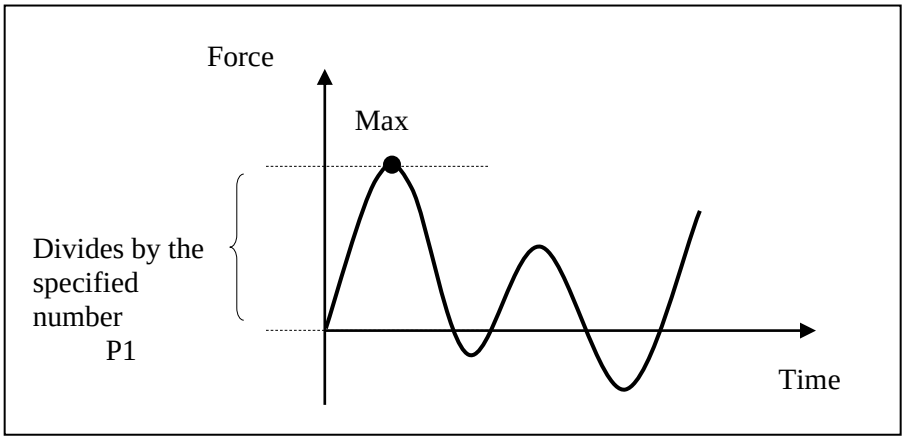
Assume P1 is set to 3 points for calculating slope.



Elastic(Devided) - Texture Mode

Determines elastic by using data until reaching [Max].

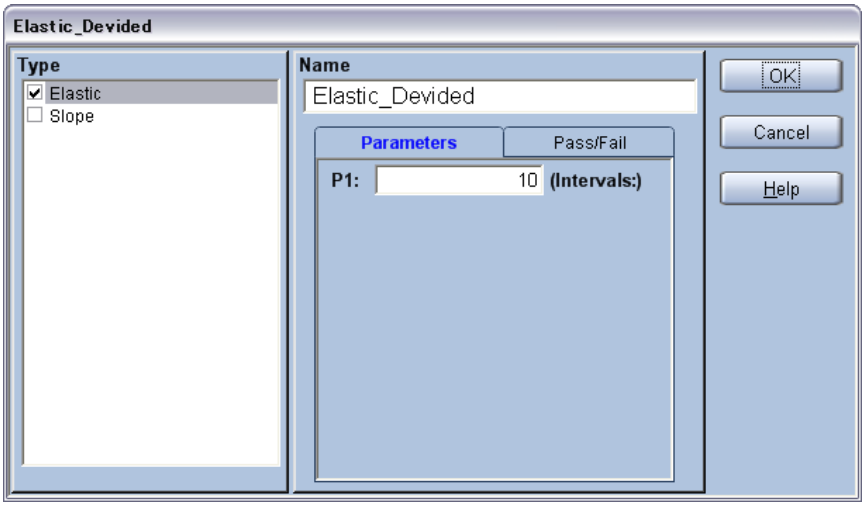
Data from [Disp. Origin] to [Max] is divided, and the inclination of "line of tangency" of each area is calculated by using the least squares. This data processing item is determined by using the value that is the maximum of these inclinations.



$$\text{Elastic} = \text{Slope} \times \frac{\text{Gauge Length}}{\text{Cross - Sectional Area of Specimen}}$$

Slope	Calculated by using the above-mentioned logic.
Gauge Length	Value specified via the [Sensor] or [Specimen] dialog box.
Cross-Sectional Area of Specimen	Calculated from dimensions specified in the [Specimen] dialog box.

Parameters



- Number of Intervals (P1)

Type

- Elastic

  Calculated from [Slope], [Gauge Length], and [Specimen Cross-Sectional Area].

- Slope

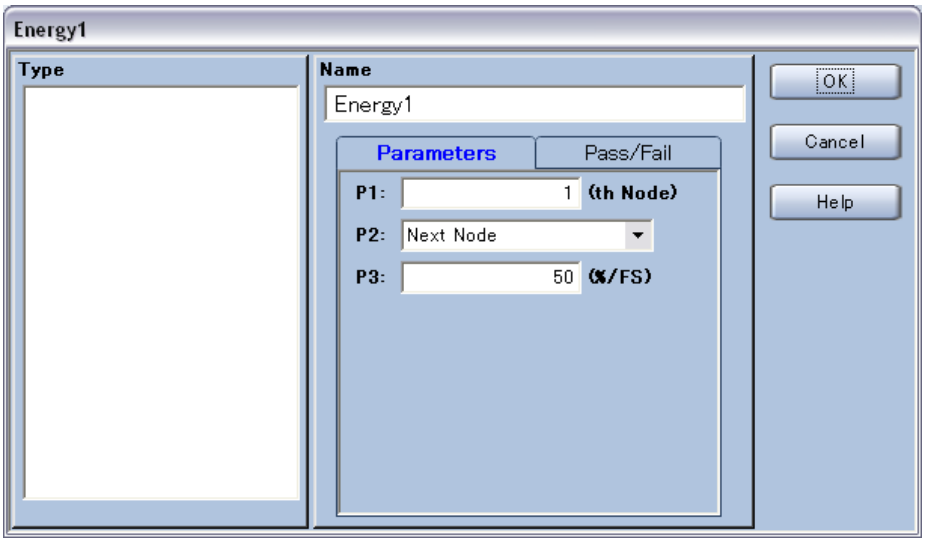
  Calculates only slope of elastic.

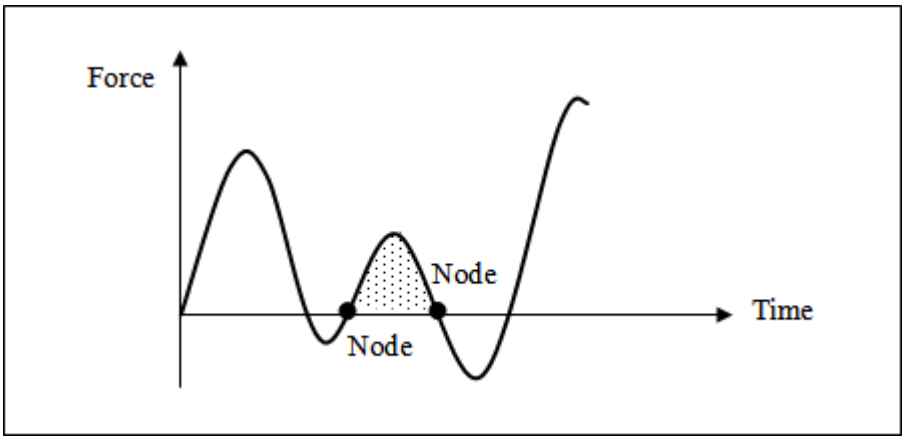
Energy - Texture Mode

Determines energy as the area formed by the graph between two specified points and the zero [Force] line. Up to five energy areas can be specified.

Parameters

- **Energy**





- Number of Intersections (P1): Starting node.

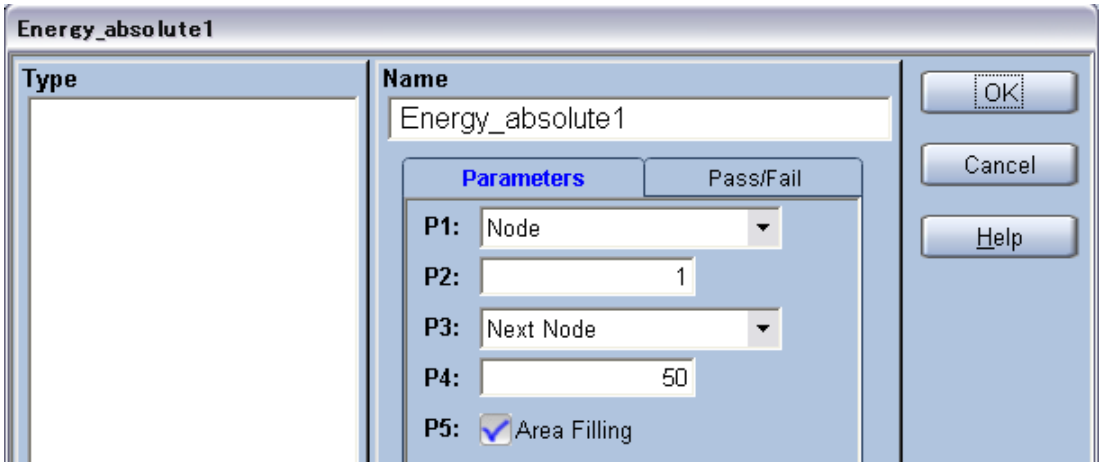
- End Point Parameter (P2): Ending point parameter is selected from the following.

Next Node	Next Peak	Next Valley
Next Max.	Next Min.	-

- Peak and Valley Detection Sensitivity (P3)

Specifies sensitivity to use for detecting [Next Peak] and [Next Valley]. Sensitivity is specified as a percent of full-scale range specified for decrease of [Force].

- **Energy_absolute**



- Start Point Parameter (P1): Select [Node], [Area] or [Auto].

  For P2 - P4 in [Auto] mode, refer to the later pages.

- Start Point (P2): Enter nuumber of intersections or control area.

- End Point Parameter (P3): Ending point parameter is selected from the following.

Next Node	Next Peak	Next Valley
Next Max.	Next Min.	Area

- End Point (P4)

  - [Next Peak], [Next Valley]

    Specifies sensitivity to use for detecting [Next Peak] and [Next Valley]. Sensitivity is specified as a percent of full-scale range specified for decrease of [Force].

  - [Area] Enter control area.

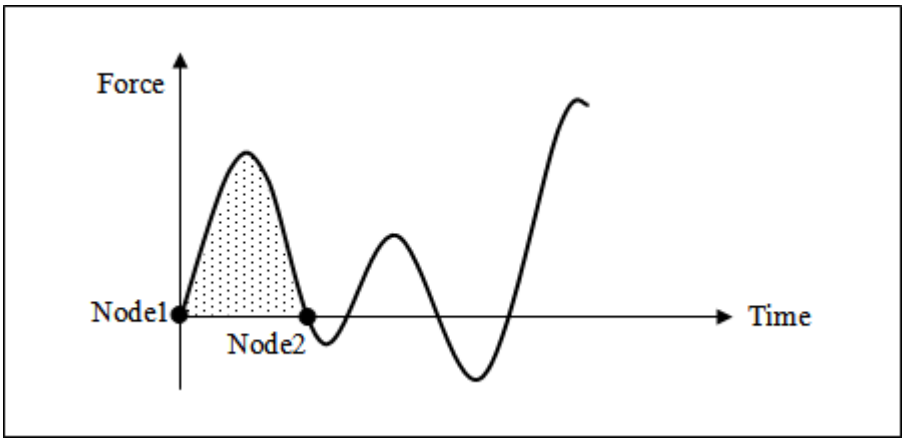
- Area Filling (P5): Specify the On / Off of the fill function of the calculation area on the chart.

Type

None

Example

Assume the first node P1 is at the first intersection, the next intersection is the next node P2, and P3 is not used.



Auto Mode

An area in which a compression plate does not attach to the sample can be removed by using [Auto] mode.

When you use [Auto] mode, register data processing items “Energy absolute N-1” previously. (Only when N >= 2)
“Energy absolute 1” is no need to register other data processing items.

In case of P1 [Auto], other parameters are following:

- Point Number (P2): Enter number of points.

- End Point Parameter (P3): Not use.

- End Point (P4): Enter the ratio (%) to load cell capcity (or range full scale).

Determine the 2 points (calculation start point and calculation end point) as following process. Determines energy as the area formed by the graph between these two specified points and the zero [Force] line

- In case of “Energy absolute 1”

  1. Find the maximum force point in the entire data.

  2. Search forward from max force point and find the point which satisfiy at least one of the following 3 rules. This point is calculation start point of “Energy absolute 1”.

     a) Point just before crossing line segment with test force 0.

     b) First point in the continueous (more than P2) of the lower force changing than P4 value.

     c) First point of the entire data.

  3. Search backward from max force point and find the point which satisfiy at least one of the following 3 rules. This point is calculation end point of “Energy absolute 1”.

     a) Point just before crossing line segment with test force 0.

     b) First point in the continueous (more than P2) of the lower force changing than P4 value.

     c) Last point of the entire data.

- In case of “Energy absolute N” (N >= 2)

  1. Find the maximum force point after the calculation end point of “Energy absolute N-1”.

  2. Search forward from max force point and find the point which satisfiy at least one of the following 3 rules. This point is calculation start point of “Energy absolute 1”.

     a) Point just before crossing line segment with force 0.

     b) First point in the continueous (more than P2) of the lower force changing than P4 value.

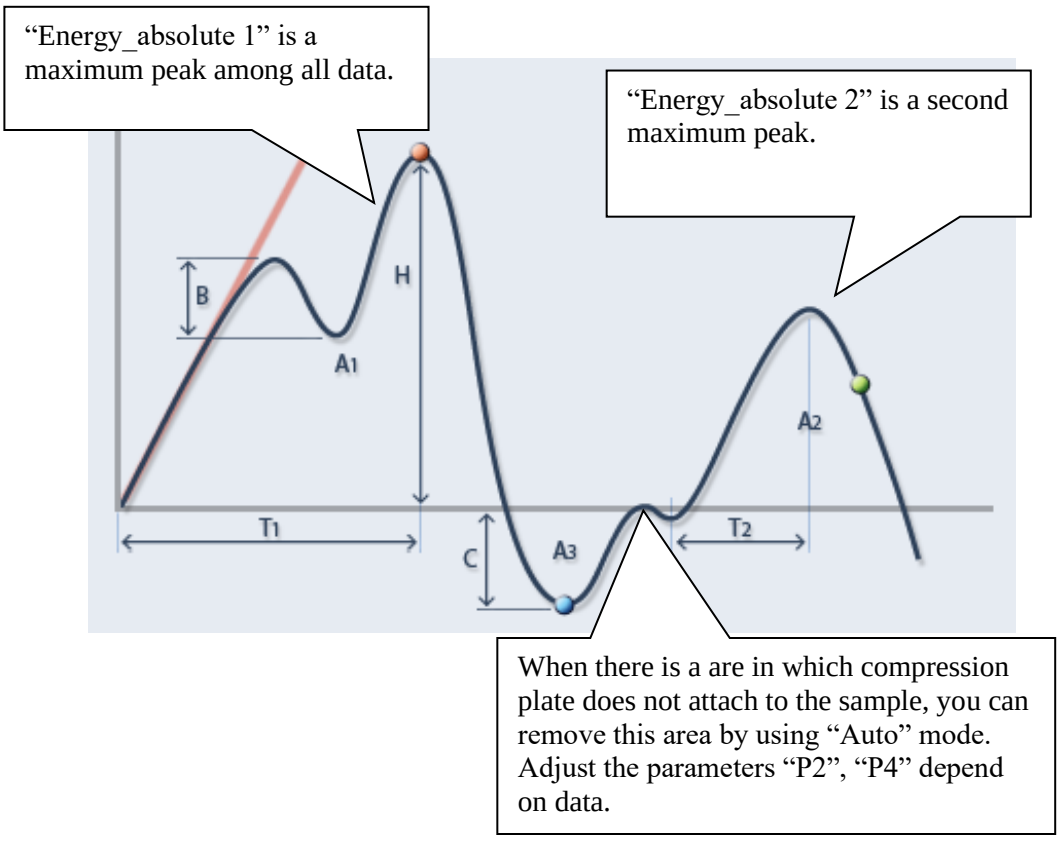
     c) Point just before the calculation end point of “Energy absolute N-1”.

  3. Search backward from max force point and find the point which satisfiy at least one of the following 3 rules. This point is calculation end point of “Energy absolute 1”.

     a) Point just before crossing line segment with force 0.

     b) First point in the continueous (more than P2) of the lower force changing than P4 value.

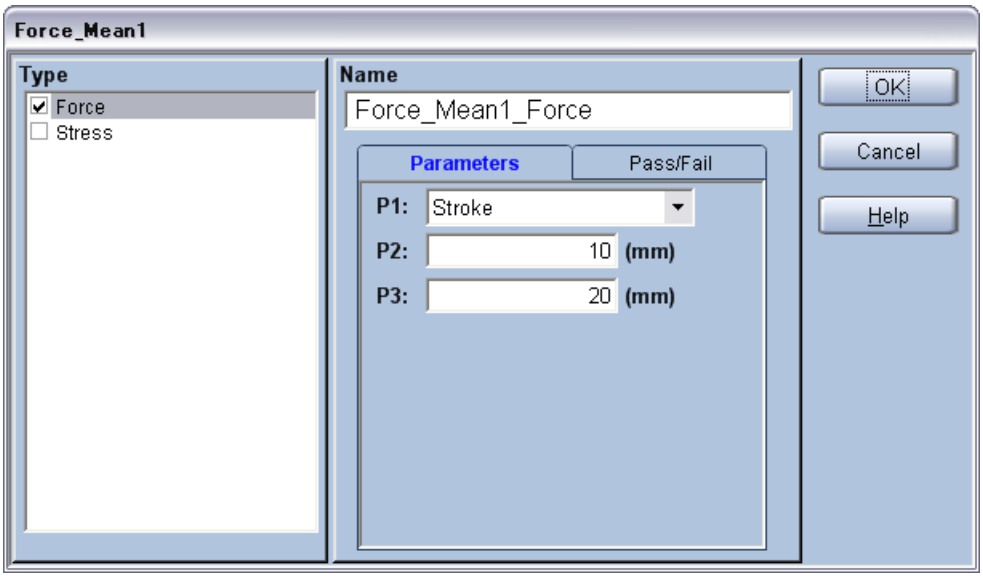
     c) Last point of the entire data.



Force Mean

This item calculates the mean value of force or stress in area specified by parameters.

Parameters



- Sensor Type (P1)

  Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

  Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Starting Point (P2)

- Ending Point (P3)

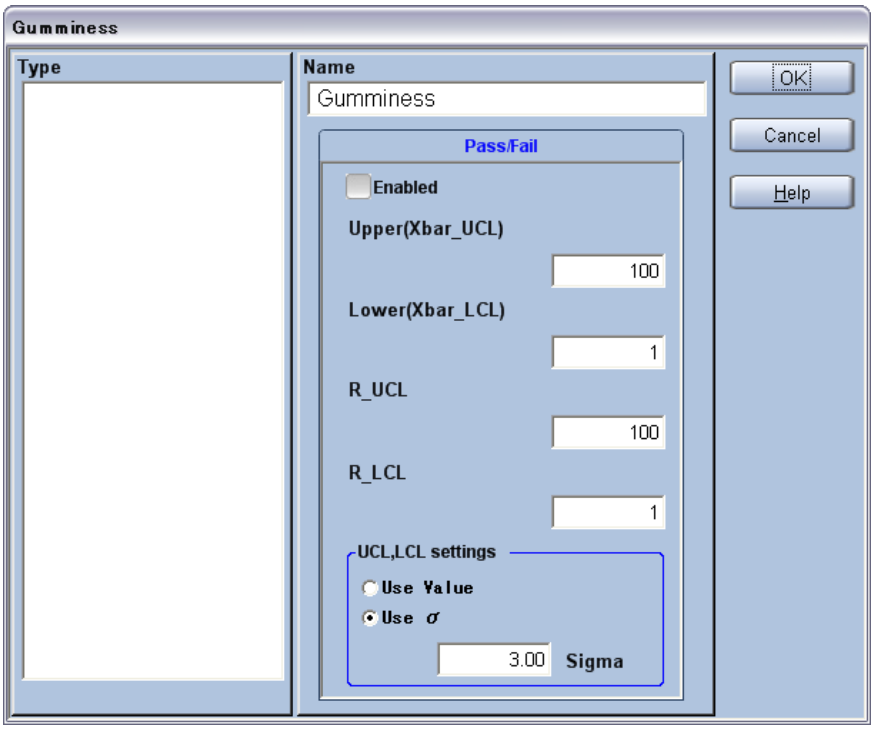
Gumminess

Guminess is expressed by following equation.

Guminess = Hardness * Cohesiveness

When you use this item, register data processing items “Hardness” and “Cohesiveness” previously.

Parameters



None

Type

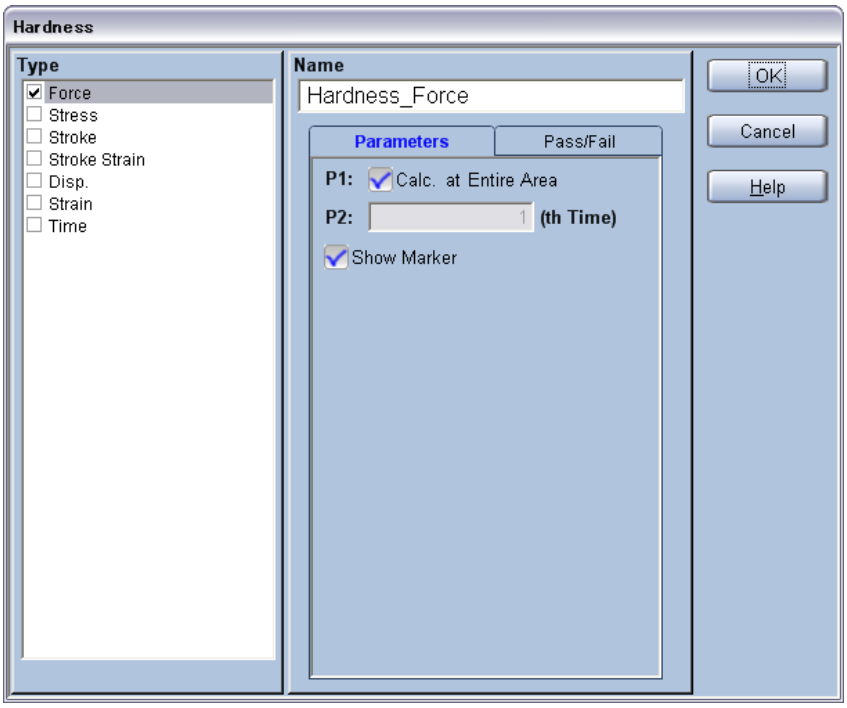
None

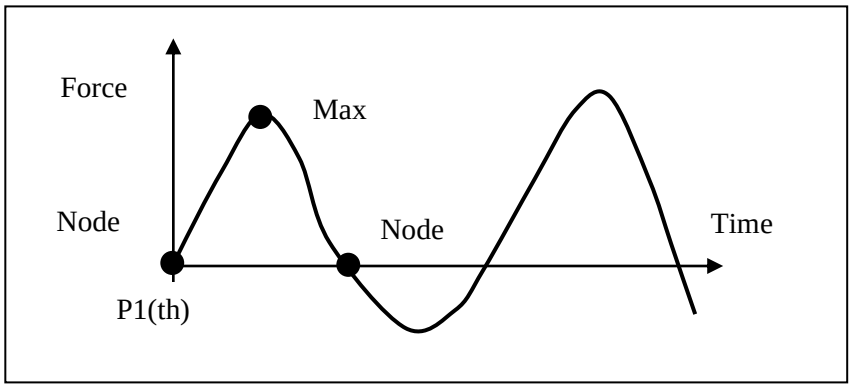
Hardness

Mark

Determines the maximum [Force] over a specified range or the entire area.

Parameters





- Specifies whether or not to search entire area (P1)
- Number of Intersections (P2):

  Data is process between the specified starting node and the next node.

Type

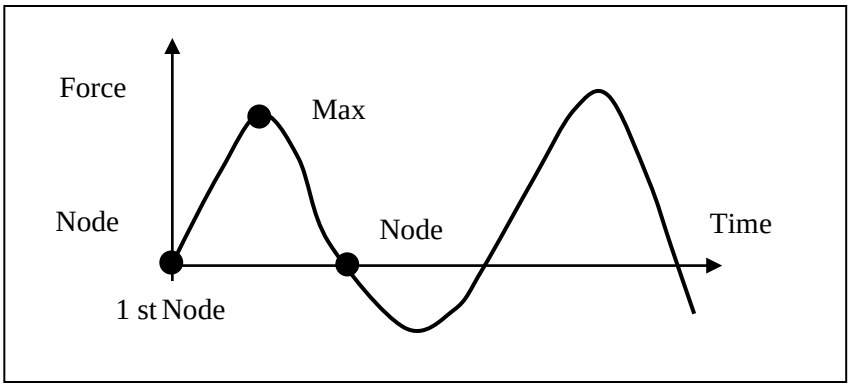
Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Example

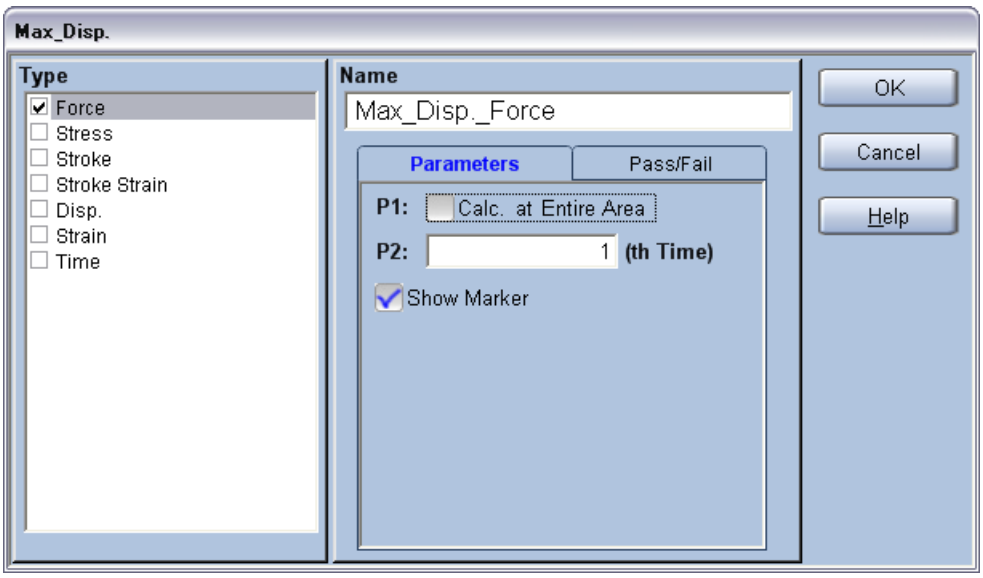
Assume [Calc. at Entire Areas] is not selected for P1 and P2 is set to 1 (first intersection).



Max Disp.

This item calculates the max value of displacement in test.

Parameters



- Specifies whether or not to search entire area (P1)

- Number of Intersections (P2):

  Data is process between the specified starting node and the next node.

Type

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

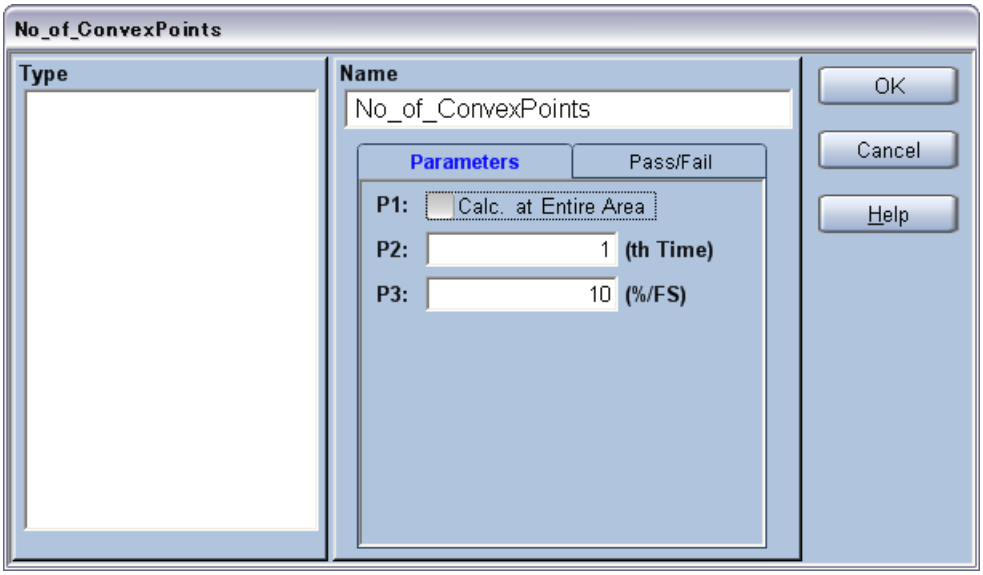
Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

No. of convex points

This item counts the convex points in area specified by parameters.

Parameters



- Specifies whether or not to search entire area (P1)

- Number of Intersections (P2):

  Data is process between the specified starting node and the next node.

- Peak and Valley Detection Sensitivity (P3)

  Specifies sensitivity to use for detecting [Next Peak] and [Next Valley]. Sensitivity is specified as a percent of full-scale range specified for decrease of [Force].

Type

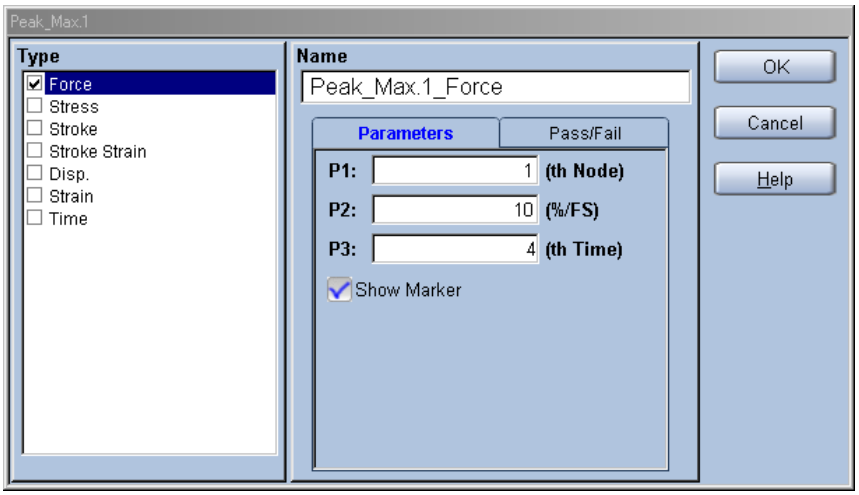
None

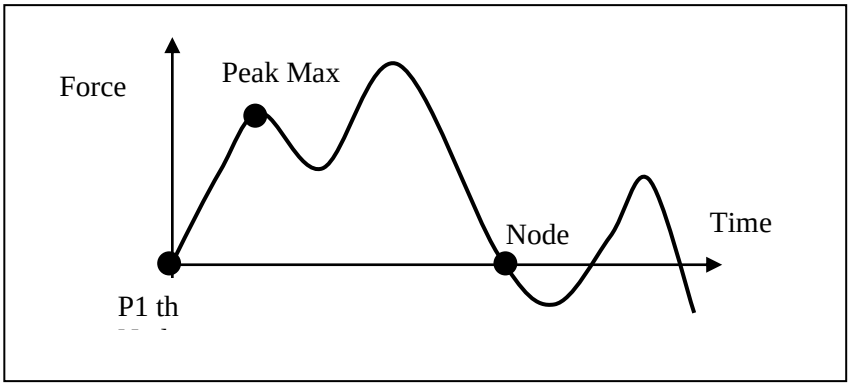
PeakMax(x) - Texture Mode

Mark

Determines the peak value between the specified node and the next node.

Parameters





- Number of Intersections (P1): Starting node.

- Peak Detection Sensitivity (P2):

  Specify the sensitivity to use for detecting peaks. Sensitivity is specified as a percent of full-scale range specified for decrease of [Force].

- “x” th largest Peak (P3):

  Specify the largest number of the peak. “1” means the largest peak. “4” meand 4th largest.

Type

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

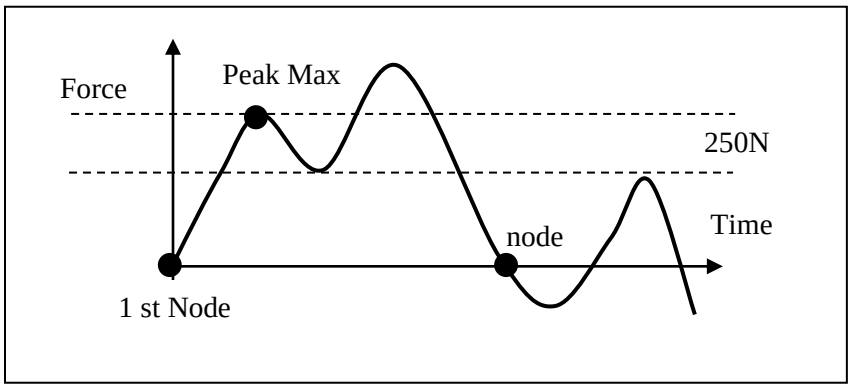
Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Example

Assume the first node P1 is at the first intersection, the detection sensitivity P2 is 5 (%FS), and P3 is 2nd and the load cell capacity (full scale) is 5 kN.

$$\text{Peak Detection Sensitivit y} = 5\text{kN} \times 5\% = 250\text{N}$$



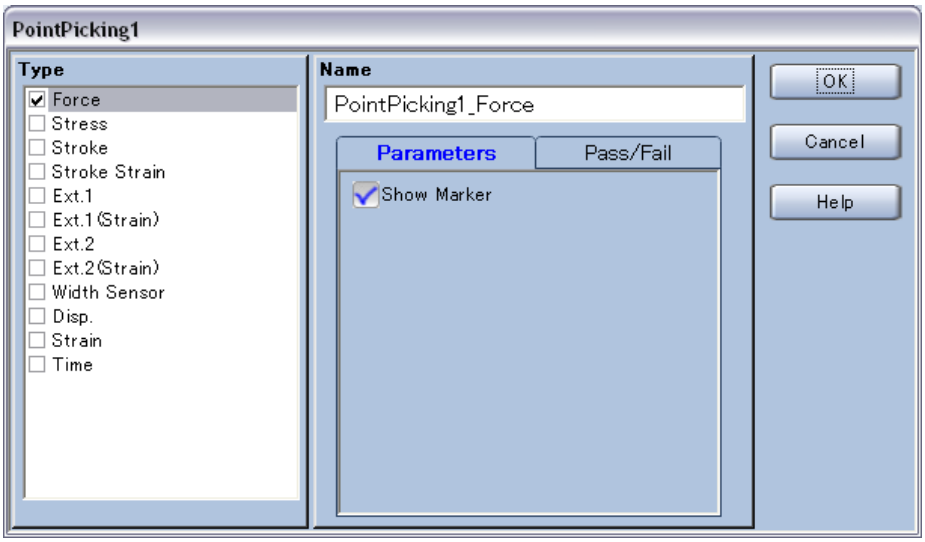
Point Picking - Texture Mode

Mark

Determines post-test information for any point on the graph specified with the mouse pointer.

Up to 20 points can be specified.

Parameters



None

Type

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

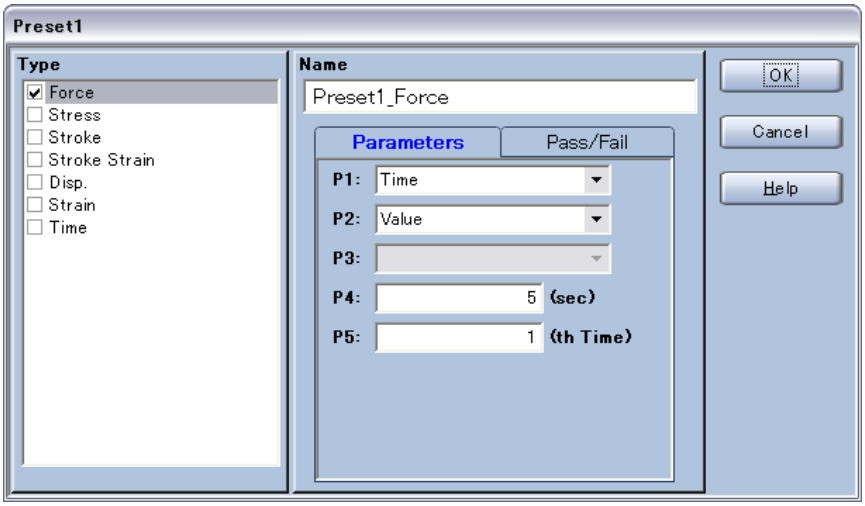
Default settings allow selecting from the following.

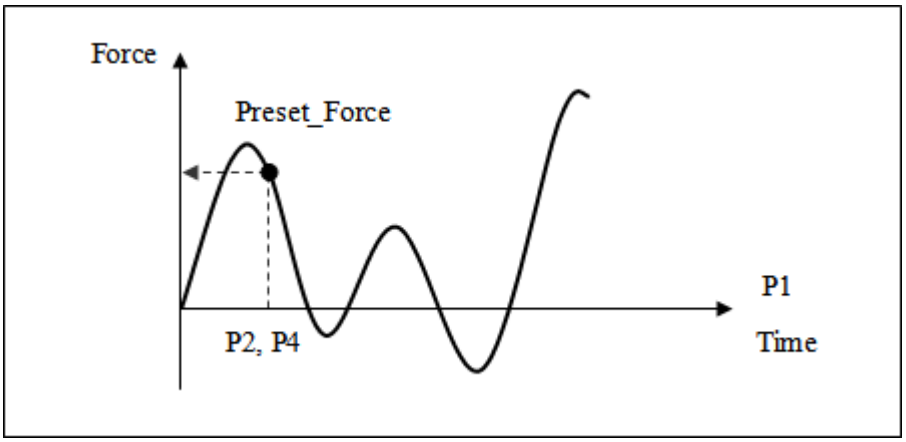
Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Preset - Texture Mode

Finds the specific points corresponding to a specified parameter. Up to 10 preset points can be specified.

Parameters





- Sensor Type (P1)

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

- Method for Specifying Parameter (P2): Select either [Formula] or [Value].

- Formula (P3): Enabled to specify only if [Formula] is specified in the [P2] box.

- Value (P4): Enabled to specify only if [Value] is specified in the [P2] box.

- Number of Preset Points (P5):

  When multiple points are specified, specify the number of points from start of test.

* Parameter values are in terms of units specified in the [System] window.

Type

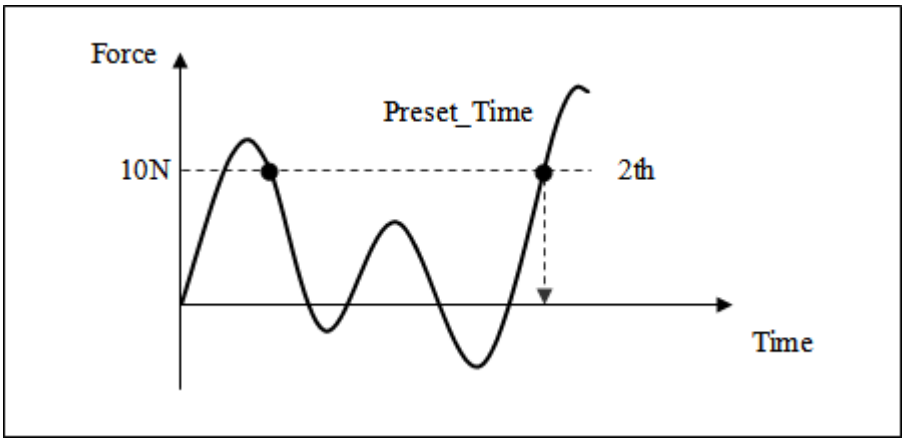
Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Example

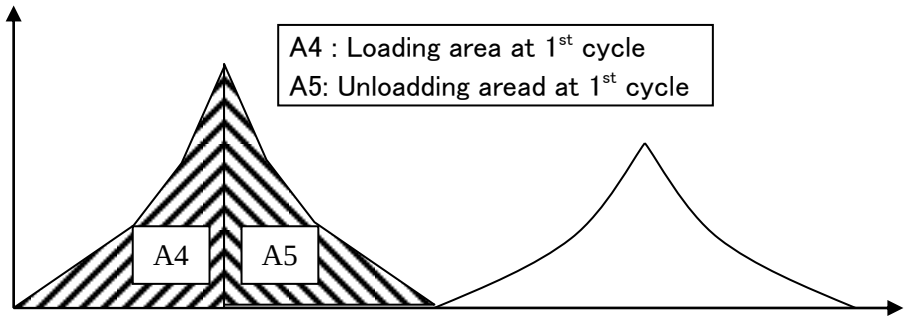
Assume the sensor type P1 is set to [Force], the specified parameter type P2 is [Value], the value P4 is 10 N, and P5 specifies that the second preset (time) is determined.



Resilience

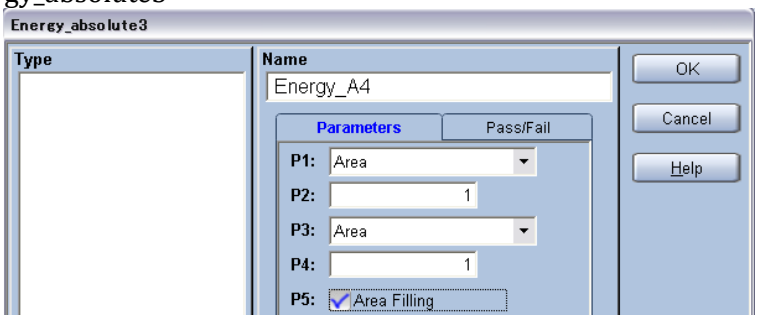
Resilience is expressed by the following calculating fomula.

Resilience = A5 / A4



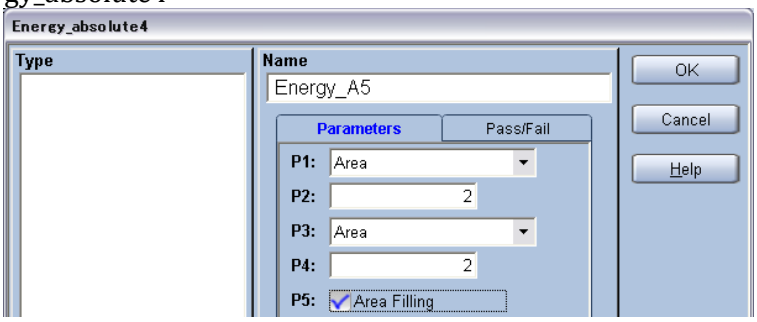
When you use this item, register data processing items “Energy_abusolute3” and “Energy_abusolute4”. These items parameter is following.

Energy_absolute3



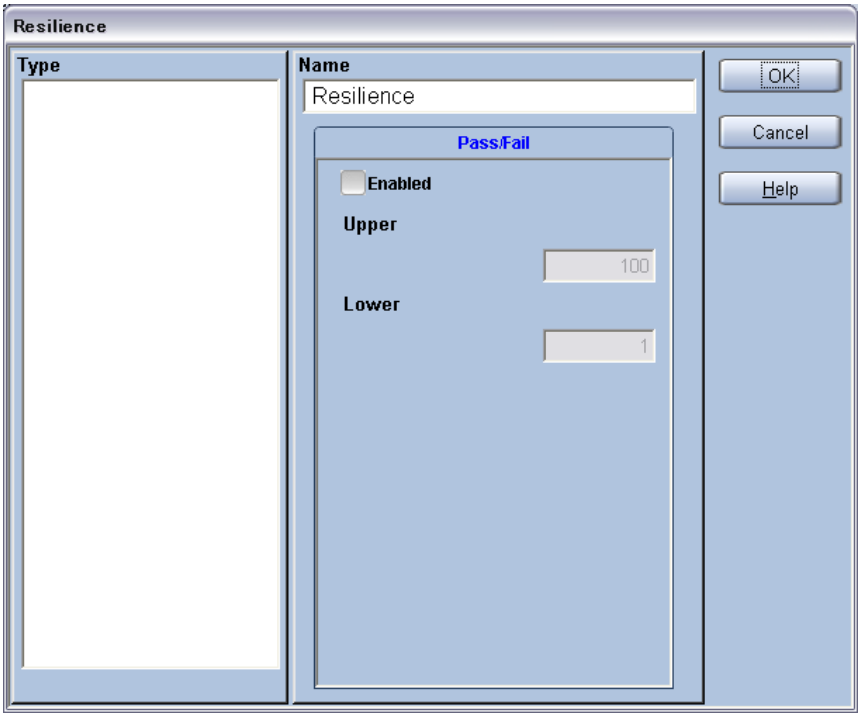
P1 : Area
P2 : 1
P3 : Area
P4 : 1

Energy_absolute4



P1 : Area
P2 : 2
P3 : Area
P4 : 2

Parameters



None

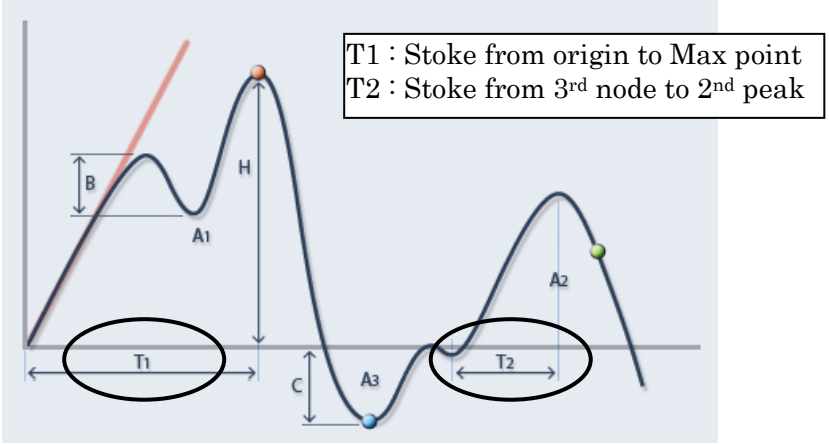
Type

None

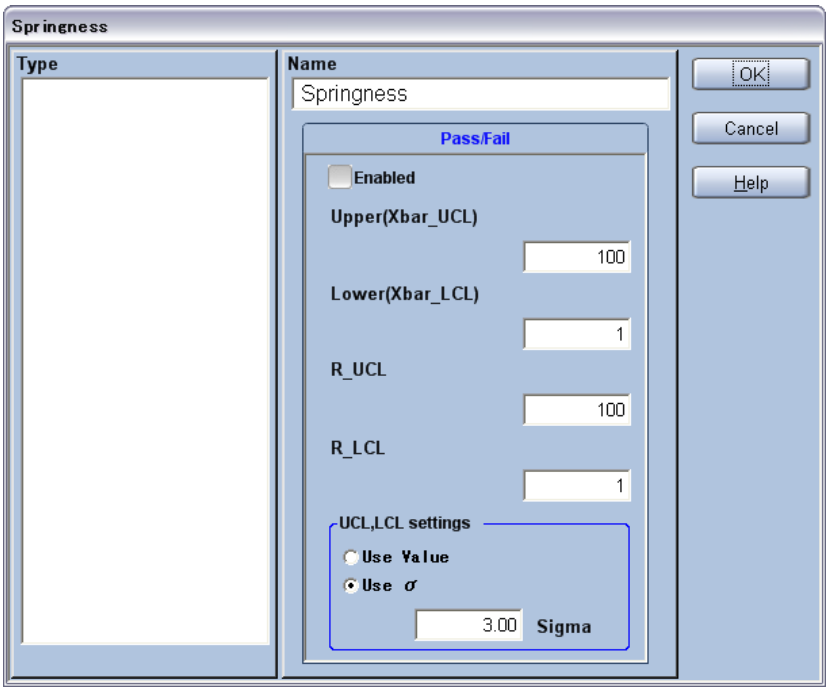
Springness

Springness is expressed by the following formula.

Springness = T2 / T1



Parameters



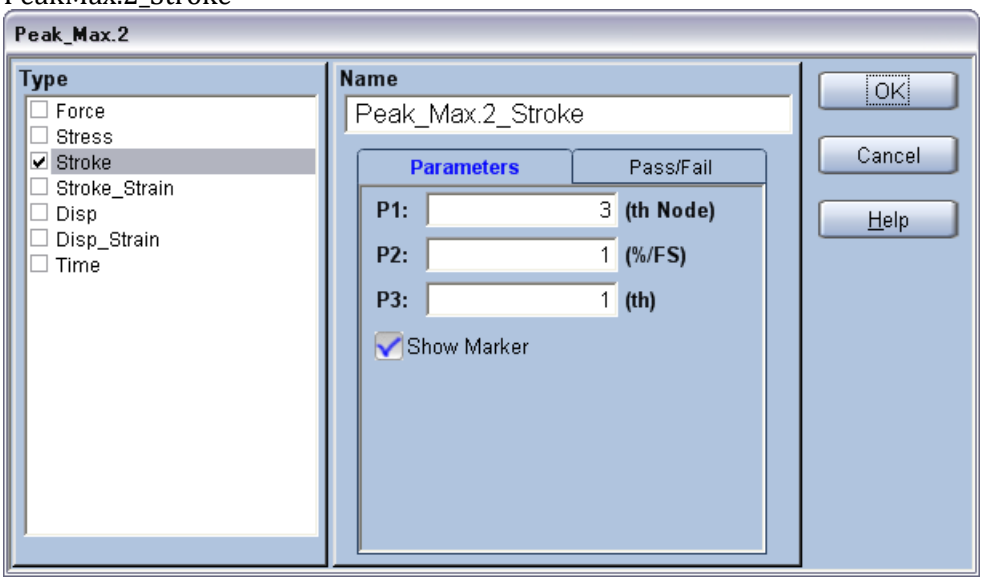
None

Type

None

When you use this item, register data processing item “Peak_Max2_Stroke” previously and set the parameter as following.

PeakMax.2_Stroke



P1:3(th Node)

P2:1(%/FS)

P3:1(th)

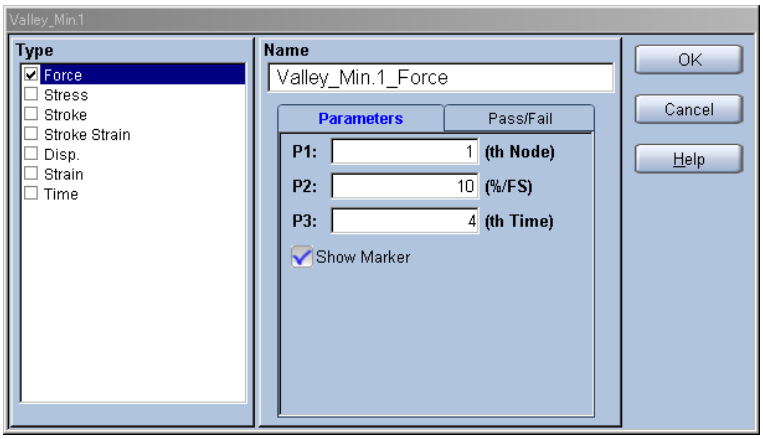
Valley Min(x) - Texture Mode

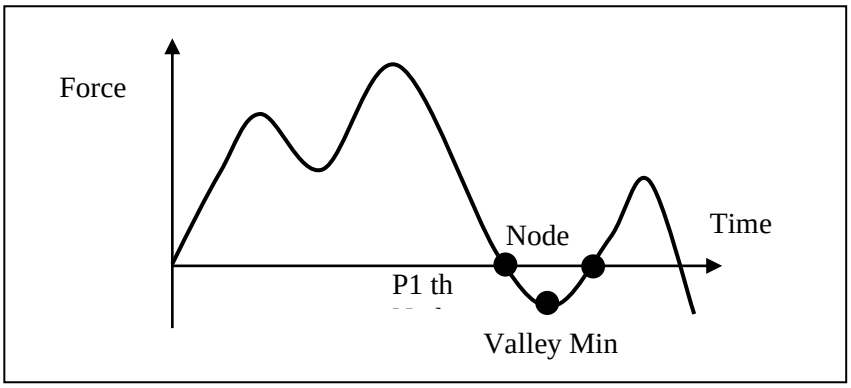
Mark

Determines the minimum valley between the specified node and the next node.

Up to 9 of the [Valley Min] can be ranked, where the minimum valley is ValleyMin1, the next minimum valley is ValleyMin2, the next minimum valley is ValleyMin3, and so on.

Parameters





- Number of Intersections (P1): Starting node.

- Valley Detection Sensitivity (P2):

  Specify the sensitivity to use for detecting valleys. Sensitivity is specified as a percent of full-scale range specified for decrease of [Force].

- “x” th smallest Valley (P3):

  Specify the smallest number of the valley. “1” means the smallest valley. “4” means 4th smallest.

Type

Selected from types specified via the [Channel] settings in the [Sensor] dialog box.

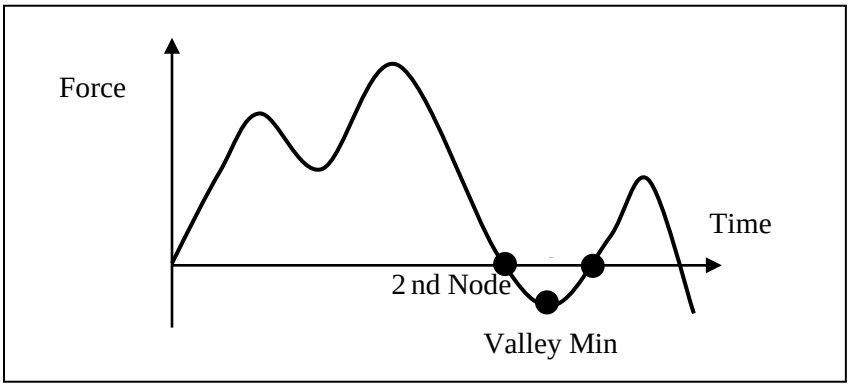
Default settings allow selecting from the following.

Force	Stress	Stroke
Stroke (strain)	Disp.	Disp. (strain)
Time	-	-

Example

Assume the first node P1 is at the 2nd intersection, the detection sensitivity P2 is 5 (%FS), P3 is 1 (th Time) and the load cell capacity (full scale) is 5 kN.

$$\text{Valley Detection Sensitivit y} = 5000\text{N} \times 5\% = 250\text{N}$$



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Chapter 5
Statistical Processing Items

This chapter describes statistical processing Items.

- Mean Value

Determines the mean value for respective data processing results.

- Standard Deviation

Determines the standard deviation for respective data processing results.

$$Std-D=\sqrt{\frac{\sum(x_i-\bar{x})^2}{N-1}}$$

x_i: Individual Data Values

$\bar{x}$: Mean Value of Data($\sum x_i / N$)

N: Number of Data Values

- Max. Value

Determines the maximum value in respective data processing results.

- Min. Value

Determines the minimum value in respective data processing results.

- Range

Determines the difference between minimum and maximum values in respective data processing results.

- Median

Determines the median value for respective data processing results.

Example:

Assuming data processing results in 5 data values (an odd number), listed in order of magnitude,

Median: Third value in list.

Assuming data processing results in 6 data values (an even number), listed in order of magnitude,

Median: Mean value of third and fourth value.

- Coefficient of Variation

Determines the coefficient of variation ($\frac{\text{standard deviation}}{\text{mean value}}$) for respective data processing results.

- 3s

Determines the value of triple standard deviations, where s = one standard deviation.

- Mean + Ns

Determines value by adding “N” times “standard deviations” to the mean value. “N” is from 1 to 6.

- Mean -Ns

Determines value by subtracting “N” times “standard deviations” from the mean value. “N” is from 1 to 6.

Chapter 6
Calculating Stress and Strain

This chapter describes calculations commonly used in material testing applications.

Calculating Stress and Strain

- Tension

$$\text{Stress} = \frac{\text{Test Force}}{S_0}, \ \text{Strain} = \frac{\text{Displacement}}{\text{Gauge Length}} \times 100$$

- Compression

$$\text{Stress} = \frac{\text{Test Force}}{S_0}, \ \text{Strain} = \frac{\text{Displacement}}{\text{Height}} \times 100$$

Shape	Dimensions		Area
Flat Plate	Thickness Width	T W	$S_0 = TW$
Rod	Diameter	D	$S_0 = \frac{\pi}{4}\mathrm{D}^2$
Tube 1	Outside Diameter Inside Diameter	Ex.D In.D	$S_0 = \frac{\pi}{4}(Ex.D^2 - In.I^2)$
Tube 2	Outside Dia. Thickness	Ex.D T	$S_0 = \pi T(D-T)$
Tube 3	Outside Dia. Thickness Width	D T W	$S_0 = \frac{W}{4}(D^2 - W^2)^{1/2} + \frac{D^2}{4}\arcsin\frac{W}{D} - \frac{W}{4}\left[(D-2T)^2 - W^2\right]^{1/2} - (\frac{D-2T}{2})^2 \arcsin(\frac{W}{D-2T})$
Balance	Length Weight Density	L W D	$S_0 = \frac{W}{DL}$
Ring	Thickness Width	T W	$S_0 = 2TW$

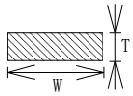
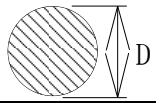
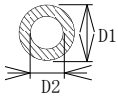
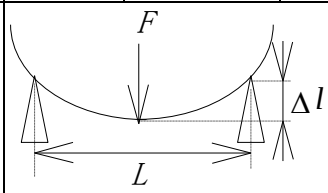
O-Ring	Thickness	T	$S_0 = \frac{\pi}{2}\mathrm{T}^2$
String/Fiber	Linear Density	D	$S_0 = D$
Hexagon	Length of 1 Face	X	$S_0 = \frac{3}{2}\sqrt{3}\mathrm{X}^2$
Cross-Section Area	Area	A	$S_0 = A$

- Peel Test (only possible in single mode)

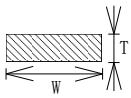
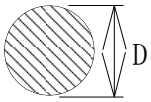
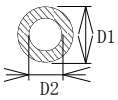
$$\text{Stress} = \frac{\text{Test Force}}{S_0}$$

Shape	Dimensions		Area
Plate 1	Thickness	T	$S_0 = T$
Plate 2	Width	W	$S_0 = W$

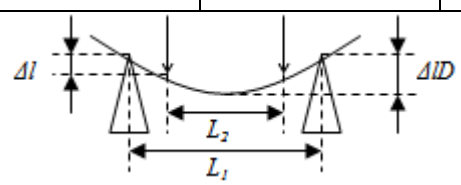
- 3-Point Bend

Shape and Input Parameters	Area Moment of Inertia	Section Modulus	Bending Moment M	Stress $\sigma = M / Z$	Strain ε	Elastic σ / ε
General (I: Area Moment of Inertia, Z: Section Modulus)	I	Z	$\frac{L}{4} \cdot F$	$\frac{L}{4Z} \cdot F$	$\frac{12}{L^2 \frac{Z}{I}} \cdot \Delta l$	$\frac{L^3}{48 \cdot I} \cdot \frac{F}{\Delta l}$
Flat Plate 	$\frac{WT^3}{12}$	$\frac{WT^2}{6}$	$\frac{LF}{4}$	$\frac{3}{2} \cdot \frac{LF}{WT^2}$	$\frac{6T}{L^2} \cdot \Delta l$	$\frac{L^3}{4WT^3} \cdot \frac{F}{\Delta l}$
Rod 	$\frac{\pi D^4}{64}$	$\frac{\pi D^3}{32}$	$\frac{LF}{4}$	$\frac{8LF}{\pi D^3}$	$\frac{6D}{L^2} \cdot \Delta l$	$\frac{4L^3}{3\pi D^4} \cdot \frac{F}{\Delta l}$
Tube 1 	$\frac{\pi}{64}(D_1^4 - D_2^4)$	$\frac{\pi}{32} \cdot \frac{(D_1^4 - D_2^4)}{D_1}$	$\frac{LF}{4}$	$\frac{8LD_1F}{\pi(D_1^4 - D_2^4)}$	$\frac{6D_1}{L^2} \cdot \Delta l$	$\frac{4L^3}{3\pi(D_1^4 - D_2^4)} \cdot \frac{F}{\Delta l}$
Note: Tube 2 requires entering thickness instead of ID, as in Tube 1.			 F : Force L : Distance Between Lower Supports Δl : Bending Deflection			

- 4-Point Bend

Shape and Input Parameters	Bending Moment	Stress $\sigma = M / Z$	Strain (**without** deflection gauge) ε	Strain (**with** deflection gauge) ε	Elastic σ / ε
General (I: Area Moment of Inertia, Z: Section Modulus)	$\frac{L_1 - L_2}{4} F$	$\frac{L_1 - L_2}{4Z} F$	$\frac{12(L_1 - L_2)\Delta l \cdot I}{(L_1^3 - 3L_1L_2^2 + 2L_2^3)Z}$	$\frac{24(L_1 - L_2)I\Delta lD}{(2L_1^3 - 3L_1L_2^2 + L_2^3)Z}$	$\frac{(L_1^3 - 3L_1L_2^2 + 2L_2^3) \cdot F}{48I\Delta l}$ (**without** deflectin gauge) $\frac{(2L_1^3 - 3L_1L_2^2 + L_2^3) \cdot F}{96I\Delta lD}$ (**with** deflectin gauge)
Flat Plate  	$\frac{L_1 - L_2}{4} F$	$\frac{3}{2} \cdot \frac{(L_1 - L_2)}{WT^2} F$	$\frac{6T(L_1 - L_2)\Delta l}{L_1^3 - 3L_1L_2^2 + 2L_2^3}$	$\frac{12T(L_1 - L_2)\Delta lD}{2L_1^3 - 3L_1L_2^2 + L_2^3}$	$\frac{(L_1^3 - 3L_1L_2^2 + 2L_2^3) \cdot F}{4WT^3\Delta l}$ (**without** deflectin gauge) $\frac{(2L_1^3 - 3L_1L_2^2 + L_2^3) \cdot F}{8WT^3\Delta lD}$ (**with** deflectin gauge)
Rod  	$\frac{L_1 - L_2}{4} F$	$\frac{8(L_1 - L_2)}{\pi D^3} F$	$\frac{6D(L_1 - L_2)\Delta l}{L_1^3 - 3L_1L_2^2 + 2L_2^3}$	$\frac{12D(L_1 - L_2)\Delta lD}{2L_1^3 - 3L_1L_2^2 + L_2^3}$	$\frac{4(L_1^3 - 3L_1L_2^2 + 2L_2^3) \cdot F}{3\pi D^4\Delta l}$ (**without** deflectin gauge) $\frac{2(2L_1^3 - 3L_1L_2^2 + L_2^3) \cdot F}{3\pi D^4\Delta lD}$ (**with** deflectin gauge)
Tube 1  	$\frac{L_1 - L_2}{4} F$	$\frac{8D_1(L_1 - L_2)}{\pi(D_1^4 - D_2^4)} F$	$\frac{6D_1(L_1 - L_2)\Delta l}{L_1^3 - 3L_1L_2^2 + 2L_2^3}$	$\frac{12D_1(L_1 - L_2)\Delta lD}{2L_1^3 - 3L_1L_2^2 + L_2^3}$	$\frac{4(L_1^3 - 3L_1L_2^2 + 2L_2^3) \cdot F}{3\pi(D_1^4 - D_2^4)\Delta l}$ (**without** deflectin gauge) $\frac{2(2L_1^3 - 3L_1L_2^2 + L_2^3) \cdot F}{3\pi(D_1^4 - D_2^4)\Delta lD}$ (**with** deflectin gauge)

Note:
Tube 2 requires entering thickness instead of ID, as in Tube 1.



F : Force
L_1 : Distance Between Lower Supports
L_2 : Distance Between Upper Supports
Δl : Bending Deflection
ΔlD : Used for Deflection Gauge

Chapter 7
Terminology

Terminology

This chapter explains terminology commonly used in material testing applications.

- Bending Test

Data is obtained by bending specimens. Bending tests include 3-point bending tests, which use two supports and applies a load in one location, and 4-point bending tests, which load the specimen in two locations.

- Break

In tensile tests, this is the point when the specimen is pulled apart and separates at the center of the specimen.

- Caliper

Used to measure specimen dimensions.

- Compression Test

Data is obtained by applying a compressive [Force] on a specimen.

- Disp.

Refers to the stroke distance on a testing machine or the amount the gauge length changes on a specimen.

If [Disp.] is output as results, it is calculated from the [Disp.] gauge reading, if used, or from the stroke distance if a gauge is not used. However, even if a [Disp.] gauge is used, [Disp.] can still be calculated from the stroke value by reanalyzing the data and specifying [Stroke Only] for the sensor.

- Break Elongation

Amount a specimen elongates from the start of test until it breaks.

- Extensometer

Used for detailed measurements of the gauge length on specimens. Many different types of displacement gauges are used, depending on the testing application, such as contact-type, non-contact type and many others.

- Elastic

Describes the stiffness of a specimen. It is especially important for metals and plastics. It is also referred to as “Young’s Modulus”.

- Energy

Indicates the energy level of a specimen.

- Fitted

Change in gauge length measured by refitting fractured parts of a specimen after a tensile test.

- Load Cell

Device for measuring the force applied to specimens. Load cells are fixed to the crosshead of the testing machine.

- Lower Yield Point

When specimens exceed their upper yield point, the [Force] may decrease temporarily before rising again. This dip in force is referred to as the lower yield point.

- Testing Machine Deflection Correction

If large loads are applied or jigs or the testing machine experience deflect, deflection correction calculates the anticipated deflection in advance and uses it to correct the results. It is not used when using a deflection gauge.

- Max Force

Maximum force applied during a test.

- Parameter

Collectively refers to parameters required for processing data or to control the testing machine.

- n-Value

The n-value is a strain hardening exponent for thin sheet materials and is calculated by dividing a given elongation between D1 and D2 into n intervals and using least squares analysis.

- Node

Intersection points between specified data processing area boundary lines and the force graph in peel tests.

- Peak

Peaks detected during peel tests. The mean value, maximum value, or other values can be determined for these peaks.

- Peel Test

The peel test peels one item from another, to which it is adhered.

- Poisson's

When a load is applied to a specimen, the [Poisson's] describes the ratio between the elastic of the specimen in the direction of load and the elastic in a direction perpendicular to the load.

- r-Value

The r-value (Lankford value) is a plastic strain ratio that describes how wide a specimen will be when elongated a specified distance P1. It is calculated using the following equation.

$$\text{r - Value} = \frac{\ln W_0 / W}{\ln L / L_0 - \ln W / W_0}$$

$W_0 = \text{Pre - test Specimen Width}$, $L_0 = \text{Gauge Length}$, and $L = L_0 + P1$

- Reduction

When a rod is pulled apart in a tensile test, reduction refers to the ratio between the cross-sectional area before and after breaking.

- Sampling Time Interval

The time interval between [Force], displacement or other data output from testing machines during tests. The sampling interval for model AG-Xplus/AG-X is 0.2, 1, 10 or 50 msec. For models AGS-X,EZ-X is 1.2, 5, 10 50 msec.For AG-IS, I , EZGraph, and MST-I models, the interval is 10 msec. For models AGS-J, AGS-H, or EZTest the interval is 50 msec. Ultra fast sampling (optional) is available on AG-IS, I, and EZGraph systems, which allows sampling at intervals of 1.25 msec or 5 msec. The sampling settings also allow configuring the sampling interval to other intermediate time increments as well.

- Strain

Ratio of pre-test gauge length to the changed post-test gauge length.

- Stress

Stress is the force applied to a specimen, per unit area.

- Stroke

Amount the specimen changes.

- Tensile Test

Data is obtained by applying a tensile [Force] to a specimen.

- Test

Using a testing machine to apply a load to a specimen.

- Upper Yield Point

The point where elastic deformation changes to plastic deformation.

- Width Sensor

Device used to measure the amount of specimen deformation in the direction perpendicular to the applied load. It is used to find values such as the r-value or [Poisson's].

- Yield Strength

A straight line having the average slope obtained for elastic and passing through the zero point for the [Force] curve is considered to be in the origin position for determining yield strength. When this line is moved horizontally a certain offset distance, the force indicated by the point where it intersects the force curve is called the yield strength. This offset is specified as a testing parameter in advance (as a percent of gauge length).

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